Causal Inference for Data Science

Lecture notes, Chapters 1–11

Based on the course of Sanghack Lee (Graduate School of Data Science, Seoul National University)

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Chapter 1. Introduction and Causal Models

1.1 Why causal inference

Motivation. Classical ML and statistics learn aspects of a *fixed* data-generating reality, which is not enough for questions about actions and unrealized worlds.

1.1.1 A brief review of machine learning

Motivation. What standard learning paradigms estimate shows what they *cannot* estimate.

1.1.1.1 Why machine learning?

Intelligent behavior needs a model of the environment; writing it down is often infeasible, so it is *learned from data through algorithms*.

Learning modifies an agent's decision mechanism to improve performance and adapt to unexpected situations. Three questions recur:

- How can the learning process be characterized?
- How should prior knowledge be traded off against performance?
- How robust and generalizable is the resulting system?

Applications: generative AI, AI for science (protein folding, drug discovery), medicine, sustainability, vision, autonomous systems and security.

1.1.1.2 The classical learning paradigm

An unknown true model M generates feature vectors x; the target is a function $f_M(x)$. From $\{x^{(i)}\}_{i=1}^N$ one selects $h \in \mathcal{H}$ with $h(x) \approx f_M(x)$; the challenge is generalization.

Supervised learning maps $\mathcal{X}$ to $\mathcal{Y}$ from pairs $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$, mimicking an oracle $f : \mathcal{X} \to \mathcal{Y}$ (trees, k-NN, Bayesian networks, CNNs, Transformers, XGBoost).

Unsupervised learning sees only $x \in \mathcal{X}$ and seeks structure: clustering, dimensionality reduction, density modeling of $P(x)$ (e.g. a phylogenetic tree).

1.1.1.3 A probabilistic view

Both estimate features of one fixed joint distribution:

- supervised learning infers $P(y \mid x)$ (a network's output activation, a tree leaf);
- unsupervised learning infers properties of $P(x)$.

We learn to imitate aspects of a static reality; $P(x, y)$ says nothing about changing the mechanism producing x.

1.1.1.4 Reinforcement learning: intervention instead of observation

In RL the agent *intervenes* in the environment and learns from its responses (A/B testing, robotics), facing the exploration/exploitation trade-off. Missing: reasoning about actions from data *not* generated by our own experiments, and about events that did not happen.

1.1.2 Conceptual foundation: the ladder of causation

Motivation. Observing, intervening and imagining structure the whole course.

Layer 1. Observation (sensors): observe the world and infer the "next" stage.

Layer 2. Agency: intervene in the world and change it.

Layer 3. Retrospection: reflect on what we did (and did not do) and infer new behaviors.

This stratification is formalized as **Pearl's Causal Hierarchy (PCH)**.

1.1.2.1 Inference about worlds that are not realized

The classical pipeline is

$$\text{Real world / nature} \xrightarrow{\text{data}} \text{AI/ML, statistics (inference)} \longrightarrow \text{new knowledge.}$$

An unrealized *alternative reality* has no data, so the only route is

$$\text{knowledge} \longrightarrow \text{causal model} \xrightarrow{\text{causal inference}} \text{new knowledge about the alternative reality,}$$

with mechanism assumptions transporting information to unobserved worlds.

1.1.2.2 Pearl's Causal Hierarchy

Level (symbol)	Activity	Question	Examples / typical models
1. Association	Seeing	What is?	What does a symptom tell us about a disease? How would seeing X change my belief in Y?
$P(y \mid x)$			(Un)supervised ML: deep nets, Bayes nets, hierarchical models, decision trees.
2. Intervention	Doing	What if?	What if I take aspirin, will my headache be cured? What if I do X?
$P(y \mid \text{do}(x), c)$			Reinforcement learning, causal Bayesian networks, MDPs, POMDPs.
3. Counterfactual	Imagining,	Why?	Was it the aspirin that stopped my headache? What if I had acted differently?
$P(y_x \mid x', y')$	retrospection		Structural causal models.

Causal inference = cross-layer inference.

- *Layer 1 data, Layer 2 query:* will performance improve if class starts earlier, from observational records?
- *Layer 2 data, Layer 2 query:* predict a new experiment from previous ones (transportability).
- *Layers 1–2 data, Layer 3 query:* would I have won with door #2 instead of door #1?

Lower layers generally *underdetermine* higher ones (proved later).

1.1.2.3 Two motivations

- **Scientific:** understanding nature and humans with data, logic, mathematics and the scientific method.
- **Pragmatic** (economics, medicine, engineering): intelligent systems and policy design; policies are interventions (Layer 2).

1.1.3 Correlation versus causation

Motivation. Strong association is compatible with no causal relation at all.

- Active Facebook users and 10-year Greek bond yields rise together, 2005–2011 (BusinessWeek, Dec. 2011).
- Lemons imported from Mexico and US highway fatality rate, 1996–2000: $R^2 = 0.97$ (negative slope).
- Organic food sales and diagnosed autism, 1998–2007: $r = 0.9971$ ($p < 0.0001$).
- Maine divorce rate and per-capita margarine consumption (2000–2009).

These reflect common trends or coincidence (cf. xkcd no. 552). The real question: *if correlation does not imply causation, what does?* Answer: correlation *plus* causal assumptions encoded in a model.

1.1.4 Example 1: Simpson's paradox in a 2×2 table

Motivation. The same data support opposite conclusions under different causal stories.

$X/\neg X$: drug/no drug; $Y/\neg Y$: recovered/not; $F/\neg F$: female/male (Simpson 1951; Pearl 2000, Ch. 6).

All	Y	$\neg Y$	total	rate
drug (X)	20	20	40	50%
no drug ($\neg X$)	16	24	40	40%
Σ	36	44	80	

Male ($\neg F$)	Y	$\neg Y$	total	rate
drug	18	12	30	60%
no drug	7	3	10	70%
Σ	25	15	40	

Female (F)	Y	$\neg Y$	total	rate
drug	2	8	10	20%
no drug	9	21	30	30%
Σ	11	29	40	

Within each sex the drug is worse,

$$P(Y \mid F, X) = 0.2 < 0.3 = P(Y \mid F, \neg X),$$
$$P(Y \mid \neg F, X) = 0.6 < 0.7 = P(Y \mid \neg F, \neg X),$$

yet overall better: $P(Y \mid X) = 0.5 > 0.4 = P(Y \mid \neg X)$.

Why this is arithmetically possible. By total probability,

$$P(Y \mid X) = P(Y \mid X, \neg F)P(\neg F \mid X) + P(Y \mid X, F)P(F \mid X),$$

with $P(F \mid X) = 10/40 = 0.25$, $P(F \mid \neg X) = 30/40 = 0.75$:

$$P(Y \mid X) = 0.75 \cdot 0.6 + 0.25 \cdot 0.2 = 0.50,$$
$$P(Y \mid \neg X) = 0.25 \cdot 0.7 + 0.75 \cdot 0.3 = 0.40.$$

Different weights: men, who recover more often anyway, are over-represented among drug takers.

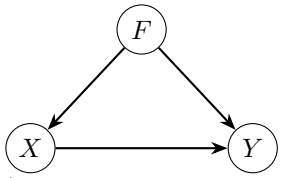
Which table should the physician consult? If sex affects both treatment and recovery, weight the strata by the *population* $P(F) = P(\neg F) = 1/2$:

$$\sum_f P(Y \mid X, f)P(f) = \tfrac{1}{2}(0.6) + \tfrac{1}{2}(0.2) = 0.40,$$
$$\sum_f P(Y \mid \neg X, f)P(f) = \tfrac{1}{2}(0.7) + \tfrac{1}{2}(0.3) = 0.50,$$

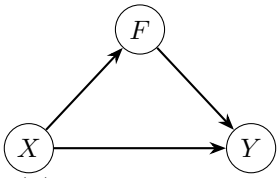
so the drug lowers recovery by 10 points (back-door adjustment, Chapter 2).

1.1.4.1 Would the result be different with a different story?

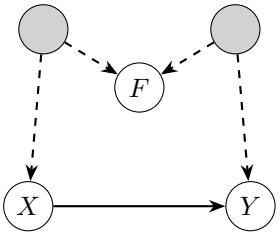
Same numbers, three graphs:



(a) F is a common cause



(b) F is a mediator



(c) F is a collider of latents

- **(a)** $F \to X$, $F \to Y$, $X \to Y$. F confounds: use the stratified tables; the drug is harmful.
- **(b)** $X \to F \to Y$, $X \to Y$. If the third variable is changed by the drug (e.g. blood pressure), stratifying blocks part of the effect; the aggregate table gives the total effect: beneficial.
- **(c)** $X \leftarrow U_1 \to F \leftarrow U_2 \to Y$, $X \to Y$, U_1, U_2 latent. Conditioning on the collider F *creates* spurious association; the aggregate table is correct.

Identical data, different answers: the decision needs the causal structure, outside the data.

1.1.5 Example 2: a "trivial" data analysis

Motivation. A perfect fit does not reveal the direction or meaning of a dependence.

Firefighters (X)	Size of fire (Y)
10	100
40	400
60	600
200	2000

$Y = 10X$ exactly, so $\beta = 10$, $\beta_0 = 0$. Read causally, each firefighter adds ten units of fire; in fact $Y \to X$. *What is the meaning of β?* A property of the joint distribution; whether it measures the effect of changing X depends on the mechanism.

1.1.6 What is going on here?

Both examples ask for the effect of an action, which is not in the joint distribution. This needs a formal language for causal assumptions (i), semantics linking assumptions, data and queries (ii), and inference rules for computability (iii). Chapters 1–3 build it: SCMs, causal graphs, identification, back-door and adjustment criteria, do-calculus, algorithmic identification.

1.2 Causal models and causal graphs

Motivation. Which object can serve as an "oracle" for causal questions, and what does it imply about observed data?

1.2.1 Reviews: notation, probability, algebra

Motivation. Notation and elementary manipulations used throughout.

1.2.1.1 Notation

- Uppercase Z: random variable with state space $\mathfrak{X}_Z$; lowercase $z \in \mathfrak{X}_Z$: a value.
- Bold $\mathbf{V}$, $\mathbf{v}$: sets of variables/values; calligraphic $\mathcal{G}$, $\mathcal{M}$: graphs, models.
- $\forall$, $\exists$; $\cup, \cap$; $\subseteq, \subset$ ($\subset$ is ambiguous); $\vee$, $\wedge$, $\oplus$ (xor); $\sum$, $\prod$; $\langle \cdots \rangle$ tuples, $\{\cdots\}$ sets, $\mid$ "given".
- "iff" = if and only if. $p \Rightarrow q$: p is *sufficient* for q; equivalently $\neg q \Rightarrow \neg p$: q is *necessary* for p.

1.2.1.2 Probability

For $P(A, B, C, D, E, F)$ write $P(a, b)$ for $P(A = a, B = b)$; variables are discrete WLOG.

Marginalization.

$$\sum_{c,e} P(a, b, c, d, e, f) = \sum_{(c,e) \in \mathfrak{X}_C \times \mathfrak{X}_E} P(a, b, c, d, e, f) = P(a, b, d, f).$$

Conditioning.

$$P(a, b, c \mid d, e) = \frac{P(a, b, c, d, e)}{P(d, e)} = \frac{\sum_f P(a, b, c, d, e, f)}{\sum_{a,b,c} \sum_f P(a, b, c, d, e, f)}.$$

Chain rule (in *any* order):

$$P(a, b, d, e) = P(a \mid b, d, e) P(b \mid d, e) P(d \mid e) P(e) = P(b \mid a, d, e) P(d \mid a, e) P(e \mid a) P(a).$$

Axioms. $0 \leq P(\cdot) \leq 1$, $\sum_{a,b,c} P(a, b, c) = 1$; often $P > 0$ so all conditionals exist.

1.2.1.3 Sums of products

Factors free of a summation index move outside the sum:

$$\sum_{a,c} f(a, b) g(b, c) h(c, d) = \sum_c h(c, d) g(b, c) \Big(\sum_a f(a, b) \Big) = \sum_a f(a, b) \Big(\sum_c g(b, c) h(c, d) \Big).$$

If $f(a, b) = P(a \mid b)$,

$$\sum_a P(a \mid b) \underbrace{\sum_c g(b, c) h(c, d)}_{\text{irrelevant to } a} = \underbrace{\Big(\sum_a P(a \mid b) \Big)}_{=1} \sum_c g(b, c) h(c, d) = \sum_c g(b, c) h(c, d),$$

the "sum out a leaf" step used constantly in identification proofs.

1.2.2 Why data alone is not enough: a prima facie example

Motivation. Every learner trained on observational data can recommend the wrong policy.

1.2.2.1 The story

A drug is *believed* to help against a contagious disease. Unknown to physicians:

- the rich always survive;
- carriers of a protective gene need no treatment, but if treated develop an allergy that is fatal under poor living conditions.

Wealth and gene are independent, $P(\text{rich}) = P(\text{poor}) = \frac{1}{2}$, and half the poor carry the gene:

$$\Pi = \text{rich} \ \cup \ \text{poor}_1 \ \cup \ \text{poor}_2,$$

$$\begin{cases} \text{rich}: & \text{alive anyway}, \\ \text{poor}_1 \ (\text{no gene}): & \text{die anyway}, \\ \text{poor}_2 \ (\text{gene}): & \text{die iff they take the drug}, \end{cases}$$

with $P(\text{poor}_1) = P(\text{poor}_2) = \frac{1}{4}$. (*Prima facie* = at first sight.)

1.2.2.2 Current protocol

The expensive drug goes to the rich only. All treated survive; among the untreated, poor_2 survive and poor_1 die:

$$P(\text{alive} \mid \text{drug}) = 1, \qquad P(\text{alive} \mid \neg\text{drug}) = 0.5.$$

Should the city give the drug to everyone for free?

1.2.2.3 Hypothetical protocols

Group (mass)	Current (drug iff rich)	Drug for all	Drug for none
rich (1/2)	alive	alive	alive
poor_1 (1/4)	dead	dead	dead
poor_2 (1/4)	alive	dead	alive
$P(\text{alive})$	3/4	1/2	3/4

Drug for all *reduces* survival from 3/4 to 1/2; the drug never helps, and the apparent benefit comes from treating only the rich.

1.2.2.4 What can a learning system see?

Rows (patient, R, D, A), e.g. (1, yes, yes, yes), (2, no, no, yes), (3, no, no, no), (4, yes, yes, yes),... At best any algorithm learns

R	D	A	$P(r, d, a)$
0	0	0	1/4
0	0	1	1/4
1	1	1	1/2
others			0

Unseen: $P(\text{gene})$, the other causes of wealth, and how the gene interacts with wealth, drug and reaction.

What is missing? In a second world the drug is inert, $A \leftarrow R \vee U_G$. Since $D = R$, it yields the same table, but drug for all gives 3/4. No algorithm using only $P(r, d, a)$ is right in both worlds; data must be supplemented by a model of how they are generated.

1.2.2.5 From statistical to causal analysis

- **Traditional statistics/ML:** data $\to P \to Q(P)$, e.g. $Q = P(B \mid A)$ for co-purchases.
- **Causal analysis:** data $\to P \xrightarrow{\text{change}} P' \to Q(P')$, e.g. sales if the price doubles, cancer if smoking is banned. P' is never observed; a formalism must represent both.
- **SCM paradigm:** data $\leftarrow P \leftarrow \mathcal{M} \to Q(\mathcal{M})$, $\mathcal{M}$ an invariant mechanism by which Nature assigns values. *P is a model of data; $\mathcal{M}$ is a model of reality.*

This refines the picture of Section 1.1 (with a grain of salt).

1.2.2.6 Modeling reality in the example

Endogenous $\mathbf{V}$: R (rich), D (drug), A (alive). Exogenous $\mathbf{U}$: U_G (gene), U_R (other causes of wealth). Mechanisms:

$$R \leftarrow U_R, \qquad D \leftarrow R, \qquad A \leftarrow R \vee (U_G \wedge \neg D),$$

$P(U_G = 1) = P(U_R = 1) = \frac{1}{2}$, $P(\mathbf{U}) = P(U_G)P(U_R)$. This fully specified model implies P and every P': the *structural causal model*.

1.2.3 Structural causal models

Motivation. Define the object representing reality and its three basic properties.

Definition (Structural causal model; Pearl 2000)

An SCM $\mathcal{M}$ is a 4-tuple $\langle \mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}) \rangle$ where

- $\mathbf{V} = \{V_1, \ldots, V_n\}$ are endogenous variables;
- $\mathbf{U}$ is a set of exogenous variables;
- $\mathcal{F} = \{f_1, \ldots, f_n\}$ are functions determining $\mathbf{V}$, $v_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i)$, with $Pa_i \subseteq \mathbf{V} \setminus \{V_i\}$ and $\mathbf{U}_i \subseteq \mathbf{U}$;
- $P(\mathbf{U})$ is a distribution over $\mathbf{U}$.

"$\leftarrow$" marks an assignment, not a symmetric equation (axiomatizations: Galles and Pearl 1998, Halpern 1998). $\mathbf{U}$ carries all randomness; $\mathcal{F}$ maps a unit $\mathbf{u}$ to its observable profile.

1.2.3.1 Property 1: an SCM induces the observational distribution

$\mathcal{F} : \mathbf{u} \mapsto (v_1, \ldots, v_n)$, so random $\mathbf{U}$ induces $P(\mathbf{v})$, Layer 1 of the PCH; each unit is an instantiation $\mathbf{U} = \mathbf{u}$.

Definition (Induced distribution)

$\mathcal{M}$ defines $P^{\mathcal{M}}(\mathbf{V})$ by, for each $\mathbf{Y} \subseteq \mathbf{V}$,

$$P^{\mathcal{M}}(\mathbf{y}) = \sum_{\{\mathbf{u}\,:\,\mathbf{Y}(\mathbf{u})=\mathbf{y}\}} P(\mathbf{u}),$$

where $\mathbf{Y}(\mathbf{u})$, the *potential response*, is the value of $\mathbf{Y}$ obtained by solving $\mathcal{F}$ at $\mathbf{u}$.

Example 1.1

In the drug example:

$P(\mathbf{u})$	(U_R, U_G)	$\xrightarrow{\mathcal{F}}$ (R, D, A)	$P(\mathbf{v})$
1/4	(1,1)	(1,1,1)	1/2
1/4	(1,0)	(1,1,1)	
1/4	(0,1)	(0,0,1)	1/4
1/4	(0,0)	(0,0,0)	1/4

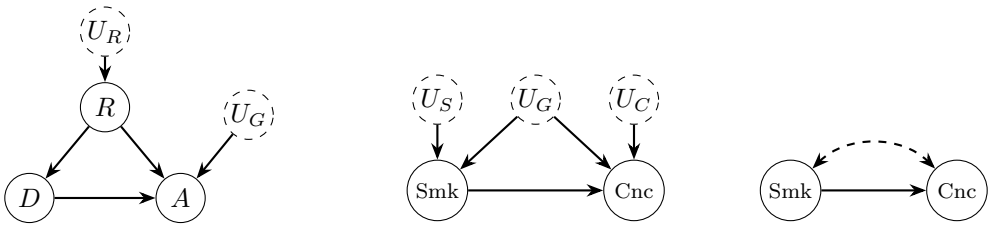
1.2.3.2 Property 2: an SCM induces a causal diagram

Definition (Causal diagram; Bareinboim et al. 2020, Def. 13)

Given $\mathcal{M} = \langle \mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}) \rangle$, the causal diagram $\mathcal{G}$ is constructed by

1. adding a vertex for every $V_i \in \mathbf{V}$;
2. adding a directed edge $V_j \to V_i$ whenever V_j is an argument of f_i;
3. adding a bidirected edge $V_j \leftrightarrow V_i$ whenever $\mathbf{U}_i$ and $\mathbf{U}_j$ are correlated or f_i, f_j share some $U \in \mathbf{U}$ as an argument.

Precisely: for every vertex set $\mathbf{W}$ without bidirected edges among them, $\{\mathbf{U}_W\}_{W \in \mathbf{W}}$ are jointly independent. With acyclic directed part (assumed throughout) $\mathcal{G}$ is an ADMG. E.g. $V_i \leftarrow f_i(A, B, U)$, $V_j \leftarrow f_j(C, U)$ give $A \to V_i \leftarrow B$, $C \to V_j$, $V_i \leftrightarrow V_j$.



drug example explicit $\mathbf{U}$ $\mathbf{U}$ implicit

Example 1.2 (Smoking and cancer)

$\mathbf{V} = \{\text{Smoking}, \text{Cancer}\}$, $\mathbf{U} = \{U_S, U_C, U_G\}$, Smoking $\leftarrow f_S(U_S, U_G)$, Cancer $\leftarrow f_C(\text{Smoking}, U_C, U_G)$; the shared U_G gives Smoking $\leftrightarrow$ Cancer.

Remark 1. A graph is compatible with infinitely many SCMs with the same function signatures and exogenous structure.

Remark 2. Researchers often know the *scope* of mechanisms but not their form; causal inference

asks what graph plus data imply.
Convention. Latents are implicit; a shared or correlated U is a dashed bidirected edge.

1.2.3.3 Graph terminology

$\mathcal{G} = \langle \mathbf{V}, \mathbf{E} \rangle$; for $V \in \mathbf{V}$:

- parents $Pa(V) = \{W : W \to V \in \mathcal{G}\}$; children $Ch(V) = \{W : V \to W \in \mathcal{G}\}$;
- ancestors $An(V) = \{W : W \to \cdots \to V \in \mathcal{G}\}$; descendants $De(V) = \{W : V \to \cdots \to W \in \mathcal{G}\}$; both include V;
- siblings $\{W : W \leftrightarrow V \in \mathcal{G}\}$;
- a *path* joins distinct vertices by edges; a *directed path* from W to V has all arrowheads toward V.

1.2.3.4 Property 3: an SCM induces Pearl's Causal Hierarchy

$\mathcal{M}$ answers $P(y \mid x)$, $P(y \mid \mathrm{do}(x), c)$ and $P(y_x \mid x', y')$. Here Layer 1; Layers 2–3 in Chapter 2.

1.2.4 The first layer: what the diagram encodes about $P(\mathbf{v})$

Motivation. Derive the factorization of the observational distribution implied by an SCM.

$\mathcal{M}$ induces

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{V} = \mathbf{v}, \mathbf{U} = \mathbf{u}) = \sum_{\mathbf{u}} P\big(\forall_{V_i \in \mathbf{V}}\, f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i,\ \mathbf{U} = \mathbf{u}\big),$$

$\mathbf{pa}_i, \mathbf{u}_i$ being the parents of V_i in $\mathcal{G}$. In the examples,

$$P(r, d, a) = \sum_{u_R, u_G} P(\underbrace{r, d, a}_{\text{endogenous}}, \underbrace{u_R, u_G}_{\text{exogenous}}),$$
$$P(s, c) = \sum_{u_S, u_G, u_C} P(s, c, u_S, u_G, u_C).$$

Topological order. $\mathcal{G}$ acyclic $\Rightarrow$ there is an order $\prec$ with $Pa_i \prec V_i$, e.g. $R \prec D \prec A$, $S \prec C$.

Factorization. By the chain rule along $\prec$,

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}),$$

and since $Pa_i \cup \mathbf{U}_i \subseteq \{V_1, \ldots, V_{i-1}\} \cup \mathbf{U}$ determine V_i,

$$P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = P\big(f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}\big) = P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

Therefore

$$\boxed{P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).}$$

Examples:

$$P(r, d, a) = \sum_{u_R, u_G} P(u_R, u_G) P(r \mid u_R) P(d \mid r) P(a \mid r, d, u_G),$$
$$P(s, c) = \sum_{u_S, u_G, u_C} P(u_S, u_G, u_C) P(s \mid u_G, u_S) P(c \mid s, u_G, u_C),$$

with $P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) \in \{0, 1\}$ (deterministic mechanisms).

1.2.5 Conditional independence and the Markov condition

Motivation. Translate the structure of $\mathcal{G}$ into testable constraints on $P(\mathbf{v})$.

1.2.5.1 Conditional independence

$$X \perp\!\!\!\perp Y \iff P(y \mid x) = P(y) \;\; \forall x, y \iff P(x, y) = P(x)P(y),$$
$$(X \perp\!\!\!\perp Y \mid Z) \iff P(x \mid z, y) = P(x \mid z) \quad \forall x, y, z \text{ with } P(y, z) > 0.$$

Intuition: *lack of functional dependence leads to probabilistic independence.*

1.2.5.2 Local Markov condition

$P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$ in any topological order, and the largest set that can precede V_i is its *non-descendants* Nd_i. Hence

$$P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = P(v_i \mid \mathbf{nd}_i, \mathbf{u}),$$

i.e.

$$V_i \perp\!\!\!\perp (Nd_i \setminus Pa_i), \; (\mathbf{U} \setminus \mathbf{U}_i) \;\mid\; Pa_i, \mathbf{U}_i.$$

1.2.5.3 Markovian models and the Bayesian factorization

If $U_1, \ldots, U_n$ are jointly independent (no bidirected edges), the model is *Markovian*:

$$\begin{aligned}
P(\mathbf{v}) &= \sum_{\mathbf{u}} P(\mathbf{u}) \prod_i P(v_i \mid \mathbf{pa}_i, u_i) \\
&= \sum_{\mathbf{u}} \prod_i P(v_i \mid \mathbf{pa}_i, u_i) P(u_i) && (U_i \text{ independent}) \\
&= \sum_{\mathbf{u}} \prod_i P(v_i \mid \mathbf{pa}_i, u_i) P(u_i \mid \mathbf{pa}_i) && (U_i \perp\!\!\!\perp Pa_i) \\
&= \sum_{\mathbf{u}} \prod_i P(v_i, u_i \mid \mathbf{pa}_i) \\
&= \prod_i \sum_{u_i} P(v_i, u_i \mid \mathbf{pa}_i) = \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i).
\end{aligned}$$

The sum distributes because the i-th factor depends only on u_i; $U_i \perp\!\!\!\perp Pa_i$ because Pa_i are functions of non-descendants' exogenous variables.

Summary. Markovian SCM $\Rightarrow P(\mathbf{v}) = \prod_i P(v_i \mid \mathbf{pa}_i) \Leftrightarrow$ local Markov $V_i \perp\!\!\!\perp Nd_i \setminus Pa_i \mid Pa_i$.

1.2.6 Reading independences from the graph: d-separation

Motivation. Three elementary structures link mechanisms to observable independences.

Binary variables, explicit mechanisms.

1.2.6.1 Chains

$X \to Z \to Y$: study habits $X \leftarrow U_X$, high Suneung score $Z \leftarrow X \vee \neg U_Z$ (U_Z: difficulty), admission $Y \leftarrow Z \wedge U_Y$. $X \perp\!\!\!\perp Y \mid Z$? **Yes**:

$$P(x, y \mid z) = \frac{P(x, y, z)}{P(z)} = \frac{P(y \mid z)P(z \mid x)P(x)}{P(z)} = \frac{P(x, z)}{P(z)} P(y \mid z) = P(x \mid z)P(y \mid z).$$

Observing Z blocks the chain (marginally X, Y are dependent).

1.2.6.2 Common causes (forks)

$X \leftarrow Z \rightarrow Y$: rain $Z \leftarrow U_Z$, traffic $X \leftarrow Z \vee \neg U_X$, umbrella $Y \leftarrow Z \wedge U_Y$. $X \perp\!\!\!\perp Y$? **No**: an umbrella makes rain, hence traffic, more likely. Given Z? **Yes**:

$$P(x, y \mid z) = \frac{P(x \mid z)P(y \mid z)P(z)}{P(z)} = P(x \mid z)P(y \mid z).$$

1.2.6.3 Common effects (colliders)

$X \rightarrow Z \leftarrow Y$: cold $X \leftarrow U_X$, holiday $Y \leftarrow U_Y$, attending class $Z \leftarrow \neg Y \wedge (\neg X \wedge U_Z)$. $X \perp\!\!\!\perp Y$? **Yes**:

$$P(x, y) = \sum_z P(x)P(y)P(z \mid x, y) = P(x)P(y)\sum_z P(z \mid x, y) = P(x)P(y).$$

Given Z? **No!** Absent ($Z = 0$) on a non-holiday ($Y = 0$) makes a cold more likely ("explaining away").

Given a descendant $W \leftarrow Z \wedge U_W$ (signing the sheet)? **No, again**: $W = 0$ makes $Z = 0$ likelier. *Watch out for descendants of colliders.*

1.2.6.4 Triplet summary

Triplet	Active (dependence flows) when	Inactive when
chain $X \rightarrow Z \rightarrow Y$	Z not observed	Z observed
fork $X \leftarrow Z \rightarrow Y$	Z not observed	Z observed
collider $X \rightarrow Z \leftarrow Y$	Z or a descendant of Z observed	neither Z nor its descendants observed

1.2.6.5 d-separation

Definition (d-separation)

To decide whether $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ (with $X, Y \notin \mathbf{Z}$):

1. look at every path between X and Y;
2. a path is *active* given $\mathbf{Z}$ if every consecutive triplet on it is active;
3. if any path is active, X and Y are d-connected (not guaranteed independent); otherwise they are d-separated, written $(X \perp_d Y \mid \mathbf{Z})_{\mathcal{G}}$.

Trivially, if $X = Y$ or X and Y are adjacent, they are dependent.

Soundness: $(X \perp_d Y \mid \mathbf{Z})_{\mathcal{G}} \Rightarrow X \perp\!\!\!\perp Y \mid \mathbf{Z}$ in every distribution induced by an SCM compatible with $\mathcal{G}$ (global Markov property, equivalent to local for DAGs).

Example 1.3 (Sprinkler network)

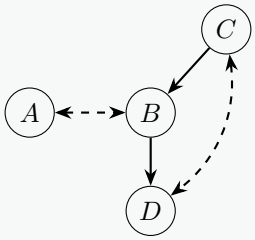
Season → Sprinkler, Season → Rain, Sprinkler → Wet, Rain → Wet, Wet → Slippery.

1. Wet $\perp\!\!\!\perp$ Sprinkler? **No**, adjacent.
2. Wet $\perp\!\!\!\perp$ Season | Sprinkler? **No**: Season → Rain → Wet is open.
3. Rain $\perp\!\!\!\perp$ Slippery | Wet? **Yes**: the only path is the chain through Wet.
4. Season $\perp\!\!\!\perp$ Wet | Sprinkler, Rain? **Yes**: both directed paths are blocked.

5. Sprinkler $\perp\!\!\!\perp$ Rain | Season, Wet? **No**: Wet is an observed collider.

Example 1.4 (Food for thought [Supplement])

Graph: $A \leftrightarrow B$, $C \to B$, $B \to D$, $C \leftrightarrow D$.



- $A \perp\!\!\!\perp D$? **No**: B is an unobserved non-collider on $A \leftrightarrow B \to D$.
- $A \perp\!\!\!\perp C$? **Yes**: colliders at B ($A \leftrightarrow B \leftarrow C$) and at D ($A \leftrightarrow B \to D \leftrightarrow C$).
- $A \perp\!\!\!\perp C \mid D$? **No**: $B \in An(D)$ opens the collider at B.
- $D \perp\!\!\!\perp C \mid B$? **No**: $D \leftrightarrow C$ are adjacent.

1.2.7 Implementation: graph algorithms in Python

Motivation. Topological ordering and d-separation in code.

1.2.7.1 Topological order

A DAG has a source; removing it leaves a DAG. Recurse:

```
def topological_order(G):
    for V in G.Vs:                  # for every node
        if not G.Pa(V):             # if there is no parent for V
            return [V] + topological_order(G - V)
    return []
```

This is quadratic and silent on cycles. Kahn's algorithm tracks in-degrees and detects cycles:

```
def topological_order(G):
    order = deque()
    in_degrees = {V: len(G.Pa(V)) for V in G.Vs}
    while in_degrees:
        for v, v_deg in in_degrees.items():
            if v_deg == 0:
                break
        else:
            raise ValueError('cyclic')
        order.append(v)
        del in_degrees[v]
        for c in G.Ch(v):
            in_degrees[c] -= 1
    return order
```

The `for`–`else` branch runs iff no zero-in-degree node remains, i.e. iff there is a cycle.

1.2.7.2 d-separation via reachability (Markovian graphs)

X, Y are d-connected given $\mathbf{Z}$ iff some path has every triplet $A - B - C$ satisfying:

$$\begin{array}{ll} A \to B \to C & \text{only if } B \notin \mathbf{Z}, \\ A \leftarrow B \leftarrow C & \text{only if } B \notin \mathbf{Z}, \\ A \leftarrow B \to C & \text{only if } B \notin \mathbf{Z}, \\ A \to B \leftarrow C & \text{only if } B \in An(\mathbf{Z}). \end{array}$$

BFS/DFS over states (arrival direction, B): into B ($\to B$) or out of B ($\leftarrow B$).

```
def d_separation(G, X, Y, Zs):
    q = visited_deque({('->', X), ('<-', X)})
    while q:
        direction, B = q.pop()
        if B == Y:
            return False
        if direction == '->':          # arrived with arrowhead into B
            if B in G.An(Zs):          # collider: open iff B in An(Z)
                q.extend({('<-', C) for C in G.Pa(B)})
            if B not in Zs:            # chain continues
                q.extend({('->', C) for C in G.Ch(B)})
        else:                          # arrived via tail of B (B -> previous)
            if B not in Zs:
                q.extend({('->', C) for C in G.Ch(B)})
                q.extend({('<-', C) for C in G.Pa(B)})
    return True
```

`visited_deque` skips visited states, so each (direction, node) is processed once: linear time. X may be left in either direction.

Chapter 2. Identification and Adjustment

2.1 Identification of causal effects

Motivation. How an SCM defines a population under intervention, and when that distribution follows from the diagram and observational data.

2.1.1 Structural causal models, revisited

Motivation. Recall the framework and introduce the do(·) operator.

2.1.1.1 Observation versus intervention

We learn about reality by *observation* (passive records) and *intervention* (trials, experiments). Questions on drug efficacy or hypothetical policies concern the system *under intervention*, even when only observational data exist.

An SCM $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ models the unknown reality: measured $\mathbf{V}$, unknown $\mathbf{U}$ (what each unit is), and $v \leftarrow f(\cdot)$ for each V. It induces a diagram and a family of distributions:

$$\mathcal{M} \longmapsto \mathcal{G},$$
$$\mathcal{M} \longmapsto \{P,\ P_{x_1},\ P_{x_2},\ P_{x_2,x_4}, \dots\}$$

for $\text{do}(\emptyset), \text{do}(x_1), \text{do}(x_2), \text{do}(x_2, x_4), \dots$

Layer 1 (observational). As in Chapter 1,

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{v}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \dots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i);$$

Layers 2 and 3 are induced as well.

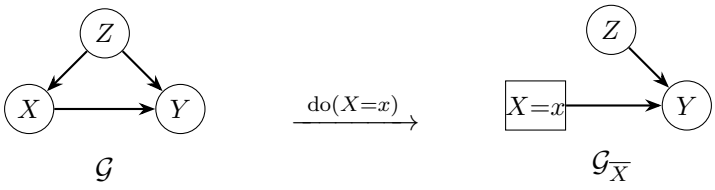
Causal diagram. $B \to V_i$ if B is an argument of f_i; $V_i \leftrightarrow V_j$ if $\mathbf{U}_i, \mathbf{U}_j$ correlate.

2.1.1.2 The do-operator

Definition (Intervention)

Given $\mathcal{M}$, fixing $X \in \mathbf{V}$ to a constant x is written $\text{do}(X = x)$. It transforms $\mathcal{M}$ into the *submodel* $\mathcal{M}_x$, i.e. $\mathcal{M}$ with $f_X \in \mathcal{F}$ replaced by the constant x.

The diagram of $\mathcal{M}_x$ is $\mathcal{G}_{\overline{X}}$: edges *into* X removed.



$\mathcal{G}$ $\qquad$ $\mathcal{G}_{\overline{X}}$

2.1.2 The second layer: causal effects

Motivation. Define causal effects and compute them in Markovian models.

2.1.2.1 Connecting different worlds

Data come from P, the query from P': would banning smoking reduce cancer? $Q = P(\text{Cancer} = \text{true} \mid \text{do}(\text{Smoking} = \text{no}))$ is *not* an aspect of P.

Observation 1. A distribution describes a population under one regime, not change.
Observation 2. We must represent how the population reacts to a change of regime.

The challenge. Age/sex Z, action X, mediator W, outcome Y: $Z \to X \to W \to Y$, $Z \to Y$. Setting $X = x$ cuts $Z \to X$; reality gives $P(z, x, w, y)$, the hypothetical world $P(z, w, y \mid \text{do}(x))$. *Causal effects* are changes of Y when X is varied between constants, free of Z. Is $P(y \mid \text{do}(x)) = P(y \mid x)$?

Definition (Causal effect; Pearl 2000, Def. 3.2.1)

Given disjoint sets $\mathbf{X}, \mathbf{Y}$, the causal effect of $\mathbf{X}$ on $\mathbf{Y}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$, is a function from $\mathfrak{X}_{\mathbf{X}}$ to the space of distributions over $\mathbf{Y}$. For each $\mathbf{x}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is the probability of $\mathbf{Y} = \mathbf{y}$ induced by deleting from the model all equations of the variables in $\mathbf{X}$ and substituting $\mathbf{X} = \mathbf{x}$ in the remaining equations:

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = P_{\mathbf{x}}(\mathbf{y}), \quad \text{the probability of } \mathbf{Y} = \mathbf{y} \text{ in } \mathcal{M}_{\mathbf{x}}.$$

2.1.2.2 Computing causal effects from observational data (Markovian case)

In the example,

$$P(\mathbf{v}) = P(z)\,P(x \mid z)\,P(w \mid x)\,P(y \mid w, z),$$
$$P_x(\mathbf{v} \setminus \{X\}) = P(z) \cdot \underbrace{1}_{X=x \text{ in } \mathcal{M}_x} \cdot P(w \mid x)\,P(y \mid w, z),$$

the other mechanisms being untouched (*modularity*). Hence

$$P(y \mid \text{do}(x)) = \sum_{z,w} P(z)P(w \mid x)P(y \mid w, z) \;\neq\; \sum_{z,w} P(z \mid x)P(w \mid x)P(y \mid w, z) = P(y \mid x)$$

in general: conditioning weights z by $P(z \mid x)$, intervening by $P(z)$.

Example 2.1 (Sprinkler: seeing versus doing)

Season $\to$ Sprinkler, Season $\to$ Rain, Sprinkler $\to$ Wet $\leftarrow$ Rain, Wet $\to$ Slippery.

$$Q_1 = P(\text{wt} \mid \text{Sp} = \text{on}) = \frac{\sum_{\text{sn,rn}} P(\text{sn})P(\text{Sp} = \text{on} \mid \text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P(\text{Sp} = \text{on} \mid \text{sn})P(\text{sn})},$$
$$Q_2 = P(\text{wt} \mid \text{do}(\text{Sp} = \text{on})) = \frac{\sum_{\text{sn,rn}} P(\text{sn})\,P'(\text{Sp} = \text{on})\,P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P'(\text{Sp} = \text{on})P(\text{sn})}$$
$$= \sum_{\text{sn,rn}} P(\text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}),$$

with $P'(\text{Sp} = \text{on}) = 1$. Seeing the sprinkler on is evidence about season and rain; turning it on is not.

2.1.3 Truncated factorization

Motivation. The general interventional formula for Markovian models, read as reweighting.

Theorem 2.1 (Truncated factorization; manipulation theorem, g-computation)

In a Markovian model, the distribution generated by $\mathrm{do}(\mathbf{X}=\mathbf{x})$ is

$$P(\mathbf{v} \mid \mathrm{do}(\mathbf{x})) = \begin{cases} \prod_{V_i \in \mathbf{V}\setminus\mathbf{X}} P(v_i \mid \mathbf{pa}_i) & \text{if } \mathbf{v} \text{ is consistent with } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases}$$

Derivation [Supplement]. $\mathcal{M}_{\mathbf{x}}$ is Markovian with the same f_i, $P(u_i)$ for $V_i \notin \mathbf{X}$ and constant mechanisms for $\mathbf{X}$; its Bayesian factorization (Chapter 1) is

$$\prod_{V_i \notin \mathbf{X}} P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) \prod_{X \in \mathbf{X}} \mathbb{1}[x \text{ agrees with } \mathbf{x}].$$

As f_i, U_i are unchanged and $U_i \perp\!\!\!\perp Pa_i$ in both models,

$$P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) = \sum_{u_i} \mathbb{1}[f_i(\mathbf{pa}_i, u_i) = v_i] P(u_i) = P(v_i \mid \mathbf{pa}_i)$$

whenever defined (positivity). ■

Reweighting form. Since $P(\mathbf{v}) = \prod_{V_i \in \mathbf{V}\setminus\mathbf{X}} P(v_i \mid \mathbf{pa}_i) \cdot \prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)$,

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \frac{P(\mathbf{v})}{\prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)}, \qquad \mathbf{v}' = \mathbf{v} \setminus \mathbf{x}.$$

For singleton X, with $\mathbf{v}'' = \mathbf{v} \setminus (\mathbf{pa}_X \cup \{x\})$,

$$P(\mathbf{v}' \mid \mathrm{do}(x)) = \frac{P(\mathbf{v})}{P(x \mid \mathbf{pa}_X)} = \frac{P(\mathbf{v}) P(\mathbf{pa}_X)}{P(x, \mathbf{pa}_X)} = P(\mathbf{v}'' \mid x, \mathbf{pa}_X)\, P(\mathbf{pa}_X).$$

Units with $X = x$ are up-weighted by $1/P(x \mid \mathbf{pa}_X)$ (origin of IPW, Chapter 6).

2.1.4 The identification problem

Motivation. When is a causal query determined by what we observe and assume?

Definition (Identifiability; Pearl 2000, Def. 3.2.2)

The causal effect of $\mathbf{X}$ on $\mathbf{Y}$ is *identifiable* from $\mathcal{G}$ if $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ can be computed uniquely from any positive distribution over the observed variables; that is, for every pair of models $\mathcal{M}_1, \mathcal{M}_2$ inducing $\mathcal{G}$,

$$P^{\mathcal{M}_1}(\mathbf{v}) = P^{\mathcal{M}_2}(\mathbf{v}) > 0 \implies P^{\mathcal{M}_1}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = P^{\mathcal{M}_2}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})).$$

Picture. The unobserved $\mathcal{M}$ induces $\mathcal{G}$ (known), $P(\mathbf{v})$ (observed) and $P(\mathbf{v} \mid \mathrm{do}(\mathbf{x}))$ (unobserved). The query is identifiable iff it is constant over all models inducing both $\mathcal{G}$ and $P(\mathbf{V})$.

Example 2.2 (A non-identifiable effect [Supplement])

$\mathcal{G}$: $X \to Y$, $X \leftrightarrow Y$, binary, shared $U \sim \mathrm{Bern}(1/2)$. $\mathcal{M}_1$: $X \leftarrow U$, $Y \leftarrow U$; $\mathcal{M}_2$: $X \leftarrow U$, $Y \leftarrow X$. Both give $P(X=Y=0) = P(X=Y=1) = 1/2$, but $P^{\mathcal{M}_1}(Y=1 \mid \mathrm{do}(X=1)) = 1/2 \neq 1 = P^{\mathcal{M}_2}(Y=1 \mid \mathrm{do}(X=1))$. (A graph permits, never requires, dependence.)

2.1.4.1 Identification in Markovian models

Theorem 2.2

Given the causal diagram $\mathcal{G}$ of any Markovian model (all relevant variables measured), $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable for all disjoint $\mathbf{X}, \mathbf{Y}$, and

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i),$$
$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v}' \setminus \mathbf{y}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i).$$

Proof [Supplement]. By the truncated factorization,

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i) \quad \text{evaluated at } \mathbf{X} = \mathbf{x},$$

each factor a conditional of $P(\mathbf{v})$, unique under positivity. Marginalizing over $\mathbf{V} \setminus (\mathbf{X} \cup \mathbf{Y})$ expresses $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ through $P(\mathbf{v})$ alone, so models agreeing on $\mathcal{G}$ and $P(\mathbf{v})$ agree on it. ■

2.1.4.2 Adjustment by direct parents (singleton)

Theorem 2.3

In a Markovian model where only a subset of variables is measured, $Q = P(\mathbf{y} \mid \mathrm{do}(x))$ is identifiable whenever X, $\mathbf{Y}$ and Pa_X are measured, and

$$P(\mathbf{y} \mid \mathrm{do}(x)) = \sum_{\mathbf{pa}_X} P(\mathbf{y} \mid x, \mathbf{pa}_X) P(\mathbf{pa}_X).$$

Proof [Supplement]. Sum the singleton reweighting identity

$$P(\mathbf{v}' \mid \mathrm{do}(x)) = P(\mathbf{v}'' \mid x, \mathbf{pa}_X) P(\mathbf{pa}_X)$$

over $\mathbf{v}'' \setminus \mathbf{y}$; unmeasured variables are summed out inside $P(\cdot \mid x, \mathbf{pa}_X)$. ■

2.1.4.3 Adjustment by direct parents for a set of treatments (optional)

For $\mathbf{X} = \{X_1, \ldots, X_k\}$ a parent of X_j may descend from an earlier X_i. Notation:

$Pa_{\mathbf{X}} = \bigcup_{X \in \mathbf{X}} Pa_X$	parents of $\mathbf{X}$ (may intersect $\mathbf{X}$),
$Pa^{-}_{\mathbf{W}} = Pa_{\mathbf{W}} \setminus \mathbf{W}$	parents of $\mathbf{W}$ excluding $\mathbf{W}$,
$\mathbf{X}_{<i} = \{X_1, \ldots, X_{i-1}\}$, $\mathbf{X}_{\le i}$	predecessors in a topological order,
$Pa'_{X_i} = Pa_{X_i} \setminus (\mathbf{X}_{<i} \cup Pa^{-}_{\mathbf{X}_{<i}})$	"new" parents of X_i.

Theorem 2.4

Let $\mathbf{X}$ be topologically ordered. If no element of Pa'_{X_j} is a descendant of X_i for every $i < j$, then

$$\prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x} \mid \mathbf{pa}^{-}_{\mathbf{X}}),$$

hence

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{pa}^{-}_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}^{-}_{\mathbf{X}}) P(\mathbf{pa}^{-}_{\mathbf{X}}).$$

Proof [Supplement]. Induction on i: $\prod_{X\in\mathbf{X}_{\le i}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$.

Base case. $P(x_1 \mid \mathbf{pa}_{X_1}) = P(x_1 \mid \mathbf{pa}_{X_1} \setminus \{X_1\}) = P(x_1 \mid \mathbf{pa}^{-}_{X_1}) = P(\mathbf{x}_{\le 1} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le 1}})$.

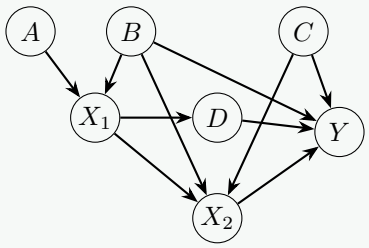
Step. Show $P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}_{X_i}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$, using $Pa^{-}_{\mathbf{X}_{\le i}} = Pa^{-}_{\mathbf{X}_{<i}} \cup Pa'_{X_i}$:

$$
\begin{aligned}
&P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}_{X_i}) \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) && \text{Markov condition} \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})\frac{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{multiply by 1} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{chain rule, } \{X_i\} \cup \mathbf{X}_{<i} = \mathbf{X}_{\le i} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \mathbf{X}_{<i} \perp\!\!\!\perp Pa'_{X_i} \mid Pa^{-}_{\mathbf{X}_{<i}} \\
&= P(\mathbf{x}_{\le i} \mid \mathbf{pa}'_{X_i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}}).
\end{aligned}
$$

Line 1: local Markov ($\mathbf{X}_{<i}$, $Pa^{-}_{\mathbf{X}_{<i}}$ are non-descendants of X_i). The key independence is d-separation: Pa'_{X_i} are non-descendants of every X_j, $j < i$, so a path from $\mathbf{X}_{<i}$ to Pa'_{X_i} either enters X_j through a parent (blocked by $Pa^{-}_{\mathbf{X}_{<i}}$) or leaves X_j along a directed edge and meets a collider that is not an ancestor of the conditioning set. ■

Example 2.3

$A \to X_1$, $B \to X_1$, $B \to X_2$, $B \to Y$, $C \to X_2$, $C \to Y$, $X_1 \to D$, $X_1 \to X_2$, $D \to Y$, $X_2 \to Y$.



$Pa_{X_1} = \{A, B\}$, $Pa_{X_2} = \{X_1, B, C\}$, $Pa^{-}_{\mathbf{X}} = \{A, B, C\}$, $Pa'_{X_2} = \{C\}$ (not a descendant of X_1). $P(y \mid \mathrm{do}(\mathbf{x})) = \sum P(\mathbf{v})/[P(x_1 \mid a, b)P(x_2 \mid x_1, b, c)]$, and

$$
\begin{aligned}
P(x_1 \mid a, b)P(x_2 \mid x_1, b, c) &= P(x_1 \mid a, b)P(x_2 \mid \underbrace{x_1}_{\mathbf{X}_{<2}}, \underbrace{a, b}_{Pa^{-}_{\mathbf{X}_{<2}}}, \underbrace{c}_{Pa'_{X_2}}) \\
&= P(x_1 \mid a, b)P(x_2 \mid x_1, a, b, c)\frac{P(c \mid x_1, a, b)}{P(c \mid x_1, a, b)} \\
&= \frac{P(x_1, x_2, c \mid a, b)}{P(c \mid x_1, a, b)} = \frac{P(x_1, x_2, c \mid a, b)}{P(c \mid a, b)} \\
&= P(x_1, x_2 \mid a, b, c) = P(\mathbf{x} \mid \mathbf{pa}^{-}_{\mathbf{X}}),
\end{aligned}
$$

so $P(y \mid \mathrm{do}(x_1, x_2)) = \sum_{a,b,c} P(y \mid x_1, x_2, a, b, c)P(a, b, c)$; D (a descendant of X_1) is correctly *not* adjusted for.

2.1.4.4 When a parent is latent

If Season is unobserved in the sprinkler model:

$$\begin{aligned} Q_2 &= \sum_{\text{sn,rn}} P(\text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}) \\ &= \sum_{\text{sn,rn}} P(\text{rn}, \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}) \\ &= \sum_{\text{rn}} P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})P(\text{rn}). \end{aligned}$$

Adjusting for Rain suffices; which sets work in general is the back-door criterion (Section 2.2).

2.1.5 Summary

- $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is defined through submodels.
- In Markovian models every effect is identifiable via the truncated factorization.
- Adjustment by direct parents: condition on Pa_X and average over their marginal.
- Identification asks whether $\mathcal{G}$ and $P(\mathbf{v})$ determine $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$.
- With latent variables it becomes non-trivial (back-door criterion and beyond).

2.2 Confounding and the back-door criterion

Motivation. How to identify an effect when some parents of the treatment are unobserved, and how to evaluate the adjustment formula from samples.

2.2.1 Recap: identifiable and non-identifiable effects

Motivation. Both sides of the identifiability definition, with concrete pairs of models.

2.2.1.1 An identifiable effect

$\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$. Any compatible model gives

$$\begin{aligned} P(\mathbf{v}) &= P(z)P(x \mid z)P(y \mid x, z), \\ P(\mathbf{v}' \mid \text{do}(x)) &= P(z)P(y \mid x, z), \\ P(y \mid \text{do}(x)) &= \sum_z P(z)P(y \mid x, z). \end{aligned}$$

For two models

$$\mathcal{M}^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow f_x^{(j)}(z, u_x) \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$\mathcal{M}_x^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow x \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$j = 1, 2,$$

agreement on $\langle \mathcal{G}, P(\mathbf{v}) \rangle$ fixes $P(z)$ and $P(y \mid x, z)$, hence $P(\mathbf{v}' \mid \text{do}(x))$.

2.2.1.2 A non-identifiable effect

$\mathcal{G}$: $X \to Y$, $X \leftrightarrow Y$ (latent U_{xy}); U_y, U_{xy} independent fair coins, $X \leftarrow U_{xy}$:

$$\mathcal{M}^{(1)}: \; Y \leftarrow (X \oplus U_{xy}) \vee U_y, \qquad \mathcal{M}^{(2)}: \; Y \leftarrow U_y.$$

Observationally $X \oplus U_{xy} = 0$, so $Y = U_y$ in both. Under $\mathrm{do}(x)$:

U_y	U_{xy}	X	$Y^{(1)}$	$Y^{(2)}$	$P(\mathbf{u})$
0	0	x	x	0	1/4
0	1	x	$\neg x$	0	1/4
1	0	x	1	1	1/4
1	1	x	1	1	1/4

Y	$P^{(1)}(Y \mid \mathrm{do}(x))$	$P^{(2)}(Y \mid \mathrm{do}(x))$
0	1/4	1/2
1	3/4	1/2

($Y^{(1)} = 0$ needs $U_y = 0$, $x = U_{xy}$: probability 1/4.) Same $\mathcal{G}$ and $P(\mathbf{v})$, different effects: $P(\mathbf{v} \mid \mathrm{do}(x)) = \sum_{u_{xy}} P(y \mid x, u_{xy}) P(u_{xy})$ depends on the latent U_{xy}.

2.2.2 Confounding bias

Motivation. A common cause makes conditioning differ from intervening.

Exercise and cholesterol, both affected by age: pooled data show cholesterol *increasing* with exercise (older people exercise more), while within each age group it decreases. So

$$P(\text{cholesterol} \mid \text{exercise}) \neq P(\text{cholesterol} \mid \mathrm{do}(\text{exercise})),$$

the **confounding bias**.

Is confounding bias removable? If some parents of X are unobserved, parent adjustment (Section 2.1) fails; which measured sets $Z_1, \ldots, Z_k$ can replace them?

2.2.3 The back-door criterion

Motivation. A graphical condition under which a covariate set substitutes for the treatment's parents.

Definition (Back-door criterion; Pearl 2000, Def. 3.3.1)

A set $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if

(i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$; and

(ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ that contains an arrow into X (a *back-door path*).

Theorem 2.5 (Back-door adjustment)

If $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$, the effect is identifiable and

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

Proof [Supplement]. We use Rules 2 and 3 of do-calculus (Chapter 3, proved from the truncated factorization alone, so no circularity). By the axioms,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})\, P(\mathbf{z} \mid \mathrm{do}(\mathbf{x})).$$

Step 1. $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x})) = P(\mathbf{z})$. In $\mathcal{G}_{\overline{\mathbf{X}}}$ all edges at $\mathbf{X}$ point out, so a collider-free path from $\mathbf{X}$ to $\mathbf{Z}$ would be directed, making part of $\mathbf{Z}$ a descendant of $\mathbf{X}$, excluded by (i); other paths contain a collider and are blocked given $\emptyset$. So $(\mathbf{Z} \perp_d \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$; apply Rule 3.

Step 2. $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$. In $\mathcal{G}_{\underline{\mathbf{X}}}$ every path between $\mathbf{X}$ and $\mathbf{Y}$ starts with an arrow into $\mathbf{X}$, i.e. is a back-door path, blocked by $\mathbf{Z}$ by (ii). So $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}$; apply Rule 2.

Step 3. Substitute both identities. ■

2.2.3.1 Back-door sets substitute for the parents: an example

In the sprinkler network, Rain is back-door admissible for (Sprinkler, Wet):

(i) Rain is not a descendant of Sprinkler;

(ii) Rain blocks the only back-door path Sprinkler ← Season → Rain → Wet. From parent adjustment,

$$\begin{aligned} P(\mathrm{wt} \mid \mathrm{do}(\mathrm{sp})) &= \sum_{\mathrm{sn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{sn}, \mathrm{rn}) P(\mathrm{rn} \mid \mathrm{sp}, \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) P(\mathrm{rn} \mid \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) \sum_{\mathrm{sn}} P(\mathrm{rn}, \mathrm{sn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) P(\mathrm{rn}), \end{aligned}$$

the third equality by Wet ⫫ Season | Sprinkler, Rain and Rain ⫫ Sprinkler | Season (d-separation).

2.2.3.2 General proof: back-door adjustment = parent adjustment

With $Pa_{\mathbf{X}}^{-}$ the parents of $\mathbf{X}$ outside $\mathbf{X}$ and $\mathbf{Z}^{-} = \mathbf{Z} \setminus Pa_{\mathbf{X}}^{-}$:

$$\begin{aligned} \text{(i)} &\implies \mathbf{X} \perp\!\!\!\perp (\mathbf{Z} \setminus Pa_{\mathbf{X}}^{-}) \mid Pa_{\mathbf{X}}^{-} && \text{(local Markov)}, \\ \text{(ii)} &\implies \mathbf{Y} \perp\!\!\!\perp (Pa_{\mathbf{X}}^{-} \setminus \mathbf{Z}) \mid \mathbf{Z}, \mathbf{X} && \text{(back-door paths blocked)}, \end{aligned}$$

the first as $\mathbf{Z}$ are non-descendants of $\mathbf{X}$, the second as a path from a parent of $\mathbf{X}$ to $\mathbf{Y}$ avoiding $\mathbf{X}$ extends to a back-door path. Then

$$\begin{aligned} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) &= \sum_{\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{parent adjustment} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}, \mathbf{z}^{-}) P(\mathbf{z}^{-} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{def. of cond. probability} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^{-} \mid \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{(ii) and (i)} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}) \end{aligned}$$

$$= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) \sum_{\mathbf{pa}_{\mathbf{X}}^- \setminus \mathbf{z}} P(\mathbf{z}^-, \mathbf{pa}_{\mathbf{X}}^-) \qquad \text{by algebra}$$
$$= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

Line 3: $\{\mathbf{x}, \mathbf{pa}_{\mathbf{X}}^-, \mathbf{z}^-\} = \{\mathbf{x}, \mathbf{z}\} \cup (\mathbf{pa}_{\mathbf{X}}^- \setminus \mathbf{z})$, so (ii) drops $\mathbf{pa}_{\mathbf{X}}^- \setminus \mathbf{z}$;

(i) drops $\mathbf{x}$ from $P(\mathbf{z}^- \mid \cdot)$. *Back-door adjustment equals parent adjustment.* (The parent formula holds in the Markovian model including latents; the result uses only measured variables.)

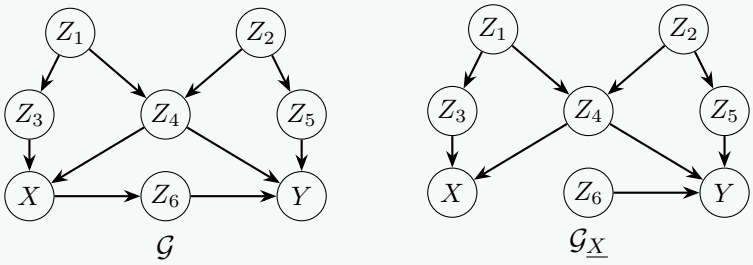
2.2.3.3 A graphical test

With $\mathcal{G}_{\underline{\mathbf{X}}}$ the graph without edges *out of* $\mathbf{X}$,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) \text{ is identified by } \mathbf{Z} \quad \text{if (i) and } (\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}.$$

Example 2.4 (Pearl's seven-variable model)

$Z_1 \to Z_3 \to X$, $Z_1 \to Z_4$, $Z_2 \to Z_4$, $Z_2 \to Z_5 \to Y$, $Z_4 \to X$, $Z_4 \to Y$, $X \to Z_6 \to Y$.



$\mathcal{G}$ $\qquad$ $\mathcal{G}_{\underline{X}}$

Back-door paths: $X \leftarrow Z_4 \to Y$, $X \leftarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \to Y$, $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \leftarrow Z_2 \to Z_5 \to Y$. Z_4 blocks the first three but opens the collider on the last, which Z_1 blocks. So $\{Z_1, Z_4\}$ is admissible:

$$P(y \mid \mathrm{do}(x)) = \sum_{z_1, z_4} P(y \mid x, z_1, z_4) P(z_1, z_4).$$

Also $\{Z_3, Z_4\}$, $\{Z_4, Z_2\}$, $\{Z_4, Z_5\}$; not $\{Z_4\}$ alone, never Z_6 (descendant of X).

Example 2.5 (More back-door examples [Supplement])

(a) $Z_2 \to Z_4$, $Z_2 \to Z_5$, $Z_4 \to X$, $Z_4 \to Y$, $X \leftrightarrow Z_4$, $X \to Z_6 \to Y$, $Z_5 \to Y$. Z_4 is needed for $X \leftarrow Z_4 \to Y$ but is a collider on $X \leftrightarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, so add Z_2 or Z_5: $\mathbf{Z} \in \{\{Z_4, Z_2\}, \{Z_4, Z_5\}, \{Z_4, Z_2, Z_5\}\}$.
(b) $X \to Z$, $X \to Y$: no back-door path, Z a descendant, so $\mathbf{Z} = \emptyset$ and $P(y \mid \mathrm{do}(x)) = P(y \mid x)$.

2.2.4 Evaluating back-door adjustment: inverse probability weighting

Motivation. An estimator whose cost does not explode with the dimension of $\mathbf{Z}$.

For admissible $\mathbf{Z}$,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}),$$
$$\mathbb{E}[Y \mid \mathrm{do}(\mathbf{x})] = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}[Y \mid \mathbf{x}, \mathbf{Z}]].$$

Naively this needs many conditionals and a sum over $\mathbf{Z}$: about $O(\exp(|\mathbf{Z}|))$ time, with sparse cells.

2.2.4.1 Rewriting with the propensity score

$$P(y \mid \mathrm{do}(x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z}) P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x, \mathbf{z})} P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})},$$

with the propensity fitted as $g(\mathbf{z}) \approx P(x \mid \mathbf{z})$.

2.2.4.2 Sample version

Replacing $P(y, x, \mathbf{z})$ by empirical frequencies,

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &\approx \sum_{\mathbf{z}} \frac{\frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \sum_{\mathbf{z}} \frac{\mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \frac{\mathbb{1}(y_i = y, x_i = x)}{g(\mathbf{z}_i)}, \end{aligned}$$

since only $\mathbf{z} = \mathbf{z}_i$ contributes; cost $O(N)$, not $O(|\mathfrak{X}_{\mathbf{Z}}|)$. Similarly,

$$\begin{aligned} \mathbb{E}[Y \mid \mathrm{do}(x)] &= \sum_{\mathbf{z}, y} y \, \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})} \\ &\approx \frac{1}{N} \sum_{i=1}^{N} \sum_{\mathbf{z}, y} \frac{y \mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \frac{y_i \mathbb{1}(x_i = x)}{g(\mathbf{z}_i)}. \end{aligned}$$

2.2.4.3 Average treatment effect

Binary X, $g(\mathbf{z}) = P(X = 1 \mid \mathbf{z})$, back-door criterion holding:

$$\begin{aligned} \widehat{\mathrm{ATE}} &= \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 1)] - \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 0)] \\ &= \frac{1}{N} \sum_{i=1}^{N} y_i \left(\frac{\mathbb{1}(x_i = 1)}{g(\mathbf{z}_i)} - \frac{\mathbb{1}(x_i = 0)}{1 - g(\mathbf{z}_i)} \right) \\ &= \frac{1}{N} \sum_{i=1}^{N} \left(\frac{y_i x_i}{g(\mathbf{z}_i)} - \frac{y_i (1 - x_i)}{1 - g(\mathbf{z}_i)} \right). \end{aligned}$$

(Elsewhere covariates are often X or L, treatment T, W or A.)

2.2.4.4 Interpretation: pseudo-causal samples

IPW builds a pseudo-population mimicking a randomized experiment: drop samples with $x_i \neq x$ and replicate each remaining one $1/g(\mathbf{z}_i)$ times (e.g. $P(x_1 \mid \mathbf{z}_1) = 1/3$: three times; $P(x_3 \mid \mathbf{z}_3) = 1/2$: twice). This removes the dependence between X and $\mathbf{Z}$.

2.2.5 Summary

- A common cause of treatment and outcome gives $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$.

- The back-door criterion tests admissibility of $\mathbf{Z}$ graphically.
- Back-door adjustment: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.
- Any back-door admissible set can substitute for the parents.
- IPW evaluates it from samples without summing over high-dimensional $\mathbf{Z}$.

2.3 The adjustment criterion

Motivation. Exactly which sets $\mathbf{Z}$ make $\sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$ equal $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, and how to test, construct and enumerate them.

2.3.1 Back-door criterion: recap and two refinements

Motivation. Back-door adjustment resolves Simpson's paradox and extends to conditional effects.

Theorem 2.6 (Back-door adjustment; Pearl 2000, Thm. 3.3.2)

If $\mathbf{Z}$ satisfies (i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$, and (ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ containing an arrow into X, then $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.

Proof [Supplement]. The proof of Section 2.2 (Rule 3 for $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}))$, Rule 2 for $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})$) applies verbatim. ■

2.3.1.1 Back to Simpson's paradox

The drug/sex/recovery tables of Chapter 1 under the three stories:

	$F \to X,\ F \to Y,\ X \to Y$	$X \to F \to Y,\ X \to Y$	$X \leftarrow U_1 \to F \leftarrow U_2 \to Y,\ X \to Y$
$\emptyset$	not admissible	admissible	admissible
$\{F\}$	admissible	not admissible ($F \in De(X)$)	not admissible (opens collider)
Use	F-specific tables	aggregated table	aggregated table

The physician's dilemma becomes a mechanical graph check.

2.3.1.2 Conditional back-door

Theorem 2.7 (Conditional back-door criterion)

The $\mathbf{w}$-specific effect $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ is identified whenever we can measure a set $\mathbf{Z}$ such that $\mathbf{W} \cup \mathbf{Z}$ satisfies the back-door criterion. Moreover,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathbf{w}).$$

Proof [Supplement]. $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$.

Step 1. All paths in $\mathcal{G}_{\underline{\mathbf{X}}}$ are back-door paths, blocked by $\mathbf{W} \cup \mathbf{Z}$: $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z}, \mathbf{W})_{\mathcal{G}_{\underline{\mathbf{X}}}}$, so Rule 2 gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w})$.

Step 2. $\mathbf{W}$ has no descendant of $\mathbf{X}$, so $\mathbf{X}(\mathbf{W}) = \mathbf{X}$ and Rule 3 uses $\mathcal{G}_{\overline{\mathbf{X}}}$. A path from $\mathbf{X}$ to $\mathbf{Z}$ there leaves $\mathbf{X}$ outward; collider-free, it would make $\mathbf{Z}$ contain a descendant of $\mathbf{X}$; otherwise its first

collider C and all descendants of C descend from $\mathbf{X}$, so are not in $\mathbf{W}$, and the path is blocked. Thus $(\mathbf{Z} \perp_d \mathbf{X} \mid \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$ and $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = P(\mathbf{z} \mid \mathbf{w})$.

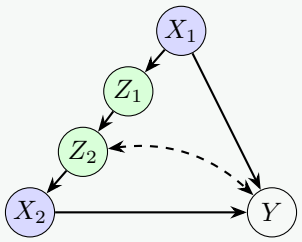
Step 3. Substitute. ■

2.3.2 Tightness of the back-door criterion

Motivation. The back-door criterion is sufficient but not necessary.

Example 2.6

$\mathbf{X} = \{X_1, X_2\}$ with $X_1 \to Z_1 \to Z_2 \to X_2$, $X_1 \to Y$, $X_2 \to Y$, $Z_2 \leftrightarrow Y$.



All other variables descend from X_1, so no back-door set exists; yet with $\mathbf{Z} = \{Z_1, Z_2\}$,

$$P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(y \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$$

(Z_2 blocks $X_2 \leftarrow Z_2 \leftrightarrow Y$; Z_1 blocks the opened $X_1 \to Z_1 \to Z_2 \leftrightarrow Y$).

Condition (i) protects causal paths, but some descendants of $\mathbf{X}$ (e.g. ones that are also ancestors of $\mathbf{X}$ off the causal paths) may, or must, be used. *Which variables must really be avoided?*

2.3.3 The adjustment criterion

Motivation. Necessary and sufficient conditions for valid adjustment sets.

Back-door ⇒ adjustment formula; we seek "???" with

$$\mathbf{Z} \text{ is ??? w.r.t. } \langle \mathbf{X}, \mathbf{Y} \rangle \iff P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

2.3.3.1 Proper causal paths

Definition (Proper causal path; Shpitser et al. 2010)

A path from $X \in \mathbf{X}$ to $Y \in \mathbf{Y}$ is *causal* if it is directed, and *proper* if it contains a member of $\mathbf{X}$ only at its starting point.

Example 2.7

$X_1 \to W_1 \to X_2 \to W_2 \to Y$, $X_1 \to W_3 \to Y$, $X_1 \leftrightarrow Y$. Proper causal: $X_1 \to W_3 \to Y$, $X_2 \to W_2 \to Y$; not proper: $X_1 \to W_1 \to X_2 \to W_2 \to Y$.

2.3.3.2 Disturbing a proper causal path

Adjusting for a variable on a proper causal path (other than $\mathbf{X}$) or a descendant of one *disturbs* it.

Example 2.8

$X \to W \to Y$, $W \to Z$: $P(y \mid \mathrm{do}(x)) = P(y \mid x)$. Adjusting for Z:

$$\begin{aligned}\sum_z P(y \mid x, z)P(z) &= \sum_{z,w} P(y \mid x, z, w)P(w \mid x, z)P(z) \\ &= \sum_w P(y \mid w) \sum_z P(w \mid x, z)P(z), \\ P(y \mid \mathrm{do}(x)) = P(y \mid x) &= \sum_{w,z} P(y \mid x, z, w)P(w \mid x, z)P(z \mid x) \\ &= \sum_w P(y \mid w) \sum_z P(w \mid x, z)P(z \mid x).\end{aligned}$$

They differ whenever $P(z) \neq P(z \mid x)$ (generic, as Z depends on X via W); likewise for adjusting for W.

2.3.3.3 The criterion

Theorem 2.8 (Adjustment criterion; Shpitser, VanderWeele and Robins 2010)

$\mathbf{Z}$ satisfies the adjustment criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ iff

(i) $\mathbf{Z}$ does not disturb any proper causal path: no element of $\mathbf{Z}$ is a descendant, in $\mathcal{G}_{\overline{\mathbf{X}}}$, of any $W \notin \mathbf{X}$ lying on a proper causal path from $\mathbf{X}$ to $\mathbf{Y}$; and

(ii) all proper non-causal paths in $\mathcal{G}$ from $\mathbf{X}$ to $\mathbf{Y}$ are blocked by $\mathbf{Z}$.

$\mathbf{Z}$ satisfies the adjustment criterion *if and only if* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})P(\mathbf{z})$ for every model compatible with $\mathcal{G}$.

Proof [Supplement]. *Sufficiency when* $\mathbf{Z}$ *has no descendant of* $\mathbf{X}$. (i) is automatic. A non-causal path either starts into $\mathbf{X}$ (back-door) or starts out of $\mathbf{X}$ and is not directed, hence has a collider descending from $\mathbf{X}$, outside $\mathbf{Z}$ with no descendant in $\mathbf{Z}$: blocked. So (ii) is the back-door criterion and the back-door theorem applies.

General case. Sufficiency and necessity are Shpitser, VanderWeele and Robins (2010, Theorem 3), via the proper back-door graph; necessity builds, for each violating $\mathbf{Z}$, two models agreeing on $P(\mathbf{v})$ but not on the formula. Not reproduced here. ■

2.3.3.4 Implementation idea

Input: $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathcal{G}$. **Output:** whether $\mathbf{Z}$ is adjustment-admissible.

Step 1 (proper causal paths).

$$\mathrm{PCP} = (De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \setminus \mathbf{X}) \cap An(\mathbf{Y})_{\mathcal{G}_{\underline{\mathbf{X}}}},$$

descendants of $\mathbf{X}$ reaching $\mathbf{Y}$ without re-entering $\mathbf{X}$. (Superset alternative: $\mathbf{W} = De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \cap An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, $\mathcal{H} = \mathcal{G}[\mathbf{W}]$ without bidirected edges; ancestors in $\mathcal{G}_{\underline{\mathbf{X}}}$ drop vertices reaching $\mathbf{Y}$ only through another treatment.)

Step 2 (forbidden variables). $\mathcal{F} = De(\mathrm{PCP})_{\mathcal{G}_{\overline{\mathbf{X}}}}$. *Check (i):* $\mathbf{Z} \cap \mathcal{F} \neq \emptyset \Rightarrow$ not admissible.

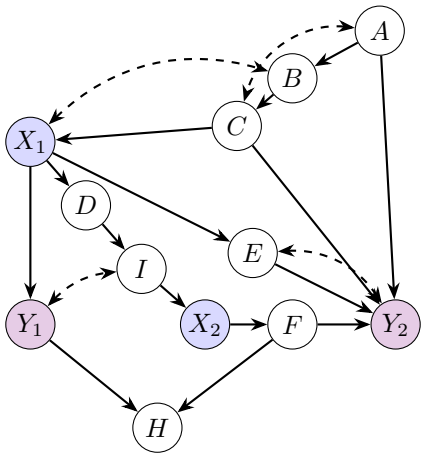
Step 3 (proper non-causal paths). Remove the first edge of every proper causal path,

$$\mathcal{G}' = \mathcal{G} \setminus \{X \to W : X \in \mathbf{X},\ W \in \mathrm{PCP}\}.$$

Check (ii): admissible iff $(\mathbf{X} \perp_d \mathbf{Y} \mid \mathbf{Z})_{\mathcal{G}'}$.

Each step is a graph search: polynomial.

2.3.3.5 An example



$\mathcal{G}$: $A \to B \to C \to X_1$, $C \to Y_2$, $A \to Y_2$, $X_1 \to D \to I \to X_2$, $X_1 \to E \to Y_2$, $X_1 \to Y_1$, $X_2 \to F \to Y_2$, $Y_1 \to H \leftarrow F$; $X_1 \leftrightarrow B$, $A \leftrightarrow C$, $Y_1 \leftrightarrow I$, $E \leftrightarrow Y_2$; $\mathbf{X} = \{X_1, X_2\}$, $\mathbf{Y} = \{Y_1, Y_2\}$.

(i) Forbidden set. Proper causal paths $X_1 \to Y_1$, $X_1 \to E \to Y_2$, $X_2 \to F \to Y_2$ ($X_1 \to D \to I \to X_2 \to \cdots$ is not proper); descendants of $\{Y_1, E, Y_2, F\}$ are $\{Y_1, E, Y_2, F, H\}$, so $\mathcal{F} = \{E, F, H\}$ besides $\mathbf{Y}$.

(ii) Evaluating $\mathbf{Z} = \{C, D, I\}$. $\mathbf{Z} \cap \mathcal{F} = \emptyset$. Cutting $X_1 \to Y_1$, $X_1 \to E$, $X_2 \to F$ gives $\mathcal{G}'$, where $X_1 \leftrightarrow B \leftarrow A \to Y_2$ is open (B a collider with descendant $C \in \mathbf{Z}$, A an unconditioned non-collider). So $(\mathbf{X} \not\perp_d \mathbf{Y} \mid C, D, I)_{\mathcal{G}'}$: *not* admissible. (Non-causal paths in $\mathcal{G}'$ are proper non-causal paths in $\mathcal{G}$.)

2.3.3.6 Explicit construction of an admissible set

Lemma 2.9 (Explicit admissible set)

Let $\mathcal{F}$ be the variables forbidden by condition (i) and $An^-(\mathbf{W}) = An(\mathbf{W}) \setminus \mathbf{W}$. If any admissible set exists, then

$$\mathbf{Z}_0 = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$$

is admissible. (Ancestors should be taken in $\mathcal{G}'$, but the result is the same in $\mathcal{G}$.)

Proof [Supplement]. By the constructive back-door criterion (van der Zander, Liśkiewicz and Textor 2014), $\mathbf{Z}$ is an adjustment set iff $\mathbf{Z} \cap \mathcal{F} = \emptyset$ and $\mathbf{Z}$ d-separates $\mathbf{X}, \mathbf{Y}$ in the proper back-door graph $\mathcal{G}'$. If an admissible $\mathbf{Z}$ exists, the inclusion–restriction lemma in $\mathcal{G}'$ with $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus (\mathcal{F} \cup \mathbf{X} \cup \mathbf{Y})$ shows that $An(\mathbf{X} \cup \mathbf{Y}) \cap \mathbf{R} = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$ also separates them in $\mathcal{G}'$ and avoids $\mathcal{F}$. ■

Here $An^-(\mathbf{X} \cup \mathbf{Y}) = \{A, B, C, D, E, F, I\}$, $\mathbf{Z}_0 = \{A, B, C, D, I\}$: A blocks $X_1 \leftrightarrow B \leftarrow A \to Y_2$ and $X_1 \leftrightarrow B \leftarrow A \leftrightarrow C \to Y_2$; C blocks $X_1 \leftarrow C \to Y_2$; D blocks $X_1 \to D \to I \leftrightarrow Y_1$ (collider I opened); I blocks $X_2 \leftarrow I \leftrightarrow Y_1$. So $\mathbf{Z}_0$ is admissible, although D, I descend from X_1 (never back-door admissible).

Testing existence. If $\mathbf{Z}_0$ is not admissible, none is: a *one-shot* test.

2.3.3.7 Recap

- The adjustment criterion characterizes all valid adjustment sets.
- Condition (i): $\mathbf{Z} \cap \mathcal{F} = \emptyset$.
- Condition (ii): $\mathbf{Z}$ blocks all proper non-causal paths.
- Testing and construction are polynomial.

2.3.4 Listing all admissible sets (optional)

Motivation. Enumerate all admissible sets with polynomial delay, to choose by cost, variance or other preferences.

2.3.4.1 Motivation

Choice may depend on availability, cost, variance, privacy or fairness. Example: $A \to X$, $A \to B \to Y$, $X \to C \to Y$: $\{A, B\}$, $\{A\}$, $\{B\}$ are admissible; which if A is a sensitive "genetic condition" and B is "headache"?

2.3.4.2 Challenge: output size

The number of solutions can be exponential: with k parallel paths $X \to W_{i,1} \to W_{i,2} \to Y$ there are 3^k separators. Total running time is the wrong yardstick.

Definition (Polynomial delay)

A listing algorithm runs with *polynomial delay* if there is a constant d such that it spends at most $O(n^d)$ time (1) between the start and either producing the first output or halting with no output, and (2) between producing one output and either producing the next or halting.

2.3.4.3 Theoretical tool: bounded separators

Lemma 2.10 (Inclusion and restriction)

Given $\mathcal{G}$ and disjoint $\mathbf{X}, \mathbf{Y}$, and $\mathbf{I} \subseteq \mathbf{R} \subseteq \mathbf{V}$: if any set $\mathbf{Z}$ with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$ separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{G}$, then

$$\mathbf{Z}_0 = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I}) \cap \mathbf{R}$$

also separates them.

Proof [Supplement]. Moralization: $\mathbf{S}$ d-separates $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}$ iff it separates them in $(\mathcal{G}_A)^m$, $A = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{S})$; for ancestral $A \subseteq A'$, $(\mathcal{G}_A)^m$ is the subgraph of $(\mathcal{G}_{A'})^m$ induced by A (parents of A lie in A).

Let $A = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I})$. Then $\mathbf{I} \subseteq \mathbf{Z}_0 \subseteq A$, so $An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z}_0) = A$. If $\mathbf{Z}_0$ fails, some path π in $(\mathcal{G}_A)^m$ from $\mathbf{X}$ to $\mathbf{Y}$ has inner nodes in $A \setminus \mathbf{R}$, hence outside $\mathbf{Z}$. With $A' = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z}) \supseteq A$, π is a path in $(\mathcal{G}_{A'})^m$ avoiding $\mathbf{Z}$, contradicting that $\mathbf{Z}$ separates. ■

$\mathbf{I}$: "include", $\mathbf{R}$: "restrict to". Testing $\mathbf{Z}_0$ in $O(n+m)$ decides existence: `exist-sep`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$. With $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$ this is the explicit construction above.

2.3.4.4 Algorithm

`list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$ outputs all separators with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$. With $\mathcal{F}^+ = \mathbf{X} \cup \mathbf{Y} \cup \mathcal{F}$, start from $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$ in $\mathcal{G}'$; every output then satisfies (ii) and (i).

Algorithm 1 `list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$

```
if exist-sep(G, X, Y, I, R) then                          ▷ O(n + m)
    if I = R then output I
    else
        pick W ∈ R \ I
        list-seps(G, X, Y, I ∪ {W}, R)                    ▷ include W
        list-seps(G, X, Y, I, R \ {W})                    ▷ exclude W
```

Example 2.9 (Running example)

$A \to X$, $A \to B \to Y$, $D \to X$, $D \leftrightarrow Y$, $X \to C \to Y$.

1. $\mathcal{F}^+ = \{C, X, Y\}$.
2. $\mathbf{I} = \emptyset$, $\mathbf{R} = \{A, B, D\}$; remove $X \to C$.
3. Branch on D: excluding D is pruned ($X \leftarrow D \leftrightarrow Y$ unblockable). Including D, branch on A, then B: leaves $\{A, B, D\}$, $\{A, D\}$, $\{B, D\}$ (outputs) and $\{D\}$ (pruned: $X \leftarrow A \to B \to Y$ open).

The admissible sets are $\{A, B, D\}, \{A, D\}, \{B, D\}$.

2.3.4.5 Analysis

- A node is expanded only if `exist-sep` holds, so every expanded node has an output leaf below it.
- Each reached leaf outputs $\mathbf{Z} = \mathbf{I} = \mathbf{R}$, satisfying (ii) and, as $\mathbf{R}$ excludes $\mathcal{F}^+$, (i).
- Depth $\leq k = n - |\mathcal{F}^+| < n$, so $O(n)$ calls between consecutive outputs.
- Each call costs $O(n + m)$, so the delay is $O(n(n + m))$.

Total time may be exponential (unavoidable), but outputs come with polynomial delay, and the algorithm halts in polynomial time if there is none.

2.3.5 Connection with conditional ignorability

Motivation. The graphical criterion justifies the key potential-outcome assumption.

The PO framework assumes *conditional ignorability*,

$$\mathbf{Y}_{\mathbf{x}} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{Z},$$

which gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$ and the adjustment formula. The adjustment criterion *justifies* it: it says when such a $\mathbf{Z}$ exists given a model of reality.

2.3.6 Summary

- Adjustment is the most popular identification technique in the data sciences.

- The back-door/adjustment criterion gives a principled validity condition.
- Admissible sets can be found systematically and efficiently from the graph.
- IPW and variants then estimate the effect from finite samples.

Chapter 3. Do-Calculus and Algorithmic Identification

3.1 Do-calculus

Motivation. Do-calculus (Pearl 1995) gives three complete graphical rules for reducing $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ to an expression in $P(\mathbf{v})$ whenever this is possible.

3.1.1 Three examples via the truncated factorization

Motivation. Eliminate the $\mathbf{U}$-terms of a semi-Markovian model by hand.

Theorem 3.1 (Generalized truncated factorization)

In a semi-Markovian model $\mathcal{M}$, the distribution generated by $\mathrm{do}(\mathbf{X} = \mathbf{x})$ is

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}),$$
$$\mathbf{V}' = \mathbf{V} \setminus \mathbf{X},$$

and for $\mathbf{Y} \subseteq \mathbf{V} \setminus \mathbf{X}$, $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v} \setminus (\mathbf{x} \cup \mathbf{y})} \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$.

Proof [Supplement]. Fix $\mathbf{u}$. In $\mathcal{M}_{\mathbf{x}}$ the mechanisms of $\mathbf{X}$ become constants and the others are unchanged, so $P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x}), \mathbf{u}) = \prod_{V_i \in \mathbf{V}'} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$, each factor the indicator of $f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i$ (acyclicity: solve in topological order). $P(\mathbf{u})$ is unchanged; average over $\mathbf{u}$. ■

Identification succeeds when the right-hand side can be rewritten without $\mathbf{U}$.

3.1.1.1 Back-door graph

$Z \to X$, $Z \to Y$, $X \to Y$, independent U_Z, U_X, U_Y. In $\mathcal{M}_x$: $Z \leftarrow f_Z(u_Z)$, $X \leftarrow x$, $Y \leftarrow f_Y(x, z, u_Y)$.

$$\begin{aligned} P(\mathbf{v}' \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, z, u_Y) P(\mathbf{u}) \\ &= \Big(\sum_{u_Z} P(z \mid u_Z) P(u_Z)\Big)\Big(\sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y)\Big)\Big(\sum_{u_X} P(u_X)\Big) \\ &= P(z) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y \mid x, z) = P(z) P(y \mid x, z), \end{aligned}$$

using $U_Y \perp\!\!\!\perp \{X, Z\}$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z) P(z)$.

3.1.1.2 Bow graph

$X \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Y \leftarrow f_Y(x, u, u_Y)$.

$$P(y \mid \mathrm{do}(x)) = \sum_{u} P(u) \sum_{u_Y} P(y \mid x, u, u_Y) P(u_Y) \sum_{u_X} P(u_X) = \sum_{u} P(y \mid x, u) P(u).$$

$P(u)$ and $P(y \mid x, u)$ are not functions of $P(x, y)$, so U cannot be eliminated (observationally u is weighted by $P(u \mid x)$ instead).

3.1.1.3 Front-door graph

$X \to Z \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Z \leftarrow f_Z(x, u_Z)$, $Y \leftarrow f_Y(z, u, u_Y)$.

$$\begin{aligned}
P(z, y \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid x, u_Z) P(y \mid z, u, u_Y) P(\mathbf{u}) \\
&= \Big(\sum_{u_Z} P(z \mid x, u_Z) P(u_Z)\Big)\Big(\sum_{u, u_Y} P(y \mid z, u, u_Y) P(u, u_Y)\Big) \\
&= P(z \mid x) \sum_{u} P(y \mid z, u) \sum_{x'} P(u \mid x') P(x') \\
&= P(z \mid x) \sum_{u, x'} P(y \mid z, u, x') P(u \mid x') P(x') && (Y \perp\!\!\!\perp X \mid Z, U) \\
&= P(z \mid x) \sum_{u, x'} P(y \mid z, u, x') P(u \mid x', z) P(x') && (U \perp\!\!\!\perp Z \mid X) \\
&= P(z \mid x) \sum_{x'} \sum_{u} P(y, u \mid z, x') P(x') \\
&= P(z \mid x) \sum_{x'} P(y \mid z, x') P(x').
\end{aligned}$$

Therefore the **front-door formula**

$$P(y \mid \mathrm{do}(x)) = \sum_{z} P(z \mid x) \sum_{x'} P(y \mid z, x') P(x').$$

3.1.2 Identification approaches

Motivation. State the syntactic goal that do-calculus serves.

Goal: reduce $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ to terms estimable from $P(\mathbf{v})$, either via a $\mathbf{U}$-free truncated factorization (as above) or by rewriting Q into a *do-free* expression with equality-preserving rules, never mentioning $\mathbf{U}$.

3.1.3 Do-calculus

Motivation. Derive the three rules from three insights.

3.1.3.1 Prelude: independence in a submodel

$\mathrm{do}(\mathbf{X} = \mathbf{x})$ replaces each f_X by a constant: graphically $\mathcal{G}_{\overline{\mathbf{X}}}$, with $\mathbf{X}$ conditioned on. Hence

$$(\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}) \text{ under } \mathrm{do}(\mathbf{x}) \quad \Longleftarrow \quad (\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C}, \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}},$$

equivalently $(\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C})$ in $\mathcal{G} \setminus \mathbf{X}$ when $\mathbf{X}$ is disjoint from $\mathbf{A}, \mathbf{B}, \mathbf{C}$. E.g. in $A \to X \to D$ the intervention severs A from X and fixed X blocks every path through it.

3.1.3.2 Insight 1: adding/removing observations

Graph: $Z \to X \to Y$, $X \leftrightarrow Y$ (via U). Then

$$P(y, z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, u_Y, u) P(\mathbf{u}) = P(z) \sum_{u} P(y \mid x, u) P(u).$$

Marginalizing, $P(z \mid \mathrm{do}(x)) = P(z)\sum_u P(u)\sum_y P(y \mid x,u) = P(z)$, so

$$P(y \mid \mathrm{do}(x), z) = \frac{P(y,z \mid \mathrm{do}(x))}{P(z \mid \mathrm{do}(x))} = \sum_u P(y \mid x,u)P(u) = \sum_z P(y,z \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x)).$$

The reason is $(Z \perp_d Y \mid X)_{\mathcal{G}_{\overline{X}}}$. In $\mathcal{G}$, $Z \not\perp\!\!\!\perp Y$ (via $Z \to X \to Y$) and $Z \not\perp\!\!\!\perp Y \mid X$ (collider on $Z \to X \leftrightarrow Y$); in $\mathcal{M}_x$ the edge $Z \to X$ is cut, and $P(y \mid \mathrm{do}(x), z) = P_x(y \mid z)$ is an ordinary conditional there.

3.1.3.3 Insight 2: exchanging action and observation

Graph: $Z \to X$, $Z \to Y$, $X \to Y$. Given Z, Y is associated with X only through causal paths, so *seeing* $X = x$ equals *doing* $X = x$:

$$(Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}} \implies P(y \mid \mathrm{do}(x), z) = P(y \mid x, z).$$

Check: $P(y,z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z)P(y \mid x,z,u_Y)P(\mathbf{u}) = P(z)P(y \mid x,z)$ and $P(z \mid \mathrm{do}(x)) = \sum_y P(z)P(y \mid x,z) = P(z)$, so $P(y \mid \mathrm{do}(x),z) = P(z)P(y \mid x,z)/P(z) = P(y \mid x,z)$.

3.1.3.4 Insight 3: adding/removing actions

Graph: $X \to Y$, $Z \leftrightarrow Y$ (latent U_{zy}), $Z \leftrightarrow X$ (latent U_{zx}). With no causal path from X to Z,

$$(Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \implies P(z \mid \mathrm{do}(x)) = P(z).$$

Check: $P(z \mid \mathrm{do}(x)) = \sum_y \sum_{u_{zy},u_{zx}} P(z \mid u_{zy},u_{zx})P(y \mid x,u_{zy})P(u_{zy},u_{zx}) = \sum_{u_{zy},u_{zx}} P(z \mid u_{zy},u_{zx})P(u_{zy},u_{zx}) = P(z)$.

3.1.3.5 The rules

Notation: $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$ is $\mathcal{G}$ with edges into $\mathbf{X}$ and out of $\mathbf{Z}$ removed.

Theorem 3.2 (Rules of do-calculus; Pearl 1995)

For any do-distribution induced by a causal model $\mathcal{M}$ with diagram $\mathcal{G}$, and disjoint $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}$:

Rule 1 (adding/removing observations) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Rule 2 (action/observation exchange) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$.

Rule 3 (adding/removing actions) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\overline{\mathbf{Z(W)}}}}$, where $\mathbf{Z(W)} = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Proof [Supplement]. Work in $\mathcal{M}_{\mathbf{x}}$ with diagram $\mathcal{G}_{\overline{\mathbf{X}}}$; by the generalized truncated factorization its distribution (latents as explicit nodes) is Markov relative to $\mathcal{G}_{\overline{\mathbf{X}}}$, so d-separation there implies conditional independence in $P(\cdot \mid \mathrm{do}(\mathbf{x}))$.

Rule 1. $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$ gives $\mathbf{Y} \perp\!\!\!\perp \mathbf{Z} \mid \mathbf{W}$ in $P(\cdot \mid \mathrm{do}(\mathbf{x}))$ ($\mathbf{X}$ constant), which is the claim.

Augmented graph. Add to $\mathcal{G}_{\overline{\mathbf{X}}}$ a parent F_Z of each $Z \in \mathbf{Z}$: $F_Z = \text{idle}$ keeps the mechanism of Z, $F_Z = z'$ replaces it by z'. Then $P(\cdot \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}))$ is the law given $F_{\mathbf{Z}} = \mathbf{z}$ and $P(\cdot \mid \mathrm{do}(\mathbf{x}))$ the law given $F_{\mathbf{Z}} = \text{idle}$; the augmented model is Markov relative to $\mathcal{G}^F$.

Rule 2. On $\mathbf{Z} = \mathbf{z}$ the regimes $F_{\mathbf{Z}} = \mathbf{z}$ and $F_{\mathbf{Z}} = \text{idle}$ differ only in the mechanism of $\mathbf{Z}$, so it suffices that $\mathbf{Y} \perp\!\!\!\perp F_{\mathbf{Z}} \mid \mathbf{X}, \mathbf{Z}, \mathbf{W}$ in $\mathcal{G}^F$. A path from F_Z enters Z via $F_Z \to Z$; continuing out of Z, Z is a conditioned non-collider and the path is blocked; continuing into Z, Z is a conditioned collider. So the open paths are $F_Z \to Z \leftarrow \cdots$, open given $(\mathbf{X}, \mathbf{Z}, \mathbf{W})$ iff the corresponding path from Z to $\mathbf{Y}$ is open in $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$. Hence $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$ implies $(F_{\mathbf{Z}} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{Z}, \mathbf{W})_{\mathcal{G}^F}$.

Rule 3. Now $\mathbf{Z}$ is unobserved and we need $\mathbf{Y} \perp\!\!\!\perp F_{\mathbf{Z}} \mid \mathbf{X}, \mathbf{W}$ in $\mathcal{G}^F$. A path from F_Z continues out of Z (unconditioned non-collider) or into Z (collider, open only if $Z \in An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$). For $Z \in \mathbf{Z}(\mathbf{W})$ the second kind is blocked and the first is open given $(\mathbf{X}, \mathbf{W})$ exactly when the corresponding path is open in a graph where Z has no incoming edges, i.e. $\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z}(\mathbf{W})}}$. For $Z \in \mathbf{Z} \setminus \mathbf{Z}(\mathbf{W})$, separation in $\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z}(\mathbf{W})}}$ blocks the paths through incoming edges too. So $(F_{\mathbf{Z}} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}^F}$ and both regimes give the same $P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{w})$. ■

With $\mathbf{W} = \emptyset$: Rule 1 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, Rule 2 $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$, Rule 3 $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{XZ}}}}$.

Simplified form. Let $\mathcal{H} = \mathcal{G} \setminus \mathbf{X}$ and $Q = P_{\mathbf{x}}$. Then

$$
\begin{aligned}
&\text{R1: } Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}},\\
&\text{R2: } Q(\mathbf{y} \mid \text{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\underline{\mathbf{Z}}}},\\
&\text{R3: } Q(\mathbf{y} \mid \text{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\overline{\mathbf{Z}(\mathbf{W})}}},\ \mathbf{Z}(\mathbf{W}) = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{H}}.
\end{aligned}
$$

Each rule concerns one fixed submodel $\mathcal{M}_{\mathbf{x}}$.

3.1.3.6 Understanding Rule 2

Since $P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{x}, \mathbf{z}, \mathbf{w})$, Rule 2 asks whether model and submodel give the same conditional, e.g. $P(\mathbf{y} \mid \text{do}(\mathbf{z}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{z}, \mathbf{w})$.

Regime indicator. Augment $\mathcal{G}$ ($W \to Z$, $W \to Y$, $Z \to Y$) with $F_Z \to Z$:

$$
f'_Z = \begin{cases} f_Z(W, U_Z) & \text{if } F_Z = 0,\\ z & \text{if } F_Z = 1. \end{cases}
$$

Then $P(y \mid \text{do}(z), z, w) = P(y \mid z, w)$ reads $P(y \mid z, w, F_Z = 1) = P(y \mid z, w, F_Z = 0)$, i.e. $(F_Z \perp\!\!\!\perp Y \mid Z, W)$ in $\mathcal{G}'$. With Z given, every d-connecting path is $F_Z \to Z \leftarrow \cdots Y$, so we check back-door paths of Z, i.e. Z versus Y in $\mathcal{G}_{\underline{Z}}$.

3.1.3.7 Understanding Rule 3

"Changing a function" subsumes "changing it to a constant", so

$$
P(\mathbf{y} \mid \text{do}(\mathbf{x}), \text{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{w}) \iff P(\mathbf{y} \mid \text{do}(\mathbf{x}), F_Z = 1, \mathbf{w}) = P(\mathbf{y} \mid \text{do}(\mathbf{x}), F_Z = 0, \mathbf{w}),
$$

which holds if $F_{\mathbf{Z}} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ in $\mathcal{G}'$. Paths $F_Z \to Z \to \cdots$ are always feasible; $F_Z \to Z \leftarrow \cdots$ only if $Z \in An(\mathbf{W})$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. So edges into Z can be cut exactly for $Z \in \mathbf{Z}(\mathbf{W})$: condition $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z}(\mathbf{W})}}}$.

Example 3.1 (Why Z(W) is needed)

Graph: $T \to Z \to Y$, $Z \to W$. Is $P(t \mid \text{do}(z), w) = P(t \mid w)$? In $\mathcal{G}_{\overline{Z}}$, T is isolated, so a naive check says yes. But $Z \in An(W)$, so $\mathbf{Z}(\mathbf{W}) = \emptyset$ and in $\mathcal{G}$ itself $T \to Z$ remains: the condition

fails. Indeed, in almost every compatible model,

$$P(t \mid \mathrm{do}(z), w) = \frac{P(t, w \mid \mathrm{do}(z))}{P(w \mid \mathrm{do}(z))} = \frac{\sum_y P(t)P(w \mid z)P(y \mid z)}{\sum_{y,t} P(t)P(w \mid z)P(y \mid z)}$$
$$= \frac{P(t)P(w \mid z)}{P(w \mid z)} = P(t) \neq P(t \mid w),$$

since observationally W informs on T through Z.

3.1.4 Do-calculus and identification

Motivation. Completeness, the front-door derivation, and tools for non-identifiability.

Theorem 3.3 (Completeness; Huang and Valtorta 2006, Shpitser and Pearl 2006)

$Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable from $P(\mathbf{v})$ and $\mathcal{G}$ iff there is a finite sequence of applications of the rules of do-calculus and the probability axioms that reduces Q to a do-free expression.

Proof [Supplement]. *If:* each rule and axiom preserves equality in every model compatible with $\mathcal{G}$, so a do-free reduction writes Q as a functional of $P(\mathbf{v})$. *Only if:* for identifiable Q the ID algorithm (Chapter 3, Section 3.2) returns an expression, each step being a do-calculus rule or probability axiom (Huang and Valtorta 2006; Shpitser and Pearl 2006); on failure it exhibits a hedge yielding two models with equal $P(\mathbf{v})$ and different Q (construction not reproduced). ■

Syntactic goal: re-express Q without do.

3.1.4.1 Front-door by do-calculus

Smoking $S \to$ Tar $T \to$ Cancer C, with $S \leftrightarrow C$.

$$\begin{aligned}
P(c \mid \mathrm{do}(s)) &= \sum_t P(c \mid \mathrm{do}(s), t)P(t \mid \mathrm{do}(s)) && \text{probability axioms} \\
&= \sum_t P(c \mid \mathrm{do}(s), \mathrm{do}(t))P(t \mid \mathrm{do}(s)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\overline{S}\underline{T}}} \\
&= \sum_t P(c \mid \mathrm{do}(t))P(t \mid \mathrm{do}(s)) && \text{Rule 3: } (S \perp_d C \mid T)_{\mathcal{G}_{\overline{T}\,\overline{S}}} \\
&= \sum_t P(c \mid \mathrm{do}(t))P(t \mid s) && \text{Rule 2: } (T \perp_d S)_{\mathcal{G}_{\underline{S}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', \mathrm{do}(t))P(s' \mid \mathrm{do}(t)) && \text{probability axioms} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t)P(s' \mid \mathrm{do}(t)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\underline{T}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t)P(s') && \text{Rule 3: } (T \perp_d S)_{\mathcal{G}_{\overline{T}}}.
\end{aligned}$$

Justifications: in $\mathcal{G}_{\overline{S}\underline{T}}$ only $S \leftrightarrow C$, $S \to T$ remain and S is given; in $\mathcal{G}_{\overline{T}\,\overline{S}}$, S connects only via $S \leftrightarrow C$, not opened by T; in $\mathcal{G}_{\underline{S}}$, $S \leftrightarrow C \leftarrow T$ is blocked at the collider C; in $\mathcal{G}_{\underline{T}}$, $T \leftarrow S \leftrightarrow C$ is blocked by S; in $\mathcal{G}_{\overline{T}}$, T reaches S only via the collider C.

3.1.4.2 Proving non-identifiability

Example 3.2

Bow graph $X \to Y$, $X \leftrightarrow Y$, U a fair coin, $\mathcal{F}$: $X \leftarrow U$, $Y \leftarrow f_Y(X,U)$. Two models agree with $\mathcal{F}$ when $U = X$ (the only observed case) and differ off the diagonal:

$$\mathcal{F}_x^{(1)}: \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 0 & \text{otherwise} \end{cases},$$

$$\mathcal{F}_x^{(2)}: \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 1 & \text{otherwise.} \end{cases}$$

Under $\mathrm{do}(X = x)$, U and X need not match:

$$P^{(i)}(Y = 1 \mid \mathrm{do}(x)) = \sum_u P^{(i)}(y \mid x, u)P(u) = P(y \mid x)P(x) + \nvDash_{i=2}\,(1 - P(x)).$$

Same $\mathcal{G}$ and $P(\mathbf{v})$, different effects whenever $P(x) < 1$.

Lemma 3.4 (Graph–subgraph; Tian and Pearl 2002)

If $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is not identifiable in $\mathcal{G}$, it is not identifiable in any graph obtained by adding a directed or bidirected edge to $\mathcal{G}$. Conversely, if Q is identifiable in $\mathcal{G}$, it remains identifiable after removing edges.

Proof [Supplement]. Every model compatible with $\mathcal{G}$ is compatible with $\mathcal{G}^+ = \mathcal{G}$ plus an edge (functions may ignore a new parent; a new latent may have no effect). So two $\mathcal{G}$-models with equal $P(\mathbf{v})$ and different Q also witness non-identifiability in $\mathcal{G}^+$. The converse is the contrapositive for $\mathcal{G}^- \subseteq \mathcal{G}$. ■

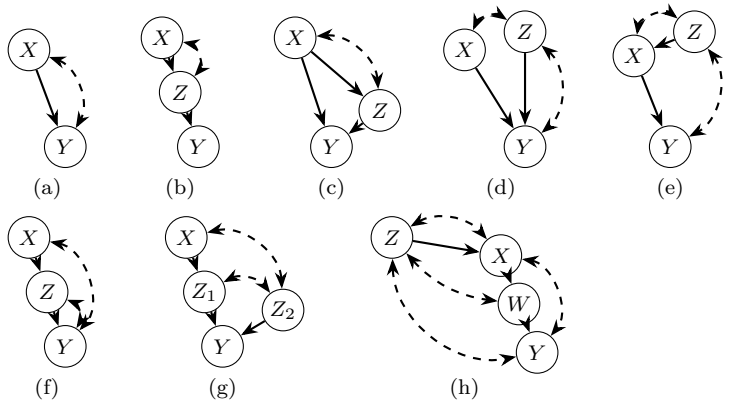
Theorem 3.5 (Non-identifiability criterion, singleton; Tian and Pearl 2002)

If there is a bidirected path connecting X to any of its children in $\mathcal{G}$, then $P(\mathbf{v}' \mid \mathrm{do}(x))$ is not identifiable from $P(\mathbf{v})$ and $\mathcal{G}$.

Proof [Supplement]. *Base case.* $X \to Y$, $X \leftrightarrow Y$, binary U, X, Y. Model 1: $X = U$, $Y = X$; model 2: $X = U$, $Y = U$. Both give $P(x, y) = P(X = Y = x) = P(U = x)$, but $P(Y = 1 \mid \mathrm{do}(X = 1))$ is 1 versus $P(U = 1)$. *General case.* For $X \leftrightarrow W_1 \leftrightarrow \cdots \leftrightarrow W_k \leftrightarrow Y$ with Y a child, let every latent on the path equal a common binary U and every W_j copy U, other variables independent noise. The two models still agree on $P(\mathbf{v})$ and disagree on $P(\mathbf{v}' \mid \mathrm{do}(x))$; by the graph–subgraph lemma the remaining edges of $\mathcal{G}$ preserve this (Tian and Pearl 2002). ■

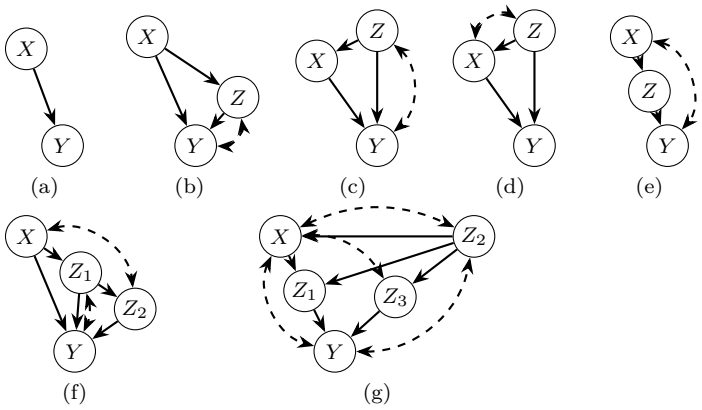
3.1.4.3 Catalogue of graphs

Graphs in which $P(y \mid \mathrm{do}(x))$ is not identifiable (Pearl 2000, Fig. 3.9):



(a) is the bow graph, while (b) and (c) contain the bow $X \to Z$, $X \leftrightarrow Z$; in (d)–(h) X is joined to a child by a bidirected path (e.g. (d) $X \leftrightarrow Z \leftrightarrow Y$; (g) $X \leftrightarrow Z_2 \leftrightarrow Z_1$, Z_1 a child of X).

Graphs in which $P(y \mid \mathrm{do}(x))$ is identifiable (Pearl 2000, Fig. 3.8):



E.g. in (b) X has no incoming edges, so $P(y \mid \mathrm{do}(x)) = P(y \mid x)$;

(c) and (d) follow by back-door adjustment for Z, and (e) by the front-door formula.

3.1.5 Back to Simpson's paradox

Motivation. The do-operator dissolves the paradox, which statistics alone cannot.

3.1.5.1 Pearson's shock

Pearson, Lee and Bramley-Moore (1899): mixing heterogeneous groups without "organic" correlation can produce a *spurious* correlation, so correlations always carry an error of unknown size. The same data give opposite conclusions aggregated or stratified; resolving this requires causal reasoning.

Examples. (1) Berkeley admissions (Bickel et al., *Science* 1975): overall 44% of men vs. 30% of women admitted, yet within departments a slight bias favoring women, who applied to more competitive departments. Whether to condition on department depends on the causal model (e.g. latent confounding of department and admission).

(2) "Reverse regression" in salary-discrimination cases: compare salaries of equally qualified men and women, or qualifications of equally paid ones? (Pearl 2009, §4.5; Bhadane et al. 2025.)

3.1.5.2 Dissolving the paradox with the do-operator

The tables of Chapter 1 give

$$P(Y \mid F, X) < P(Y \mid F, \neg X), \qquad P(Y \mid \neg F, X) < P(Y \mid \neg F, \neg X), \qquad P(Y \mid X) > P(Y \mid \neg X).$$

If instead

$$P(Y \mid F, \mathrm{do}(X)) < P(Y \mid F, \mathrm{do}(\neg X)), \qquad P(Y \mid \neg F, \mathrm{do}(X)) < P(Y \mid \neg F, \mathrm{do}(\neg X)),$$

and the action does not affect F (sex), then $P(Y \mid \mathrm{do}(X)) < P(Y \mid \mathrm{do}(\neg X))$, since by Rule 3 ($F$ not a descendant of X)

$$P(y \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f),$$

and both sides average the stratum effects with the *same* weights $P(f)$.

3.1.5.3 The paradox stems from causal interpretation

Surprise arises only about efficacy, not prediction: when two causal models fit the same data but recommend differently, no statistical criterion picks the right table.

Theorem 3.6 (Sure-thing principle; Savage 1972, Pearl 2000 Thm. 6.1.1)

An action X that increases the probability of an event Y in each subpopulation must also increase the probability of Y in the population as a whole, provided that the action does not change the distribution of the subpopulations.

Proof [Supplement]. Let F index the subpopulations, $P(f \mid \mathrm{do}(x)) = P(f \mid \mathrm{do}(\neg x)) = P(f)$. Then

$$P(Y \mid \mathrm{do}(x)) = \sum_f P(Y \mid \mathrm{do}(x), f) P(f) > \sum_f P(Y \mid \mathrm{do}(\neg x), f) P(f) = P(Y \mid \mathrm{do}(\neg x)),$$

term by term with identical weights, so no reversal is possible. ■

Reversal is possible in the calculus of proportions, impossible in the calculus of causation. Applications: adjustment, salary discrimination, optimal treatment design, fairness in ML.

3.1.5.4 Appendix: the Berkeley data

Department	Men applied	Men admitted	Women applied	Women admitted
A	825	62%	108	82%
B	560	56%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%
Total	2691	44%	1835	30%

Women applied disproportionately to competitive departments (C–F). Which graph?

- $\mathcal{G}_1$: Sex → Dept → Admit, Sex → Admit (Dept a mediator). The total effect is the aggregate rate; adjusting for Dept gives the controlled direct effect (discrimination within departments), the indirect effect being sex's influence on department choice.

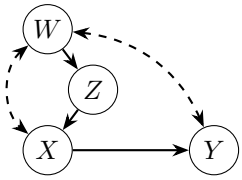
- $\mathcal{G}_2$: $U \to$ Dept, $U \to$ Sex, Dept $\to$ Admit, Sex $\to$ Admit, U latent (Dept a confounder). Dept blocks Sex $\leftarrow U \to$ Dept $\to$ Admit, so $P(\text{admit} \mid \text{do(sex)}) = \sum_d P(\text{admit} \mid \text{sex}, d)P(d)$, showing no significant discrimination.

Same data and adjustment, different causal interpretations.

3.1.6 Summary and food for thought

- Do-calculus characterizes policy evaluation for atomic interventions syntactically.
- Non-parametric confounding and identification are essentially solved.
- Simpson's paradox is mathematised and dissolved.
- Applications: social and health sciences, statistics, ML, AI.

Food for thought: the napkin graph. $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$. Is $P(y \mid \text{do}(x))$ identifiable?



Yes, though neither back-door nor front-door applies: intervening on Z makes X unconfounded given W,

$$P(y \mid \text{do}(x)) = \frac{\sum_w P(x, y \mid z, w)P(w)}{\sum_w P(x \mid z, w)P(w)} \qquad \text{(for any } z\text{)}.$$

(Rule 3: $P(y \mid \text{do}(x)) = P(y \mid \text{do}(x), \text{do}(z))$; back-door through W: $P(x, y \mid \text{do}(z)) = \sum_w P(x, y \mid z, w)P(w)$; Rule 2: $P(y \mid \text{do}(z), \text{do}(x)) = P(y \mid \text{do}(z), x)$, since in $\mathcal{G}_{\overline{Z}\underline{X}}$ every X–Y path goes through W, cut from X except via Z. Section 3.2 does this systematically.)

3.2 Algorithmic identification

Motivation. Decide identifiability of $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ mechanically, without searching for do-calculus steps.

Roadmap. (1) Factorize the distribution along the causal diagram (c-components, c-factors).

(2) Operations that identify particular factors.

(3) Write the target in factors and identify each one.

3.2.1 Factorizing observational distributions

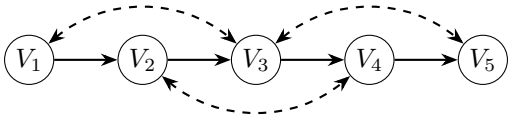
Motivation. Group variables sharing latent confounders; the joint factorizes over these groups.

3.2.1.1 Confounded components

Definition (C-component)

When V_i and V_j share an unobservable parent U, they are in the same *confounded component* (c-component). Closing under chaining: two variables are in the same c-component iff they are connected by a path of bidirected edges.

Running example: $V_1 \to V_2 \to V_3 \to V_4 \to V_5$ with U_1: $V_1 \leftrightarrow V_3$, U_2: $V_3 \leftrightarrow V_5$, U_3: $V_2 \leftrightarrow V_4$.



The relation is an equivalence (read it off the bidirected edges alone), so it *partitions* the observed variables: here $\{V_1, V_3, V_5\}$ and $\{V_2, V_4\}$.

3.2.1.2 Markovian case

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

For $Z \to X$, $Z \to Y$, $X \to Y$:

$$\begin{aligned} P(z, x, y) &= \sum_{u_z} P(z \mid u_z) P(u_z) \sum_{u_x} P(x \mid z, u_x) P(u_x) \sum_{u_y} P(y \mid x, z, u_y) P(u_y) \\ &= P(z) P(x \mid z) P(y \mid x, z). \end{aligned}$$

Each canonical factor $P(v_i \mid \mathbf{pa}_i)$ is computable from $P(\mathbf{v})$:

$$P(v_i \mid \mathbf{pa}_i) = \frac{\sum_{\mathbf{v} \setminus (\{v_i\} \cup \mathbf{pa}_i)} P(\mathbf{v})}{\sum_{\mathbf{v} \setminus \mathbf{pa}_i} P(\mathbf{v})},$$

$$\text{e.g. } P(z) = \sum_{x,y} P(\mathbf{v}), \; P(x \mid z) = \frac{\sum_y P(\mathbf{v})}{\sum_{x,y} P(\mathbf{v})}, \; P(y \mid x, z) = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})}.$$

3.2.1.3 Semi-Markovian case

With only U_1 ($V_1 \leftrightarrow V_3$) and U_2 ($V_3 \leftrightarrow V_5$):

$$P(\mathbf{v}) = P(v_2 \mid v_1) P(v_4 \mid v_3) \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2).$$

Adding U_3 ($V_2 \leftrightarrow V_4$):

$$\begin{aligned} P(\mathbf{v}) = &\Big[\sum_{u_3} P(v_2 \mid v_1, u_3) P(v_4 \mid v_3, u_3) P(u_3) \Big] \\ &\cdot \Big[\sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2) \Big]. \end{aligned}$$

The sums separate because different c-components share no latent variable.

3.2.1.4 C-factors

For $C \subseteq \mathbf{V}$ define

$$Q[C](\mathbf{c}, \mathbf{pa}_C) = \sum_{\mathbf{u}(C)} P(\mathbf{u}(C)) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i),$$
$$\mathbf{U}(C) = \bigcup_{V_i \in C} \mathbf{U}_i.$$

Then

$$P(\mathbf{v}) = Q[V_2, V_4](v_2, v_4, v_1, v_3) \cdot Q[V_1, V_3, V_5](v_1, v_3, v_5, v_2, v_4),$$

abbreviated $P(\mathbf{v}) = Q[V_1, V_3, V_5]\, Q[V_2, V_4]$, with no $\mathbf{U}$. $Q[\mathbf{V}] = P(\mathbf{v})$ and $Q[\emptyset] = 1$.

3.2.1.5 C-factors are causal effects

For $C \subseteq \mathbf{V}$, by the truncated factorization,

$$P(\mathbf{c} \mid \mathrm{do}(\mathbf{v} \setminus \mathbf{c})) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = \sum_{\mathbf{u}(C)} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}(C)) = Q[C],$$

since latents that are not parents of C sum out. *C-factors are interventional distributions.* E.g.

$$Q[V_1, V_3, V_5](v_1, v_3, v_5, v_2, v_4) = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2)$$

is the law of (V_1, V_3, V_5) under $\mathrm{do}(V_2 = v_2, V_4 = v_4)$.

3.2.2 C-factor algebra

Motivation. Two operations drive identification: marginalizing to ancestral sets and factorizing into c-components.

3.2.2.1 Marginalizing variables in c-factors

In $Q[V_1, V_3, V_5]$ no member is a parent of another, so each appears in one term and

$$\sum_{v_3} Q[V_1, V_3, V_5] = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) \Big(\sum_{v_3} P(v_3 \mid v_2, u_1, u_2) \Big) P(v_5 \mid v_4, u_2) = Q[V_1, V_5].$$

In $Q[V_1, V_2, V_3]$, V_1 is a parent of V_2:

$$\sum_{v_1} Q[V_1, V_2, V_3] = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) \sum_{v_1} P(v_1 \mid u_1) P(v_2 \mid v_1, u_3),$$

whereas

$$Q[V_2, V_3](v_2, v_3, v_1) = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) P(v_2 \mid v_1, u_3)$$

keeps v_1 fixed. The coupling of v_1 with u_1 (shared with V_3) gives $\sum_{v_1} Q[V_1, V_2, V_3] \neq Q[V_2, V_3]$ in general: only variables not feeding the rest can be summed out.

Lemma 3.7 (Ancestral reduction)

Let $W \subseteq C \subseteq \mathbf{V}$. If W is ancestral in $\mathcal{G}[C]$ (it contains all its ancestors within the subgraph induced by C), then

$$Q[W] = \sum_{\mathbf{c}\setminus\mathbf{w}} Q[C].$$

Proof [Supplement]. Sum out $C \setminus W$ in reverse topological order. When $V \in C \setminus W$ is summed it has no remaining child: a child in W would make $V \in W$ (W ancestral), and children in $C \setminus W$ are already summed. So v appears in $Q[C] = \sum_{\mathbf{u}} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$ only in its own factor, which sums to one; latents attached only to summed variables integrate out. What remains is $Q[W]$. ■

Example: in $\mathcal{G}[C]$, $C = \{V_1, V_2, V_3\}$ ($V_1 \to V_2 \to V_3$), $\{V_1, V_2\}$ and $\{V_1\}$ are ancestral, $\{V_2, V_3\}$ is not.

3.2.2.2 C-component factorization

Lemma 3.8 (Tian 2002)

For any $H \subseteq \mathbf{V}$, let $H_1, \ldots, H_k$ be the c-components of $\mathcal{G}[H]$. Then

$$Q[H] = \prod_{j=1}^{k} Q[H_j].$$

Moreover, let $V^{(1)} < \cdots < V^{(|H|)}$ be a topological order of $\mathcal{G}[H]$ and $H^{\le i}$ the variables of H up to and including $V^{(i)}$. Then each factor is computable from $Q[H]$:

$$Q[H_j] = \prod_{V^{(i)} \in H_j} \frac{Q[H^{\le i}]}{Q[H^{\le i-1}]}, \qquad Q[H^{\le i}] = \sum_{\mathbf{h}\setminus\mathbf{h}^{\le i}} Q[H].$$

Proof [Supplement]. A latent with children in two c-components would join them, so the latents split by c-component. In $Q[H] = \sum_{\mathbf{u}} \prod_{V_i \in H} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$ the latents of H_j appear only in factors of H_j and the groups are independent under $P(\mathbf{u})$, so the sum factorizes into $\prod_j Q[H_j]$. For the second part, $H^{\le i}$ is ancestral in $\mathcal{G}[H]$, so $Q[H^{\le i}] = \sum_{h\setminus h^{\le i}} Q[H]$ by ancestral reduction; the ratios $Q[H^{\le i}]/Q[H^{\le i-1}]$ are functions of $Q[H]$, and their product over H_j reproduces $Q[H_j]$ (each latent group integrated once). ■

Example 3.3 (Factorizing $P(\mathbf{v})$ by c-components)

Graph: $Z_2 \to X$, $Z_2 \to Z_1$, $Z_2 \to Z_3$, $X \to Z_1$, $Z_1 \to Y$, $Z_3 \to Y$, $Z_2 \leftrightarrow X$, $X \leftrightarrow Z_3$, $X \leftrightarrow Y$, $Z_2 \leftrightarrow Y$. C-components $\{Z_2, Z_3, X, Y\}$ and $\{Z_1\}$:

$$Q[\mathbf{V}] = Q[Z_2, Z_3, X, Y]\, Q[Z_1],$$

$$P(\mathbf{v}) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, z_3, x, y)) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, x)),$$

the last step by Rule 3 (only the parents Z_2, X of Z_1 matter).

Example 3.4 (Running example)

With $H = \mathbf{V}$ and the order $V_1 < \cdots < V_5$, $Q[\mathbf{V}] = Q[V_1, V_3, V_5]Q[V_2, V_4]$ and

$$\begin{aligned} Q[V_1, V_3, V_5] &= \frac{Q[V_1]}{Q[\emptyset]} \cdot \frac{Q[V_1, V_2, V_3]}{Q[V_1, V_2]} \cdot \frac{Q[V_1, \ldots, V_5]}{Q[V_1, \ldots, V_4]} \\ &= \frac{\sum_{v_2,\ldots,v_5} P(\mathbf{v})}{\sum_{\mathbf{v}} P(\mathbf{v})} \cdot \frac{\sum_{v_4,v_5} P(\mathbf{v})}{\sum_{v_3,v_4,v_5} P(\mathbf{v})} \cdot \frac{P(\mathbf{v})}{\sum_{v_5} P(\mathbf{v})} \\ &= \frac{P(v_1)}{1} \cdot \frac{P(v_1, v_2, v_3)}{P(v_1, v_2)} \cdot \frac{P(\mathbf{v})}{P(v_1, \ldots, v_4)} = P(v_1)P(v_3 \mid v_1, v_2)P(v_5 \mid v_1, \ldots, v_4), \\ Q[V_2, V_4] &= \frac{Q[V_1, V_2]}{Q[V_1]} \cdot \frac{Q[V_1, \ldots, V_4]}{Q[V_1, V_2, V_3]} = P(v_2 \mid v_1)P(v_4 \mid v_1, v_2, v_3). \end{aligned}$$

$\mathcal{G}[\{V_2, V_4\}]$ has no directed edges, so $\{V_2\}$ and $\{V_4\}$ are ancestral there:

$$Q[V_2] = \sum_{v_4} Q[V_2, V_4] = P(v_2 \mid v_1) \sum_{v_4} P(v_4 \mid v_1, v_2, v_3) = P(v_2 \mid v_1),$$
$$Q[V_4] = \sum_{v_2} P(v_2 \mid v_1)P(v_4 \mid v_1, v_2, v_3).$$

3.2.2.3 Visualization

From $Q[\mathbf{V}]$ one moves down a lattice of c-factors by summing out a "last" variable ($Q[\mathbf{V}] \to Q[V_{1234}] \to Q[V_{123}] \to Q[V_{12}] \to Q[V_1]$) or splitting along c-components ($Q[\mathbf{V}] \to Q[V_{135}] \cdot Q[V_{24}]$, $Q[V_{1234}] \to Q[V_{13}] \cdot Q[V_{24}]$, $Q[V_{123}] \to Q[V_{13}] \cdot Q[V_2]$). From $Q[V_{135}]$ every subset is reachable by summation (no directed edges in $\mathcal{G}[\{V_1, V_3, V_5\}]$), and $Q[V_{15}] = Q[V_1]Q[V_5]$ since V_1, V_5 are not bidirected-connected within $\{V_1, V_5\}$.

Summary of the algebra.

1. *Reduce to an ancestral set:* $Q[W] = \sum_{\mathbf{c} \setminus \mathbf{w}} Q[C]$ if W is ancestral in $\mathcal{G}[C]$.
2. *Factorize into c-components:* $Q[H] = \prod_j Q[H_j]$, with $H_1, \ldots, H_k$ the c-components of $\mathcal{G}[H]$, each computable from $Q[H]$.

Together they yield an if-and-only-if condition for identifiability.

3.2.3 Causal queries in terms of c-factors: examples

Motivation. Rederive the back-door, front-door and napkin formulas with c-factor algebra.

3.2.3.1 Back-door

$Z \to X, Z \to Y, X \to Y$ (Markovian): $P(\mathbf{v}) = Q[X]Q[Y]Q[Z]$. In $\mathcal{G}_{\overline{X}}$, $P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$ with

$$Q[Y] = \frac{Q[X, Y, Z]}{Q[X, Z]} = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})} = P(y \mid x, z), \qquad Q[Z] = \sum_{x,y} Q[X, Y, Z] = P(z),$$

so $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid z, x)P(z)$.

3.2.3.2 Front-door

$X \to Z \to Y$, $X \leftrightarrow Y$: $P(\mathbf{v}) = Q[X,Y] \cdot Q[Z]$ with (order $X < Z < Y$)

$$Q[X,Y] = \frac{Q[X]}{Q[\emptyset]} \cdot \frac{Q[X,Z,Y]}{Q[X,Z]} = P(x)P(y \mid x,z), \qquad Q[Z] = \frac{Q[X,Z]}{Q[X]} = P(z \mid x).$$

Then

$$P(y \mid \mathrm{do}(x)) = \sum_z P(y,z \mid \mathrm{do}(x)) = \sum_z Q[Y,Z] = \sum_z Q[Y]Q[Z],$$

and $\{Y\}$ is ancestral in $\mathcal{G}[\{X,Y\}]$, so $Q[Y] = \sum_{x'} Q[X,Y] = \sum_{x'} P(x')P(y \mid x',z)$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(z \mid x) \sum_{x'} P(x')P(y \mid x',z)$.

3.2.3.3 Napkin

$W_1 \to W_2 \to X \to Y$, $W_1 \leftrightarrow X$, $W_1 \leftrightarrow Y$. Then $P(\mathbf{v}) = Q[W_1,X,Y]\,Q[W_2]$ and

$$P(y \mid \mathrm{do}(x)) = \sum_{w_1,w_2} Q[W_1,W_2,Y] = Q[Y],$$

as $\{Y\}$ is ancestral in $\mathcal{G}[\{W_1,W_2,Y\}]$. With the order $W_1 < W_2 < X < Y$,

$$Q[W_1,X,Y] = \frac{Q[W_1]}{Q[\emptyset]} \cdot \frac{Q[W_1,W_2,X]}{Q[W_1,W_2]} \cdot \frac{Q[W_1,W_2,X,Y]}{Q[W_1,W_2,X]} = P(w_1)P(x \mid w_1,w_2)P(y \mid w_1,w_2,x).$$

$\{X,Y\}$ is ancestral in $\mathcal{G}[\{W_1,X,Y\}]$, so $Q[X,Y] = \sum_{w_1} P(w_1)P(x \mid w_1,w_2)P(y \mid w_1,w_2,x)$; in $\mathcal{G}[\{X,Y\}]$, X and Y are not bidirected-connected, so $Q[X,Y] = Q[X]Q[Y]$ and

$$P(y \mid \mathrm{do}(x)) = Q[Y] = \frac{Q[X,Y]}{Q[X]} = \frac{Q[X,Y]}{\sum_y Q[X,Y]} = \frac{\sum_{w_1} P(w_1)P(x \mid w_1,w_2)P(y \mid w_1,w_2,x)}{\sum_{w_1} P(w_1)P(x \mid w_1,w_2)}.$$

The same result by do-calculus.

$$\begin{aligned}
P(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x,w_1,w_2)) && \text{Rule 3}\\
&= P(y \mid x, \mathrm{do}(w_1,w_2)) && \text{Rule 2}\\
&= \frac{P(y,x \mid \mathrm{do}(w_1,w_2))}{P(x \mid \mathrm{do}(w_1,w_2))} = \frac{P(y,x \mid \mathrm{do}(w_2))}{P(x \mid \mathrm{do}(w_2))} && \text{Rule 3}\\
&= \frac{\sum_{w_1} P(y,x \mid \mathrm{do}(w_2),w_1)P(w_1 \mid \mathrm{do}(w_2))}{\sum_{w_1} P(x \mid \mathrm{do}(w_2),w_1)P(w_1 \mid \mathrm{do}(w_2))} &&\\
&= \frac{\sum_{w_1} P(y,x \mid w_2,w_1)P(w_1)}{\sum_{w_1} P(x \mid w_2,w_1)P(w_1)} && \text{Rules 2 and 3.}
\end{aligned}$$

3.2.3.4 Overview of the three derivations

Back-door	Front-door	Napkin
$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X,Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[W_1,X,Y]\,Q[W_2]$
	$Q[X,Y] \xrightarrow{\text{sum}} Q[Y]$	$Q[W_1,X,Y] \xrightarrow{\text{sum}} Q[X,Y] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]$
$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = Q[Y]$

3.2.4 A general identification algorithm

Motivation. Combine the two operations into a sound and complete procedure.

3.2.4.1 Reducing the query to c-factors

By Rule 3, every variable that is not an ancestor of $\mathbf{Y}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$ can be intervened on without changing the query. Let $C = An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$. Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c}\setminus\mathbf{y}} P(\mathbf{c} \mid \mathrm{do}(\mathbf{v}\setminus\mathbf{c})) = \sum_{\mathbf{v}\setminus(\mathbf{x}\cup\mathbf{y})} Q[\mathbf{V}\setminus\mathbf{X}] = \sum_{\mathbf{c}\setminus\mathbf{y}} Q[C].$$

If $\mathcal{G}[C]$ has c-components $C_1, \dots, C_k$, then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c}\setminus\mathbf{y}} \prod_{i=1}^{k} Q[C_i],$$

while $P(\mathbf{v}) = \prod_j Q[T_j]$ over the c-components T_j of $\mathcal{G}$, each $Q[T_j]$ computable. Each C_i lies in a single T_j; it remains to decide whether $Q[C_i]$ is computable from $Q[T_j]$.

3.2.4.2 The Identify procedure

Algorithm 2 IDENTIFY$(C, T, Q[T], \mathcal{G})$ (decides whether $Q[C]$ is computable from $Q[T]$, $C \subseteq T$)

$A \leftarrow An(C)_{\mathcal{G}[T]}$
if $A = C$ **then** ▷ C is ancestral in T
  return $Q[C] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q[T]$ ▷ marginalize
else if $A = T$ **then** ▷ nothing to marginalize, unable to factorize
  return FAIL
else
  $A_i \leftarrow$ the c-component of $\mathcal{G}[A]$ containing C
  obtain $Q[A] = \sum_{\mathbf{t}\setminus\mathbf{a}} Q[T]$, then $Q[A_i]$ by c-component factorization ▷ marginalize, then factorize
  return IDENTIFY$(C, A_i, Q[A_i], \mathcal{G})$

The three cases. (1) $A = C$: marginalize, $Q[C] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q[T]$.

(2) $A = T$: T is its own ancestral closure and a single c-component, so it cannot shrink: fail.

(3) $C \subsetneq A \subsetneq T$: marginalize to A, factorize, keep the c-component $A_i \supseteq C$ (C is a c-component of $\mathcal{G}[C]$), and recurse. $T \supsetneq A_i \supsetneq \cdots$, so at most $|T|$ steps.

Theorem 3.9 (Completeness; Tian 2002)

$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable from $\mathcal{G}$ and $P(\mathbf{v})$ if and only if each c-factor $Q[C_i]$ is identified by IDENTIFY$(C_i, T_i, Q[T_i], \mathcal{G})$, where T_i is the c-component of $\mathcal{G}$ containing C_i.

Proof [Supplement]. *If:* each call returns an expression in $Q[T_i]$, identified from $P(\mathbf{v})$ by c-component factorization ($H = \mathbf{V}$); combining the factors by the generalized truncated factorization gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$. *Only if:* a failure $A = T$, $C_i \subsetneq T$ contains a hedge, from which Shpitser and Pearl (2006) build two models agreeing on $P(\mathbf{v})$ but not on the query (construction not reproduced). ■

Huang and Valtorta (2006) and Shpitser and Pearl (2006) proved completeness of the equivalent ID algorithm and, through it, of do-calculus.

Example 3.5 (Bow graph fails)

$X \to Y$, $X \leftrightarrow Y$, query $P(y \mid \text{do}(x))$: $C = An(Y)_{\mathcal{G}_{\overline{X}}} = \{Y\}$, $T = \{X, Y\}$, and $A = An(Y)_{\mathcal{G}[T]} = \{X, Y\} = T$: fail, matching Chapter 2.

Example 3.6 (Front-door succeeds)

$X \to Z \to Y$, $X \leftrightarrow Y$: $C = \{Z, Y\}$ with c-components $\{Z\}$, $\{Y\}$. $Q[Z]$ is a c-component of $\mathcal{G}$. For $Q[Y]$, $T = \{X, Y\}$ and $A = An(Y)_{\mathcal{G}[T]} = \{Y\} = C$ ($Z \notin T$), so $Q[Y] = \sum_x Q[X, Y]$.

Chapter 4. Linear Structural Causal Models

Motivation. When can a structural coefficient of a linear Gaussian SCM be computed from the covariance matrix, and when does regression return it?

Outline:

(1) linear SCMs and Wright's path-tracing rules;

(2) when regression finds causal effects (single-door, back-door);

(3) algorithmic identification (polynomial systems, IVs, conditional IVs, instrumental sets); appendices on complexity, reversed regression and Gröbner bases.

4.1 Linear structural causal models

Motivation. Specialize SCMs to linear mechanisms and read covariances off the graph.

4.1.1 Definition and conventions

A linear SCM writes each variable as a linear function of its direct causes,

$$Y = \sum_i \lambda_{x_i y} X_i + \mathcal{E}_y,$$

with $\lambda_{x_i y}$ the change in Y per unit change of X_i and all error covariances specified. WLOG $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 1$ and $\mathcal{E} \sim \mathcal{N}$, so P is fixed by $\Sigma = (\sigma_{ij})$. Only f_i changes from the non-parametric case:

$$V_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i) \quad \rightsquigarrow \quad V_i \leftarrow \sum_{V_j \in Pa_i} \lambda_{ji} V_j + \mathcal{E}_i.$$

(1) λ_{ji} is the *structural coefficient*;

(2) U_i is renamed $\mathcal{E}_i$;

(3) all λ_{ji} give all causal effects.

Example 4.1

$X \leftarrow f_X(U_X)$, $W \leftarrow f_W(U_W)$, $Y \leftarrow f_Y(X, W, U_Y)$ becomes $X \leftarrow \mathcal{E}_x$, $W \leftarrow \mathcal{E}_w$, $Y \leftarrow \lambda_{xy} X + \lambda_{wy} W + \mathcal{E}_y$; coefficients on the edges specify the model.

Latent confounding. $\epsilon_{ij} = \mathbb{E}[\mathcal{E}_i \mathcal{E}_j]$ labels a bidirected edge $V_i \leftrightarrow V_j$; here it means *error covariance*, not necessarily a shared latent.

Interventions. For $X \xrightarrow{\lambda} Y$,

$$\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \mathbb{E}[\lambda x + \mathcal{E}_y] = \lambda x + \mathbb{E}[\mathcal{E}_y] = \lambda x.$$

4.1.2 The identification problem

Given (i) a graph $\mathcal{G}$ and (ii) data $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$ on all observed variables, but (iii) neither the λ (the targets) nor the ϵ.

4.1.3 Connecting observed and unobserved quantities

For $X \xrightarrow{\lambda} Y$ without confounding,

$$\sigma_{xy} = \mathbb{E}[XY] = \mathbb{E}[X(\lambda X + \mathcal{E}_y)] = \lambda \underbrace{\mathbb{E}[XX]}_{1} + \underbrace{\mathbb{E}[X\mathcal{E}_y]}_{0} = \lambda.$$

With $X \leftrightarrow Y$ (ϵ_{xy}), $X = \mathcal{E}_x$ and

$$\sigma_{xy} = \lambda\mathbb{E}[XX] + \mathbb{E}[X\mathcal{E}_y] = \lambda + \mathbb{E}[\mathcal{E}_x\mathcal{E}_y] = \lambda + \epsilon_{xy}.$$

A curious property. For $X \xrightarrow{\lambda_{xz}} Z \xrightarrow{\lambda_{zy}} Y$:

$$\begin{aligned}\sigma_{xy} &= \mathbb{E}[X(\lambda_{zy}Z + \mathcal{E}_y)] = \lambda_{zy}\mathbb{E}[XZ] + \mathbb{E}[X\mathcal{E}_y] = \lambda_{zy}\mathbb{E}[X(\lambda_{xz}X + \mathcal{E}_z)] \\ &= \lambda_{zy}\lambda_{xz}\mathbb{E}[XX] + \lambda_{zy}\mathbb{E}[X\mathcal{E}_z] = \lambda_{zy}\lambda_{xz} + \lambda_{zy}\mathbb{E}[\mathcal{E}_x\mathcal{E}_z].\end{aligned}$$

Without confounding $\sigma_{xy} = \lambda_{zy}\lambda_{xz}$; with $X \leftrightarrow Z$ (ϵ_{xz}), $\sigma_{xy} = \lambda_{zy}\lambda_{xz} + \lambda_{zy}\epsilon_{xz}$.

4.1.4 Treks and Wright's rule

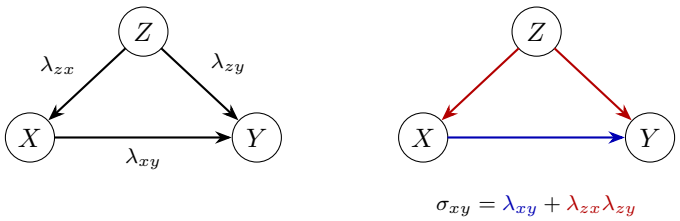


Figure. *Wright's rule: σ_{xy} sums the coefficient products over the treks, here the direct path (blue) and the trek through Z (red).*

For $Z \to X \to Y$, $Z \to Y$: $\sigma_{xy} = \lambda_{xy} + \lambda_{zx}\lambda_{zy}$, from $X \to Y$ and $X \leftarrow Z \to Y$.

Theorem 4.1 (Wright's rules, 1921)

For an acyclic linear SCM with standardized variables, σ_{xy} equals the sum, over all *open* paths between X and Y (non-self-intersecting paths without colliding arrowheads, i.e. treks; a bidirected edge counts as one step contributing ϵ), of the product of the path coefficients along the path. In particular $\sigma_{xy} = 0$ when X and Y are d-separated by $\emptyset$.

Proof [Supplement]. Write $\mathbf{V} = \Lambda^\top\mathbf{V} + \boldsymbol{\epsilon}$ (Λ strictly triangular in a topological order) and $\Omega = \mathrm{Cov}(\boldsymbol{\epsilon})$ (off-diagonal entries: bidirected coefficients). Then $\mathbf{V} = (I - \Lambda^\top)^{-1}\boldsymbol{\epsilon}$, $\Sigma = (I - \Lambda)^{-\top}\Omega(I - \Lambda)^{-1}$, and since Λ is nilpotent, $(I - \Lambda)^{-1} = \sum_{k\geq 0}\Lambda^k$ with $(\Lambda^k)_{ij}$ the sum over length-k directed paths $i \to j$ of coefficient products. Hence

$$\sigma_{xy} = \sum_{a,b}\Big(\sum_{\text{directed } a\to x}\prod\lambda\Big)\,\omega_{ab}\,\Big(\sum_{\text{directed } b\to y}\prod\lambda\Big),$$

a sum over pairs of directed paths to x and y joined at a common source $a = b$ (weight ω_{aa}) or by $a \leftrightarrow b$ (weight ω_{ab}): exactly the treks. With standardized variables, the error variance at a common-cause source combines with its other contributions to the variance 1, giving Wright's factor 1 at the top of each trek. If $\emptyset$ d-separates X, Y there are no treks, so $\sigma_{xy} = 0$. ■

So if $X \xrightarrow{\lambda} Y$, $\sigma_{xy} = \lambda +$ (other paths), and $\sigma_{xy} = \lambda$ if X, Y are d-separated in $\mathcal{G}_\lambda$ (edge removed).

Example 4.2

$W \to X$, $W \to Y$, $Z \to W$, $Z \to X$, $X \to Y$: $\sigma_{xy} = \lambda_{xy} + \lambda_{wx}\lambda_{wy} + \lambda_{zx}\lambda_{wz}\lambda_{wy}$ (paths $X \to Y$, $X \leftarrow W \to Y$, $X \leftarrow Z \to W \to Y$).

Example 4.3

$W \to X$, $W \to V \to Y$, $X \to Z \to Y$, $X \leftrightarrow Z$: $\sigma_{xy} = (\lambda_{xz} + \epsilon_{xz})\lambda_{zy} + \lambda_{wx}\lambda_{wv}\lambda_{vy}$.

4.2 Linear regression and causal effects

Motivation. What does regression compute, and when is it a structural coefficient?

4.2.1 The medical researcher

Does a new drug help? Doctors nationwide will follow the recommendation.

Step 1. data on drug dose X and biomarker (antibodies) Y.

Step 2. regression $Y = \beta X + e$ gives $\beta = 0.375$; the drug is authorized.

Step 3. given to everyone, the drug shows slope -1: it hurts, and not for lack of data. Why did regression (red line) fail, and can the causal effect (green line) be recovered?

4.2.2 The implicit assumption: no confounding

A causal reading of β presumes $X \leftarrow \mathcal{E}_x$, $Y \leftarrow \lambda_{xy}X + \mathcal{E}_y$, $\mathcal{E}_x \perp\!\!\!\perp \mathcal{E}_y$; then $\sigma_{xy} = \lambda_{xy}\mathbb{E}[XX] + \mathbb{E}[X\mathcal{E}_y] = \lambda_{xy}$.

The ground-truth model. If confounding cannot be excluded, $\mathcal{E}_x, \mathcal{E}_y$ are *correlated* ($X \leftrightarrow Y$) and regression is biased:

$$\sigma_{xy} = \lambda_{xy}\mathbb{E}[XX] + \mathbb{E}[\mathcal{E}_x\mathcal{E}_y] = \lambda_{xy} + \epsilon_{xy};$$

λ_{xy} is not identifiable, which explains 0.375 versus -1.

What does regression compute? $\mathbb{E}[(Y - \beta X)^2] = 1 - 2\beta\sigma_{xy} + \beta^2$ has derivative $-2\sigma_{xy} + 2\beta$, so $\beta = \sigma_{xy}$: *for standardized variables the regression coefficient is the covariance.*

4.2.3 Regression equation versus structural equation

- **Regression equation** $Y = \beta X + e$, $e \perp\!\!\!\perp X$ by construction; $\beta = \sigma_{xy} =: r_{yx}$. *No* causal claim.
- **Structural equation** $Y \leftarrow \lambda X + \mathcal{E}_y$ ($X \to Y$): claims $\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \lambda x$, testable and falsifiable.

Confusing them is the "confusion of the century". Causal claims from regression implicitly assume $X_1 \xrightarrow{b_1} Y \xleftarrow{b_2} X_2$ with regressors unconfounded with Y; then OLS of $Y = \beta_1 X_1 + \beta_2 X_2 + e$ gives $\beta_1 = b_1$, $\beta_2 = b_2$.

Warning: make sure you have the right graph. If $X_1 \xleftarrow{b_1} Y \xrightarrow{b_2} X_2$, simple regressions give $r_{yx_1} = b_1$, $r_{yx_2} = b_2$, but multiple regression gives

$$\beta_1 = \frac{b_1(1 - b_2^2)}{1 - b_1^2 b_2^2}, \qquad \beta_2 = \frac{b_2(1 - b_1^2)}{1 - b_1^2 b_2^2}$$

(Appendix 4.6).

4.2.4 Testing conditional independence with regression

For Gaussian variables, $(X \perp\!\!\!\perp Y \mid Z)$ iff $r_{yx\cdot z} = 0$, the coefficient α of X in $Y = \alpha X + \beta Z$.

4.2.5 Single-door criterion: direct effects

Model $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $Z \leftrightarrow Y$ (ϵ_{zy}); regress $Y = \alpha X + \beta Z + e$:

$$\mathbb{E}[(Y - \alpha X - \beta Z)^2] = 1 - 2\alpha\sigma_{xy} - 2\beta\sigma_{yz} + 2\alpha\beta\sigma_{xz} + \alpha^2 + \beta^2,$$

$$\frac{\partial}{\partial\alpha} = 2\alpha - 2\sigma_{xy} + 2\beta\sigma_{xz}, \qquad \frac{\partial}{\partial\beta} = 2\beta - 2\sigma_{yz} + 2\alpha\sigma_{xz}.$$

With Wright $(\sigma_{xz} = \lambda_{zx},\ \sigma_{xy} = \lambda_{xy} + \lambda_{zx}\epsilon_{zy},\ \sigma_{yz} = \lambda_{xy}\lambda_{zx} + \epsilon_{zy})$:

$$\begin{aligned} \lambda_{xy} + \lambda_{zx}\epsilon_{zy} &= \alpha + \beta\lambda_{zx}, \\ \lambda_{xy}\lambda_{zx} + \epsilon_{zy} &= \alpha\lambda_{zx} + \beta, \end{aligned} \qquad \Longrightarrow \qquad \alpha = \lambda_{xy}, \quad \beta = \epsilon_{zy}$$

(unique solution when $\lambda_{zx}^2 \neq 1$). The coefficient of X is causal although Z is confounded with Y.

Theorem 4.2 (Single-door criterion; Pearl 2000)

Let λ be the coefficient of $X \to Y$ and $\mathcal{G}_\lambda$ the diagram with $X \to Y$ removed. λ is identifiable if there is a set $\mathbf{Z}$ such that (i) $\mathbf{Z}$ contains no descendant of Y, and (ii) $\mathbf{Z}$ d-separates X from Y in $\mathcal{G}_\lambda$. Moreover $\lambda = r_{yx\cdot\mathbf{Z}}$.

Proof [Supplement]. The coefficient of X in the regression of Y on $X, \mathbf{Z}$ is $r_{yx\cdot\mathbf{Z}}$. Write $Y = \lambda X + R$ (R: other parents and error). Then $r_{yx\cdot\mathbf{Z}} = \lambda + r_{rx\cdot\mathbf{Z}}$, and $r_{rx\cdot\mathbf{Z}} = 0$ iff R, X are uncorrelated given $\mathbf{Z}$. Treks between X and R are paths between X and Y in $\mathcal{G}_\lambda$; condition (ii) blocks them given $\mathbf{Z}$, and (i) prevents opening a path through a descendant of Y (which would involve R). So $\lambda = r_{yx\cdot\mathbf{Z}}$. ■

Example 4.4

$A \to B$, $A \to Y$, $B \to Y$: $\lambda_{ay} = r_{ya\cdot b}$, $\lambda_{by} = r_{yb\cdot a}$.

Example 4.5

$X \to Y$, $X \to W$, $Y \to W$, $Y \to Z$, $W \to Z$, $X \leftrightarrow Z$, $Y \leftrightarrow Z$: $\lambda_{xy} = r_{yx}$ (without $X \to Y$ every path meets a collider), and $\lambda_{wz} = r_{zw\cdot yx}$ ($\{Y, X\}$ blocks $W \leftarrow Y \to Z$, $W \leftarrow Y \leftrightarrow Z$, $W \leftarrow X \leftrightarrow Z$).

4.2.6 Back-door criterion: total effects

Theorem 4.3 (Back-door, linear; Pearl 2000)

The total effect of X on Y is identifiable if there is $\mathbf{Z}$ with no descendant of X that d-separates X from Y in $\mathcal{G}_{\underline{X}}$; it equals $r_{yx\cdot\mathbf{Z}}$.

Proof [Supplement]. In a linear Gaussian model $\mathbb{E}[Y \mid \mathrm{do}(x)]$ is linear in x with slope the total effect τ. By back-door adjustment (Chapter 2), $\mathbb{E}[Y \mid \mathrm{do}(x)] = \sum_{\mathbf{z}} \mathbb{E}[Y \mid x, \mathbf{z}]P(\mathbf{z})$, and $\mathbb{E}[Y \mid x, \mathbf{z}] = \alpha + r_{yx\cdot\mathbf{Z}}x + \boldsymbol{\beta}^{\mathsf{T}}\mathbf{z}$, so $\mathbb{E}[Y \mid \mathrm{do}(x)] = \alpha + r_{yx\cdot\mathbf{Z}}x + \boldsymbol{\beta}^{\mathsf{T}}\mathbb{E}\mathbf{Z}$ and $\tau = r_{yx\cdot\mathbf{Z}}$. ■

Example 4.6

$W \to X$, $W \leftrightarrow X$, $W \to Y$, $X \to Z \to Y$, $X \to Y$, $Z \leftrightarrow Y$. The total effect $\lambda_{xz}\lambda_{zy} + \lambda_{xy}$ equals $r_{yx\cdot w}$ (back-door with $\{W\}$); λ_{xy} alone is not obtained by also regressing on Z, because $Z \leftrightarrow Y$.

4.3 Algorithmic identification methods

Motivation. Identification means solving the polynomial system linking Σ to (λ, ϵ).

4.3.1 The equations of linear identification

For $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $Z \leftrightarrow Y$:

$$\Sigma = \underbrace{\begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}}_{\text{computable}} = \underbrace{\begin{bmatrix} 1 & \lambda_{xy} + \lambda_{zx}\epsilon_{zy} & \lambda_{zx} \\ \lambda_{xy} + \lambda_{zx}\epsilon_{zy} & 1 & \lambda_{zx}\lambda_{xy} + \epsilon_{zy} \\ \lambda_{zx} & \lambda_{zx}\lambda_{xy} + \epsilon_{zy} & 1 \end{bmatrix}}_{\text{expressions with unobservables}}.$$

With $\sigma_{xz} = \lambda_{zx}$,

$$\sigma_{xy} = \lambda_{xy} + \sigma_{xz}\epsilon_{zy}, \qquad \sigma_{zy} = \sigma_{xz}\lambda_{xy} + \epsilon_{zy}$$

is a full-rank 2×2 linear system:

$$\lambda_{xy} = \frac{\sigma_{xy} - \sigma_{xz}\sigma_{zy}}{1 - \sigma_{xz}^2}, \qquad \epsilon_{zy} = \frac{\sigma_{zy} - \sigma_{xz}\sigma_{xy}}{1 - \sigma_{xz}^2}.$$

Non-identifiability. $X \to Y$, $X \leftrightarrow Y$: one equation $\sigma_{xy} = \lambda_{xy} + \epsilon_{xy}$, two unknowns.

Finite identifiability. $X \to W$, $X \to Z$, $X \to Y$, each also $\leftrightarrow X$: six equations, six unknowns,

$$\begin{aligned} \sigma_{xw} &= \lambda_{xw} + \epsilon_{xw}, & \sigma_{wz} &= \lambda_{xw}\lambda_{xz} + \lambda_{xz}\epsilon_{xw} + \lambda_{xw}\epsilon_{xz}, \\ \sigma_{xz} &= \lambda_{xz} + \epsilon_{xz}, & \sigma_{wy} &= \lambda_{xw}\lambda_{xy} + \lambda_{xw}\epsilon_{xy} + \lambda_{xy}\epsilon_{xw}, \\ \sigma_{xy} &= \lambda_{xy} + \epsilon_{xy}, & \sigma_{zy} &= \lambda_{xz}\lambda_{xy} + \lambda_{xz}\epsilon_{xy} + \lambda_{xy}\epsilon_{xz}. \end{aligned}$$

Substitute $\epsilon = \sigma - \lambda$: $\sigma_{wz} = \lambda_{xz}\sigma_{xw} + \lambda_{xw}\sigma_{xz} - \lambda_{xw}\lambda_{xz}$ (likewise σ_{wy}, σ_{zy}); the last two give $\lambda_{xw} = (\sigma_{wy} - \lambda_{xy}\sigma_{xw})/(\sigma_{xy} - \lambda_{xy})$, $\lambda_{xz} = (\sigma_{zy} - \lambda_{xy}\sigma_{xz})/(\sigma_{xy} - \lambda_{xy})$; plugging into the first and clearing denominators,

$$0 = (\sigma_{xw}\sigma_{xz} - \sigma_{wz})\lambda_{xy}^2 + 2(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy})\lambda_{xy} + (\sigma_{yw}\sigma_{xz}\sigma_{xy} + \sigma_{yz}\sigma_{xy}\sigma_{xw} - \sigma_{xy}^2\sigma_{wz} - \sigma_{yz}\sigma_{yw}),$$

$$\lambda_{xy} = \frac{-(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy}) \pm \sqrt{(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy})^2 - (\sigma_{xw}\sigma_{xz} - \sigma_{wz})\,c}}{\sigma_{xw}\sigma_{xz} - \sigma_{wz}},$$

c the constant term: at most two values, *finite identifiability*.

The three cases. *Identifiable*: one consistent value of λ_{xy}; *finitely identifiable*: finitely many; *not identifiable*: infinitely many.

Example 4.7

$W \to Z$, $W \to X$, $Z \to X$, $Z \leftrightarrow X$, $X \to Y$, $W \leftrightarrow Y$:

$$\begin{aligned} \sigma_{wz} &= \lambda_{wz}, & \sigma_{wy} &= (\lambda_{wx} + \lambda_{wz}\lambda_{zx})\lambda_{xy} + \epsilon_{wy}, \\ \sigma_{wx} &= \lambda_{wx} + \lambda_{wz}\lambda_{zx}, & \sigma_{zy} &= (\lambda_{zx} + \epsilon_{zx} + \lambda_{wz}\lambda_{wx})\lambda_{xy} + \lambda_{wz}\epsilon_{wy}, \\ \sigma_{zx} &= \lambda_{zx} + \epsilon_{zx} + \lambda_{wz}\lambda_{wx}, & \sigma_{xy} &= (\lambda_{wz}\lambda_{zx} + \lambda_{wx})\epsilon_{wy} + \lambda_{xy}. \end{aligned}$$

Here $\lambda_{xy} = (\sigma_{zy} - \sigma_{wz}\sigma_{wy})/(\sigma_{zx} - \sigma_{wz}\sigma_{wx})$ (Z is an IV conditional on W). ID algorithms find solvable subsystems efficiently; generic computer algebra is doubly exponential.

4.4 From instrumental variables to instrumental sets

Motivation. Variables affecting Y only through X yield solvable linear subsystems.

4.4.1 Instrumental variables

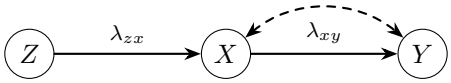


Figure. *The IV graph:* $\sigma_{zy} = \lambda_{zx}\lambda_{xy}$ *and* $\sigma_{zx} = \lambda_{zx}$ *give* $\lambda_{xy} = \sigma_{zy}/\sigma_{zx}$ *despite* $X \leftrightarrow Y$.

Add an independent doctor's recommendation Z: $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $X \leftrightarrow Y$. (Non-parametrically $P(y \mid \mathrm{do}(x))$ is *not* identifiable; linearity helps.)

Definition (Instrumental variable; Pearl 2000, p. 248)

Z is an IV for λ_{xy} if Z is d-separated from Y in $\mathcal{G}_{\lambda_{xy}}$ and Z is not d-separated from X in $\mathcal{G}_{\lambda_{xy}}$.

By Wright, $\sigma_{zx} = \lambda_{zx}$, $\sigma_{zy} = \lambda_{zx}\lambda_{xy}$, so

$$\lambda_{xy} = \frac{\sigma_{zy}}{\sigma_{zx}} = \frac{r_{zy}}{r_{zx}}.$$

Two-stage least squares. $X = \alpha Z + e$ gives $\alpha = \sigma_{zx}$; regressing Y on $\hat{X} = \sigma_{zx}Z$ gives $\beta = \sigma_{zy}/\sigma_{zx}$ ($\mathrm{Cov}(Y, \hat{X}) = \sigma_{zx}\sigma_{zy}$, $\mathrm{Var}(\hat{X}) = \sigma_{zx}^2$).

4.4.2 Conditional instrumental variables

Definition (Conditional IV; Brito and Pearl 2002)

Z is a conditional IV given $\mathbf{W}$ for λ_{xy} if (i) $\mathbf{W}$ contains only non-descendants of Y;
(ii) $\mathbf{W}$ d-separates Z from Y in $\mathcal{G}_{\lambda_{xy}}$;
(iii) $\mathbf{W}$ does not d-separate Z from X in $\mathcal{G}_{\lambda_{xy}}$.

Then $\lambda_{xy} = r_{yz \cdot \mathbf{W}}/r_{xz \cdot \mathbf{W}}$. Example: $W \to Z \to X \to Y$, $X \leftrightarrow Y$, $W \leftrightarrow Y$: Z is no IV ($Z \leftarrow W \leftrightarrow Y$ is open) but a conditional IV given W.

4.4.3 Instrumental sets

$Z_i \to X_j$ $(i, j \in \{1, 2\})$, $X_j \to Y$, $X_j \leftrightarrow Y$. No single IV exists (every path from Z_i to Y passes through both X_j). With $\sigma_{z_i x_j} = \lambda_{z_i x_j}$,

$$\sigma_{z_1 y} = \sigma_{z_1 x_1}\lambda_{x_1 y} + \sigma_{z_1 x_2}\lambda_{x_2 y}, \qquad \sigma_{z_2 y} = \sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y},$$

solvable by Cramer's rule if the matrix is non-singular.

When does the system fail? If $Z_1 \to Z_2 \to \{X_1, X_2\}$, paths from Z_1 to Y are those from Z_2 times $\lambda_{z_1 z_2}$:

$$\lambda_{z_1 z_2}\sigma_{z_2 y} = \lambda_{z_1 z_2}(\sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y}), \qquad \sigma_{z_2 y} = \sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y},$$

proportional equations: degenerate.

Theorem 4.4 (Non-intersecting paths)

The system is non-degenerate (generically) iff there is a full matching between $\{Z_1, Z_2\}$ and $\{X_1, X_2\}$ by vertex-disjoint (non-intersecting) paths.

Proof [Supplement]. The equations $\sigma_{z_i y} = \sum_j \sigma_{z_i x_j}\lambda_{x_j y}$ have a unique solution iff $\det[\sigma_{z_i x_j}] \neq 0$. Each $\sigma_{z_i x_j}$ is a trek sum, and by Lindström–Gessel–Viennot the determinant is a signed sum over vertex-disjoint path systems $\{Z_1, Z_2\} \to \{X_1, X_2\}$: none $\Rightarrow$ all terms cancel, determinant $\equiv 0$; one $\Rightarrow$ a nonzero polynomial, nonzero for generic parameters. ($k \times k$: Brito and Pearl 2002.) ■

In the first graph $Z_1 \to X_1$, $Z_2 \to X_2$ are disjoint; in the degenerate one both paths pass through Z_2. Max-flow finds such paths: sources Z_i, sinks X_j, unit vertex capacities; flow $|\mathbf{Z}|$ certifies a full matching.

4.4.4 Conclusions

1. Linear SCMs are the most common causal models in practice.
2. Know what regression actually computes.
3. A structural equation is not a regression equation, though regression sometimes recovers structural coefficients.
4. The diagram can be leveraged to solve for causal effects in general settings.

4.5 Appendix: Complexity of identification in linear SCMs

Landscape. Single-door, IV and instrumental sets are sufficient but incomplete; the half-trek criterion and auxiliary cutsets/ACID (Chen et al. 2016, Kumor et al. 2019) cover more, not all, identifiable instances.

Limits. Graphical criteria are polynomial but incomplete; Gröbner bases are complete but doubly exponential (a 4-node graph can take over 75 days; García-Puente, Spielvogel and Sullivant 2010; Bardet 2002).

Recent result (Dörfler, van der Zander, Bläser and Liśkiewicz, NeurIPS 2024). The general problem is harder than NP (real-valued unknowns) but lies in PSPACE, hence $2^{\text{poly}(n)}$ time instead of $2^{2^{\text{poly}(n)}}$. Use graphical criteria when they work; otherwise the problem is hard but solvable.

4.6 Appendix: Multiple regression when Y is the cause

$X_1 \xleftarrow{b_1} Y \xrightarrow{b_2} X_2$: $\sigma_{yx_1} = b_1$, $\sigma_{yx_2} = b_2$, $\sigma_{x_1x_2} = b_1b_2$. The normal equations of $Y = \beta_1 X_1 + \beta_2 X_2 + e$ are

$$b_1 = \beta_1 + \beta_2 b_1 b_2, \qquad b_2 = \beta_2 + \beta_1 b_1 b_2.$$

Substituting $\beta_2 = b_2 - \beta_1 b_1 b_2$: $b_1 = \beta_1 + b_1 b_2^2 - \beta_1 b_1^2 b_2^2$, so

$$\beta_1 = \frac{b_1(1 - b_2^2)}{1 - b_1^2 b_2^2}, \qquad \beta_2 = \frac{b_2(1 - b_1^2)}{1 - b_1^2 b_2^2}.$$

4.7 Appendix: The Gröbner basis approach

Identification as polynomial elimination. Covariances are polynomials in λ, ϵ. For $X_1 \xrightarrow{\lambda_{12}} X_2 \xrightarrow{\lambda_{23}} X_3$, $X_2 \leftrightarrow X_3$ (ϵ_{23}):

$$\sigma_{12} = \lambda_{12}, \qquad \sigma_{13} = \lambda_{12}\lambda_{23}, \qquad \sigma_{23} = \lambda_{23} + \epsilon_{23},$$

so $\lambda_{23} = \sigma_{13}/\sigma_{12}$ (X_1 is an instrument), while σ_{23} alone cannot separate λ_{23} from ϵ_{23}. Gröbner bases automate this elimination.

How it works.

1. *Set up:* $\sigma_{ij} -$ (polynomial in λ, ϵ) $= 0$ for every pair (i, j).
2. *Eliminate:* elimination term orders keep only the target λ_{xy} and the σ_{ij}.
3. *Check:* degree 1 in λ_{xy}: generically identifiable; degree d: algebraically d-identifiable (d possible values); λ_{xy} absent: not generically identifiable.

Computational results (García-Puente et al. 2010; all linear SCMs on 3 and 4 nodes):

	3 nodes ($2^6 = 64$)	4 nodes ($2^{12} = 4096$)
Generically identifiable	22	1246
Not generically identifiable	42	2844
Algebraically k-identified	0	6
Solved by single-door / IV	22/22	1093/1246
Only by Gröbner basis	0	153

On 4 nodes, 153 cases need the algebraic method; runtimes range from seconds to over 75 days per graph.

Chapter 5. Partial Identification and Soft Interventions

5.1 Partial identification

Motivation. When $P(y \mid \mathrm{do}(x))$ is not identifiable, we seek the tightest interval $[a(y;x), b(y;x)] \subseteq [0,1]$ consistent with the data and the graph.

5.1.1 The partial identification problem

Motivation. Extend the yes/no identification question to bounds.

When $P(y \mid \mathrm{do}(x))$ is not determined by $P(\mathbf{v})$ and $\mathcal{G}$, we ask for a non-trivial bound

$$P(y \mid \mathrm{do}(x)) \in [a(y;x),\ b(y;x)] \subseteq [0,1].$$

Definition (Partial identifiability)

The effect of X on Y is partially identifiable given $\mathcal{G}$ if $P(\mathbf{V})$ determines a bound $P(y \mid \mathrm{do}(x)) \in [a,b]$ that is a proper subset of $[0,1]$.

Identifiable $\Rightarrow$ partially identifiable ($a = b$), not conversely.

5.1.2 Natural bounds

Motivation. The simplest bounds hold for any confounded pair and cannot be improved without more structure.

Theorem 5.1 (Natural bounds; Manski 1990, Robins 1989)

For an SCM compatible with $X \to Y$, $X \leftrightarrow Y$ (latent U), given $P(x,y)$,

$$a(y;x) = P(y,x) \ \le\ P(y \mid \mathrm{do}(x)) \ \le\ P(y,x) + 1 - P(x) = b(y;x).$$

Proof [Supplement]. With $X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x,u)$, the truncated product gives

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &= \sum_u P(y \mid x,u)P(u) = \sum_u P(y \mid x,u)\big(P(u,x) + P(u) - P(u,x)\big) \\ &= \sum_u P(y,x,u) + \sum_u P(y \mid x,u)\big(P(u) - P(u,x)\big) \\ &= P(y,x) + \sum_u P(y \mid x,u)\big(P(u) - P(u,x)\big). \end{aligned}$$

As $0 \le P(y \mid x,u) \le 1$ and $P(u) - P(u,x) \ge 0$, the last sum lies in $[0, \sum_u (P(u) - P(u,x))] = [0, 1 - P(x)]$. ∎

The width $1 - P(x)$ is the mass of units not receiving x; equivalently $b(y;x) = 1 - P(x,y')$.

Example 5.1

$X, Y, U \in \{0,1\}$, $X \leftarrow U$, $Y \leftarrow X \oplus U$, $P(U=0) = 0.9$. Observationally $Y = 0$: $P(X = 0, Y = 0) = 0.9$, $P(X = 1, Y = 0) = 0.1$. Under $\mathrm{do}(X = x)$, $Y = x \oplus U$:

	$P(Y=1 \mid \mathrm{do}(x))$	$a(Y=1;x)$	$b(Y=1;x)$
$x=0$	0.1	0	0.1
$x=1$	0.9	0	0.9

The true effects lie on the upper bounds.

5.1.2.1 Tightness

Theorem 5.2

For binary X, Y and any $P(x,y)$ there exist SCMs $\mathcal{M}^{(1)}, \mathcal{M}^{(2)}$ with $P^{(1)}(x,y) = P^{(2)}(x,y) = P(x,y)$ and $P^{(1)}(y \mid \mathrm{do}(x)) = a(y;x)$, $P^{(2)}(y \mid \mathrm{do}(x)) = b(y;x)$.

Proof [Supplement]. Take any compatible $\mathcal{M}$ ($X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x,u)$) inducing P, and define, with the same $P(U)$ and $X \leftarrow f_X(u)$,

$$\mathcal{F}^{(1)}: \; Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 0 & \text{otherwise} \end{cases}, \qquad \mathcal{F}^{(2)}: \; Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 1 & \text{otherwise} \end{cases}.$$

Without intervention $X = f_X(U)$, so $Y = f_Y(X,U)$ and $P^{(i)}(x,y) = \sum_u P(x \mid u)P(y \mid x,u)P(u) = P(y,x)$. Under $\mathrm{do}(X=x)$,

$$\begin{aligned} P^{(i)}(Y=1 \mid \mathrm{do}(x)) &= P(Y=1 \mid x, f_X(U)=x)P(f_X(U)=x) \\ &\quad + P^{(i)}(Y=1 \mid x, f_X(U)\neq x)P(f_X(U)\neq x) \\ &= P(x, Y=1) + \mathbb{1}[i=2](1-P(x)). \end{aligned}$$

So $\mathcal{M}^{(1)}$ attains a, $\mathcal{M}^{(2)}$ attains b ($Y=0$ is symmetric). ■

5.1.2.2 Limitations

Natural bounds are tight for the binary bow graph but not when more structure (e.g. an instrument Z) is available; canonical type models and linear programming exploit it.

5.1.3 Canonical type models and LP bounds

Motivation. Replace unknown functional forms by finitely many response types, turning bounding into a linear program.

5.1.3.1 Experiments with non-compliance

Assigned treatment Z (randomized instrument), received treatment X, outcome Y, latent U: $Z \to X \to Y$, $X \leftrightarrow Y$. Given $P(x,y \mid z)$, bound $P(y \mid \mathrm{do}(x))$.

Theorem 5.3 (Natural bounds in the IV model)

$$a(y;x) = \max_z P(y,x \mid z), \qquad b(y;x) = \min_z \left[P(y,x \mid z) + 1 - P(x \mid z)\right].$$

Proof [Supplement]. Fix z. Natural bounds in $\mathcal{M}_z$: $P(y,x \mid \mathrm{do}(z)) \le P(y \mid \mathrm{do}(x,z)) \le P(y,x \mid \mathrm{do}(z)) + 1 - P(x \mid \mathrm{do}(z))$. Exclusion gives $P(y \mid \mathrm{do}(x,z)) = P(y \mid \mathrm{do}(x))$, randomization $P(\cdot \mid \mathrm{do}(z)) = P(\cdot \mid z)$. Take $\max_z$ of the lower and $\min_z$ of the upper bounds. ■

They are not tight in IV models.

5.1.3.2 Optimization perspective

With $\mathcal{M}$ the SCMs compatible with $Z \leftarrow f_Z(U_Z)$, $X \leftarrow f_X(Z,U)$, $Y \leftarrow f_Y(X,U)$, the sharp bounds are

$$a(y;x) = \min_{M \in \mathcal{M}:\, P_M(x,y|z) = P(x,y|z)} P_M(y \mid \mathrm{do}(x)), \qquad b(y;x) = \max_{M \in \mathcal{M}:\, P_M(x,y|z) = P(x,y|z)} P_M(y \mid \mathrm{do}(x)),$$

hard to solve directly since $\mathcal{F}$ and $P(\mathbf{U})$ are unspecified.

5.1.3.3 U as a selector and response variables

For fixed $U = u$, $f_X(\cdot, u)$ is a function $Z \to X$: U *selects* the mapping, so $\mathfrak{X}_U$ is partitioned by the induced function. Binary Z, X give $|\mathfrak{X}_X|^{|\mathfrak{X}_Z|} = 4$ cells:

$$\{u : f_X(0,u) = 0,\ f_X(1,u) = 0\} \quad (X = 0), \qquad \{u : f_X(0,u) = 0,\ f_X(1,u) = 1\} \quad (X = Z),$$
$$\{u : f_X(0,u) = 1,\ f_X(1,u) = 0\} \quad (X = 1 - Z), \qquad \{u : f_X(0,u) = 1,\ f_X(1,u) = 1\} \quad (X = 1)$$

(ternary Z: 2^3). A *response variable* indexes the partition:

$R_X = 0$: $X = 0$ (never-taker) $R_X = 1$: $X = Z$ (complier) $R_X = 2$: $X = 1 - Z$ (defier)
$R_X = 3$: $X = 1$ (always-taker)

Likewise $R_Y \in \{0,1,2,3\}$: never-recover ($Y = 0$), helped ($Y = X$), hurt ($Y = 1 - X$), always-recover ($Y = 1$). The 16 cells have probabilities

$$q_{j,k} = P(R_X = j, R_Y = k), \qquad \sum_{j=0}^{3} \sum_{k=0}^{3} q_{j,k} = 1.$$

Definition (Canonical type model)

An SCM with $X, Y, Z \in \{0,1\}$ and $\mathbf{U} = \{R_X, R_Y, R_Z\}$, $R_X, R_Y \in \{0,1,2,3\}$, $R_Z \in \{0,1\}$:

$$Z \leftarrow r_Z, \qquad X \leftarrow f_X(z, r_X) = \begin{cases} 0 & r_X = 0 \\ z & r_X = 1 \\ 1 - z & r_X = 2 \\ 1 & r_X = 3 \end{cases}, \qquad Y \leftarrow f_Y(x, r_Y) = \begin{cases} 0 & r_Y = 0 \\ x & r_Y = 1 \\ 1 - x & r_Y = 2 \\ 1 & r_Y = 3 \end{cases}.$$

Theorem 5.4 (Correspondence; Balke and Pearl 1997)

For any IV model $\mathcal{M}_1$ there is a canonical type model $\mathcal{M}_2$ with $P^{\mathcal{M}_1}(x,y \mid z) = P^{\mathcal{M}_2}(x,y \mid z)$ and $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x)) = P^{\mathcal{M}_2}(y \mid \mathrm{do}(x))$.

Proof [Supplement]. Partition $\mathfrak{X}_U$ by $u \mapsto (x(\cdot), y(\cdot))$, $x(z) = f_X(z,u)$, $y(x) = f_Y(x,u)$: $4 \times 4 = 16$ cells. Let $\mathcal{M}_2$ have $R = (R_X, R_Y) = (j,k)$ with probability $q_{j,k}$, the $P(U)$-mass of cell (j,k), and $X = \mathrm{response}_j(Z)$, $Y = \mathrm{response}_k(X)$. Both $P^{\mathcal{M}_1}(x,y \mid z)$ (mass of $\{u : f_X(z,u) = x, f_Y(x,u) =$

$y\}$) and $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x))$ (mass of $\{u : f_Y(x,u) = y\}$) are masses of unions of cells, so they agree with $\mathcal{M}_2$. ■

Hence, with $\mathcal{M}$ now the canonical models,

$$a(y;x) = \min_{M\in\mathcal{M}} P_M(y \mid \mathrm{do}(x)), \quad b(y;x) = \max_{M\in\mathcal{M}} P_M(y \mid \mathrm{do}(x)) \quad \text{s.t. } P_M(x,y \mid z) = P(x,y \mid z),$$

a linear program over $\mathbf{q} = (q_{j,k}) \in \mathbb{R}^{16}$.

5.1.3.4 LP formulation

Observational constraints. Each $p_{yx.z} = P(Y = y, X = x \mid Z = z)$ is linear in $\mathbf{q}$, e.g. ($Z = 1$, $X = 0$ needs $R_X \in \{0, 2\}$; $Y = 1$ at $X = 0$ needs $R_Y \in \{2, 3\}$)

$$\begin{aligned} p_{10.1} = P(Y = 1, X = 0 \mid Z = 1) &= P(\text{never-taker, hurt}) + P(\text{never-taker, always-recover}) \\ &\quad + P(\text{defier, hurt}) + P(\text{defier, always-recover}) \\ &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}. \end{aligned}$$

In full:

$$\begin{aligned} p_{00.0} &= q_{0,0} + q_{0,1} + q_{1,0} + q_{1,1}, & p_{00.1} &= q_{0,0} + q_{0,1} + q_{2,0} + q_{2,1}, \\ p_{01.0} &= q_{2,0} + q_{2,2} + q_{3,0} + q_{3,2}, & p_{01.1} &= q_{1,0} + q_{1,2} + q_{3,0} + q_{3,2}, \\ p_{10.0} &= q_{0,2} + q_{0,3} + q_{1,2} + q_{1,3}, & p_{10.1} &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}, \\ p_{11.0} &= q_{2,1} + q_{2,3} + q_{3,1} + q_{3,3}, & p_{11.1} &= q_{1,1} + q_{1,3} + q_{3,1} + q_{3,3}. \end{aligned}$$

Causal effect. Under $\mathrm{do}(X = 0)$, $Y = 1$ iff $R_Y \in \{\text{hurt}, \text{always-recover}\}$:

$$P(Y = 1 \mid \mathrm{do}(X = 0)) = P(R_Y = 2) + P(R_Y = 3) = \textstyle\sum_j (q_{j,2} + q_{j,3}),$$

$$P(Y = 1 \mid \mathrm{do}(X = 1)) = \textstyle\sum_j (q_{j,1} + q_{j,3}).$$

Optimization. $P(Y = 1 \mid \mathrm{do}(X = 0)) \in [a_{10}, b_{10}]$ with

$$a_{10} = \min_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3}), \qquad b_{10} = \max_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3})$$

subject to $\sum_{j,k} q_{j,k} = 1$, $q_{j,k} \geq 0$ and the eight equalities above.

5.1.3.5 Closed-form solution

Each bound is a linear combination of the $p_{yx.z}$ whose gap from the objective is non-negative:

$$P(Y = 1 \mid \mathrm{do}(X = 0)) \;\geq\; \max \begin{cases} p_{10.0} \\ p_{10.1} \\ p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} \\ p_{01.0} + p_{10.0} - p_{00.1} - p_{01.1} \end{cases}$$

$$P(Y = 1 \mid \mathrm{do}(X = 0)) \;\leq\; \min \begin{cases} 1 - p_{00.0} \\ 1 - p_{00.1} \\ p_{10.0} + p_{11.0} + p_{01.1} + p_{10.1} \\ p_{01.0} + p_{10.0} + p_{10.1} + p_{11.1} \end{cases}$$

The first two entries are the natural bounds; the others tighten them.

Validity of $p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}$. Substituting,

$$p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} = q_{0,2} + q_{0,3} + q_{1,2} + q_{2,3} - q_{0,0} - q_{0,1} - q_{2,0} - q_{1,1},$$

$$\sum_j (q_{j,2} + q_{j,3}) - (p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}) = q_{2,2} + q_{3,2} + q_{1,3} + q_{3,3} + q_{0,0} + q_{0,1} + q_{2,0} + q_{1,1} \geq 0.$$

Every closed-form bound is certified this way (LP duality: the coefficients are a feasible dual solution).

5.1.4 Back-door with partially observed covariates

Motivation. Bound an effect when the admissible set is partly measured but the missing part's marginal is known.

Let $\{W, U\}$ be back-door admissible ($W, U \to X$, $W, U \to Y$, W–U related, $X \to Y$), W observed, U not:

$$P(y \mid \mathrm{do}(x)) = \sum_{w,u} P(y \mid x, w, u) P(w, u) = \sum_{w,u} \frac{P(x, y, w, u) P(w, u)}{P(x, w, u)}.$$

Given $P(X, Y, W)$ and $P(U)$, the natural bound is $P(x, y) \leq P(y \mid \mathrm{do}(x)) \leq 1 - P(x, y')$ ($y' = 1 - y$, $x' = 1 - x$). Li and Pearl (2022) tighten it:

$$a(y; x) = \min_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}}, \qquad b(y; x) = \max_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}},$$

with $a_{w,u}, b_{w,u}, c_{w,u}$ the unknown $P(x, y, w, u)$, $P(w, u)$, $P(x, w, u)$, subject to, for all $w \in \mathfrak{X}_W$, $u \in \mathfrak{X}_U$,

$$\begin{gathered}
\textstyle\sum_u a_{w,u} = P(x, y, w), \qquad \sum_u b_{w,u} = P(w), \qquad \sum_u c_{w,u} = P(x, w), \\
b_{w,u} \geq c_{w,u} \geq a_{w,u} \quad (\text{since } P(x, y, w, u) \leq P(x, w, u) \leq P(w, u)), \\
\max\{0, P(x, y, w) + P(u) - 1\} \leq a_{w,u} \leq \min\{P(x, y, w), P(u)\}, \\
\max\{0, P(w) + P(u) - 1\} \leq b_{w,u} \leq \min\{P(w), P(u)\}, \\
\max\{0, P(x, w) + P(u) - 1\} \leq c_{w,u} \leq \min\{P(x, w), P(u)\}
\end{gathered}$$

(the last three lines are Fréchet bounds; $a_{w,u}$ alone suffices as parametrization).

5.2 Soft interventions

Motivation. Realistic policies install conditional or stochastic rules instead of constants, and the do-calculus extends to them.

5.2.1 Motivation

Motivation. Atomic $\mathrm{do}(x)$ is often not an implementable policy.

A tutoring program. Previous GPA W, motivation Z (low/high), tutoring X, final GPA Y: $W \to Z \to X \to Y$, $W \to Y$, $Z \to Y$.

ML can predict $P(y \mid w, z, x)$ well, but only in the current *natural regime*; decisions require predictions under an unrealised regime.

What we know so far. From $P(W, Z, X, Y)$ and $\mathcal{G}$,

$$P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P(z \mid \mathrm{do}(x)) \qquad \text{condition on } Z$$

$$= \sum_z P(y \mid \text{do}(x), z)P(z) \qquad \text{Rule 3: } (Z \perp_d X)_{\mathcal{G}_{\overline{X}}}$$
$$= \sum_z P(y \mid x, z)P(z) \qquad \text{Rule 2: } (Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}}.$$

If $P(Y = 1 \mid \text{do}(X = 1)) > P(Y = 1)$, $\text{do}(X = 1)$ helps, but it means mandatory tutoring for every student.

Implementing do-interventions. Resources may be insufficient and compliance not guaranteed, so we need realizable interventions:

Do-like intervention	Realistic intervention
make sure no one smokes	reduce tobacco consumption to 20% of current
provide treatment to all patients	treat iff a patient is in critical condition
move a robotic arm to coordinate (x, y, z)	move it to (x, y, z) with a small error
mark all applicants as males	mark all applicants as males on paper

5.2.2 Canonical types of regimes

Motivation. Classify interventions by the mechanism they install.

- **Idle/natural:** $X = f_X(\mathbf{pa}_X, u_X)$.
- **Hard/atomic:** $X = x$, $\text{do}(X = x)$: every student is tutored.
- **Conditional:** $X = g(\mathbf{w})$ for observables $\mathbf{W}$: tutoring iff low GPA.
- **Stochastic:** $X \sim P^*(x \mid \mathbf{w})$: low-GPA students enter a raffle for 80% of the spots, other interested students for 20%.

Write σ_X for the intervention and $P(\cdot; \sigma_X)$ for the induced distribution.

Changes in regime. Data come from the natural regime with graph $\mathcal{G}$. The new regime's graph $\mathcal{G}_{\sigma_X}$ has a regime indicator (small square σ_X) pointing into the intervened variables, whose incoming edges follow the new mechanism.

5.2.3 Reduction to known machinery

Motivation. When do soft-intervention effects follow from hard-intervention results?

A reducible example. Let $P(X = 1 \mid Z = 1) = 0.3$ and double it, σ_X: $P^*(X = 1 \mid Z = 1) = 0.6$; X's parents stay $\{Z\}$. Then

$$\begin{aligned} P(y; \sigma_X) &= \sum_z P(y \mid z; \sigma_X)P(z; \sigma_X) = \sum_z P(y \mid z; \sigma_X)P(z) \\ &= \sum_{z,x} P(y \mid z, x; \sigma_X)P(x \mid z; \sigma_X)P(z) = \sum_{z,x} P(y \mid z, x; \sigma_X)P^*(x \mid z)P(z) \\ &= \sum_{z,x} P(y \mid z, x)P^*(x \mid z)P(z), \end{aligned}$$

since Z is unaffected, X follows $P^*(x \mid z)$, and Y given Z, X responds as before. Is a do-calculus-like derivation always enough?

An intentionally wrong derivation. Treatment X_1; a physician observes Z and picks X_2; survival Y. Graph: $X_1 \to Z$, $X_1 \to X_2$, $X_1 \to Y$, $Z \to X_2$, $X_2 \to Y$, $X_1 \leftrightarrow Z$, $Z \leftrightarrow Y$; policy $\sigma_{\mathbf{X}} = \{X_1 = x_1, X_2 = g(x_1, z)\}$. One might write

$$P(y \mid \text{do}(x_1), \text{do}(X_2 = g(x_1, z))) = P(y \mid x_1, \text{do}(X_2 = g(x_1, z))),$$

citing Rule 2 with $(Y \perp_d X_1 \mid X_2)$ in $\mathcal{G}_{\overline{X_2}\underline{X_1}}$. Wrong: edges into X_2 stay active, so the check must use $\mathcal{G}_{\sigma_{X_2}\underline{X_1}}$, where $X_1 \leftrightarrow Z \to X_2 \to Y$ is open.

Where do-calculus intuition breaks. Do-interventions cut all dependence on parents; soft ones may keep some, change their distribution, or add new ones.

5.2.4 σ-calculus

Motivation. Inference rules for soft interventions consult both pre- and post-interventional graphs.

Theorem 5.5 (σ-calculus, simplified; Correa and Bareinboim 2020)

For disjoint $\mathbf{X}, \mathbf{Y}, \mathbf{Z} \subseteq \mathbf{V}$ and disjoint $\mathbf{T}, \mathbf{W} \subseteq \mathbf{V} \setminus (\mathbf{Z} \cup \mathbf{Y})$:

Rule 1 (insertion/deletion of observations) $P(\mathbf{y} \mid \mathbf{w}, \mathbf{t}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$.

Rule 2 (change of regimes under observation) $P(\mathbf{y} \mid \mathbf{w}, \mathbf{x}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}, \mathbf{x})$ if $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\underline{\mathbf{X}}}$ and in $\mathcal{G}_{\underline{\mathbf{X}}}$.

Rule 3 (change of regimes without observation) $P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w})$ if $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\overline{\mathbf{X(W)}}}$ and in $\mathcal{G}_{\overline{\mathbf{X(W)}}}$.

Proof [Supplement]. As for do-calculus (Chapter 3), with the regime indicator selecting between the natural mechanism and $\sigma_{\mathbf{X}}$. **Rule 1**: the post-regime distribution is Markov to $\mathcal{G}_{\sigma_{\mathbf{X}}}$. **Rules 2 and 3**: add a switch $F_X \to X$ with values σ and idle; as in do-calculus, $F_X \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ (Rule 2) or $F_X \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}$ (Rule 3) follows from the separation with outgoing (resp. incoming) edges of $\mathbf{X}$ removed. A soft regime may add parents of $\mathbf{X}$, so the separation must hold in the graph of each regime, hence both $\mathcal{G}_{\sigma_{\mathbf{X}}\cdots}$ and $\mathcal{G}_{\cdots}$. ■

Full version. With outer regime $\sigma_{\mathbf{X}}$ and a change on $\mathbf{Z}$ from $\sigma_{\mathbf{Z}}$ to $\sigma'_{\mathbf{Z}}$:

- Rule 1: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{t}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$;
- Rule 2: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\underline{\mathbf{Z}}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\underline{\mathbf{Z}}}$;
- Rule 3: $P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\overline{\mathbf{Z(W)}}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\overline{\mathbf{Z(W)}}}$.

5.2.4.1 Comparison with do-calculus

Remark. Under $\mathrm{do}(x)$, $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x), w, x)$; under a non-atomic intervention, in general $P(y; \sigma_X) \neq P(y \mid x; \sigma_X)$ for $X \to Y$.

	do-calculus	σ-calculus
Rule 1	$P(y \mid \mathrm{do}(x), w, t) = P(y \mid \mathrm{do}(x), w)$ if $(Y \perp_d T \mid W, X)_{\mathcal{G}_{\overline{X}}}$	$P(y \mid w, t; \sigma_X) = P(y \mid w; \sigma_X)$ if $(Y \perp_d T \mid W)_{\mathcal{G}_{\sigma_X}}$
Rule 2	$P(y \mid \mathrm{do}(x), w) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\underline{X}}}$	$P(y \mid x, w; \sigma_X) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X\underline{X}}$ and $\mathcal{G}_{\underline{X}}$
Rule 3	$P(y \mid \mathrm{do}(x), w) = P(y \mid w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\overline{X(W)}}}$	$P(y \mid w; \sigma_X) = P(y \mid w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X\overline{X(W)}}$ and $\mathcal{G}_{\overline{X(W)}}$

Rule 1's graph depends on the intervention; Rules 2 and 3 need both graphs.

Revisiting Rule 2. For the sequential treatment,

$$P(y \mid \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) = P(y \mid x_1, \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) \\ = P(y \mid x_1; \sigma_{X_1, X_2}) \stackrel{?}{=} P(y \mid x_1; \sigma_{X_2}).$$

Rule 2 needs no back-door path $X_1 \to Y$ both when $X_2 = g$ ignores Z and when X_2 listens to X_1, Z; only the first holds, so the rule does not apply.

5.2.4.2 Another intervention

Conditional policy: only low-GPA students ($W = 0$) are tutored, $P^*(X = 1 \mid W = 0) = 1$, $P^*(X = 1 \mid W = 1) = 0$. In $\mathcal{G}_{\sigma_X}$, X depends on W (not Z):

$$\begin{aligned} P(y;\sigma_X) &= \sum_{w,z,x} P(y \mid x,w,z;\sigma_X)P(x \mid w,z;\sigma_X)P(w,z;\sigma_X) \\ &= \sum_{w,z,x} P(y \mid x,w,z;\sigma_X)P(x \mid w;\sigma_X)P(w,z;\sigma_X) && \text{Rule 1} \\ &= \sum_{w,z,x} P(y \mid x,w,z)P(x \mid w;\sigma_X)P(w,z;\sigma_X) && \text{Rule 2} \\ &= \sum_{w,z,x} P(y \mid x,w,z)P^*(x \mid w)P(w,z) && \text{Rule 3.} \end{aligned}$$

5.2.5 C-factors and soft interventions

Motivation. Identify soft-intervention effects systematically with c-factors.

For $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$,

$$P(z,x,y) = \sum_{u_Z} P(z \mid u_Z)P(u_Z) \sum_{u_X} P(x \mid z,u_X)P(u_X) \sum_{u_Y} P(y \mid x,z,u_Y)P(u_Y) = Q[Z]\,Q[X]\,Q[Y].$$

Under $\mathcal{M}' = \mathcal{M}_{\sigma_X}$ (diagram $\mathcal{G}' = \mathcal{G}_{\sigma_X}$) only X's mechanism changes:

$$\begin{aligned} P(z,x,y;\sigma_X) &= \sum_{u_Z} P(z \mid u_Z)P(u_Z) \sum_{u'_X} P'(x \mid z,u'_X)P(u'_X) \sum_{u_Y} P(y \mid x,z,u_Y)P(u_Y) \\ &= Q[Z]\,Q[X;\sigma_X]\,Q[Y]. \end{aligned}$$

In general $P(\mathbf{y};\sigma_{\mathbf{X}}) = \sum_{\mathbf{d}\setminus\mathbf{y}} \prod_{\mathbf{D}_i \in \mathcal{C}(\mathbf{D})} Q[\mathbf{D}_i]$ over the c-components of $\mathcal{G}_{\sigma_{\mathbf{X}}}$ restricted to the relevant ancestors; $Q[\mathbf{X};\sigma_{\mathbf{X}}]$ is given by $\sigma_{\mathbf{X}}$, the other factors are identified from $P(\mathbf{V})$, e.g. by IDENTIFY (Chapter 3).

Example 5.2 (Systematic identification 1)

$\mathcal{G}$: $W \to Z$, $X \to Z \to Y$, $X \leftrightarrow Y$; $\sigma_X = P^*(X \mid W)$, so $\mathcal{G}_{\sigma_X}$ gains $W \to X$ and loses $X \leftrightarrow Y$. C-components: $\mathcal{G}$: $\mathbf{C}_1 = \{X,Y\}$, $\mathbf{C}_2 = \{W\}$, $\mathbf{C}_3 = \{Z\}$; $\mathcal{G}_{\sigma_X}$: $\mathbf{D}_1 = \{Y\}$, $\mathbf{D}_2 = \{W\}$, $\mathbf{D}_3 = \{Z\}$, $\mathbf{D}_4 = \{X\}$, with $Q[\mathbf{D}_4] = Q[X;\sigma_X]$ given. IDENTIFY$(\{Y\},\{X,Y\},Q[T],\mathcal{G})$: $A = An(\{Y\})_{\mathcal{G}[\{X,Y\}]} = \{Y\} = C$, so $Q[Y] = \sum_x Q[X,Y]$; $\{W\}$ and $\{Z\}$ return immediately. Hence

$$\begin{aligned} P(y;\sigma_X) &= \sum_{w,x,z} Q[X;\sigma_X]\,Q[Z]\,Q[W]\,Q[Y] = \sum_{w,x,z} Q[X;\sigma_X]\,Q[Z]\,Q[W] \sum_{x'} Q[X,Y] \\ &= \sum_{w,x,z} P^*(x \mid w)\,P(z \mid x,w)\,P(w) \sum_{x'} P(x')\,P(y \mid x',z,w). \end{aligned}$$

Example 5.3 (Systematic identification 2: failure)

$\mathcal{G}$: $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$; $\sigma_X = P^*(X \mid Z)$. C-components: $\mathcal{G}$: $\mathbf{C}_1 = \{W, X, Y\}$, $\mathbf{C}_2 = \{Z\}$; $\mathcal{G}_{\sigma_X}$: $\mathbf{D}_1 = \{Y, W\}$, $\mathbf{D}_2 = \{X\}$, $\mathbf{D}_3 = \{Z\}$. IDENTIFY$(\{W, Y\}, \{W, X, Y\}, Q[T], \mathcal{G})$: $A = An(\{W, Y\})_{\mathcal{G}[\{W,X,Y\}]} = \{W, X, Y\} \neq C$ and $A = T$, so FAIL. (Recall $Q[\mathbf{C}_j] = \prod_{V_i \in \mathbf{C}_j} Q[\mathbf{V}^{\leq i}]/Q[\mathbf{V}^{\leq i-1}]$, $Q[\mathbf{V}^{\leq i}] = \sum_{\mathbf{v}>i} Q[\mathbf{V}]$.)

5.2.6 Summary

- Implementable policies are often conditional or stochastic, not atomic.
- σ-calculus rules consult pre- and post-interventional graphs.
- C-factor procedures identify soft-intervention effects systematically.

Chapter 6. Potential Outcomes and Effect Estimation

6.1 Potential outcomes

Motivation. Define and identify causal effects as comparisons of potential outcomes, a missing-data problem, instead of structural equations and graphs.

6.1.1 Frameworks for causal inference

Motivation. Place the potential outcome (PO) framework next to the causal-diagram framework.

Two frameworks are in common use:

- the **potential outcome framework** (counterfactual framework, Neyman–Rubin causal model; Neyman 1923, Rubin 1974);
- the **causal diagram framework** (Pearl 2009).

They are mathematically connected (Richardson and Robins 2013; see the SWIG lecture).

The PO framework dominates medicine, policy and the social sciences. Causal inference is a *missing-data problem*, with explicit modeling of the *assignment mechanism* that reveals the data; it grew from Fisher (1918, 1925) and Neyman (1923) on randomized experiments, extended by Rubin (1974, 1978) to non-randomized studies.

6.1.2 Basic concepts

Motivation. Units, treatments, potential outcomes, causal effects, and the fundamental problem.

- **Unit:** the person, place or thing a treatment operates on, at a particular time (the same person at two times is two units).
- **Treatment:** an intervention whose effect on the units is assessed relative to no intervention (control).
- **Potential outcomes:** the unit's measurement under active treatment, $Y(1)$, and under control, $Y(0)$; also *counterfactuals*.
- **Causal effect:** a per-unit comparison of $Y(1)$ and $Y(0)$, e.g. $Y_i(1) - Y_i(0)$.

6.1.2.1 The fundamental problem of causal inference

At most one potential outcome is observed per unit; the other is counterfactual (Holland 1986).

Unit	$Y(x{=}0)$	$Y(x{=}1)$	X	Y
1	?	1	1	1
2	0	?	0	0
3	?	0	1	0
4	?	1	1	1
5	1	?	0	1

Neyman's urn model. Each unit is a ticket carrying Y_1 and Y_0; assignment reveals only Y_1 for treated units and Y_0 for controls.

6.1.2.2 God's eye view: individual versus average effects

Y: death (1) or survival (0), both potential outcomes visible (adapted from Hernán and Robins, *What If*, Table 1.1):

	$Y(0)$	$Y(1)$	effect		$Y(0)$	$Y(1)$	effect
Rheia	1	0	-1	Hestia	1	0	-1
Zeus	0	0	0	Poseidon	1	0	-1
Kronos	0	1	$+1$	Hera	1	0	-1
Demeter	0	0	0	Artemis	1	1	0
Hades	0	0	0	Apollo	0	1	$+1$
Risk: $\mathbb{E}[Y(0)] = 0.5$, $\mathbb{E}[Y(1)] = 0.3$							

The average causal effect is $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = 0.3 - 0.5 = -0.2$, but individual effects vary: 4 helped, 2 harmed, 4 unaffected.

6.1.2.3 What the researcher actually sees

(Adapted from *What If*, Table 1.2.)

	$Y(0)$	$Y(1)$	X	Y^{obs}		$Y(0)$	$Y(1)$	X	Y^{obs}
Rheia	?	0	1	0	Hestia	?	0	1	0
Zeus	0	?	0	0	Poseidon	1	?	0	1
Kronos	?	1	1	1	Hera	?	0	1	0
Demeter	0	?	0	0	Artemis	1	?	0	1
Hades	0	?	0	0	Apollo	?	1	1	1

By consistency $Y^{\text{obs}} = Y(X)$. The naive comparison gives $2/5 = 0.4$ in both groups, a difference $0 \neq -0.2$: sicker patients ($Y(0) = 1$) chose treatment more often (confounding), hence the need for exchangeability.

6.1.3 Assumptions linking potential and observed outcomes

Motivation. SUTVA and consistency, and their failures.

6.1.3.1 SUTVA

Assumption 1 (Stable Unit Treatment Value Assumption; Rubin 1980). (1) *No interference* and (2) *no different versions* of a treatment (e.g. the same surgery by different surgeons). It links the observed X with the intervention x:

$$X_i = 1 \Rightarrow Y_i = Y_i(1), \qquad X_i = 0 \Rightarrow Y_i = Y_i(0),$$

equivalently

$$Y_i = X_i Y_i(1) + (1 - X_i) Y_i(0).$$

Violation by interference: $Y_i(x)$ depends on others' treatments (vaccination, advertising, infectious disease, social networks, agricultural plots); other assumptions are then needed (Rosenbaum 2007; Hudgens and Halloran 2008), and spillovers are often of interest themselves.

On the history of SUTVA (VanderWeele and Hernán, "Causal inference under multiple versions of treatment"). The notation $Y_j(a)$ presupposes no interference (Cox 1958) and no multiple versions of a (Neyman 1935) (Rubin 1980), plus *consistency* (Robins 1986): $Y_j(a) = Y_j$ when $A_j = a$. These govern randomized trials as well as observational studies.

6.1.3.2 Consistency and well-defined interventions

Consistency: if $X_i = x$ then $Y_i = Y_i(x)$, which requires "$X = x$" to mean the same for everyone. Example: "losing 20 kg" via diet and exercise ($Y(x)$ = survived) versus via starvation or illness ($Y(x)$ = died) is one nominal treatment with different outcomes, so $Y(x)$ is ill-defined. Fix: specify the intervention ("20 kg loss via a supervised diet and exercise program"; *What If*, Fine Point 3.4).

6.1.4 Causal estimands

Motivation. The quantities typically targeted.

Differences.

$$\begin{aligned} \text{ATE:} \quad & \tau = \mathbb{E}[Y(1) - Y(0)], \\ \text{CATE:} \quad & \tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z], \\ \text{ATT:} \quad & \text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid X = 1]. \end{aligned}$$

Ratios. Individual $r_i = Y_i(1)/Y_i(0)$; population $r = \mathbb{E}[Y(1)]/\mathbb{E}[Y(0)]$, not an average of the r_i unless the effect is homogeneous; lift $l = \mathbb{E}[Y(1) - Y(0)]/\mathbb{E}[Y(0)]$. Continuous outcomes dominate the social sciences, binary ones medicine.

6.1.5 Randomization and observational studies

Motivation. Why randomization identifies causal effects, and what replaces it in observational data.

6.1.5.1 Unconfounded assignment

An assignment mechanism is *unconfounded* if it depends only on pre-treatment covariates:

$$p(X = 1 \mid Z, Y(0), Y(1)) = p(X = 1 \mid Z).$$

$p(X = 1 \mid Z)$ is the *propensity score* (Rosenbaum and Rubin 1983). In randomized experiments it is unconfounded and also known.

6.1.5.2 The role of randomization

Randomization balances observed covariates ($X \perp\!\!\!\perp Z$), unobserved ones ($X \perp\!\!\!\perp U$) and potential outcomes, i.e. ignorability

$$X \perp\!\!\!\perp (Y(1), Y(0)).$$

Then, by ignorability and consistency,

$$P(Y(x)) = P(Y(x) \mid X = x) = P(Y \mid X = x), \qquad x \in \{0, 1\},$$

so association is causation:

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1) \mid X = 1] - \mathbb{E}[Y(0) \mid X = 0] = \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0].$$

As graphs. Nothing points into X (e.g. $X \to Y \leftarrow Z$). The SWIG splits the treatment node into observed X and intervened x: $X \mid x \to Y(x) \leftarrow Z$; X and $Y(x)$ are d-separated, so $Y(x) \perp\!\!\!\perp X$.

6.1.5.3 Observational studies

The most common assumptions:

1. **No unmeasured confounding:** $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$ (unconfoundedness, ignorability, selection on observables, exogeneity, conditional exchangeability): given covariates, X_i is as if randomized.
2. **Positivity (overlap):** $0 < P(X_i = 1 \mid Z_i) < 1$.

Choosing Z is the adjustment-criterion problem. Graphically $Z \to X$, $Z \to Y$, $X \to Y$; in the SWIG $Z \to X$, $Z \to Y(x)$, $x \to Y(x)$, and $Y(x) \perp\!\!\!\perp X \mid Z$ by d-separation.

6.1.5.4 Identification of the ATE

Under positivity and no unmeasured confounding,

$$\begin{aligned}\tau &= \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}_Z\{\mathbb{E}[Y(1) - Y(0) \mid Z]\} = \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid Z] - \mathbb{E}[Y(0) \mid Z]\} \\ &= \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid X = 1, Z] - \mathbb{E}[Y(0) \mid X = 0, Z]\} && \text{(ignorability)} \\ &= \mathbb{E}_Z\{\mathbb{E}[Y \mid X = 1, Z] - \mathbb{E}[Y \mid X = 0, Z]\} && \text{(consistency)},\end{aligned}$$

positivity making the last line defined for every z. With the conditional expectation functions (CEFs)

$$\mu_1(z) = \mathbb{E}[Y(1) \mid Z = z], \qquad \mu_0(z) = \mathbb{E}[Y(0) \mid Z = z],$$

the key step is

$$\underbrace{\mu_1(z)}_{\text{counterfactual}} = \underbrace{\mathbb{E}[Y \mid X = 1, Z = z]}_{\text{observational}}, \qquad \mu_0(z) = \mathbb{E}[Y \mid X = 0, Z = z].$$

6.1.5.5 Regression estimation of the ATE

Fit $\hat{\mu}_1(z)$, $\hat{\mu}_0(z)$, preferably separately in each treatment group; predict both for all units and average the differences:

$$\hat{\tau}_{\text{reg}} = \frac{1}{N} \sum_{i=1}^{N} (\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)).$$

6.1.5.6 Summary

Potential outcomes	$Y(1)$, $Y(0)$
Conditional ignorability (back-door)	$Y(1), Y(0) \perp\!\!\!\perp X \mid Z$

Identification: $\mathbb{E}[Y(1)] = \mathbb{E}[\mathbb{E}[Y(1) \mid Z]] = \mathbb{E}[\mathbb{E}[Y(1) \mid X = 1, Z]] = \mathbb{E}[\mathbb{E}[Y \mid X = 1, Z]]$.

6.1.6 Appendix: Positivity and consistency in causal diagrams

Positivity concerns arrows into the treatment node. A deterministic edge $Z \to X$ violates positivity by design; otherwise positivity is implicit in diagrams.

Consistency concerns arrows out of the treatment node. $X \to Y$ needs a well-defined intervention; if $X = x$ can be reached in ways with different effects (diet, exercise, surgery), the effect is ill-defined (*What If*, §6.4).

6.1.7 Appendix: Confounding and exchangeability

- **Marginal exchangeability** $Y(x) \perp\!\!\!\perp X$: no open back-door paths, as in a marginally randomized experiment ($X \to Y$ only).
- **Conditional exchangeability** $Y(x) \perp\!\!\!\perp X \mid Z$: Z blocks all back-door paths, as in a conditionally randomized experiment ($Z \to X \to Y$, $Z \to Y$).
- The back-door criterion implies conditional exchangeability; the converse needs the adjustment criterion (Perković et al. 2018).
- Confounding (structural) = lack of exchangeability (counterfactual) (*What If*, §7.2, Technical Point 7.1).

6.2 Inverse probability weighting and doubly robust estimation

Motivation. Estimate $\mathbb{E}[Y(x)]$ and the ATE under unconfoundedness and positivity, robustly to misspecified nuisance models.

Recall:

(1) unconfoundedness $\{Y(1), Y(0)\} \perp\!\!\!\perp X \mid Z$;

(2) positivity $0 < P(X = 1 \mid Z) < 1$;

(3) hence identification by adjustment, $\mathbb{E}[Y(x)] = \mathbb{E}_Z[\mathbb{E}[Y \mid X = x, Z]]$.

6.2.1 The propensity score

Motivation. A scalar summary of the covariates that suffices for confounding control.

Definition (Propensity score; Rosenbaum and Rubin 1983)

$e(Z) = P(X = 1 \mid Z) = \mathbb{E}[X \mid Z]$, where $Z = (Z_1, \ldots, Z_p)$ collects the p pre-treatment covariates.

Proposition 6.1 (Property 1: balancing)

$X \perp\!\!\!\perp Z \mid e(Z)$. More generally a *balancing score* is any $b(Z)$ with $X \perp\!\!\!\perp Z \mid b(Z)$; $e(Z)$ is one (Z itself trivially is another), and Rosenbaum and Rubin (1983) show that $e(Z)$ is the *coarsest* balancing score.

Proof [Supplement]. It suffices that $P(X = 1 \mid Z, e(Z)) = P(X = 1 \mid e(Z))$. Left: $P(X = 1 \mid Z) = e(Z)$. Right: $\mathbb{E}[\mathbb{E}[X \mid Z] \mid e(Z)] = \mathbb{E}[e(Z) \mid e(Z)] = e(Z)$. ■

Proposition 6.2 (Property 2: unconfoundedness given the PS)

If $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$, then $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid e(Z_i)$.

Proof [Supplement].

$$\begin{aligned} P(X = 1 \mid Y(1), Y(0), e(Z)) &= \mathbb{E}\big[P(X = 1 \mid Y(1), Y(0), Z) \mid Y(1), Y(0), e(Z)\big] \\ &= \mathbb{E}\big[e(Z) \mid Y(1), Y(0), e(Z)\big] = e(Z), \end{aligned}$$

free of the potential outcomes. ■

So adjusting for $e(Z)$ removes all covariate-induced bias.

6.2.2 Inverse probability weighting

Motivation. Estimate potential-outcome means by reweighting observed outcomes.

6.2.2.1 The IPW identity

$$\text{ATE} = \mathbb{E}\left[\frac{XY}{e(Z)} - \frac{(1-X)Y}{1-e(Z)}\right].$$

By consistency ($XY = XY(1)$) and unconfoundedness ($X \perp\!\!\!\perp Y(1) \mid Z$),

$$\mathbb{E}\left[\frac{XY}{e(Z)}\right] = \mathbb{E}\left[\frac{\mathbb{E}[XY(1) \mid Z]}{e(Z)}\right] \overset{\text{uncf.}}{=} \mathbb{E}\left[\frac{\mathbb{E}[X \mid Z]\,\mathbb{E}[Y(1) \mid Z]}{e(Z)}\right] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid Z]\big] = \mathbb{E}[Y(1)],$$

and symmetrically for controls: the Horvitz–Thompson (HT) estimator, or IPTW.

6.2.2.2 Estimators

With weights $\hat{w}_1(Z_i) = 1/\hat{e}(Z_i)$ $(X_i = 1)$ and $\hat{w}_0(Z_i) = 1/(1-\hat{e}(Z_i))$ $(X_i = 0)$,

$$\hat{\tau}_{\text{ipw},1} = \frac{1}{N}\left(\sum_{i=1}^{N}\frac{Y_i X_i}{\hat{e}(Z_i)} - \sum_{i=1}^{N}\frac{Y_i(1-X_i)}{1-\hat{e}(Z_i)}\right) = \frac{1}{N}\sum_{i=1}^{N}\{Y_i X_i \hat{w}_1(Z_i) - Y_i(1-X_i)\hat{w}_0(Z_i)\}$$

is unbiased when $\hat{e} = e$.

6.2.2.3 Normalizing the weights

$\mathbb{E}[\mathbb{E}[X \mid Z]/e(Z)] = 1$, but $\frac{1}{N}\sum_i X_i/\hat{e}(Z_i) \neq 1$ in general. Normalizing within each group (divide $\hat{w}_i$ by $\sum_{j:X_j=x}\hat{w}_j$) gives the Hájek estimator:

$$\hat{\tau}_{\text{ipw},2} = \frac{\sum_i Y_i X_i \hat{w}_1(Z_i)}{\sum_i X_i \hat{w}_1(Z_i)} - \frac{\sum_i Y_i(1-X_i)\hat{w}_0(Z_i)}{\sum_i (1-X_i)\hat{w}_0(Z_i)},$$

typically with $V(\hat{\tau}_{\text{ipw},2}) \leq V(\hat{\tau}_{\text{ipw},1})$ (Hirano, Imbens and Ridder 2003).

6.2.2.4 The weighted population

In the IPW-weighted population covariates are balanced:

$$\mathbb{E}\left[\frac{X}{e(Z)}Z\right] = \mathbb{E}\left[\frac{1-X}{1-e(Z)}Z\right] \ (= \mathbb{E}[Z]),$$

since $\mathbb{E}[XZ/e(Z)] = \mathbb{E}[\mathbb{E}[X \mid Z]Z/e(Z)] = \mathbb{E}[Z]$. The less likely a subject was treated, the more of the population it represents.

6.2.2.5 Pros and cons

Advantages: simple, Horvitz–Thompson foundation; no outcome model; target population (ATE, ATT, ...) set by the weights; extends to time-varying treatments and marginal structural models.
Disadvantages: sensitive to PS misspecification; PS near 0 or 1 gives extreme weights, handled ad hoc.

6.2.2.6 Trimming and truncation

- **Symmetric trimming** (Crump et al. 2009): drop units with $\hat{e} \notin [\alpha, 1-\alpha]$, typically $\alpha = 0.1$.
- **Asymmetric trimming** (Stürmer et al. 2010): drop units outside the common PS range, then treated units below the q quantile of the treated PS and controls above the $(1-q)$ quantile of the control PS.
- **Truncation:** set $\hat{e} < \alpha$ to α and $\hat{e} > 1-\alpha$ to $1-\alpha$.

Range restriction thus removes the non-overlapping tails of the untreated (left-heavy) and treated (right-heavy) PS histograms.

6.2.2.7 PS weighting in randomized trials

In RCTs PS weighting (with an estimated PS) adjusts for chance imbalances and improves precision.

6.2.2.8 Weighting versus balancing

IPW balances covariates only *in expectation*. Balancing methods choose w_i directly so that

$$\frac{1}{N_1}\sum_{i=1}^{N} w_i X_i Z_i \approx \frac{1}{N_0}\sum_{i=1}^{N} w_i (1-X_i) Z_i,$$

minimizing a dispersion measure (e.g. entropy) subject to balance: entropy balancing (Hainmueller 2011), CBPS (Imai and Ratkovic 2014), stabilized balancing weights (Zubizarreta 2015). Only the specified moments are balanced: if $Y \sim Z_1 + Z_2 + Z_1 Z_2$ but $Z_1 Z_2$ is not balanced, the result may be badly biased.

6.2.3 Doubly robust estimation

Motivation. Combine outcome regression and IPW so that one correct model suffices.

6.2.3.1 Terminology

- **Correctly specified model:** the truth lies in the assumed class (e.g. $\mathbb{E}[Y \mid X, Z] = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$ fitted with that linear model).
- **Misspecified model:** it does not (e.g. truth non-linear in Z); the fit converges to the best in-class approximation, generally biased.
- **Consistent estimator:** $\hat{\theta}_N \xrightarrow{p} \theta$; correct specification gives consistency (under regularity), misspecification generally not.

6.2.3.2 Two building blocks

By the symmetry $X \leftrightarrow 1-X$, $e \leftrightarrow 1-e$, focus on $\mu_1 = \mathbb{E}[Y(1)]$. Let $m_x(Z) = \mathbb{E}[Y(x) \mid Z]$ ($\mu_x(z)$ in Section 6.1).

- **Outcome regression:** $\hat{\mu}_{1,\text{adj}} = N^{-1}\sum_i \hat{m}_1(Z_i)$, consistent if m_1 is correct.
- **IPW:** $\hat{\mu}_{1,\text{ipw}} = N^{-1}\sum_i Y_i X_i / \hat{e}(Z_i)$, consistent if the PS model is correct, possibly inefficient.

Combined:

$$\hat{\mu}_{1,\text{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right].$$

6.2.3.3 Verification with the true nuisance functions

$$\mathbb{E}\left[\frac{XY}{e(Z)} - \frac{X - e(Z)}{e(Z)} m_1(Z)\right] = \mathbb{E}\left[\frac{XY - Xm_1(Z) + e(Z)m_1(Z)}{e(Z)}\right]$$
$$= \mathbb{E}\left[m_1(Z) + \frac{X}{e(Z)}\{Y - m_1(Z)\}\right] = \mu_1,$$

since $\mathbb{E}[X\{Y(1) - m_1(Z)\}/e(Z)] = \mathbb{E}[\mathbb{E}[Y(1) - m_1(Z) \mid Z]] = 0$ (iterated expectations, $\mathbb{E}[X \mid Z] = e(Z)$).

6.2.3.4 Two equivalent forms

$$\hat{\mu}_{1,\mathrm{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right] \qquad \text{(IPW augmented by OR)}$$
$$= \frac{1}{N}\sum_{i=1}^{N}\left[\hat{m}_1(Z_i) + \frac{X_i\{Y_i - \hat{m}_1(Z_i)\}}{\hat{e}(Z_i)}\right] \qquad \text{(OR augmented by IPW residual correction).}$$

6.2.3.5 What does it estimate?

With arbitrary limits $\hat{e}$, $\hat{m}_1$ and $XY = XY(1)$:

$$\mathbb{E}\left[\frac{XY}{\hat{e}(Z)} - \frac{X - \hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right] = \mathbb{E}\left[Y(1) + \frac{X - \hat{e}(Z)}{\hat{e}(Z)}Y(1) - \frac{X - \hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right]$$
$$= \mathbb{E}[Y(1)] + \underbrace{\mathbb{E}\left[\frac{X - \hat{e}(Z)}{\hat{e}(Z)}\{Y(1) - \hat{m}_1(Z)\}\right]}_{R}.$$

$\hat{\mu}_{1,\mathrm{dr}}$ estimates $\mathbb{E}[Y(1)]$ iff $R = 0$.

Scenario 1: PS wrong, OR right ($\hat{m}_1 = m_1$). Conditioning on Z and using unconfoundedness,

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right], \qquad \mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z] = m_1(Z) - \hat{m}_1(Z) = 0.$$

Scenario 2: PS right, OR wrong ($\hat{e} = e$).

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right] = 0 \quad \text{since } e(Z) - \hat{e}(Z) = 0.$$

So $\hat{\mu}_{1,\mathrm{dr}}$ is consistent in both cases (and when both are correct). With $X \to 1 - X$, $e \to 1 - e$, $m_1 \to m_0$,

$$\hat{\mu}_{0,\mathrm{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1 - X_i)Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)}\hat{m}_0(Z_i)\right]$$

is consistent for $\mathbb{E}[Y(0)]$.

6.2.3.6 The doubly robust ATE estimator

$$\hat{\tau}_{\mathrm{dr}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right]}_{\hat{\mu}_{1,\mathrm{dr}}}$$
$$- \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1 - X_i)Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)}\hat{m}_0(Z_i)\right]}_{\hat{\mu}_{0,\mathrm{dr}}}.$$

6.2.3.7 Concluding remarks

- **Double robustness:** $\hat{\tau}_{\text{dr}}$ is consistent if either the PS or the outcome model is correct; a large-sample property.
- Both correct: smaller asymptotic variance than IPW. Outcome model correct: larger asymptotic variance than direct regression.
- Empirically the outcome model matters more: a misspecified $\hat{m}$ gives notable bias even with correct $\hat{e}$, so PS augmentation helps only somewhat.
- Kang and Schafer (2007, *Stat. Sci.*): moderate misspecification of both models (a likely scenario) gives DR large bias and variance.

6.2.3.8 Summary

1. $e(Z) = P(X = 1 \mid Z)$ is a scalar balancing summary of the covariates.
2. IPW reweights by $1/e(Z)$: simple, outcome-model-free; extreme weights.
3. Outcome regression models $\mathbb{E}[Y \mid X, Z]$: efficient when correct; sensitive to misspecification.
4. DR combines both and is consistent if either model is correct.

6.3 Conditional average treatment effects

Motivation. How effects vary with covariates, and how ML meta-learners estimate $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$.

6.3.1 Why heterogeneous treatment effects?

Motivation. A single average can hide opposite subgroup effects.

6.3.1.1 The average can be misleading

20 individuals, half women ($Z = 1$), half men ($Z = 0$), null average effect:

$$P(Y(1) = 1) - P(Y(0) = 1) = 0.5 - 0.5 = 0.$$

Yet the causal risk difference is $0.6 - 0.4 = +0.2$ among women and $0.4 - 0.6 = -0.2$ among men, $\frac{1}{2}(+0.2) + \frac{1}{2}(-0.2) = 0$. *There is generally no such thing as "the causal effect of treatment"; it depends on the population* (Hernán and Robins, *What If*, §4.1).

6.3.1.2 Effect modification

Definition

Z is a *modifier* of the effect of X on Y when the average causal effect of X on Y varies across levels of Z.

- **Additive** effect modification: $\mathbb{E}[Y(1) - Y(0) \mid Z = 1] \neq \mathbb{E}[Y(1) - Y(0) \mid Z = 0]$.
- **Multiplicative** effect modification: $\dfrac{\mathbb{E}[Y(1) \mid Z = 1]}{\mathbb{E}[Y(0) \mid Z = 1]} \neq \dfrac{\mathbb{E}[Y(1) \mid Z = 0]}{\mathbb{E}[Y(0) \mid Z = 0]}$.

Modification can be scale-specific (risks $0.2 \to 0.4$ and $0.4 \to 0.8$: equal ratios, different differences); *qualitative* modification (opposite directions) appears on both scales.

6.3.1.3 Why care?

1. **Policy targeting:** if treatment harms women and helps men, treat only men.
2. **Transportability:** a different modifier distribution in the target population changes the ATE.
3. **Mechanisms:** modification may reveal biological or social mechanisms.

Conditioning on all covariates gives the CATE $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$ (*What If*, §4.3).

6.3.1.4 From stratification to CATE estimation

Stratum-wise effects require discretizing continuous modifiers, suffer the curse of dimensionality, and leave strata with few observations. Meta-learners and causal forests estimate $\tau(z)$ with flexible ML instead (*What If*, §4.2).

6.3.2 Meta-learners for CATE

Motivation. CATE estimators from arbitrary regression algorithms (Künzel et al. 2019, PNAS).

Under unconfoundedness and positivity, $\mu_x(z) := \mathbb{E}[Y(x) \mid Z = z] = \mathbb{E}[Y \mid X = x, Z = z]$, so any regression method applies.

6.3.2.1 T-learner and S-learner

T-learner: fit $\hat{\mu}_1(z)$ on the treated and $\hat{\mu}_0(z)$ on the controls;

$$\hat{\tau}_T(z) = \hat{\mu}_1(z) - \hat{\mu}_0(z), \qquad \widehat{\mathrm{ATE}}_T = \frac{1}{n}\sum_{i=1}^{n} \big(\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)\big).$$

S-learner: one model $\hat{\mu}(z, x) \approx \mathbb{E}[Y \mid Z = z, X = x]$ with X as a feature;

$$\hat{\tau}_S(z) = \hat{\mu}(z, 1) - \hat{\mu}(z, 0), \qquad \widehat{\mathrm{ATE}}_S = \frac{1}{n}\sum_{i=1}^{n} \big(\hat{\mu}(Z_i, 1) - \hat{\mu}(Z_i, 0)\big).$$

T is noisy when one arm is small; S may regularize the effect of X towards zero.

6.3.2.2 Regression imputation

Impute only the *unobserved* counterfactual:

$$\hat{\tau}_{RI} = \frac{1}{n}\sum_{i=1}^{n} \Big[\big(X_i Y_i + (1 - X_i)\hat{\mu}_1(Z_i)\big) - \big((1 - X_i)Y_i + X_i \hat{\mu}_0(Z_i)\big)\Big].$$

6.3.2.3 X-learner

1. Estimate $\hat{\mu}_0(z) \approx \mathbb{E}[Y(0) \mid Z = z]$ and $\hat{\mu}_1(z) \approx \mathbb{E}[Y(1) \mid Z = z]$.
2. Impute effects $\tilde{D}_i^1 := Y_i - \hat{\mu}_0(z_i)$ (treated) and $\tilde{D}_i^0 := \hat{\mu}_1(z_i) - Y_i$ (controls); regress to get $\hat{\tau}_1(z) \approx \mathbb{E}[\tilde{D}_i^1 \mid Z = z]$ and $\hat{\tau}_0(z) \approx \mathbb{E}[\tilde{D}_i^0 \mid Z = z]$.

3. Combine with $g(z) \in [0,1]$, e.g. $g(z) = \hat{e}(z)$:

$$\hat{\tau}(z) = g(z)\,\hat{\tau}_0(z) + (1 - g(z))\,\hat{\tau}_1(z).$$

Illustration. One covariate $x \in [-1,1]$, true $\tau = 1$, rare treatment ($g = \hat{e} \approx 0.05$). $\hat{\mu}_0$ (many controls) is flexible with jumps; $\hat{\mu}_1$ (few treated) is nearly linear. $\hat{\tau}_T = \hat{\mu}_1 - \hat{\mu}_0$ inherits the jumps, while $\hat{\tau}_X \approx (1-g)\hat{\tau}_1$ is nearly linear and close to τ, since $\hat{\tau}_1$ uses the accurate $\hat{\mu}_0$ and the effect itself is simple.

6.3.2.4 DR-learner

Apply the DR score of Section 6.2 to $\tau(z)$:

1. Estimate the nuisances $\hat{\mu}_x(z)$ and $\hat{e}(z)$.
2. Construct the pseudo-outcome

$$\tilde{Y}_i = \frac{X_i(Y_i - \hat{\mu}_1(Z_i))}{\hat{e}(Z_i)} - \frac{(1-X_i)(Y_i - \hat{\mu}_0(Z_i))}{1 - \hat{e}(Z_i)} + \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i).$$

3. Regress $\tilde{Y}_i$ on Z_i flexibly to obtain $\hat{\tau}(z)$.

Why the DR-learner?

- $\mathbb{E}[\tilde{Y}_i \mid Z_i = z] = \tau(z)$ if one nuisance is correct (the DR computation, conditional on Z).
- *Neyman orthogonality:* nuisance errors enter $\hat{\tau}$ only through their product.
- It adapts to the complexity of $\tau(z)$, even with more complex nuisances (Kennedy 2023).
- Use K-fold cross-fitting: nuisances from folds $\neq k$, $\tilde{Y}_i$ on fold k.

6.4 The R-learner

Motivation. A loss whose minimizer is the CATE itself, so nuisance errors matter only at second order.

6.4.1 Motivation and identification

Motivation. The target and its residualised representation.

We observe i.i.d. $O_i = (X_i, Z_i, Y_i)$, $X_i \in \{0,1\}$, and want $\tau(z) = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = z]$. S-, T-, X-learners can pass nuisance bias into $\hat{\tau}$; DR-learners regress an orthogonal pseudo-outcome; the R-learner builds a *loss whose target is $\tau(\cdot)$ itself* (Nie and Wager 2021).

Assumptions: consistency $Y_i = Y_i(X_i)$, unconfoundedness $\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp X_i \mid Z_i$, overlap $0 < e(Z_i) < 1$. **Nuisances:**

$$e(z) = \Pr(X = 1 \mid Z = z), \qquad m(z) = \mathbb{E}[Y \mid Z = z],$$

m being the Z-only mean, not an arm-specific μ_x. With residuals $\tilde{Y}_i = Y_i - m(Z_i)$, $\tilde{X}_i = X_i - e(Z_i)$,

$$\tilde{Y}_i = \tilde{X}_i\,\tau(Z_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0,$$

Robinson's partially linear decomposition with a varying coefficient.

Derivation. With $\mu_x(z) = \mathbb{E}[Y(x) \mid Z = z]$, $\tau(z) = \mu_1(z) - \mu_0(z)$, consistency and unconfoundedness give $\mathbb{E}[Y \mid X = x, Z = z] = \mu_0(z) + x\,\tau(z)$ and

$$m(z) = \mathbb{E}[Y \mid Z = z] = e(z)\mu_1(z) + \{1 - e(z)\}\mu_0(z) = \mu_0(z) + e(z)\tau(z),$$

so $\mathbb{E}[\tilde{Y} \mid X, Z = z] = \{X - e(z)\}\tau(z) = \tilde{X}\tau(z)$; the remainder is ε.

6.4.2 The R-loss

Motivation. Turn the representation into a loss and interpret its minimizer.

Oracle R-loss. Knowing m and e, minimize

$$L_R(\tau) = \mathbb{E}\big[\{\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\}^2\big], \qquad \tilde{\tau} = \arg\min_{\tau}\Big\{\frac{1}{n}\sum_{i=1}^{n}\big(\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\}\Big\},$$

with regularizer Λ_n (Robinson 1988; Nie and Wager 2021, §2).

A residual regression. Given $Z = z$, $Q_z(t) = \mathbb{E}[(\tilde{Y} - \tilde{X}t)^2 \mid Z = z]$ is least squares of $\tilde{Y}$ on $\tilde{X}$ through the origin; $0 = \mathbb{E}[\tilde{X}(\tilde{Y} - \tilde{X}t_z) \mid Z = z]$ gives

$$t_z = \frac{\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z]}{\mathbb{E}[\tilde{X}^2 \mid Z = z]}.$$

Why this slope is the CATE. $\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z] = \mathbb{E}[\tilde{X}\,\mathbb{E}[\tilde{Y} \mid X, Z = z] \mid Z = z] = \mathbb{E}[\tilde{X}^2 \mid Z = z]\tau(z)$, so $t_z = \tau(z)$: *the CATE is the slope of outcome residuals on treatment residuals.*

6.4.3 Algorithm

Motivation. Implement the R-learner with cross-fitting.

1. Split the data into K folds.
2. For each fold k, fit on the other folds $\hat{m}^{(-k)}(z) \approx \mathbb{E}[Y \mid Z = z]$, $\hat{e}^{(-k)}(z) \approx \Pr(X = 1 \mid Z = z)$.
3. Out-of-fold residuals: $Y_i^r = Y_i - \hat{m}^{(-k_i)}(Z_i)$, $X_i^r = X_i - \hat{e}^{(-k_i)}(Z_i)$.
4. Estimate $\hat{\tau} = \arg\min_{\tau}\Big\{\frac{1}{n}\sum_{i=1}^{n}\big(Y_i^r - X_i^r\tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\}\Big\}$.

The R-loss weights residual treatment variation.

$$\sum_i \big(Y_i^r - X_i^r\tau(Z_i)\big)^2 = \sum_i (X_i^r)^2\Big(\frac{Y_i^r}{X_i^r} - \tau(Z_i)\Big)^2.$$

Units with large $|X_i^r|$ (treatment poorly predicted by Z_i) inform $\tau(Z_i)$; those with treatment nearly determined by Z_i barely do.

What can optimize the R-loss? The second stage is empirical risk minimization,

$$\min_{\tau\in\mathcal{F}} \frac{1}{n}\sum_i (Y_i^r - X_i^r\tau(Z_i))^2 + \Lambda_n\{\tau\} :$$

(sparse) linear τ, splines, kernels, or boosting, forests, neural nets (any squared-loss learner with weights $(X_i^r)^2$, targets Y_i^r/X_i^r), tuned by *validation R-loss*, not outcome prediction loss.

6.4.4 Orthogonality and the quasi-oracle property

Motivation. Why nuisance errors matter only at second order.

For a candidate τ use the moment

$$\psi(O;\tau,m,e) = \{Y - m(Z) - (X - e(Z))\tau(Z)\}\{X - e(Z)\}, \qquad \mathbb{E}[\psi(O;\tau^*,m,e) \mid Z = z] = 0$$

Proposition 6.3 (Neyman orthogonality)

Let $\hat{m} = m + \delta_m$, $\hat{e} = e + \delta_e$ and hold $\tau = \tau^*$. Then

$$\mathbb{E}[\psi(O; \tau^*, \hat{m}, \hat{e}) \mid Z = z] = \underbrace{\delta_m(z)\delta_e(z)}_{\text{product}} - \underbrace{\tau^*(z)\delta_e(z)^2}_{\text{square}}.$$

Proof [Supplement]. With $\tilde{Y} = Y - m(Z)$, $\tilde{X} = X - e(Z)$, the moment is $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^* - \delta_m + \delta_e\tau^*)(\tilde{X} - \delta_e) \mid Z]$. Here $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^*)\tilde{X} \mid Z] = \mathbb{E}[\varepsilon\tilde{X} \mid Z] = 0$, $-\delta_e\mathbb{E}[\tilde{Y} - \tilde{X}\tau^* \mid Z] = -\delta_e\mathbb{E}[\varepsilon \mid Z] = 0$, and $(-\delta_m + \delta_e\tau^*)\mathbb{E}[\tilde{X} - \delta_e \mid Z] = (-\delta_m + \delta_e\tau^*)(-\delta_e)$ as $\mathbb{E}[\tilde{X} \mid Z] = 0$. ■

No first-order term remains, so flexible nuisance fits are usable (Nie and Wager 2021; Chernozhukov et al. 2018).

The quasi-oracle property. With the R-risk $R(\tau) = \mathbb{E}[\{Y - m(Z) - (X - e(Z))\tau(Z)\}^2]$, $\hat{\tau}$ fitted on $\hat{m}, \hat{e}$ residuals and $\tilde{\tau}$ the oracle, Nie and Wager show

$$R(\hat{\tau}) - R(\tau^*) \approx R(\tilde{\tau}) - R(\tau^*),$$

provided nuisance RMSEs are $o(n^{-1/4})$ (product $o(n^{-1/2})$). The rate is governed by the complexity of τ, not of m and e.

Why cross-fitting matters. If unit i trains $\hat{m}, \hat{e}$, they can memorize its noise, shrinking residuals, and the second stage may read overfitting artefacts as heterogeneity. Out-of-fold residuals avoid this: orthogonality handles nuisance error, cross-fitting prevents reusing the same noise.

6.4.5 Comparison and summary

Motivation. Contrast the R-learner with the DR-learner.

	DR-learner	R-learner
Second stage	regress $\hat{\phi}_i^{DR}$ on Z_i	minimize a residualised weighted loss
Orthogonality	via the EIF / DR score	via Robinson residualisation
Main nuisances	μ_0, μ_1, e	m, e
Model selection	validation MSE for $\hat{\phi}^{DR}$	validation R-loss

DR pseudo-outcome:

$$\hat{\phi}_i^{DR} = \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i) + \frac{X_i\{Y_i - \hat{\mu}_1(Z_i)\}}{\hat{e}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{\mu}_0(Z_i)\}}{1 - \hat{e}(Z_i)}.$$

Validation losses on a held-out V: DR, $\sum_{i \in V}\{\hat{\phi}_i^{DR} - \hat{\tau}(Z_i)\}^2$; R, $\sum_{i \in V}\{Y_i^r - X_i^r\hat{\tau}(Z_i)\}^2$.

Uses of the R-learner: CATE after removing prognosis $m(z)$ and assignment $e(z)$; model selection by validation R-loss; shrinkage or structure placed directly on τ when the signal is weak.

Summary.

- Start from Robinson's decomposition $Y - m(Z) = \{X - e(Z)\}\tau(Z) + \varepsilon$.
- Cross-fit m and e, then minimize the R-loss.
- The R-loss is orthogonal to nuisance error and fits many ML tools.
- Quasi-oracle: difficulty lies mainly in the complexity of τ.

Chapter 7. Matching and Instrumental Variables

7.1 Matching

Motivation. How to estimate the ATT by imputing each treated unit's missing control outcome from similar controls, with little dependence on modeling choices.

7.1.1 Matching as nonparametric imputation

Motivation. Matching imputes counterfactual control outcomes without a model.

7.1.1.1 Setup

N units, treatment $X_i \in \{0,1\}$, covariates Z_i, potential outcomes $Y_i(0), Y_i(1)$; $N_1 = |\{i : X_i = 1\}|$, $N_0 = N - N_1$. Assume ignorability $(Y(0), Y(1)) \perp\!\!\!\perp X \mid Z$ and positivity $0 < P(X = 1 \mid Z) < 1$. Target: the ATT.

7.1.1.2 From regression to matching

With $\mu_t(Z) = \mathbb{E}[Y \mid X = t, Z]$,

$$\text{ATT} = \mathbb{E}_{Z|X=1}\big[\underbrace{\mu_1(Z)}_{\text{observed}} - \underbrace{\mu_0(Z)}_{\text{impute}}\big].$$

Regression imputes $\mu_0(Z_i)$ by a model:

$$\hat{\tau}_{\text{reg}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{\mu}_0(Z_i)).$$

If the groups differ in Z, the result depends on the form of $\hat{\mu}_0$ (Section 7.1.5); matching imputes *nonparametrically*.

7.1.1.3 The matching idea

Given a large reservoir of controls, match each treated unit to one or more controls with similar covariates.

Matching as study design. Matching can be done before outcomes are measured: select treated units and controls, then measure outcomes. Design (whom to compare) is thus separated from analysis and made blind to outcomes (Stuart 2010, §1).

Bias–variance trade-off in M. With M controls per treated unit: $M = 1$ gives lower bias, higher variance; $M = 3$ uses farther matches (more bias) but averages three outcomes (less variance) (Abadie and Imbens 2006).

7.1.1.4 The matching estimator

With matched set $\mathcal{M}_i$ of treated unit i, under ignorability

$$\hat{\tau}_{\text{match}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\Big(Y_i - \underbrace{\frac{1}{|\mathcal{M}_i|}\sum_{j\in\mathcal{M}_i} Y_j}_{\approx Y_i(0)}\Big).$$

Matching may be one-to-one or one-to-many, with or without replacement.

7.1.1.5 Exact matching and coarsened exact matching

Exact matching: match i to a control with $Z_j = Z_i$; perfect balance, no model dependence, but infeasible with many or continuous covariates.
Coarsened exact matching (CEM; Iacus et al. 2012): discretize covariates, then match exactly on the coarsened values; some treated units may remain unmatched (no overlap).

7.1.2 Nearest-neighbor matching and its theory

Motivation. NN matching: definition, algorithms and large-sample behavior.

7.1.2.1 Definition

Let $\mathcal{M}_i$ index the M closest units of the opposite group in a distance $\|\cdot\|$, and

$$\hat{Y}_i(t) = \begin{cases} Y_i, & X_i = t, \\ \dfrac{1}{M}\sum_{j\in\mathcal{M}_i} Y_j, & X_i = 1-t, \end{cases} \qquad t \in \{0,1\},$$

with $\hat{\tau}_{\text{ATT}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{Y}_i(0))$.

7.1.2.2 Algorithms (matching without replacement)

- **Greedy:** order the treated units (e.g. hardest first); give each the M closest remaining controls, remove them, repeat.
- **Optimal:** in the $N_1 \times N_0$ distance matrix D choose exactly M entries per row and at most one per column, minimizing total distance (Hungarian algorithm, polynomial time).

Why order matters. With $M = 1$, an arbitrary order lets i_2 take the only control near the isolated i_3 (distance 0.9), forcing i_3 to a control 2.9 away. Hardest-first: i_3 gets its control (1.2), i_2 another (1.1); the worst distance drops from 2.9 to 1.2.

7.1.2.3 The curse of dimensionality

N uniform points in p dimensions have spacing about $N^{-1/p}$ per axis, so the NN distance is $O(N^{-1/p})$. If μ_0 is L-Lipschitz, the mismatch bias satisfies

$$|\mu_0(Z_i) - \mu_0(Z_j)| \le L\,\|Z_i - Z_j\| = O(N^{-1/p}).$$

p	1	5	10	20
$N^{-1/p}$ for $N = 10{,}000$	0.0001	0.16	0.40	0.63

Hence dimension reduction, e.g. propensity-score matching (Abadie and Imbens 2006, Thm. 1).

7.1.2.4 ATT matching: independence and consistency

With fixed M and replacement, each treated unit appears once as a subject, so the imputed $\hat{Y}_i(0)$ are independent across treated units. With the LLN and negligible bias (large control reservoir), the ATT estimator is $\sqrt{N}$-consistent; matching is most natural for the ATT (Abadie and Imbens 2006, Thms. 1–4).

7.1.2.5 A toy example

$N_1 = 3$, $N_0 = 6$, $p = 2$, $M = 1$, distance $\|\cdot\|_1$:

Treated				Control			
Unit	Z_1	Z_2	Y	Unit	Z_1	Z_2	Y
$i=1$	3	5	8	$j=1$	1	2	2
$i=2$	6	2	12	$j=2$	2	6	4
$i=3$	7	7	10	$j=3$	4	1	6
				$j=4$	3	4	5
				$j=5$	5	3	7
				$j=6$	8	6	9

Matches: $i=1 \to j=4$ (distance 1), $i=2 \to j=5$ (2), $i=3 \to j=6$ (2), so

$$\hat{\tau}_{\text{ATT}} = \tfrac{1}{3}\big[(8-5)+(12-7)+(10-9)\big] = 3.0.$$

The matches are imperfect ($Z_2 - Z_5 \neq 0$); the resulting bias is corrected next.

7.1.2.6 Bias-corrected matching (Abadie and Imbens 2011)

Y_j estimates $\mu_0(Z_j)$, not $\mu_0(Z_i)$; shift it by the predicted mismatch:

$$\hat{\tau}_{\text{bc}} = \frac{1}{N_1}\sum_{i:X_i=1}\Big(Y_i - \frac{1}{M}\sum_{j\in\mathcal{M}_i}\big[Y_j + \hat{\mu}_0(Z_i) - \hat{\mu}_0(Z_j)\big]\Big),$$

with $\hat{\mu}_0$ a regression fit of $\mathbb{E}[Y \mid Z, X=0]$. Default in the R packages `MatchIt` and `Matching`.

7.1.2.7 Inference after matching

ATT variance estimator (Abadie and Imbens 2006):

$$\widehat{\text{Var}}(\hat{\tau}_{\text{ATT}}) = \frac{1}{N_1^2}\sum_{i=1}^{N}\big(1 + K_M(i)\big)^2\hat{\sigma}_i^2,$$

$K_M(i)$ the number of times i is used as a match, $\hat{\sigma}_i^2$ an estimate of $\text{Var}(Y_i \mid Z_i, X_i)$. Frequently used units inflate the variance, like extreme IPW weights. **The bootstrap is invalid** for NN matching: the matched unit's identity is a discontinuous function of the data (Abadie and Imbens 2008).

7.1.2.8 Kernel matching

All controls within bandwidth h get weight, closer ones more; with kernel K (e.g. $K(u) = e^{-u^2/2}$),

$$w_i(j) = \frac{K\Big(\frac{Z_j - Z_i}{h}\Big)}{\sum_{k:X_k=0} K\Big(\frac{Z_k - Z_i}{h}\Big)}, \qquad \hat{\tau}_{\text{kernel}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\Big(Y_i - \sum_{j:X_j=0} w_i(j)Y_j\Big).$$

It suffers the same curse of dimensionality.

7.1.3 Propensity-score matching

Motivation. Reduce matching to one dimension, with caveats.

Dimension reduction. With 20 binary covariates there are $2^{20} \approx 10^6$ patterns. The propensity score $e(Z) = P(X = 1 \mid Z)$ is one-dimensional, and $Y(0), Y(1) \perp\!\!\!\perp X \mid Z$ implies $Y(0), Y(1) \perp\!\!\!\perp X \mid e(Z)$.

Distance. $d(Z_i, Z_j) = |\hat{e}(Z_i) - \hat{e}(Z_j)|$, or (preferred, as it stretches differences near 0 and 1)

$$d(Z_i, Z_j) = |\operatorname{logit} \hat{e}(Z_i) - \operatorname{logit} \hat{e}(Z_j)|, \qquad \operatorname{logit}(p) = \ln \frac{p}{1-p}.$$

Specification: balance, not fit. Estimate the PS by logistic regression, GBM, random forest, etc.; the goal is covariate balance, not prediction. Collinearity is irrelevant and overfitting can help (Appendix). Iterate: check balance on covariates, squares and interactions, add terms if imbalanced. Impute missing covariates and add missingness indicators (Stuart 2010; Rosenbaum and Rubin 1984).

Common support. Trim units outside the overlap of the two PS distributions.

Caveats. Matching on the *estimated* PS can be more efficient than on the true PS under correct specification (Abadie and Imbens 2016). The "PS paradox" (King and Nielsen 2019): a narrower caliper improves PS balance but can worsen covariate balance. Always check multivariate covariate balance.

7.1.4 Practical matching

Motivation. The iterative matching workflow and its design choices.

Workflow. Distance → match → check balance → if imbalanced, change metric, M or caliper and repeat; else estimate and infer. Regression or weighting on the matched sample corrects residual imbalance.

Design choices.

- Covariates (adjustment criterion; never post-treatment variables).
- Distance: Mahalanobis, propensity score, or hybrid.
- M fixed or caliper-based; smaller M: less bias, more variance. A large control group suits 1-to-many matching.
- With or without replacement.

Mahalanobis distance $d(x, y) = \sqrt{(x-y)^\top S^{-1}(x-y)}$, S the sample covariance (treated or pooled), works for few continuous covariates ($p \lesssim 5$–8); otherwise the PS reduces dimension. Common practice: PS caliper, then Mahalanobis within it.

Caliper matching. All controls within distance $< c$ are matches, so M varies by unit (e.g. $M_1 = 2$, $M_2 = 3$, $M_3 = 1$). Common choice $c = 0.2 \times \mathrm{SD}(\operatorname{logit} \hat{e}(Z))$ (Austin 2011).

Full matching (Rosenbaum 1991; Hansen 2004). Every unit is placed in a matched set with variable ratio (1:k or k:1), minimizing total within-set distance. It discards no controls in the overlap and yields a weighted sample, between matching and IPW; often with a caliper. R: `MatchIt`, `method = "full"`.

With versus without replacement. With: easier, order-free, but units with extreme PS may be reused and dominate. Without: no reuse, but order-dependent (greedy vs. optimal). Software: `MatchIt` (Ho et al.), `Matching` (Sekhon).

Checking balance. Standardized mean difference of covariate k:

$$\mathrm{SMD}_k = \frac{1}{s_t} \cdot \frac{1}{N_1} \sum_{i=1}^{N} X_i \Big(Z_{ik} - \frac{1}{|\mathcal{M}_i|} \sum_{j \in \mathcal{M}_i} Z_{jk} \Big),$$

s_t the pre-matching treated-group SD. Compare distributions by empirical CDFs and Kolmogorov–Smirnov statistics. **Do not use p-values**: they mix balance with sample size, and balance is an in-sample property (Ho et al. 2007).

Love plot. Dot plot of SMDs per covariate before and after adjustment, lines at ± 0.1. In the LaLonde (1986) data (PS, age, education, married, no degree, race, 1974/1975 earnings), unadjusted |SMD| reach 1.5–2; after matching all lie within ± 0.1.

7.1.5 Model dependence and empirical evidence

Motivation. Matching reduces sensitivity to modeling assumptions; the LaLonde benchmark tests it.

The extrapolation problem. When groups differ on covariates, regression extrapolates into sparse regions: linear and quadratic fits agreeing on the controls can differ widely on the treated (Ho, Imai, King and Stuart 2007).

Matching as preprocessing. After matching, regression interpolates, and specifications agree:

(1) match for balance, (2) regress/weight on the matched sample, (3) obtain estimates robust to model choice.

The LaLonde benchmark. The NSW job-training experiment (LaLonde 1986) has experimental ATT $\approx$ \$1,794. With observational controls (CPS/PSID), which differ sharply in demographics and prior earnings, unadjusted estimates range from $-$\$15,000 to $+$\$2,000.

Lessons. Dehejia and Wahba (1999): PS matching on a subsample with pre-treatment earnings recovers the benchmark. Smith and Todd (2005): fragile across subsamples, methods and metrics. Without pre-treatment earnings, the key confounder, ignorability itself is doubtful. Covariate sufficiency comes before method choice.

Practical takeaways.

- Matching is nonparametric preprocessing; follow it by regression or weighting.
- Check multivariate covariate balance, not just PS balance.
- In high dimensions combine matching with regression.
- Complement with sensitivity analysis for unmeasured confounding.

7.1.6 Appendix: ATE matching: why it is harder

The ATE needs $\hat{Y}_i(0)$ for treated and $\hat{Y}_i(1)$ for controls, so each unit is both subject and match, creating cross-group dependence. For $p > 2$, $\mathrm{Bias}(\hat{\tau}_{\mathrm{ATE}}) = O(N^{-1/p}) \gg O(N^{-1/2})$, so simple matching is not $\sqrt{N}$-consistent; use bias correction (Abadie and Imbens 2011) or IPW/DR (Abadie and Imbens 2006).

7.1.7 Appendix: Why the estimated PS can beat the true PS

Units with $e(Z_i) = e(Z_j)$ can have very different covariates. The fitted $\hat{e}(Z) = g(Z^\top \hat{\beta})$ adapts to the sample, so $\hat{e}(Z_i) \approx \hat{e}(Z_j)$ means closeness in the directions separating treated from control here: implicit covariate refinement. Under correct specification $\mathrm{Var}(\hat{\tau}_{\mathrm{ATT}} \mid \hat{e}) \leq \mathrm{Var}(\hat{\tau}_{\mathrm{ATT}} \mid e)$ (Abadie

and Imbens 2016); similarly IPW with $\hat{e}$ attains the efficiency bound and with e does not (Hirano, Imbens and Ridder 2003). The advantage disappears under misspecification.

7.1.8 Appendix: The PS paradox (King and Nielsen 2019)

Treated unit A (high income, low education) and control B (low income, high education) may both have $\hat{e} = 0.50$: perfect PS balance, opposite covariates. A tighter caliper yields more such pairs, so covariate balance can worsen, requiring more extrapolation than the raw data.

7.1.9 Appendix: PS overfitting and the broader picture

Overfitting can help, since the goal is in-sample balance; the limit is overlap ($\hat{e} \to 0$ or 1). The estimated PS beats the true PS and overfitting helps by the same sample adaptation, yet the paradox shows 1D compression loses information. Use the PS for dimension reduction, always verify multivariate balance, and consider Mahalanobis matching within a PS caliper.

7.1.10 Appendix: Beyond binary treatment

For multiple levels, compare pairs by binary matching or use the generalized propensity score (GPS; Imai and van Dyk 2004). For continuous treatment the GPS is the conditional density of treatment given covariates (Hirano and Imbens 2004); subclassification or weighting on it is more common than matching.

7.1.11 Appendix: Bias-corrected matching versus post-matching regression

Bias correction: matching is primary, regression corrects the $Z_i \neq Z_j$ gap within pairs. Post-matching regression: matching is preprocessing and $Y \sim X + Z$ on the matched data estimates. Full-sample regression uses all controls and is fine if the model is correct, but extrapolates if it is wrong.

	Model correct	Model wrong
Full-sample regression	optimal (more data)	risky (extrapolation)
Post-matching regression	less efficient (data discarded)	safe (interpolation)

Example: controls in $Z \in [0, 3]$, treated in $[5, 8]$; linear and quadratic fits give different $\hat{\mu}_0(6)$, but after matching both agree (Ho et al. 2007).

7.1.12 Appendix: Stratification

For a discrete covariate Z (levels k) making assignment ignorable, $\mathbb{E}[Y(1)] = \sum_k \mathbb{E}(Y \mid Z = k, X = 1)P(Z = k)$, estimated by $\sum_k \bar{Y}_{k,1} n_k/n$ (n_k units, cell means $\bar{Y}_{k,x}$); hence

$$\hat{\tau} = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) \frac{n_k}{n}.$$

For continuous Z split into k classes: $\hat{\tau}_k = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) n_k/n$, generally biased. With $R_k = 1 - \frac{\mathbb{E}(\hat{\tau}_k) - \tau}{\mathbb{E}(\hat{\tau}_1) - \tau}$ the fraction of bias removed, Cochran (1968): $R_k \geq 90\%$ for $k \geq 5$ over a broad class of models.

7.2 Instrumental variables

Motivation. How to learn a causal effect under unmeasured confounding from a variable that moves the treatment but affects the outcome only through it.

7.2.1 Motivation

Motivation. Instruments in practice, and what makes one good.

7.2.1.1 Instrumental variables in action

Study	Instrument Z	Treatment X	Outcome Y
Angrist and Krueger (1991)	quarter of birth	years of schooling	earnings
Card (1995)	proximity to college	years of schooling	earnings
Acemoglu et al. (2001)	settler mortality	institutional quality	GDP per capita
Miguel et al. (2004)	rainfall shocks	economic growth	civil conflict
Levitt (1996)	overcrowding litigation	prison population	crime rate

Returns to education. Ability U confounds schooling X and earnings Y. Angrist–Krueger: under compulsory-schooling laws, those born early in the year can drop out sooner. Card: living near a four-year college lowers the cost of attending.
Institutions and development. Geography and culture U confound institutions X and GDP Y. Acemoglu–Johnson–Robinson: high settler mortality led to extractive, persistent institutions. Miguel et al.: rainfall instruments growth in studying African civil conflict.

7.2.1.2 Why instruments?

Back-door adjustment needs all confounders measured. Otherwise, use Z affecting Y only through X ($Z \to X \to Y$, $X \leftarrow U \to Y$, $Z \perp\!\!\!\perp U$): an *instrumental variable*.

7.2.1.3 Conditions for a good instrument

1. **Relevance:** large $\mathrm{Cov}(X, Z)$, else the IV estimate has large variance;
2. **Exclusion restriction:** no direct effect on the outcome;

and (implicitly) no confounding between Z and Y.

7.2.2 IV under linearity

Motivation. Classical IV estimators in the linear constant-effect model.

7.2.2.1 The econometric setup

$Y = \beta_0 + \beta_1 X + \beta_2 W + \varepsilon$, scalar treatment X, covariates W, estimand β_1. X is *endogenous* (correlated with ε), W *exogenous*, so OLS of β_1 is biased. An instrument is a K-vector Z with $\mathrm{Cov}[\varepsilon, Z] = \mathrm{Cov}[\varepsilon, W] = 0$; $K > 1$: over-identified, $K = 1$: just-identified.

7.2.2.2 Just-identified, no covariates

IV estimator.

$$\hat{\beta}_{1,\text{iv}} = \frac{\widehat{\text{Cov}}(Y_i, Z_i)}{\widehat{\text{Cov}}(X_i, Z_i)} = \frac{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \bar{Y})(Z_i - \bar{Z})}{\frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X})(Z_i - \bar{Z})}.$$

For binary Z, since $\widehat{\text{Cov}}(Y, Z) = \hat{p}(1-\hat{p})(\bar{Y}_1 - \bar{Y}_0)$ ($\hat{p}$ the fraction with $Z = 1$) and likewise for X, this is the *Wald estimator*

$$\hat{\beta}_{1,\text{iv}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0},$$

$\bar{Y}_z, \bar{X}_z$ the means in group $Z = z$.

Indirect least squares (ILS). Fit the reduced forms

$$Y = \pi_{10} + \pi_{11} Z + \varepsilon_1, \qquad X = \pi_{20} + \pi_{21} Z + \varepsilon_2;$$

each slope is $\widehat{\text{Cov}}(\cdot, Z)/\widehat{\text{Var}}(Z)$, so $\hat{\beta}_{1,\text{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21} = \hat{\beta}_{1,\text{iv}}$.

Two-stage least squares (2SLS). Stage 1: $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z$; stage 2: OLS of $Y = \beta_0 + \beta_1 \hat{X} + \varepsilon$. ($\hat{X}$ is the "unconfounded portion" of X.) Since $\widehat{\text{Cov}}(Y, \hat{X}) = \hat{\pi}_{21}\widehat{\text{Cov}}(Y, Z)$ and $\widehat{\text{Var}}(\hat{X}) = \hat{\pi}_{21}^2 \widehat{\text{Var}}(Z)$, $\hat{\beta}_{1,\text{2sls}} = \widehat{\text{Cov}}(Y, Z)/(\hat{\pi}_{21}\widehat{\text{Var}}(Z)) = \hat{\beta}_{1,\text{iv}}$. Hence $\hat{\beta}_{1,\text{iv}} = \hat{\beta}_{1,\text{ils}} = \hat{\beta}_{1,\text{2sls}}$.

7.2.2.3 Just-identified, with covariates

ILS: $Y = \pi_{10} + \pi_{11} Z + \pi_{12} W + \varepsilon_1$, $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$, $\hat{\beta}_{1,\text{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21}$. 2SLS: $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z + \hat{\pi}_{22} W$, then $Y = \beta_0 + \beta_1 \hat{X} + \beta_2 W + \varepsilon$; $\hat{\beta}_{1,\text{2sls}}$ is the OLS coefficient of $\hat{X}$.

IV derivation. Let $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$, $Y = \beta_0 + \beta_1 X + \underbrace{\beta_2 W + \varepsilon}_{\eta}$, and assume $\text{Cov}[W, Z] = 0$. Then $\text{Cov}[X, Z] \neq 0$, $\text{Cov}[\eta, Z] = 0$, so

$$\text{Cov}[Y, Z] = \text{Cov}[\beta_0 + \beta_1 X + \eta, Z] = \beta_1 \,\text{Cov}[X, Z],$$

$$\beta_1 = \frac{\text{Cov}[Y, Z]}{\text{Cov}[X, Z]} = \frac{\text{Cov}[Y, Z]/\,\text{Var}[Z]}{\text{Cov}[X, Z]/\,\text{Var}[Z]} = \frac{\text{reduced form}}{\text{first stage}}.$$

2SLS derivation. Substituting X,

$$\begin{aligned} Y &= \beta_1(\pi_{22} W + \pi_{21} Z + \pi_{20} + \varepsilon_2) + \beta_2 W + \beta_0 + \varepsilon \\ &= \beta_1 \pi_{21} Z + (\beta_2 + \beta_1 \pi_{22}) W + \underbrace{(\beta_1(\pi_{20} + \varepsilon_2) + \beta_0 + \varepsilon)}_{\xi}, \end{aligned}$$

hence

$$Y = \beta_1(\pi_{22} W + \pi_{21} Z) + \beta_2 W + \xi = \beta_1 \underbrace{\mathbb{E}[X \mid W, Z]}_{\text{population analogue of } \hat{X}} + \beta_2 W + \xi,$$

with π_{20} absorbed into ξ ($\mathbb{E}[X \mid W, Z] - \pi_{20} = \pi_{21} Z + \pi_{22} W$). Regressing Y on $\mathbb{E}[X \mid W, Z]$ and W recovers β_1; the sample version uses $\hat{X}$.

7.2.2.4 From linear to nonparametric IV

With a constant effect β_1, IV = ILS = 2SLS. What does IV identify when $Y_i(1) - Y_i(0)$ varies? Key case: a randomized experiment with non-compliance, assignment Z instrumenting treatment received X (Angrist, Imbens and Rubin 1996).

7.2.3 Randomized experiments with non-compliance

Motivation. Identify the complier effect under random assignment, monotonicity and exclusion.

7.2.3.1 Setting

RCT of a drug: Z = assignment, X = treatment received, Y = outcome. Some assigned patients refuse; some controls obtain the drug. Comparing $X = 1$ with $X = 0$ is confounded by self-selected compliance; Z is randomized and acts only through X, so it is an instrument.

Potential outcomes (Angrist, Imbens and Rubin 1996). Outcomes $Y(z)$ and treatment received $X(z)$ are indexed by assignment $z = 0, 1$; observed Z, $X = X(Z)$, $Y = Y(Z)$. Latent *compliance types* $S = (X(0), X(1))$:

	$X(0) = 0$	$X(0) = 1$
$X(1) = 0$	never-taker ($X = 0$ always)	defier (X opposes Z)
$X(1) = 1$	complier (X follows Z)	always-taker ($X = 1$ always)

Each observed (Z, X) cell is a mixture: $(0,0)$: {C, NT}; $(0,1)$: {AT, D}; $(1,0)$: {NT, D}; $(1,1)$: {C, AT}; extra assumptions are needed. Goal: $\text{CACE} = \mathbb{E}[Y(1) - Y(0) \mid \text{complier}] = \text{ITT}_y/\text{ITT}_x$.

7.2.3.2 Intent-to-treat effects

S is a baseline characteristic. With $\text{ITT}_s = \mathbb{E}[Y(1) - Y(0) \mid S = s]$, $s \in \{c, n, a, d\}$, and proportions π_s, the global ITT $\text{ITT}_y := \mathbb{E}[Y(1) - Y(0)]$ is

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \text{ITT}_n + \pi_a \text{ITT}_a + \pi_d \text{ITT}_d.$$

7.2.3.3 Identification assumptions

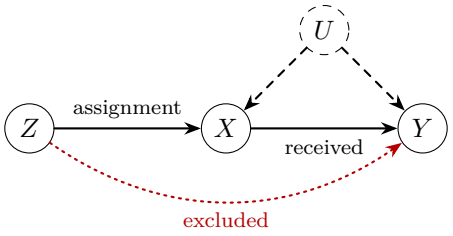


Figure. *Randomized encouragement with non-compliance: Z randomized (A1), no defiers (A2), no direct $Z \to Y$ path (exclusion, A3: dotted arrow absent); U confounds X and Y.*

A1 (random assignment) $\{Y(0), Y(1), X(0), X(1)\} \perp\!\!\!\perp Z$.

A2 (strong monotonicity) (1) no defiers: $X(1) \geq X(0)$ for all units;

(2) compliers exist: $\mathbb{E}[X \mid Z = 1] \neq \mathbb{E}[X \mid Z = 0]$. Often true by design; one-sided compliance means $X(0) = 0$.

A3 (exclusion restriction for non-compliers) $Y(0) = Y(1)$ for all units with $S \in \{n, a\}$: Z has no direct effect (e.g. placebo) on Y, as in linear IV.

7.2.3.4 Identification

A2 gives $\pi_d = 0$; A3 gives $\text{ITT}_n = \text{ITT}_a = 0$. Hence

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \cdot 0 + \pi_a \cdot 0 + 0 \cdot \text{ITT}_d = \pi_c \text{ITT}_c, \qquad \text{ITT}_c = \text{ITT}_y/\pi_c.$$

With $0 < \pi_c < 1$, $|\text{ITT}_y| < |\text{ITT}_c|$: the global ITT is attenuated.

Compliance-type proportions. Under A1–A2, $X = 1$ at $Z = 0$ only for always-takers and $X = 0$ at $Z = 1$ only for never-takers:

$$\pi_a = P(X = 1 \mid Z = 0) = \mathbb{E}[X \mid Z = 0], \qquad \pi_n = P(X = 0 \mid Z = 1) = 1 - \mathbb{E}[X \mid Z = 1],$$
$$\pi_c = 1 - \pi_a - \pi_n = \mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0].$$

Observed cells under monotonicity. $(0,0)$: compliers and never-takers; $(0,1)$: always-takers; $(1,0)$: never-takers; $(1,1)$: compliers and always-takers. The pure cells identify non-complier means; with the known proportions the mixed cells give the complier means:

$$\mathbb{E}[Y \mid X = 0, Z = 0] = \tfrac{\pi_c}{\pi_c+\pi_n}\mathbb{E}[Y(0) \mid \text{complier}] + \tfrac{\pi_n}{\pi_c+\pi_n}\mathbb{E}[Y(0) \mid \text{never-taker}],$$
$$\mathbb{E}[Y \mid X = 0, Z = 1] = \mathbb{E}[Y(0) \mid \text{never-taker}], \qquad \mathbb{E}[Y \mid X = 1, Z = 0] = \mathbb{E}[Y(1) \mid \text{always-taker}],$$
$$\mathbb{E}[Y \mid X = 1, Z = 1] = \tfrac{\pi_c}{\pi_c+\pi_a}\mathbb{E}[Y(1) \mid \text{complier}] + \tfrac{\pi_a}{\pi_c+\pi_a}\mathbb{E}[Y(1) \mid \text{always-taker}].$$

(Here $Y(0)$, $Y(1)$ are outcomes under treatment received; for compliers they equal the assignment-indexed ones.)

ITT on treatment. $\text{ITT}_x = \mathbb{E}[X(1) - X(0)] = \pi_c$: under monotonicity only compliers switch.

Theorem 7.1 (CACE / LATE; Imbens and Angrist 1994, Angrist et al. 1996)

Under A1–A3,

$$\text{CACE} \equiv \mathbb{E}[Y(1) - Y(0) \mid S = c] = \frac{\text{ITT}_y}{\text{ITT}_x} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]}.$$

Proof [Supplement]. By A1, $\text{ITT}_y = \mathbb{E}[Y(1, X(1)) - Y(0, X(0))]$; split by $S \in \{c, a, n, d\}$. Always- and never-takers have $X(1) = X(0)$, so by exclusion (A3) they contribute zero; by monotonicity (A2) there are no defiers. Hence $\text{ITT}_y = P(S = c)\,\mathbb{E}[Y(1) - Y(0) \mid S = c]$. Also $\text{ITT}_x = \mathbb{E}[X(1) - X(0)] = P(S = c)$, as $X(1) - X(0)$ is 1 for compliers and 0 otherwise. Dividing $(P(S = c) > 0)$ gives $\text{CACE} = \text{ITT}_y/\text{ITT}_x$, and by A1 each ITT is a difference of observed means. ■

The CACE (LATE) equals the Wald estimand with assignment as instrument.

7.2.3.5 A worked example

$N = 1{,}000$, 500 with $Z = 1$ and 500 with $Z = 0$.

1. **Proportions.** 100 of the $Z = 0$ group take the drug: $\hat{\pi}_a = 0.20$; 150 of the $Z = 1$ group refuse: $\hat{\pi}_n = 0.30$; $\hat{\pi}_c = 0.50$.
2. **ITT effects.** $\bar{Y}_{Z=1} = 7.0$, $\bar{Y}_{Z=0} = 6.0$: $\text{ITT}_y = 1.0$; $\bar{X}_{Z=1} = 0.70$, $\bar{X}_{Z=0} = 0.20$: $\text{ITT}_x = 0.50 = \hat{\pi}_c$.
3. **CACE** $= 1.0/0.50 = 2.0$.

The ITT (1.0) is attenuated; the naive $X = 1$ vs. $X = 0$ comparison is confounded.

7.2.3.6 CACE estimator

For two-sided non-compliance,

$$\widehat{\text{CACE}} = \frac{\widehat{\text{ITT}}_y}{\hat{\pi}_c} = \frac{\sum_i Y_i Z_i / \sum_i Z_i - \sum_i Y_i(1 - Z_i) / \sum_i (1 - Z_i)}{1 - \sum_i X_i(1 - Z_i) / \sum_i (1 - Z_i) - \sum_i (1 - X_i) Z_i / \sum_i Z_i},$$

denominator $\hat{\pi}_c = 1 - \hat{\pi}_a - \hat{\pi}_n$. Standard errors by asymptotics or bootstrap. Without A1–A3 only nonparametric bounds remain (Chapter 5), often too wide.

7.2.3.7 The "fourth condition": homogeneity

Homogeneity makes the IV estimand the population ATE. Any one of these suffices:

(a) constant effect, $Y(1) - Y(0) = \beta_1$;

(b) equal average effects across Z, $\mathbb{E}[Y(1) - Y(0) \mid Z = 0] = \mathbb{E}[Y(1) - Y(0) \mid Z = 1]$;

(c) no effect modification by U, $\mathbb{E}[Y(1) - Y(0) \mid U]$ constant.

(a) implies (b) implies (c).

(a) is rarely plausible, and (b) and (c) are still strong; applied work mostly relies on monotonicity and the complier-only LATE (Hernán and Robins 2020, §§16.3–16.4).

7.2.3.8 Summary

- **Linear IV:** $\beta_1 = \mathrm{Cov}(Y, Z)/\mathrm{Cov}(X, Z)$; IV = ILS = 2SLS (just-identified); constant effect.
- **LATE:** under A1–A3, CACE = $\mathrm{ITT}_y/\mathrm{ITT}_x$; heterogeneity allowed, local estimand.
- Under constant effects, CACE = ATE = β_1.

7.2.4 Appendix: Candidate instruments in observational studies

Type	Example	Key concern
Genetic variant (Mendelian randomization)	ALDH2 polymorphism, alcohol metabolism	pleiotropy (direct effect on Y?)
Provider preference	physician's prescribing tendency	unmeasured patient factors drive referral
Geographic/temporal access	distance to facility, policy change date	correlated with SES, urban/rural

Only relevance is testable; exclusion and no confounding need subject-matter knowledge (*What If*, Fine Point 16.1).

7.2.5 Appendix: IV versus other causal methods

IV estimation	IPW / g-methods
Does not require conditional exchangeability $Y(x) \perp\!\!\!\perp X \mid W$; works with unmeasured confounding.	Requires conditional exchangeability (all confounders measured).
Sensitive to violations of exclusion/no confounding, with bias amplification.	Minor violations give minor bias (no amplification).
Identifies only a local effect (compliers) under monotonicity.	Identifies the population ATE; extends to time-varying treatments.

Prefer IV with likely unmeasured confounding, a strong instrument, dichotomous treatment, and plausible homogeneity or monotonicity (*What If*, §16.6).

7.2.6 Appendix: Practical issues with classic IV studies

Quarter of birth (Angrist and Krueger 1991): weak (QOB explains $< 0.1\%$ of schooling variation); many-instruments bias (180 instruments push 2SLS toward OLS; Bound et al. 1995 got similar results with random data); exclusion (season of birth may affect health, school readiness); exogeneity

(birth timing tracks family background; Buckles and Hungerman 2013). Still foundational for IV/LATE.

Proximity to college (Card 1995): exclusion (correlated with urban/rural status, SES, school quality); monotonicity (may not uniformly raise schooling); LATE applies to marginal, likely lower-income compliers, so LATE $\neq$ ATE (Carneiro et al. 2011).

7.2.7 Appendix: Weak instruments

Three problems. (1) Wide confidence intervals: weak Z–X association inflates $\text{Var}(\hat{\beta}_{1,\text{iv}})$.

(2) Bias amplification: in $\hat{\beta}_1 = \text{ITT}_y/\text{ITT}_x$ a small numerator bias (minor exclusion violation) is divided by a small denominator.

(3) Finite-sample bias toward OLS, even with a valid instrument. Rule of thumb: first-stage $F < 10$ (test of $H_0 : \alpha = 0$ in $X = \alpha_0 + \alpha Z + \epsilon$) signals weakness (Staiger and Stock 1997; *What If*, §16.5).

Example (*What If*, §16.2): smoking cessation X, weight gain Y, cigarette price in birth state Z. $P(X = 1 \mid Z = 1) - P(X = 1 \mid Z = 0) = 25.8\% - 19.5\% = 6.3\%$, $F = 0.8 \ll 10$; IV estimate 2.4 kg, 95% CI $[-36.5, 41.3]$: uninformative.

What to do. Report the first-stage F against the threshold for K instruments (Stock and Yogo 2005). Use weak-instrument-robust sets: the Anderson–Rubin set bypasses the first stage and is optimal with one instrument. With many instruments prefer LIML to 2SLS (less biased) (Anderson and Rubin 1949; Andrews, Stock and Sun 2019; Montiel Olea and Pflueger 2013).

What not to do. Screen on $F > 10$ and report only passing specifications (worsens size distortion); shop for the instrument set with highest F; bootstrap IV inference; trust $\hat{\beta} \pm 1.96\,\text{SE}$ when F is small. The real fix is a stronger instrument.

7.2.8 Appendix: Many instruments and robust IV

Each of K instruments gives $\hat{\beta}_k = \text{Cov}(Y, Z_k)/\,\text{Cov}(X, Z_k)$; valid ones agree, invalid ones (e.g. pleiotropy) diverge and contaminate 2SLS. Robust alternatives: weighted median (Bowden et al. 2016), consistent if over 50% are valid; mode-based (Hartwig et al. 2017), consistent if a plurality agree; MR-Egger (Bowden et al. 2015), an intercept absorbing directional pleiotropy. Standard in Mendelian randomization with hundreds of genetic variants.

Chapter 8. Causal Data Science

Motivation. How to answer a causal query about a target population from data collected under different conditions, populations, sampling, missingness or measurements.

8.1 Motivation and big picture

Motivation. Real data rarely match the inferential target; the mismatches fall into a few dimensions.

Examples. *Genetics:* carcinogenic in rats, hence in humans? *Advertisement:* what transfers from product A's campaign to product B? *Robotics:* how can a rover trained in the California desert use Earth experience on Mars?

"Not all data are created equal." Data are collected under conditions differing from the target:

(1) different experimental conditions, (2) different populations, (3) non-random sampling, (4) non-random treatment assignment, (5) unmeasured relevant variables. These dimensions have been formalized, with conditions and algorithms.

Heterogeneous data sets. For $Q = P(y \mid \mathrm{do}(x))$, data sets may differ along four dimensions:

		Dataset 1	Dataset 2	Dataset n
d_1	population	Los Angeles	New York	Seoul
d_2	obs./exp.	experimental (randomized Z_1)	observational	experimental (randomized Z_2)
d_3	sampling	selection on age	selection on SES	–
d_4	measured	$\{X_1, Z_1, W, M, Y_1\}$	$\{X_1, X_2, Z_1, N, Y_2\}$	$\{X_2, Z_1, W, L, M, Y_1\}$

Goal. Make the most of multiple heterogeneous data sets with formal machinery (language, conditions, algorithms) for causal inference and *fusion* across studies.

Classic identifiability takes a query $P_x(y) = P(y \mid \mathrm{do}(x))$, a diagram and $P(\mathbf{V})$, and returns "no" or $P_x(y) = f_{\mathcal{G}}(P(\mathbf{V}))$. In data fusion the target may be the US with sources NY, LA, Boston, ..., each with its own diagram $(\mathcal{G}_{US}, \mathcal{G}_{NYC}, \dots)$ and distribution $(P_{NYC} + P_{LA} + \cdots)$:

(a) US: census data available (the target population);

(b) NY: survey data resembling the target;

(c) LA: survey data, younger subjects;

(d) Boston: age not recorded, mostly successful lawyers;

(e) SF: high post-treatment blood pressure;

(f) Texas: mostly Spanish subjects, high attrition;

(g) Arkansas: randomized trial on college students;

(h) Utah: RCT with paid volunteers, mainly unemployed;

(i) Wyoming: natural experiment on young athletes.

Tasks. With data collections as tuples (d_1, d_2, d_3, d_4):

1. Descriptive statistics: $(d_1, \mathrm{Samples(Obs)}, d_3, d_4) \to (d_1, \mathrm{Distribution(Obs)}, d_3, d_4)$.
2. Experimental statistics: $(d_1, \mathrm{Samples(do}(x)), d_3, d_4) \to (d_1, \mathrm{Distribution(do}(x)), d_3, d_4)$.
3. Causal inference from observational studies: $(d_1, \mathrm{Obs}, d_3, d_4) \to (d_1, \mathrm{do}(x), d_3, d_4)$ (mainly confounding bias, 1974–2011: Rubin, Robins, Dawid, Pearl, ...).

4. Generalized identification (experiments on other variables): $(d_1, \text{do}(Z), d_3, d_4) \to (d_1, \text{do}(x), d_3, d_4)$.
5. Sampling selection bias: $(d_1, d_2, \text{Select}(\text{Age}), d_4) \to (d_1, d_2, \{\}, d_4)$.
6. Transportability (external validity): $(\text{Bonobos}, d_2, d_3, d_4) \to (\text{Humans}, d_2, d_3, d_4)$.

Usually treated in isolation under parametric assumptions; together they form "causal data science".

Four dimensions. (1) Experimental conditions: generalized identification.

(2) Environmental conditions: transportability.

(3) Sampling conditions: recovering from selection bias.

(4) Responding conditions: recovering from missingness. Plus measurement (partial observation).

8.2 Experimental conditions: general identifiability

Motivation. Identify an effect from observational data plus experiments on other variables.

8.2.1 The challenge

Diet Z, cholesterol X, heart attack Y: $Z \to X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow Y$. Given $P(x, y, z)$ and $P(x, y \mid \text{do}(z))$:

$$Q = P(y \mid \text{do}(x)) = \frac{P(x, y \mid \text{do}(z))}{P(x \mid \text{do}(z))} = P(y \mid \text{do}(z), x)$$

(Rule 3 adds $\text{do}(z)$; Rule 2 exchanges $\text{do}(x)$ for x, as $\mathcal{G}_{\overline{Z}\underline{X}}$ has no back-door path). From $P(x, y, z)$ alone Q is not identifiable ($X \leftrightarrow Z \leftrightarrow Y$), so the experiment is essential.

8.2.2 Subtleties of z-ID

Same query, three graphs ($\mathcal{G}$ with $P(x, y, z)$; $\mathcal{G}_{\overline{Z}}$ with $P(x, y \mid \text{do}(z))$; $\mathcal{G}_{\overline{X}}$ for the query):

- $Z \to X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow Y$: **z-ID**.
- $Z \to X \to Y$, $X \leftrightarrow Y$, $Z \leftrightarrow Y$: **not z-ID**; $X \leftrightarrow Y$ persists in $\mathcal{G}_{\overline{Z}}$ (linear IV would help, nonparametric IV does not).
- $Z \leftrightarrow X$, $Z \to Y$, $Z \leftrightarrow Y$, $X \to Y$: **not z-ID**; Z does not cause X.

8.2.3 z-ID reduced to calculus

$Z \to X \to W \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow W$, $Z \leftrightarrow Y$, $X \leftrightarrow Y$, with $P(\mathbf{v})$ and $P(\mathbf{v} \mid \text{do}(z))$:

$$\begin{aligned} P(y \mid \text{do}(x)) &= \sum_w P(y \mid \text{do}(x), w) P(w \mid \text{do}(x)) = \sum_w P(y \mid \text{do}(x, w)) P(w \mid \text{do}(x)) \\ &= \sum_w \underbrace{P(y \mid \text{do}(w))}_{Q[Y]} \underbrace{P(w \mid \text{do}(x))}_{Q[W]}. \end{aligned}$$

Neither factor is identifiable from $P(\mathbf{V})$; both are from $P(\mathbf{v} \mid \text{do}(z))$:

$$\begin{aligned} Q[Y] &= P(y \mid \text{do}(w)) = P(y \mid \text{do}(w, z)) = \sum_x P(y \mid \text{do}(w, z), x) P(x \mid \text{do}(w, z)) \\ &= \sum_x P(y \mid \text{do}(w, z), x) P(x \mid \text{do}(z)) = \sum_x P(y \mid \text{do}(z), w, x) P(x \mid \text{do}(z)), \\ Q[W] &= P(w \mid \text{do}(x)) = P(w \mid \text{do}(x, z)) = P(w \mid \text{do}(z), x). \end{aligned}$$

8.2.4 Example: drug–drug interactions

Y: cardiovascular disease, B: blood pressure, X_1: antihypertensive, X_2: anti-diabetic. Graph (a): $X_1 \to B \to Y$, $X_2 \to Y$, $X_1 \leftrightarrow B$, $X_1 \leftrightarrow X_2$, $X_2 \leftrightarrow Y$. Effect of *both* drugs from single-drug experiments $\mathbb{Z} = \{\{X_1\}, \{X_2\}\}$:

$$P_{x_1,x_2}(y) = \sum_b P_{x_1,x_2}(y,b) = \sum_b P_{x_1}(b) P_{x_2,b}(y) = \sum_b P_{x_1}(b) P_{x_2}(y \mid b),$$

$P_{x_1}(b)$ from the X_1 trial, $P_{x_2}(y \mid b)$ from the X_2 trial (B unconfounded with Y in $\mathcal{M}_{x_2}$). Adding $B \leftrightarrow X_2$ (b) keeps identifiability; $B \leftrightarrow Y$ (c) or $X_1 \leftrightarrow Y$ (d) destroys it.

8.2.5 Algorithm and summary

Two phases, using probability axioms and do-calculus:

(1) modify and decompose the query into factors;

(2) identify each factor from some $P_{\mathbf{Z}_j} \in \mathbb{P} = \{P_{\mathbf{Z}_1}, \ldots, P_{\mathbf{Z}_m}\}$; fail only if some factor is identifiable from none.

Identifiability of any $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})$ from observational plus experimental studies is decidable for any graph, estimands are found in polynomial time, and do-calculus is complete. References: Pearl (1993, 1994); Tian and Pearl (2002); Shpitser and Pearl, Huang and Valtorta (2006); Bareinboim and Pearl (2012) z-ID; Lee, Correa and Bareinboim (2019) g-ID; Correa and Bareinboim (2019, 2020) σ-calculus.

8.3 Environmental conditions: transportability

Motivation. Transfer causal knowledge from source populations to a different target.

8.3.1 The problem

Can the effect of $\mathbf{X}$ on $\mathbf{Y}$ in a target Π^* be computed from findings in other populations (e.g. US education policy applied to South Korea)? Elsewhere this is external validity (Shadish, Cook and Campbell 2002; Cox 1958 asks for "some understanding of the reasons for the differences"; Manski 2007). Internal validity concerns the study's own population.

From lab to real world. H_0: all mechanisms agree ($f_z = f_z^*, \ldots$), trivially transportable; H_a: all differ, nothing transports. Real problems lie between.

8.3.2 What can an experiment in LA tell us about NYC?

Z (age) $\to X$, $Z \to Y$, $X \to Y$; only the age distribution differs. LA: $P(x,y,z)$ and $P(y \mid \mathrm{do}(x), z)$; NYC: $P^*(x,y,z)$. Then

$$Q = P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z)\, P^*(z)\,:$$

LA's z-specific effects reweighted by NYC's ages.

Transport formulas depend on the causal story. (a) Z = age: $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z)$.

(b) Z = language skill, a causally irrelevant proxy: $P^*(y \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x))$.

(c) Z = bio-marker on $X \to Z \to Y$: $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z \mid x)$. ($P^*$: target, P: source.)

8.3.3 Selection diagrams

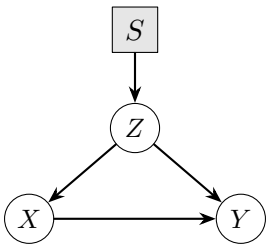


Figure. *Selection diagram for LA/NYC: $S \to Z$ marks the only differing mechanism, so* $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z)P^*(z)$.

Overlay source $\mathcal{G}$ and target $\mathcal{G}^*$ and add a *selection node* S (small square) into each variable whose mechanism or exogenous distribution may differ. E.g. $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to W \to Y$, $\mathcal{G}^*$ without $Z \to X$; with $f_X \neq f_X^*$, $f_Y \neq f_Y^*$ and the rest equal, $S \to X$, $S \to Y$ (edges present in only one graph: use the union). Target queries: $P^*(\cdot) = P(\cdot \mid S = s^*)$.

Theorem 8.1 (Transportability; Pearl and Bareinboim 2011)

Q is transportable from Π to Π^* iff there is a do-calculus reduction of $Q(\Pi^*)$ to an estimand that is a function of the observed distributions.

Proof [Supplement]. *If:* the selection diagram is one causal model with nodes S, and $Q(\Pi^*) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), S = s^*)$. Each do-calculus step is valid there, so a reduction to observables (with S only in factors estimable in Π^*) expresses $Q(\Pi^*)$ through available data. *Only if:* Bareinboim and Pearl (2012) construct, for non-transportable queries, two pairs of domains agreeing on all available distributions but not on $Q(\Pi^*)$; not reproduced here. ■

Example 8.1

$S \to Z$, $Z \to X$, $Z \to W \to Y$, $X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow W$, $X \leftrightarrow Y$:

$$\begin{aligned} Q = P^*(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x), S) = \sum_w P(y \mid \mathrm{do}(x), S, w)P(w \mid \mathrm{do}(x), S) \\ &= \sum_w P(y \mid \mathrm{do}(x), w)P(w \mid \mathrm{do}(x), S) = \sum_w P(y \mid \mathrm{do}(x), w)P(w \mid S) \\ &= \sum_w P(y \mid \mathrm{do}(x), w)P^*(w). \end{aligned}$$

Step 3: $(Y \perp_d S \mid X, W)_{\mathcal{G}_{\overline{X}}}$ (Rule 1); step 4: $X \notin An(W)$ (Rule 3). First factor from the source experiment, second from target observations.

Algorithm. Input: a selection diagram. Output: transportability and, if so, a formula saying what to measure where, e.g.

$$P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) \sum_w P^*(z \mid w) \sum_t P(w \mid \mathrm{do}(x), t)P^*(t).$$

A c-factor not identifiable from $P^*(\mathbf{V})$ must come from source experiments, possible only if no selection node points into it.

8.3.4 General transportability

Experiments over $\{\mathbf{Z}^1, \ldots, \mathbf{Z}^n, \mathbf{Z}^*\}$ in domains $\Pi = \{\Pi^1, \ldots, \Pi^n, \Pi^*\}$ sometimes suffice; the theory says when and how.

Example 8.2

$Q = P^*(y \mid \text{do}(x))$ from Π^a (LA; experiments on X) and Π^b (NYC; experiments on Z). Both: $X \to Z \to Y$, $X \leftrightarrow Z$, $X \leftrightarrow Y$; selection nodes into X, Y (Π^a) and X, Z (Π^b). Neither source alone suffices. Combined (S_{xzy}: selection nodes),

$$
\begin{aligned}
Q = P(y \mid \text{do}(x), S_{xzy}) &= \sum_z P(y \mid \text{do}(x), z, S_{xzy}) P(z \mid \text{do}(x), S_{xzy}) && \text{def., axioms} \\
&= \sum_z P(y \mid \text{do}(x), \text{do}(z), S_{xzy}) P(z \mid \text{do}(x), S_{xzy}) && \text{Rule 2} \\
&= \sum_z P(y \mid \text{do}(z), S_{xzy}) P(z \mid \text{do}(x), S_{xzy}) && \text{Rule 3} \\
&= \sum_z P(y \mid \text{do}(z), S_y) P(z \mid \text{do}(x), S_z) && \text{Rule 1} \\
&= \sum_z \underbrace{P^{(b)}(y \mid \text{do}(z))}_{\text{NYC}} \underbrace{P^{(a)}(z \mid \text{do}(x))}_{\text{LA}} && \text{def.}
\end{aligned}
$$

($P(y \mid \text{do}(z))$ from where S does not point to Y, $P(z \mid \text{do}(x))$ from where it does not point to Z; selection diagrams extend to multiple sources.)

8.3.5 Is the gold standard golden?

$Z \to X$, $Z \to Y$, $X \to Y$ with confounding and $S \to Z$. Randomization removes edges into X, but $S \to Z \to Y$ remains: even a perfect RCT needs a transportability argument.

Summary. Nonparametric transportability is decidable from selection diagrams, formulas are derived in polynomial time, and calculus and algorithm are complete, addressing external validity, meta-analysis and heterogeneity. References: Pearl and Bareinboim (2011); Bareinboim and Pearl (2012, 2013a,b, 2014); Lee, Correa and Bareinboim (2020).

8.4 Sampling conditions: recovering from selection bias

Motivation. Recover population quantities from samples whose inclusion depends on the variables.

8.4.1 The problem

Preferential inclusion of samples biases inference. Add a selection indicator S ($S = 1$: sampled). In Pearl's seven-variable model ($Z_1 \to W_1 \to X$, $Z_1 \to Z_3$, $Z_2 \to Z_3$, $Z_2 \to W_2 \to Y$, $Z_3 \to X$, $Z_3 \to Y$, $X \to W_3 \to Y$), isolated S is random sampling; $W_1 \to S$, $W_2 \to S$ make it covariate-dependent. When is, e.g., $P(y \mid x)$ estimable from $P(\mathbf{v} \mid S = 1)$?

8.4.2 The origin of selection bias

$X = U_x$, $Y = U_y$, $S = \neg(X \oplus Y)$, $P(u_x, u_y) = 1/4$. In the sample $P(X = 1, Y = 1 \mid S = 1) = P(X = 0, Y = 0 \mid S = 1) = 1/2$:

$$P(Y = 1 \mid X = 1, S = 1) - P(Y = 1 \mid X = 0, S = 1) = 1,$$

while in the population the difference is 0. Conditioning on a common effect ($X \to S \leftarrow Y$) creates dependence.

Confounding versus selection bias. Both are unblockable spurious flows: through a common cause $X \leftarrow Z \rightarrow Y$, or through a conditioned common effect $X \rightarrow S \leftarrow Y$.

Transportability versus selection. Nature-made versus man-made: in transportability S marks difference-producing factors; in selection its parents are sample-producing factors.

8.4.3 Variants

In statistics:

(i) without external data, is the odds ratio $OR(y,x) = \frac{P(y|x)P(y'|x')}{P(y'|x)P(y|x')}$ recoverable from $P(\mathbf{v} \mid S=1)$? (ii) without external data, is $P(y \mid x)$ recoverable? (iii) with external $P(\mathbf{t})$, $\mathbf{T} \subseteq \mathbf{V}$, is $P(y \mid x)$ recoverable? In causal inference: is $P(y \mid \text{do}(x))$ recoverable from $P(\mathbf{v} \mid S=1)$ and $P(\mathbf{t})$, or, with instruments, bounds on it?

8.4.4 Recoverability without external data

Theorem 8.2 (Bareinboim, Tian and Pearl 2014)

$Q = P(y \mid x)$ is recoverable from selection-biased data iff $(S \perp\!\!\!\perp Y \mid X)$.

Proof [Supplement]. *If:* $P(y \mid x) = P(y \mid x, S=1)$, estimable from the sample. *Only if:* if $S \not\perp\!\!\!\perp Y \mid X$, Bareinboim, Tian and Pearl (2014) construct two models compatible with $\mathcal{G}$ agreeing on $P(x,y \mid S=1)$ but not on $P(y \mid x)$; not reproduced here. ■

$Y \rightarrow S$: not recoverable; only $X \rightarrow S$: $P(y \mid x) = P(y \mid x, S=1)$.

8.4.5 Recoverability with external data

With $W_1 \rightarrow S \leftarrow W_2$, $P(y \mid x)$ is not recoverable from the sample alone; what if unbiased $P(W_1, W_2)$ is available?

Theorem 8.3 (Sufficiency)

$P(y \mid x)$ is recoverable if there is a set C with $(Y \perp\!\!\!\perp S \mid C, X)$ in $\mathcal{G}$ and $P(C, X)$ estimable; then

$$P(y \mid x) = \sum_c P(y \mid x, c, S=1) P(c \mid x).$$

Proof [Supplement]. $P(y \mid x) = \sum_c P(y \mid x, c) P(c \mid x)$, and $(Y \perp\!\!\!\perp S \mid C, X)$ gives $P(y \mid x, c) = P(y \mid x, c, S=1)$, estimable; $P(c \mid x)$ is available by assumption. ■

$C = \{W_1, W_2\}$: yes (parents of S). $C = \{W_1, Z_1, Z_2\}$: no ($S \leftarrow W_2 \rightarrow Y$ open). $C = \{W_2, Z_3\}$: yes.

Confounding and selection together (C given externally):

(1) $W_1 \rightarrow X$, $W_1 \rightarrow W_2 \rightarrow Y$, $X \rightarrow Y$, $W_2 \rightarrow S$: $C = \{W_2\}$ recovers $P(y \mid \text{do}(x))$; $C = \{W_1\}$ does not.

(2) $X \rightarrow Y$, $X \rightarrow W \leftarrow Y$, $W \rightarrow S$: $C = \{\}$ and $C = \{W\}$ fail, $C = \{W, X\}$ works.

Theorem 8.4 (Generalized adjustment; Correa, Tian and Bareinboim 2018)

$(\mathbf{Z}, \mathbf{Z}^T)$ with $\mathbf{Z}^T \subseteq \mathbf{Z}$ (the part measured without bias) is admissible for generalized adjustment relative to $(\mathbf{X}, \mathbf{Y})$ iff

1. $\mathbf{Z}$ does not disturb any proper causal path;
2. $\mathbf{Z} \cup \{S\}$ blocks all non-causal paths (cut the first edge of every proper causal path; then $\mathbf{Z} \cup \{S\}$ must separate $\mathbf{X}$ and $\mathbf{Y}$);
3. $\mathbf{Z}^T$ d-separates $\mathbf{Y}$ from S in the graph where proper causal paths have been severed.

Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} \underbrace{P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, S=1) P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T, S=1)}_{\text{biased data}} \quad \underbrace{P(\mathbf{z}^T)}_{\text{unbiased data}} .$$

Proof [Supplement]. *Sufficiency.* Condition 3 gives $\mathbf{Y} \perp\!\!\!\perp S \mid \mathbf{X}, \mathbf{Z}$ in the relevant graph, so $P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, S=1)$. Conditions 1–2 make $\mathbf{Z}$ an adjustment set: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$. Finally $P(\mathbf{z}) = P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T) P(\mathbf{z}^T)$ with $P(\mathbf{z}^T)$ unbiased and $P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T) = P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T, S=1)$ by the remaining condition. *Necessity*: Correa, Tian and Bareinboim (2018), not reproduced here. ■

Example 8.3

With $\mathbf{Z} = \{Z_1, Z_2, Z_3\}$ satisfying the conditions and $\mathbf{Z}^T = \{Z_2, Z_3\}$: $P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{z_1, z_2, z_3} P(y \mid \mathbf{x}, z_1, z_2, z_3, S=1) P(z_1 \mid z_2, z_3, S=1) P(z_2, z_3)$.

Summary. Recoverability is decidable from a diagram augmented with the selection mechanism, estimands are polynomial-time, results are complete for pure recoverability and sufficient with external data (Correa, Tian and Bareinboim 2019), and the back-door criterion generalizes to confounding plus selection. References: Bareinboim, Tian and Pearl (2014); Bareinboim and Tian (2015); Correa, Tian and Bareinboim (2018, 2019a,b).

8.5 Responding conditions: missing data

Motivation. Missingness graphs decide when quantities are recoverable from incomplete data.

(Partly following Mohan and Pearl 2021, "Graphical models for processing missing data", JASA.)

8.5.1 Motivation

Sensors fail, respondents skip questions, patients forget. The Rubin-rooted literature mostly assumes MAR, needed for EM (Dempster et al. 1977) and multiple imputation (Rubin 1978) to be consistent; yet MAR is almost impossible to verify, and beyond-MAR results are scarce. Graphs encode the *reasons* for missingness.

Why graphs? *Transparency:* MCAR/MAR independences are invisible in data but visible in the graph. *Recoverability:* decide whether and how any method can consistently estimate a parameter. *Testability:* check the model against the (corrupted) data via implied independences.

8.5.2 A motivating example

Age A, Gender G, Obesity O ($G \to O \leftarrow A$); A, G from school records, O missing (m) for students hiding their weight. The proxy is

$$O^* = f(R_O, O) = \begin{cases} O & R_O = 0, \\ m & R_O = 1. \end{cases}$$

Missingness may be random (lost questionnaires), caused by a fully observed variable (rebellious teenagers, $A \to R_O$), or by a partially observed one (embarrassed obese students, $O \to R_O$).

8.5.3 m-graphs

Nodes: $\mathbf{U} \cup \mathbf{V}_o \cup \mathbf{V}_m \cup \mathbf{V}^* \cup \mathbf{R}$, i.e. latents, always-observed variables, sometimes-missing variables, proxies (V_i^* observed for $V_i \in \mathbf{V}_m$), and missingness mechanisms. No variable in $\mathbf{V}_o \cup \mathbf{V}_m \cup \mathbf{U}$ is a child of an R unless stated. *Observed-data distribution*: $P(\mathbf{V}_o, \mathbf{V}^*, \mathbf{R})$; *underlying distribution*: $P(\mathbf{V}_o, \mathbf{V}_m, \mathbf{R})$.

8.5.4 Three categories

- **MCAR:** $\mathbf{V}_m, \mathbf{V}_o, \mathbf{U} \perp\!\!\!\perp \mathbf{R}$; no edges between $\mathbf{R}$ and $\mathbf{V}_o \cup \mathbf{V}_m$. Example: $O, A, G \perp\!\!\!\perp R_O$.
- **MAR:** $\mathbf{V}_m, \mathbf{U} \perp\!\!\!\perp \mathbf{R} \mid \mathbf{V}_o$;

  (i) no edge between any R and a partially observed variable;

  (ii) no bidirected edge between an R and a fully observed variable. Example: $A \to R_O$, $O \perp\!\!\!\perp R_O \mid A, G$.
- **MNAR:** neither. Example: $O \to R_O$, $O \not\perp\!\!\!\perp R_O \mid A, G$.

Recovering under MCAR. $P(G, O, A) = P(G, O, A \mid R_O = 0) = P(G, O^*, A \mid R_O = 0)$: list-wise deletion, which wastes data.

Recovering under MAR. $P(G, O, A) = P(G, O \mid A)P(A) = P(G, O \mid A, R_O = 0)P(A) = P(G, O^* \mid A, R_O = 0)P(A)$ (A separates (G, O) from R_O); alternatively $P(G, O, A) = P(O^* \mid A, G, R_O = 0)P(A, G)$.

MCAR implies MAR, so MAR methods apply; compare the efficiency of $P(G, O^*, A \mid R_O = 0)$ with $P(O^* \mid A, G, R_O = 0)P(A, G)$, which uses all records for $P(A, G)$.

Non-recoverability under MNAR. With $O \to R_O$, $P(G, O, A)$ is provably non-recoverable here.

A recoverable MNAR example. $Z_1 \to Z_2$, $Z_1 \to X$, $Z_1 \to Y$, $Z_2 \to X$, $X \to Y$ ($Z_2 \to Y$), Z_1 into R_{Z_2}, R_X, R_Y, $Z_2 \to R_{Z_2}$ etc. $P(Z_1) = P(Z_1 \mid R_{Z_1} = 0) = P(Z_1^* \mid R_{Z_1} = 0)$ since $R_{Z_1} \perp\!\!\!\perp Z_1$, but *not* from complete cases ($Z_1 \not\perp\!\!\!\perp R_X$, so $P(Z_1) \neq P(Z_1 \mid \mathbf{R} = 0)$). $P(Y \mid X, Z_1, Z_2) = P(Y \mid X, Z_1, Z_2, \mathbf{R} = 0) = P(Y^* \mid X^*, Z_1^*, Z_2^*, \mathbf{R} = 0)$ is recoverable from complete cases. (Proxies are often omitted.)

8.5.5 General results

Theorem 8.5 (Sequential factorization)

Given an m-graph and $P(\mathbf{V}^*, \mathbf{V}_o, \mathbf{R})$, a target Q is recoverable if it can be decomposed into an ordered (chain-rule) factorization, or a sum of such, in which every factor $Q_i = P(Y_i \mid X_i)$ satisfies $Y_i \perp\!\!\!\perp (R_{y_i}, R_{x_i}) \mid X_i$. Then each $Q_i = P(Y_i \mid X_i, R_{Y_i} = 0, R_{X_i} = 0)$.

Proof [Supplement]. Write $Q = \prod_i P(Y_i \mid X_i)$ (or a sum of such). The independence gives $P(Y_i \mid X_i) = P(Y_i \mid X_i, R_{Y_i} = 0, R_{X_i} = 0) = P(Y_i^* \mid X_i^*, R_{Y_i} = 0, R_{X_i} = 0)$ (proxies equal true values when $R = 0$), a functional of the observed distribution. ■

Theorem 8.6 (R-factorization: recoverability of $P(\mathbf{V})$)

Given an m-graph with no edges between R variables, $P(\mathbf{V})$ is recoverable iff there is no $X \in \mathbf{V}_m$ such that (1) X and R_x are neighbors, or (2) X and R_x are connected by a path whose intermediate nodes are all colliders in $\mathbf{V}_m \cup \mathbf{V}_o$. When recoverable,

$$P(\mathbf{v}) = \frac{P(\mathbf{R} = 0, \mathbf{v})}{\prod_i P(R_i = 0 \mid \mathrm{Mb}^o_{r_i}, \mathrm{Mb}^m_{r_i}, R_{\mathrm{Mb}^m_{r_i}} = 0)},$$

where $\mathrm{Mb}^o_{r_i} \subseteq \mathbf{V}_o$ and $\mathrm{Mb}^m_{r_i} \subseteq \mathbf{V}_m$ form the Markov blanket of R_i.

Proof [Supplement]. *Formula.* $P(\mathbf{v}) = P(\mathbf{v}, \mathbf{R} = 0)/P(\mathbf{R} = 0 \mid \mathbf{v})$. With no edges among the R's, $P(\mathbf{R} = 0 \mid \mathbf{v}) = \prod_i P(R_i = 0 \mid \mathbf{v}, \mathbf{R}_{-i} = 0)$, each factor equal to $P(R_i = 0 \mid \mathrm{Mb}^o_{r_i}, \mathrm{Mb}^m_{r_i}, R_{\mathrm{Mb}^m_{r_i}} = 0)$. The graphical conditions keep the variable indicated by R_i out of its Markov blanket, so each factor uses only variables observed when the relevant indicators are 0; the numerator is observed directly. *Necessity*: Mohan and Pearl (2014, 2021), not reproduced here. ■

8.5.6 Recovering a causal effect

$W \to Z \to Y$, $W \leftrightarrow Y$, $R_Y \leftrightarrow W$, Y partially observed ($Y \to Y^* \leftarrow R_Y$):

$$\begin{aligned} P(y \mid \mathrm{do}(z)) &= P(y \mid \mathrm{do}(z), r_y) = P(y^* \mid \mathrm{do}(z), r_y) \\ &= \sum_w P(y^* \mid w, \mathrm{do}(z), r_y) P(w \mid \mathrm{do}(z), r_y) = \sum_w P(y^* \mid w, z, r_y) P(w \mid r_y), \end{aligned}$$

with r_y denoting $R_Y = 0$.

Conclusion. Statisticians should model the missingness process far more often; such models have received little attention.

8.6 Measurement: general identifiability with partial observation

Motivation. Combine experiments that each measure only some variables.

Studies measure only accessible variables (oximeter for COVID vs. carbon-monoxide poisoning), so *GID-PO* uses $\mathbb{P} = \{P_{\mathbf{Z}_i}(\mathbf{V}_i)\}_i$ with $\mathbf{V}_i \subseteq \mathbf{V}$.

Example 8.4 (Exercise and stroke)

Age $A \to$ exercise X, $A \to$ BMI B, $X \to B$, $B \to$ blood pressure C, $C \to$ stroke Y, $X \leftrightarrow Y$. Given $P_X(A, C)$ (exercise experiment) and $P(B, C, Y)$ (observational):

$$\begin{aligned} P_x(y) &= \sum_{a,b,c} P_x(a, b, c, y) = \sum_{a,b,c} P(a) P_{a,x}(b) P_b(c) P_c(y) && \text{factorization} \\ &= \sum_{a,c} P(a) \Big[\sum_b P_{a,x}(b) P_b(c) \Big] P_c(y) = \sum_{a,c} P(a) P_{a,x}(c) P_c(y) && \text{proposition} \\ &= \sum_{a,c} P_x(a) P_x(c \mid a) \sum_b P(y \mid b, c) P(b) && \text{GID of each factor.} \end{aligned}$$

$P(a)P_{a,x}(c)$ comes from $P_X(A, C)$ ($P(a) = P_x(a)$, as X does not affect A); $P_c(y) = \sum_b P(y \mid b, c)P(b)$ from $P(B, C, Y)$ by adjusting for B (blocking $C \leftarrow B \leftarrow X \leftrightarrow Y$).

8.7 Conclusions

- Using abundant data requires representing the causal mechanisms, coarsely and parsimoniously.
- Necessary and sufficient conditions and algorithms exist for confounding, selection bias and generalizability (sufficient ones for missing data and partial measurement).
- Together they give a framework for pooling observational and experimental knowledge across heterogeneous domains.

Chapter 9. Regression Discontinuity, Difference-in-Differences and Synthetic Control

9.1 Regression discontinuity

Motivation. How does a known cutoff rule identify a causal effect at the cutoff, and how does the bandwidth drive bias, variance and inference?

9.1.1 Where RDD fits

Motivation. Place RDD among empirical designs.

Types of empirical studies, from more to less researcher control:

Experiment (RCT)	Natural experiment	Quasi-experiment	Observational study
researcher randomizes treatment	as-if random assignment by nature/policy (lottery, IV)	known non-random rule exploited (RD, DiD, SCM)	confounders must be assumed away (matching, IPW)

Method	Design type	Key source of variation
Back-door / matching / IPW	observational	adjusted confounders
IV	natural / quasi-experiment	exogenous instrument
RDD	quasi-experiment	cutoff rule
DiD	quasi-experiment	temporal variation across groups
SCM	quasi-experiment	weighted donor pool

IV straddles the boundary (lottery vs. policy threshold). Here treatment is determined by whether a pre-treatment variable crosses a known threshold.

Notation. $Z_i \in \mathbb{R}$: forcing (running) variable (not an instrument as in Chapter 7); c: cutoff; $X_i = \mathbb{1}\{Z_i \geq c\}$: treatment in the sharp design; $Y_i(1), Y_i(0)$: potential outcomes; τ_{SRD}: sharp-RD effect at the cutoff; $W_i = \mathbb{1}\{Z_i \geq c\}$: the instrument in the fuzzy design.

9.1.2 Concept and identification

Motivation. Define the sharp RD effect and show that continuity identifies it.

9.1.2.1 Examples

Incumbency advantage (Lee 2008; Lee et al. 2004). Running variable: Democratic vote-share margin at t; cutoff 50%; outcome: Democratic vote share at $t+1$. Near the cutoff the election outcome is as good as random; barely winning raises next-election vote share by 6–8 points.

Scholarships and test scores (Thistlethwaite and Campbell 1960, the first RDD): eligibility depends only on an exam score exceeding a threshold. **Financial aid**: eligibility if family income is below a threshold.

9.1.2.2 The design

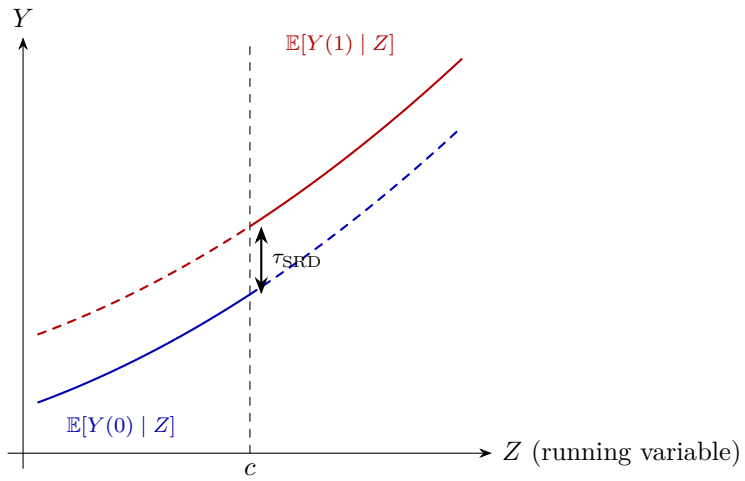


Figure. *Sharp RD:* $\mathbb{E}[Y(0) \mid Z]$ *is observed only left of* c, $\mathbb{E}[Y(1) \mid Z]$ *only right (solid). Under continuity the jump at* c *is the effect.*

Treatment changes discontinuously in a pre-treatment forcing variable while confounders vary smoothly; comparing units with similar Z_i but different treatment identifies a *local* effect at the cutoff.

Sharp RD. Z_i continuous (discrete running variables: Cattaneo et al. 2020) and $X_i = \mathbb{1}\{Z_i \geq c\}$ deterministic. Positivity fails by construction:

$$P[X_i = 1 \mid Z_i = c - \varepsilon] = 0, \qquad P[X_i = 1 \mid Z_i = c + \varepsilon] = 1,$$

so back-door adjustment and IPW fail; continuity-based extrapolation is used instead.

Fuzzy RD. The *probability* of treatment jumps at c, but not from 0 to 1 (Section 9.1.5).

9.1.2.3 Estimand and identification

$$\tau_{\text{SRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = c] = \mathbb{E}[Y_i(1) \mid Z_i = c] - \mathbb{E}[Y_i(0) \mid Z_i = c].$$

No unit has $Z_i = c$ exactly, so local extrapolation needs an assumption.

Continuity assumption. $\mu_1(z) = \mathbb{E}[Y_i(1) \mid Z_i = z]$ and $\mu_0(z) = \mathbb{E}[Y_i(0) \mid Z_i = z]$ are continuous in z. Then

$$\begin{aligned}
\mathbb{E}[Y_i(0) \mid Z_i = c] &= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid Z_i = z] && \text{(continuity)} \\
&= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid X_i = 0, Z_i = z] && \text{(SRD: } z < c \Rightarrow X_i = 0\text{)} \\
&= \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z] && \text{(consistency),}
\end{aligned}$$

and likewise $\mathbb{E}[Y_i(1) \mid Z_i = c] = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$. Hence

$$\boxed{\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z].}$$

Estimation means two regression functions at a boundary point; nonparametric regression is consistent but slow.

What can go wrong? Sorting: richer students retake an exam more often to pass the aid threshold, so those just above differ from those just below.

9.1.3 Validity and diagnostics

Motivation. Check the design's assumptions graphically and with falsification tests.

Three standard figures:

1. **Outcomes by running variable:** a visible jump at c ("hard to sell" otherwise; Imbens and Lemieux 2008).
2. **Covariates by running variable:** should be smooth across c; a jump suggests invalid RD.
3. **Density of the running variable:** a discontinuity suggests sorting.

Manipulation and the McCrary test. If units can precisely manipulate Z_i, smoothness likely fails. McCrary (2008) tests for a jump of $f_Z(z)$ at c.

Bunching. Administrative data often show units locating just below thresholds (taxable earnings, full-time hours; World Bank blog "Ready, set, bunch"), e.g. reported wealth in Colombia bunching below a new wealth-tax threshold in 2010 (Londoño-Vélez).

Falsification (placebo) tests. A valid RDD shows an effect only at the true cutoff.

- *Placebo cutoffs:* re-estimate at fake cutoffs; in a simulation with an effect only at $c = 0$, placebo estimates on $[-2, 2]$ are near zero.
- *Placebo outcomes:* pre-treatment covariates as outcomes; a jump signals sorting.

9.1.4 Estimation

Motivation. Estimate boundary limits with local polynomial regression and choose the bandwidth.

9.1.4.1 General strategy

The target is a single *boundary* point with data on one side only, so boundary bias shrinks slowly; the naive difference in means uses data too far away. Write $\mu_+(c) = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$, $\mu_-(c) = \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]$, $\tau_{\text{SRD}} = \mu_+(c) - \mu_-(c)$.

Kernel (local average) regression. With a uniform kernel and bandwidth h,

$$\hat{\mu}_-(c) = \frac{\sum_{i=1}^n Y_i \mathbb{1}\{c - h \le Z_i < c\}}{\sum_{i=1}^n \mathbb{1}\{c - h \le Z_i < c\}}$$

(similarly above), or equivalently

$$(\hat{\alpha}, \hat{\tau}_{\text{SRD}}) = \arg\min_{\alpha, \tau} \sum_{i: Z_i \in [c-h, c+h]} (Y_i - \alpha - \tau X_i)^2,$$

the difference in means, assuming locally constant outcomes. Large h: high bias, low variance; small h: the reverse; bias shrinks slowly. Kernels: uniform $K(u) = \frac{1}{2}\mathbb{1}\{|u| < 1\}$, triangular $K(u) = (1 - |u|)\mathbb{1}\{|u| < 1\}$.

9.1.4.2 Local linear regression

Allow a slope. On $Z_i \in [c - h, c)$,

$$(\hat{\alpha}_-, \hat{\beta}_-) = \arg\min_{\alpha, \beta} \sum_{i: Z_i \in [c-h, c)} \big(Y_i - \alpha - \beta(Z_i - c)\big)^2,$$

and the same on $[c, c+h]$. Then

$$\hat{\tau}_{\text{SRD}} = \hat{\mu}_+(c) - \hat{\mu}_-(c) = \hat{\alpha}_+ + \hat{\beta}_+(c-c) - \hat{\alpha}_- - \hat{\beta}_-(c-c) = \hat{\alpha}_+ - \hat{\alpha}_-.$$

Single regression. Equivalently,

$$\underset{(\alpha,\beta,\tau,\gamma)}{\arg\min} \sum_{i:Z_i\in[c-h,c+h]} \{Y_i - \alpha - \beta(Z_i-c) - \tau X_i - \gamma(Z_i-c)X_i\}^2,$$

with $\hat{\tau}_{\text{SRD}} = \hat{\tau}$: the fit is $\alpha + \beta(Z_i - c)$ for $X_i = 0$ and $(\alpha+\tau) + (\beta+\gamma)(Z_i - c)$ for $X_i = 1$.
Kernel weighting (typically triangular):

$$\underset{(\alpha,\beta,\tau,\gamma)}{\arg\min} \sum_{i=1}^{n} K\left(\frac{Z_i - c}{h}\right) \{Y_i - \alpha - \beta(Z_i-c) - \tau X_i - \gamma(Z_i-c)X_i\}^2.$$

9.1.4.3 Bias–variance trade-off and the MSE-optimal bandwidth

Let $\mu(z) = \mathbb{E}[Y \mid Z = z]$ with one-sided second derivatives $\mu_\pm^{(2)}(c)$, density $f(c)$, and $\sigma_\pm^2(c) = \text{Var}(Y \mid Z = c^\pm)$. For local linear regression,

$$\text{Bias} \approx C_1 h^2 \underbrace{(\mu_+^{(2)}(c) - \mu_-^{(2)}(c))}_{\text{curvature difference}}, \qquad \text{Variance} \approx \frac{C_2}{Nh} \cdot \underbrace{\frac{\sigma_+^2(c) + \sigma_-^2(c)}{f(c)}}_{\text{noise / density at } c}.$$

Minimizing MSE = Bias^2 + Variance (Imbens and Kalyanaraman 2012):

$$h_{\text{MSE}} = C_K \left(\frac{\sigma_+^2(c)/f_+(c) + \sigma_-^2(c)/f_-(c)}{\left(\mu_+^{(2)}(c) - \mu_-^{(2)}(c)\right)^2} \right)^{1/5} N^{-1/5},$$

since $\frac{d}{dh}[C_1^2 h^4 D^2 + C_2 V/(Nh)] = 0$ gives $h^5 \propto V/(ND^2)$. All inputs are estimated, curvature by local quadratic regression.

9.1.4.4 Why the MSE-optimal bandwidth may fail for inference

$h_{\text{MSE}} \propto N^{-1/5}$ is wide enough that bias is not negligible relative to the SE: a nominal 95% CI may cover about 80%. Remedies: *robust bias correction* (RBC; Calonico, Cattaneo and Titiunik 2014) subtracts an estimate of the leading bias and accounts for its variability; the *CE-optimal bandwidth* (Calonico, Cattaneo and Farrell 2020) minimizes coverage error, $h_{\text{RBC}} \propto N^{-1/4}$. Rule of thumb $\hat{h}_{\text{RBC}} \approx (0.7\text{–}0.8) \times \hat{h}_{\text{MSE}}$.

Bandwidth	Rate	Bias	Variance	CI length	Coverage
MSE-optimal	$N^{-1/5}$	high	low	short	fails
Undersmoothing	$N^{-1/3}$	very low	high	too long	valid
CE-optimal	$N^{-1/4}$	moderate	moderate	efficient	optimal

CE-optimal with RBC is the `rdrobust` default.

Bandwidth sensitivity. Credible estimates are stable across bandwidths (simulation with $\tau = 3$: small h gives wide CIs, large h bias).

9.1.4.5 Covariates and software

Covariates are not needed for identification but improve precision:

$$Y_i = \alpha + \beta(Z_i - c) + \tau X_i + \gamma(Z_i - c)X_i + \delta^\top V_i + \varepsilon_i,$$

with V_i pre-treatment (Calonico et al. 2019); a jump in V_i at c is evidence against validity.

`rdrobust` (R/Stata) implements MSE- and CE-optimal bandwidths, RBC CIs, covariates, clustering and fuzzy RD: `library(rdrobust); rd <- rdrobust(Y, Z, c = 1500); summary(rd)`.

9.1.5 Fuzzy RD

Motivation. Treat imperfect compliance with the cutoff as a local instrumental-variable problem.

Concept. Crossing the cutoff changes the probability of treatment,

$$\lim_{z \downarrow c} P[X_i = 1 \mid Z_i = z] \neq \lim_{z \uparrow c} P[X_i = 1 \mid Z_i = z],$$

but not from 0 to 1 (exemptions, imperfect compliance, administrative errors; e.g. not all eligible students attend summer school).

Fuzzy RD as IV: the Wald estimator. With instrument $W_i = \mathbb{1}\{Z_i \geq c\}$ for X_i, under continuity

$$\tau_{\text{FRD}} = \frac{\lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]}{\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z]} = \frac{\text{ITT effect on } Y}{\text{first-stage effect on } X}.$$

Assumptions. (1) Continuity of $\mathbb{E}[Y_i(x) \mid Z_i = z]$ and $\mathbb{E}[X_i(w) \mid Z_i = z]$ at c;

(2) first stage: $\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z] \neq 0$;

(3) monotonicity: no defiers. Exclusion is built in (W_i affects Y_i only through X_i near c).

Interpretation. $\tau_{\text{FRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}, Z_i = c]$: a LATE local to compliers *and* to the cutoff; strong internal, limited external validity.

9.1.6 Case study: remedial education in Chicago

Motivation. See a fuzzy RD in action (Jacob and Lefgren 2004, REStat).

Background. From 1996 Chicago Public Schools required minimum ITBS reading and math scores to advance (third graders: 2.8 grade equivalents); students below had to attend summer school and could be retained. *Question:* the effect of summer school and retention on achievement.

Design. Fuzzy (exemptions, discretion), but the treatment probability jumps sharply at the cutoff. Data: CPS records 1993–1999, about 350,000 third and sixth graders; outcomes: changes in ITBS scores (logits) one and two years later.

IV estimation. With $W_i = \mathbb{1}\{\text{Score}_i \geq \text{cutoff}\}$ (falling below triggers treatment, so the first-stage jump is negative),

$$\begin{aligned} \text{First stage:} \quad & \text{Treat}_i = \alpha_0 + \alpha_1 W_i + \alpha_2 \text{Score}_i + \alpha_3 (W_i \times \text{Score}_i) + \nu_i, \\ \text{Second stage:} \quad & Y_i = \beta_0 + \tau \widehat{\text{Treat}}_i + \beta_1 \text{Score}_i + \beta_2 (W_i \times \text{Score}_i) + \varepsilon_i, \end{aligned}$$

with τ the LATE near the cutoff; the interactions allow different slopes on each side.

Results. Third graders: about 0.11–0.13 logits in year one (about 20% of a year's learning), persisting with some fade-out; sixth graders: no reading effect, small insignificant math effect.

Retention alone: short-term gains in grade 3, negative reading effects in grade 6. Summer school alone: positive for third graders, especially low performers.

Strengths and limitations. A policy cutoff with fuzzy-RD IV and heterogeneity by grade, gender and SES; but LATE only, no 3+ year outcomes, and manual bandwidth choice.

9.1.7 Summary

- RDD exploits a known cutoff rule for treatment assignment.
- Identification: continuity at the cutoff, $\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]$.
- Diagnostics: outcome plot, covariate balance, McCrary test, placebo tests.
- Estimation: kernel-weighted local linear regression.
- Bandwidth: MSE-optimal ($N^{-1/5}$) for estimation, CE-optimal ($N^{-1/4}$) for inference; shrink h_{MSE} by 20–30%.
- Fuzzy RD = IV at the cutoff: LATE for compliers at c.
- Chicago: positive effects for third graders, little for sixth graders.

9.1.8 Appendix: Kink RD

Sharp kink RD looks for a jump in the slope of $\mathbb{E}[Y_i \mid Z_i = z]$ at c (Card, Lee, Pei and Weber 2015); local quadratic regression is preferable.

9.1.9 Appendix: Assumptions for the MSE-optimal bandwidth

(Imbens and Kalyanaraman 2012.) (1) $\{Y_i, Z_i\}$ i.i.d.;

(2) the density $f(z)$ is continuous with $f(c) > 0$;

(3) $\mu(z)$ is three times differentiable near c;

(4) K is symmetric, bounded, with compact support;

(5) $\sigma^2(z)$ is continuous near c with $\sigma^2(c) > 0$.

9.1.10 Appendix: Regularization of the bandwidth

If the estimated curvature difference is near zero, h explodes; IK (2012) add a penalty $\hat{r}$ (the variance of the curvature estimate):

$$\hat{h}_{\text{reg}} \propto \left(\frac{\text{sum of variances}}{f(c)\,[(\text{curvature})^2 + \text{regularization}]} \right)^{1/5},$$

with $\hat{\mu}_{\pm}^{(2)}(c)$ from local quadratic regressions and

$$\hat{r}_+ = \frac{2160\,\hat{\sigma}_+^2(c)}{N_{h,+}h^4}, \qquad \hat{r}_- = \frac{2160\,\hat{\sigma}_-^2(c)}{N_{h,-}h^4},$$

growing with less data or more noise.

9.1.11 Appendix: Why local linear? Polynomial order

Higher order p reduces bias but increases variance near c; in finite samples the variance cost dominates. Local linear ($p = 1$) gives the main estimate; local quadratic ($p = 2$) estimates the bias for the MSE-optimal bandwidth (IK 2012) and RBC (CCT 2014): "estimate with p, correct bias with $p + 1$" (`rdrobust` default).

9.1.12 Appendix: Global polynomials: a cautionary tale

Global high-order polynomials over the whole range (Gelman and Imbens 2019) are sensitive to far observations, oscillate (Runge-type behavior) and give poor CI coverage; local linear fits with a data-driven bandwidth avoid this. (Further reading: *The Effect*; *The Mixtape*.)

9.2 Difference-in-differences

Motivation. How do repeated measurements identify the ATT when the untreated trend is a valid proxy for the treated group's counterfactual trend?

9.2.1 Motivation

Motivation. Introduce the two sources of variation in panel data.

Panel data give within-unit variation over time and cross-sectional variation in treatment. Before–after uses the treated group's own past as counterfactual; DiD uses the control group's change as the treated group's counterfactual change (parallel paths).

9.2.1.1 Case study: minimum wages (Card and Krueger 1994)

In 1992 New Jersey raised its minimum wage from \$4.25 to \$5.05 while neighboring Pennsylvania stayed at \$4.25. Employment at fast-food restaurants in eastern PA and NJ (similar prices, wages, products) is observed before and after; the ATT is the gap between NJ's actual and counterfactual (parallel to PA) post-period levels.

9.2.2 The DiD design

Motivation. Formalize the two-group, two-period design and identify the ATT.

9.2.2.1 Setup

Two groups, two periods; only the group $G_i = 1$ is treated, at $t = 1$: $X_{i0} = 0$, $X_{i1} = G_i$.

	Pre-period ($t = 0$)	Post-period ($t = 1$)
Treated group ($G_i = 1$)	$X_{i0} = 0$	$X_{i1} = 1$
Control group ($G_i = 0$)	$X_{i0} = 0$	$X_{i1} = 0$

Potential outcomes. $Y_{it}(x)$; consistency $Y_{it} = X_{it}Y_{it}(1) + (1 - X_{it})Y_{it}(0)$. Hence $Y_{i0}(0) = Y_{i0}$ for all; $Y_{i1} = Y_{i1}(1)$ if $G_i = 1$; $Y_{i1} = Y_{i1}(0)$ if $G_i = 0$.

9.2.2.2 The identification problem

$$\text{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1] = \underbrace{\mathbb{E}[Y_{i1} \mid G_i = 1]}_{(a)} - \underbrace{\mathbb{E}[Y_{i1}(0) \mid G_i = 1]}_{(b)}.$$

(a) is estimable;

(b) is counterfactual.

9.2.2.3 Three control strategies

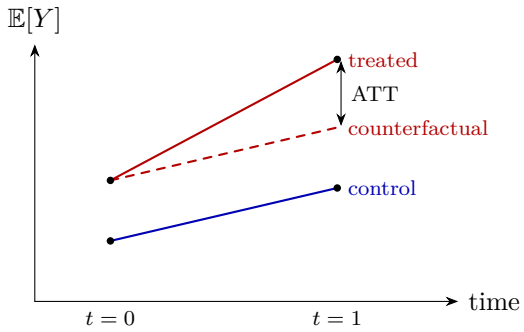


Figure. *DiD: absent treatment the treated group would change like the control group (dashed); the ATT is the difference of the two changes.*

1. **Before–after:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i0} \mid G_i = 1]$ (no change over time absent treatment).
2. **Cross-sectional comparison:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1} \mid G_i = 0]$ (no group differences).
3. **Difference-in-differences:** assume parallel trends; then

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i0} \mid G = 1]) - (\mathbb{E}[Y_{i1} \mid G = 0] - \mathbb{E}[Y_{i0} \mid G = 0]).$$

9.2.2.4 Parallel trends

Definition (Parallel trends)

$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1]$.

It allows time-invariant unobserved group differences and common trends, but is *not invariant to nonlinear transformations*: parallel trends in Y_{it} imply non-parallel trends in $\log Y_{it}$, and vice versa.

Theorem 9.1 (Identification)

Under consistency and parallel trends,

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 1]) - (\mathbb{E}[Y_{i1} \mid G_i = 0] - \mathbb{E}[Y_{i0} \mid G_i = 0]).$$

Proof [Supplement]. By parallel trends, $\mathbb{E}[Y_{i1}(0) \mid G = 1] = \mathbb{E}[Y_{i0}(0) \mid G = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G = 0]$. Consistency gives $Y_{i0}(0) = Y_{i0}$ for everyone and $Y_{i1}(0) = Y_{i1}$ for controls, so $\mathbb{E}[Y_{i1}(0) \mid G = 1] = \mathbb{E}[Y_{i0} \mid G = 1] + \mathbb{E}[Y_{i1} - Y_{i0} \mid G = 0]$. Substitute into $\text{ATT} = \mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i1}(0) \mid G = 1]$. ■

Threats: non-parallel trends (time-varying confounding). *Ashenfelter's dip:* earnings fall just before training-program entry, lowering the treated pre-period outcome and overestimating the effect. *Falsification test:* parallel pre-treatment trends; necessary, not sufficient.

9.2.2.5 Estimation

With panel data,

$$\widehat{\text{ATT}} = \underbrace{\frac{1}{N_1}\sum_{i=1}^{N} G_i\{Y_{i1} - Y_{i0}\}}_{\text{average trend, treated}} - \underbrace{\frac{1}{N_0}\sum_{i=1}^{N}(1-G_i)\{Y_{i1} - Y_{i0}\}}_{\text{average trend, control}},$$

with difference-in-means standard errors. Equivalently, $\Delta Y_i = Y_{i1} - Y_{i0} = \alpha + \tau G_i + \varepsilon_i$ gives $\hat{\tau}_{\text{OLS}} = \widehat{\text{ATT}}$, convenient for covariates. DiD also works with repeated cross-sections.

Card and Krueger with numbers (average full-time-equivalent employment per restaurant):

	Before (Feb 1992)	After (Nov 1992)	Difference
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	−2.16
DiD estimate			+2.76

DiD $= (21.03 - 20.44) - (21.17 - 23.33) = 0.59 - (-2.16) = 2.76$: no evidence of employment loss.

9.2.2.6 DiD and two-way fixed effects (2×2)

$$Y_{it} = \mu + \gamma G_i + \delta t + \tau X_{it} + \varepsilon_{it}.$$

Since $X_{it} = G_i \cdot t$, the four cell means are μ, $\mu+\gamma$, $\mu+\delta$, $\mu+\gamma+\delta+\tau$, so $\hat{\tau}$ equals the DiD estimate.

9.2.3 Conditional and nonlinear DiD

Motivation. Weaken parallel trends with covariates, and replace mean trends by rank preservation.

9.2.3.1 Conditional parallel trends

For all x,

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1, X_i = x] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0, X_i = x]:$$

trends need only be parallel within covariate strata. Estimators:

- **Outcome regression:** include X_i in the (TWFE) model.
- **IPW-DiD:** reweight controls by $\hat{e}(X_i)/(1-\hat{e}(X_i))$.
- **Doubly robust DiD:** consistent if either model is correct (Sant'Anna and Zhao 2020; Callaway and Sant'Anna 2021), as in Chapter 6.

9.2.3.2 Nonlinear DiD (changes-in-changes)

Assume parallel trends in *ranks*: absent treatment, a unit at the q-th quantile of its group in period 0 stays at the q-th quantile in period 1. This is invariant to monotone transformations.

Formalization (Athey and Imbens 2006). Let $F_{gt}(y) = P(Y_{it}(0) \le y \mid G_i = g)$. Observed: F_{00}, F_{01}, F_{10}, and the CDF $F_{11}^{(1)}$ of $Y_{i1}(1)$ for the treated. Goal: the counterfactual F_{11}. Quantile treatment effect:

$$\tau(q) = (F_{11}^{(1)})^{-1}(q) - F_{11}^{-1}(q).$$

Let $h_0(y) = F_{01}^{-1}(F_{00}(y))$ (controls) and $h_1(y) = F_{11}^{-1}(F_{10}(y))$ (treated). *Identification assumption:* $h_0 = h_1$. Then $F_{11}^{-1}(F_{10}(y)) = F_{01}^{-1}(F_{00}(y))$, and solving,

$$F_{11}(y) = F_{10}(F_{00}^{-1}(F_{01}(y))).$$

(Take $y' = F_{01}^{-1}(F_{00}(y''))$; the assumption says $F_{11}(y') = F_{10}(y'')$ with $y'' = F_{00}^{-1}(F_{01}(y'))$.)

9.2.3.3 Key takeaways

- Parallel trends are the core of DiD identification.
- DiD handles time-invariant confounding but not time-varying confounders (Ashenfelter's dip).
- Conditional parallel trends lead to IPW-DiD and doubly robust DiD.
- Pre-trend tests do not validate parallel trends; sensitivity analysis is preferred.
- Linear DiD is not invariant to nonlinear transformations; nonlinear DiD needs distributional assumptions.
- TWFE equals DiD in the 2×2 case but needs care with staggered timing.

9.2.4 Appendix: Pre-testing bias

Proceeding only after an insignificant pre-trend test biases the estimate: low-power tests select samples whose pre-trends happen to look flat, and that chance variation persists into the post-period. Under-rejection of the pre-trend test leads to over-rejection of "no effect" (Roth 2022).

9.2.5 Appendix: Sensitivity analysis for parallel trends

Allow bounded violations,

$$\left|\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0]\right| \leq M,$$

build a confidence set for the ATT for each M, and report the *breakdown value*, the smallest M whose set includes 0. Bounds can be calibrated from pre-treatment trend differences (Rambachan and Roth 2023).

9.2.6 Appendix: Staggered treatment timing

With staggered entry, TWFE is a weighted average of all 2×2 DiDs (Goodman-Bacon): early vs. never-treated and late vs. never-treated (good), and late vs. early-treated (dangerous). If effects change over time, the early-treated are invalid controls, and the estimate is attenuated or even reversed.

Modern solutions. Callaway and Sant'Anna (2021): group-time ATTs with never- or not-yet-treated controls. Event-study plots by $t - t^*$ show pre-trends and dynamics. (See the lecture on staggered DiD.)

9.3 Staggered difference-in-differences

Motivation. How should DiD be done when adoption times differ and effects vary across units and over time, where TWFE can mislead?

9.3.1 Staggered adoption

Motivation. Describe the setting and why it complicates DiD.

Units are treated at different times (groups A, B, C, and a never-treated group); after adoption treated paths jump and may keep changing (dynamic effects); with treatment reversal effects switch on and off.

Staggered DiD handles gradual roll-outs: a *staggered adoption design* (SAD). Not-yet-treated units serve as controls, blurring treatment/control and pre/post. Once treated, a unit remains treated.

9.3.1.1 TWFE as is is not sufficient

TWFE $Y_{it} = \alpha_i + \beta_t + \tau D_{it} + \varepsilon_{it}$ recovers the ATT (under parallel trends and no anticipation) only if (i) effects are homogeneous or (ii) there are two periods. Otherwise already-treated units act as controls and their dynamics leak into τ. Fix: use only "good" comparisons.

9.3.1.2 Terminology

- **Static effects:** constant across post-treatment periods.
- **Dynamic effects:** change with time since adoption.
- **Homogeneous effects:** identical across units.
- **Heterogeneous effects:** vary across units, constant over time for each unit.
- **Carryover effects:** a past treatment keeps acting after withdrawal (a dynamic effect).

9.3.1.3 Valid and invalid comparisons

- **Valid:** treated vs. never-treated; treated vs. not-yet-treated.
- **Invalid:** treated vs. already-treated; later-treated vs. earlier-treated using the earlier group's post-treatment periods.

Bias arises under dynamic or heterogeneous effects (Goodman-Bacon decomposition, Section 9.2).

Modern alternatives to TWFE: Callaway and Sant'Anna (2021), group-time ATTs; Sun and Abraham (2021), event studies with proper controls; stacked DiD, one DiD per cohort.

9.3.2 Callaway–Sant'Anna group-time ATTs

Motivation. Define, identify and aggregate effects by adoption cohort and calendar time.

9.3.2.1 Notation

G_i: period in which i becomes treated; $Y_{it}(0)$: untreated potential outcome; $Y_{it}(g)$: potential outcome if first treated in g. Never-treated units have $Y_{it} = Y_{it}(0)$; for others

$$Y_{it} = \mathbb{1}\{G_i > t\} Y_{it}(0) + \mathbb{1}\{G_i \leq t\} Y_{it}(G_i),$$

which embeds no anticipation. C_i: never-treated indicator; D_{it}: treated by t; X_i: pre-treatment covariates.

Staggered adoption assumption: $D_{it} = 1 \Rightarrow D_{i,t+1} = 1$ for $t = 1, \ldots, T-1$. Effects may be heterogeneous across time, groups, sequences.

9.3.2.2 Parallel trends

Based on never-treated units: for all $g = 2, \ldots, T$ and $t = 2, \ldots, T$ with $t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid C = 1],$$

presuming a large and similar enough never-treated group.

Based on not-yet-treated units: for all g and s, t with $s \geq t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid D_s = 0, G \neq g].$$

This uses more data but restricts some pre-treatment trends (Marcus and Sant'Anna 2021). $s \geq t$ guarantees, by staggered adoption, that comparison units are untreated in $t-1$ and t.

9.3.2.3 Group-time ATT and identification

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g],$$

e.g. $\mathrm{ATT}(2, 3)$ is the period-3 effect for units first treated in period 2. For $t \geq g$, with never-treated controls,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid C = 1];$$

with not-yet-treated controls,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid D_t = 0, G \neq g].$$

Derivation (never-treated): $\mathbb{E}[Y_t(g) \mid G = g] = \mathbb{E}[Y_t \mid G = g]$; telescoping parallel trends, $\mathbb{E}[Y_t(0) - Y_{g-1}(0) \mid G = g] = \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$; and $Y_{g-1}(0) = Y_{g-1}$ for group g (no anticipation). Each $\mathrm{ATT}(g, t)$ is a clean 2×2 DiD with base period $g - 1$.

9.3.2.4 Conditional parallel trends

Never-treated: for $t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, C = 1]$. *Not-yet-treated:* for $s \geq t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, D_s = 0, G \neq g]$. Estimation by outcome regression, IPW or DR DiD (Section 9.2).

9.3.2.5 Aggregation

By group.

$$\theta_S(g) = \frac{1}{T - g + 1} \sum_{t=2}^{T} \mathbb{1}\{g \leq t\} \, \mathrm{ATT}(g, t),$$

showing heterogeneity in adoption time, and

$$\theta_S^O := \sum_{g=2}^{T} \theta_S(g) \, P(G = g),$$

the overall effect among ever-treated groups, the recommended single summary.

By exposure length (event study).

$$\theta_D(e) := \sum_{g=2}^{T} \mathbb{1}\{g + e \leq T\} \, \mathrm{ATT}(g, g + e) \, P(G = g \mid g + e \leq T),$$

the average effect after exactly e periods of exposure.

9.3.2.6 Summary

- Group-time ATTs ATT(g,t): effect in period t for units first treated in g.
- Clean controls: not-yet-treated or never-treated units.
- Pre-treatment covariates via conditional parallel trends.
- Anticipation: shift the base period before any behavioral adjustment.

References: Athey and Imbens (2022); Baker, Larcker and Wang (2022); Callaway and Sant'Anna (2021); Chiu, Lan, Liu and Xu (2023); Goodman-Bacon (2021); Wing et al. (2024).

9.4 Synthetic control

Motivation. How can the effect on a *single* treated unit be estimated by a weighted combination of untreated units that tracks it before the intervention?

9.4.1 From DiD to synthetic controls

Motivation. Motivate a data-driven comparison unit when there is only one treated unit.

With a single treated unit, no single control may be comparable. The **synthetic control method** (SCM; Abadie and Gardeazabal 2003) uses a *convex combination* of controls whose weights minimize pre-treatment differences from the treated unit.

9.4.1.1 Case study: California Proposition 99

In 1988 California raised its cigarette tax by \$0.25 per pack (then the largest state increase). SCM builds "synthetic California" from 38 control states, mainly Utah (33.4%), Nevada (23.4%), Montana (19.9%) and Colorado (16.4%), matching pre-1988 sales, income, beer consumption, retail price and the share aged 15–24. By 2000 California's per-capita sales were about 26 packs lower than synthetic California's (Abadie, Diamond and Hainmueller 2010); the two track closely in 1970–1988, whereas the average of other states differs already before 1988. Like matching, SCM forces the researcher to show the affinity of treated and synthetic units and guards against "extreme counterfactuals" outside the convex hull.

9.4.2 The estimator

Motivation. Define the setting, estimand and weights.

9.4.2.1 Setup

Time period	1	2	$\cdots$	T_0	T_0+1	$\cdots$	T
Treated unit ($i=1$)	0	0	0	0	1	1	1
Control group ($i=2,\ldots,J+1$)	0	0	0	0	0	0	0

Unit 1 is treated from T_0+1 to T; observed $Y_{it}=Y_{it}(X_{it})$. Estimand:

$$\tau_{1t}=Y_{1t}(1)-Y_{1t}(0), \qquad t>T_0.$$

9.4.2.2 Missing counterfactuals and synthetic weights

For $t > T_0$, $\tau_{1t} = Y_{1t} - Y_{1t}(0)$, so $Y_{1t}(0)$ must be imputed. Choose weights with

$$w_j \geq 0, \qquad \sum_j w_j = 1 \quad \text{(convex)},$$

minimizing, over $t \leq T_0$,

$$\left\| Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt} \right\| \quad \text{and} \quad \left\| Z_1 - \sum_{j=2}^{J+1} w_j Z_j \right\|,$$

with pre-treatment predictors Z_i, combined via predictor-importance weights V (Appendix 9.4.6), optionally with a dispersion penalty. Then $\hat{\tau}_{1t} = Y_{1t} - \sum_j w_j Y_{jt}$, hoping $\sum_j w_j Y_{jt} \approx Y_{1t}(0)$.

9.4.2.3 Model-based justifications

Model 1: interactive factor model.

$$Y_{it}(0) = Z_i^\top \theta_t + \alpha_i + \delta_t + \lambda_t \mu_i + \varepsilon_{it},$$

with θ_t time-varying covariate coefficients, α_i unit and δ_t time effects, λ_t a $1 \times F$ vector of common factors and μ_i an $F \times 1$ vector of loadings. $\lambda_t \mu_i$ allows structured time-varying confounding; DiD omits it.

Model 2: autoregressive model (no fixed effects):

$$Y_{i,t+1}(0) = \alpha_t Y_{it}(0) + \beta_{t+1}^\top W_{i,t+1} + u_{i,t+1}, \qquad W_{i,t+1} = \gamma_t Y_{it}(0) + \Pi_t W_{it} + v_{i,t+1},$$

with time-varying covariates W_{it}, AR coefficient α_t, feedback γ_t and AR matrix Π_t.

Properties. Suppose perfect balance: $\sum_j w_j Y_{jt} = Y_{1t}$ for $t \leq T_0$ and $\sum_j w_j Z_j = Z_1$; let $\hat{Y}_{1t}(0) = \sum_j w_j Y_{jt}$ for $t > T_0$.

- Model 1: $\hat{Y}_{1t}(0) \to Y_{1t}(0)$ as $T_0 \to \infty$ (matching many pre-period outcomes forces $\sum_j w_j \mu_j \approx \mu_1$).
- Model 2: $\mathbb{E}[\hat{Y}_{1t}(0)] = \mathbb{E}[Y_{1t}(0)]$, unbiased even with one pre-period.

9.4.2.4 Convexity: no extrapolation

Example: California's pre-treatment smoking is 120 packs per capita, donors lie in $[60, 110]$; convex weights reach at most 110, so the poor fit is visible. OLS may assign negative weights (e.g. $w_j = -0.3$) and extrapolate far outside the observed range, invisibly. SCM trades flexibility for transparency.

Sparsity: geometric intuition. The treated unit is approximated by a point on the boundary of the donors' convex hull, so only a few "vertex" donors get $w_j > 0$. Weights use no post-treatment data, protecting against specification searching.

9.4.2.5 Practical concerns

- Restrict the donor pool to similar units (less interpolation bias, overfitting).
- Donors must be unaffected by treatment (no spillovers).
- "Although we define our synthetic controls as convex combinations of unexposed units, negative weights or weights larger than one can be used at the cost of allowing extrapolation."

- Convexity may prevent a perfect pre-treatment fit outside the convex hull; a deliberate trade-off.
- With $N = 1$ treated unit, inference uses placebo tests.

9.4.3 Inference and bias

Motivation. Assess significance by placebo permutation, and understand bias from imperfect fit.

9.4.3.1 Placebo tests

Apply SCM to each of the 38 control states as if treated. Under the null, California's gap should look like a typical placebo gap; the "p-value" is the fraction of placebo gaps at least as large. In Abadie et al. (2010) only 1 of 34 states had a larger gap, $p \approx 0.03$.

In the placebo plot, gaps $Y_{jt} - \hat{Y}_{jt}^{\text{SC}}$ of control states (gray) center at zero while California's (red) diverges after 1988. Exclude states with high pre-treatment RMSPE, whose gaps are uninformative.

9.4.3.2 Why imperfect pre-treatment fit causes bias

Under Model 1, a perfect pre-treatment fit gives $\sum_j w_j \mu_j \approx \mu_1$. If it fails, $\sum_j w_j \mu_j \neq \mu_1$ and

$$\sum_j w_j Y_{jt}(0) - Y_{1t}(0) \approx \lambda_t \Big(\sum_j w_j \mu_j - \mu_1 \Big) \neq 0,$$

the terms $Z_i^\top \theta_t$, α_i, δ_t cancelling by balance and ε_{it} averaging out: predictions drift after treatment, especially outside the convex hull.

9.4.3.3 Bias correction: augmented SCM

(Ben-Michael, Feller and Rothstein 2021.) Model $Y_{it}(0) = m_{it} + \varepsilon_{it}$, fit on controls ($j = 2, \ldots, J+1$), with predictions $\hat{m}_{it}$ also for $i = 1$:

$$\hat{Y}_{1t}^{\text{aug}}(0) = \underbrace{\sum_{j=2}^{J+1} w_j Y_{jt}}_{\text{SC estimate}} + \underbrace{\Big(\hat{m}_{1t} - \sum_{j=2}^{J+1} w_j \hat{m}_{jt} \Big)}_{\text{bias correction}} = \underbrace{\hat{m}_{1t}}_{\text{outcome model}} + \underbrace{\sum_{j=2}^{J+1} w_j (Y_{jt} - \hat{m}_{jt})}_{\text{SC-weighted residuals}}.$$

Doubly robust: consistent if either the SC weights or the outcome model is approximately correct (cf. AIPW, DR DiD).

9.4.4 Discussion

Motivation. Weigh strengths and limitations and position SCM among alternatives.

Strengths	Limitations
Transparency: weights are explicit; one can inspect which donors contribute and how much.	Needs a good donor pool: similar units with no spillovers.
Data-driven: no need to pick a single comparison unit by hand.	Designed for a single treated unit; extensions needed for several.
No parallel-trends assumption (uses a factor model instead).	Sensitive to donor-pool and predictor choice.
Pre-treatment fit is a built-in diagnostic of credibility.	Formal inference is limited (permutation-based, not sampling-based); convexity can prevent good fit when the treated unit is an outlier.

(Abadie 2021.)

When to use SCM: few treated units, a rich pre-period, factor structure more plausible than parallel trends. Otherwise DiD, or synthetic DiD in larger panels. Open areas: conformal inference, sensitivity analysis, experimental design, external validity.

Extensions. Augmented SCM; synthetic DiD (Arkhangelsky et al. 2021: SCM unit weights plus DiD time weights and unit fixed effects); penalized SCM (Abadie and L'Hour 2021: L_1/L_2 penalties allow partial extrapolation); conformal inference (Chernozhukov, Wüthrich and Zhu 2021: permutation of residuals); matrix completion (Athey et al. 2021: low-rank imputation).

DiD versus synthetic control.

	Difference-in-differences	Synthetic control
Setup	two groups, two (or more) periods	one treated unit, many donors
Key assumption	parallel trends	pre-treatment fit (factor model)
Estimand	ATT (group level)	τ_{1t} (unit level)
Strength	simple, transparent	handles one treated unit well
Weakness	sensitive to nonlinear transforms	requires a good donor pool
Extensions	staggered DiD, nonlinear DiD	augmented SCM, matrix completion

9.4.5 Appendix: More treated units

Interactive fixed effects: estimate Model 1 directly by iterating:

(1) fix the IFE terms and fit the parametric part on untreated observations to get $\hat{\theta}$;

(2) fix the covariate coefficients and estimate the IFE terms by principal components of residuals;

(3) repeat until convergence. *Matrix completion* (Athey et al. 2021): estimate a regularized low-rank $\mathbf{L}$ approximating the observed entries of $\mathbf{Y}(0)$.

9.4.6 Appendix: Predictor importance weights V

$$w(V) = \underset{w\geq 0,\ \sum_j w_j = 1}{\arg\min} \sum_{m=1}^{k} v_m \Big(X_{1m} - \sum_{j=2}^{J+1} w_j X_{jm} \Big)^2,$$

with $v_m \geq 0$ the importance of predictor m (covariate or pre-period outcome): equal $v_m = 1$; $v_m = 1/\operatorname{Var}(X_m)$; or $v_m \propto |\hat{\beta}_m|$ from an outcome regression. Abadie's standard choice is nested: $V^* = \arg\min_V \mathrm{MSPE}_{\mathrm{pre}}(w(V))$, the inner loop solving for $w(V)$.

Caveats. (1) *Outcome-lag collapse* (Kaul, Klößner, Pfeifer and Schieler 2022): with all pre-period lags, $v_m \to 0$ for other predictors; use a few lags.

(2) *Sensitivity:* V depends on predictor and lag choice; check robustness across V methods and report donor weights.

9.5 Synthetic difference-in-differences

Motivation. How can DiD's invariance to unit-level shifts and SC's re-weighting to pre-treatment trends be combined?

9.5.1 DiD, SC and SDiD

Motivation. Recall DiD and SCM and preview what SDiD changes.

Picture (Proposition 99, 1970–2000). *DiD*: California vs. the unweighted control average, all pre-periods equal; trends visibly not parallel. *SC*: re-weighted controls match California's pre-1988 level and trend; no time weights. *SDiD*: re-weighted controls parallel (not level-matched) to California, with time weights concentrated just before treatment.

Recap. DiD allows time-invariant level differences under parallel trends; SCM builds a synthetic control for one treated unit. They differ:

(1) SCM adds unit weights;

(2) SCM has time fixed effects but no unit fixed effects and no overall intercept.

SDiD (Arkhangelsky et al. 2021) re-weights to match pre-treatment trends like SCM and is invariant to additive unit shifts like DiD.

9.5.2 Formulations

Motivation. Write all three estimators as (weighted) TWFE regressions.

With unit effects α_i, time effects β_t, intercept μ and treatment X_{it},

$$\hat{\tau}^{\text{DiD}} = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2, \tag{9.1}$$

$$\hat{\tau}^{\text{SCM}} = \arg\min_{\tau,\mu,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i, \tag{9.2}$$

$$\hat{\tau}^{\text{SDiD}} = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i\hat{\lambda}_t. \tag{9.3}$$

9.5.2.1 Setting and algorithm

Balanced panel, N units, T periods, $X_{it} \in \{0,1\}$. The first N_{co} units are never exposed; the last $N_{tr} = N - N_{co}$ are exposed after T_{pre} ($T_{post} = T - T_{pre}$). Unit weights $\hat{\omega}$ align pre-exposure trends; time weights $\hat{\lambda}$ balance pre- with post-exposure periods.

1. Compute unit weights $\hat{\omega}$.
2. Compute time weights $\hat{\lambda}$.
3. Compute $(\hat{\tau}^{\text{SDiD}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i,t}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\hat{\omega}_i\hat{\lambda}_t$.

9.5.2.2 Unit weights

$$(\hat{\omega}_0, \hat{\omega}) = \underset{\omega_0\in\mathbb{R},\ \omega\in\Omega}{\arg\min} \left\|\bar{y}_{pre,tr} - (\omega_0 + \omega_{co}^{\top}Y_{pre,co})\right\|_2^2 + \zeta^2 T_{pre}\|\omega_{co}\|_2^2,$$

with $\bar{y}_{pre,tr} = \frac{1}{N_{tr}}\sum_{i=N_{co}+1}^{N} Y_{it}$ the treated pre-period path, $\omega_{co}^{\top}Y_{pre,co} = \sum_{i=1}^{N_{co}}\omega_i Y_{it}$ the synthetic path, and Ω the simplex. Differences from SCM:

1. The intercept ω_0: only a *parallel trend* is needed, not the same level.
2. The L_2 penalty spreads weight over more control units.

Regularization (optional). $\zeta = (N_{tr}T_{post})^{1/4}\,\hat{\sigma}(\Delta_{it})$, with $\Delta_{it} = Y_{i(t+1)} - Y_{it}$ the pre-period first differences of controls,

$$\bar{\Delta} = \frac{1}{N_{co}(T_{pre}-1)}\sum_{i=1}^{N_{co}}\sum_{t=1}^{T_{pre}-1}\Delta_{it}, \qquad \hat{\sigma}^2(\Delta_{it}) = \frac{1}{N_{co}(T_{pre}-1)}\sum_{i=1}^{N_{co}}\sum_{t=1}^{T_{pre}-1}(\Delta_{it} - \bar{\Delta})^2.$$

9.5.2.3 Time weights

$$(\hat{\lambda}_0, \hat{\lambda}) = \underset{\lambda_0 \in \mathbb{R},\ \lambda \in \Lambda}{\arg\min} \left\| \bar{y}_{post,co} - (\lambda_0 + \lambda_{pre}^\top Y_{pre,co}) \right\|_2^2,$$

with $\bar{y}_{post,co} = \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it}$ the $1 \times N_{co}$ post-period control averages, $Y_{pre,co}$ the $T_{pre} \times N_{co}$ pre-period control outcomes, $\lambda_{pre} \in \Lambda$ (simplex) a $1 \times T_{pre}$ vector, and λ_0 an intercept letting the post-period lie above or below all pre-periods.

9.5.3 Comparisons and takeaways

Motivation. Explain what the weights and fixed effects buy.

SDiD versus DiD. DiD has no weights. The weights make the regression "local", emphasizing units and periods similar to the treated ones: more robust, and more precise by removing predictable parts of the outcome.

SDiD versus SCM. SCM omits unit fixed effects and time weights:

$$\hat{\tau}^{\text{SCM}} = \underset{\mu,\beta,\tau}{\arg\min} \sum_{i,t} (Y_{it} - \mu - \beta_t - X_{it}\tau)^2 \hat{\omega}_i^{\text{SCM}}.$$

Time weights drop periods unlike the post-period (less bias, more precision); unit fixed effects add flexibility and explain much outcome variation.

Key takeaways.

- Like SCM, SDiD works when pre-treatment trends are not parallel.
- Unlike SCM, its weights ω give a control only *parallel* to the treated group.
- Like DiD, it uses time (β) and unit (α) fixed effects, reducing variance.
- Its unit weights use an L_2 penalty and an intercept ω_0.
- It adds time weights, which neither DiD nor SCM has.

Chapter 10. Causal Discovery

10.1 Causal discovery

Motivation. How much of the causal diagram can be recovered from data, under which assumptions and algorithms?

10.1.1 The structure-learning problem

Motivation. Input, output and assumptions of causal discovery.

Causal discovery goes from $P(\mathbf{V})$ (or samples) to $\mathcal{G}$. Does the rooster's crow cause sunrise? Binary (rooster, sun) observations are almost always equal, but data alone do not give the arrow's direction.

Structure learning. *Input:* data i.i.d. from $P(\mathbf{V})$. *Output:* the graphs consistent with the data (hopefully including $\mathcal{G}$); an unsupervised problem. Extensions: multiple data sets, interventional data, unknown interventions.

10.1.1.1 DAG models

A DAG model has one node per variable and an edge per direct functional relationship; it corresponds to a Markovian model (*causal sufficiency*: no unmeasured confounder). Three equivalent definitions, for $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$:

- **Factorization:** $P(A,B,C,D,E) = P(E \mid D)P(D \mid B,C)P(C \mid A)P(B \mid A)P(A)$.
- **Local Markov property:** $(C \perp\!\!\!\perp B \mid A)$, $(D \perp\!\!\!\perp A \mid B,C)$, $(E \perp\!\!\!\perp A,B,C \mid D)$.
- **Global Markov property:** $\mathbf{A}$ d-separated from $\mathbf{B}$ by $\mathbf{C}$ in $\mathcal{G} \Rightarrow \mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

10.1.1.2 Faithfulness

The global Markov property is one-directional, and a complete graph is compatible with every $P(\mathbf{V})$; minimality needs the converse.

Definition (Faithfulness)

For all $\mathbf{A},\mathbf{B},\mathbf{C}$: $\mathbf{A}$ is d-separated from $\mathbf{B}$ given $\mathbf{C}$ in $\mathcal{G}$ *iff* $\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

Unfaithful distributions have measure zero, but nearly unfaithful ones are many and hard to distinguish in finite samples, and nature may evolve unfaithful configurations (homeostasis, equilibria). Justify faithfulness for the problem.

Example 10.1 (Violation by cancellation)

$X = \epsilon_X$, $Z = X + \epsilon_Z$, $Y = X - Z + \epsilon_Y$ (edges $X \xrightarrow{1} Z$, $Z \xrightarrow{-1} Y$, $X \xrightarrow{1} Y$). X, Y are d-connected, yet $Y = -\epsilon_Z + \epsilon_Y$, so $X \perp\!\!\!\perp Y$: the paths cancel.

10.1.2 Constraint-based structure learning

Motivation. Find the CIs that hold and keep the graphs consistent with them.

10.1.2.1 What constraints a DAG encodes

For $X \to W \to Y$, $X \to Z \leftarrow Y$, of

$$(1)\ X \perp\!\!\!\perp Y \mid W, \quad (2)\ X \perp\!\!\!\perp Y, \quad (3)\ X \perp\!\!\!\perp Y \mid W, Z, \quad (4)\ X \perp\!\!\!\perp Y \mid Z,$$

only (1) is forced; (2)–(4) are dependences ($X \to W \to Y$ open; collider Z opened). Under faithfulness CIs refine the possible DAGs; with no CI information the best guess is the *complete undirected* graph.

Adjacency. If $A \perp\!\!\!\perp B \mid \mathbf{S}$ for some $\mathbf{S} \subseteq \mathbf{V} \setminus \{A, B\}$, then A, B are not adjacent (adjacent variables are dependent given every set). Here only X–Y is removed (separator $\{W\}$), giving the *skeleton* $X - W - Y$, $X - Z - Y$.

Orientation. For an *unshielded triple* $A - C - B$ with separator $\mathbf{S}$ of A, B: if $C \notin \mathbf{S}$, orient the *unshielded collider* (v-structure) $A \to C \leftarrow B$; otherwise C is a non-collider. Here $Z \notin \{W\}$ gives $X \to Z \leftarrow Y$, ruling out $X \to Z \to Y$, $X \leftarrow Z \leftarrow Y$, $X \leftarrow Z \to Y$; $X - W - Y$ is a non-collider of undetermined direction.

10.1.2.2 Markov equivalence

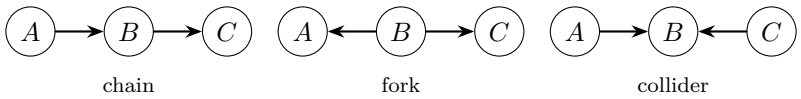


Figure. *Chain and fork both imply $A \perp\!\!\!\perp C \mid B$ (Markov equivalent); the collider implies $A \perp\!\!\!\perp C$, $A \not\perp\!\!\!\perp C \mid B$, so only colliders are orientable.*

$X \to W \to Y$, $X \leftarrow W \leftarrow Y$, $X \leftarrow W \to Y$ (each with $X \to Z \leftarrow Y$) entail the same CIs: they are *observationally* (Markov) *equivalent*. A *Markov equivalence class* (MEC) is a set of such DAGs.

Theorem 10.1 (Verma and Pearl 1990)

Two DAGs represent the same statistical model iff they have the same skeleton and the same unshielded colliders.

Proof [Supplement]. *If.* Whether a path is open given $\mathbf{S}$ depends only on which inner nodes are colliders and their relation to $\mathbf{S}$. Unshielded triples agree by assumption; a shielded triple can be bypassed by the shielding edge, so a shortest open path has no differing shielded colliders. Hence the d-separations, and so the Markov models, coincide. *Only if.* Different skeletons: an edge $A - B$ in one DAG only; in the other A, B are separated by some set (e.g. parents of the later one), never in the first. Same skeleton, but $A \to C \leftarrow B$ unshielded collider in one only: in the first some separator of A, B excludes C; in the second every separator contains C ($A - C - B$ is open otherwise). Either way the d-separations differ. ■

A class is represented by one mixed graph, the *pattern* or CPDAG (completed partially directed acyclic graph). On three variables the 25 DAGs form 11 classes (Eberhardt 2017): chain, reversed chain and fork on X, Y, Z share one; the collider $X \to Y \leftarrow Z$ is alone.

10.1.2.3 The PC algorithm

Basic idea:

(1) non-adjacent vertices have a separator;

(2) for an unshielded collider $X \to Z \leftarrow Y$, the separator excludes Z.

Meek's rules.

Algorithm 3 PC algorithm (Spirtes and Glymour)

start with the complete undirected graph $\mathcal{G}$ on n vertices
for $i = 0, \ldots, n-2$ **do** ▷ skeleton
  for every adjacent pair A, B in $\mathcal{G}$ **do**
    if $\exists \mathbf{S}$ with $|\mathbf{S}| = i$, $\mathbf{S} \subseteq \mathrm{adj}_{\mathcal{G}}(A) \cup \mathrm{adj}_{\mathcal{G}}(B) \setminus \{A, B\}$, and $A \perp\!\!\!\perp B \mid \mathbf{S}$ **then**
      remove $A - B$; sepset$[\{A, B\}] \leftarrow \mathbf{S}$
for every non-adjacent A, B with $A - C - B$ **do** ▷ v-structures
  if $C \notin$ sepset$[\{A, B\}]$ **then** orient $A \to C \leftarrow B$
orient as many remaining edges as possible without creating new v-structures or directed cycles (Meek's rules)

1. $b - c \Rightarrow b \to c$ if $a \to b$ with a, c non-adjacent (else new v-structure $a \to b \leftarrow c$).
2. $a - b \Rightarrow a \to b$ if $a \to c \to b$ (else a cycle).
3. $a - b \Rightarrow a \to b$ if $a - c \to b$ and $a - d \to b$ with c, d non-adjacent (if $b \to a$, Rule 2 forces $c \to a$, $d \to a$: new v-structure).
4. $a - b \Rightarrow a \to b$ if $a - c \to d$ and $c \to d \to b$ with c, b non-adjacent and a, d adjacent.

Example 10.2 (Walk-through)

True DAG $A \to B$, $A \to C$, $B \to D$, $C \to D$; CIs $B \perp\!\!\!\perp C \mid A$ and $A \perp\!\!\!\perp D \mid \{B, C\}$.

- $|\mathbf{S}| = 0$: all pairs dependent.
- $|\mathbf{S}| = 1$: remove $B - C$, sepset$[\{B, C\}] = \{A\}$.
- $|\mathbf{S}| = 2$: remove $A - D$, sepset$[\{A, D\}] = \{B, C\}$. Skeleton $A - B$, $A - C$, $B - D$, $C - D$.
- V-structures: $D \notin \{A\}$ gives $B \to D \leftarrow C$; $B - A - C$, $A - B - D$, $A - C - D$ are non-colliders.
- Meek's rules: one might apply Rule 1 to $A - B$ (via $B \to D$) and $A - C$, giving $A \to B$, $A \to C$, $B \to D \leftarrow C$, the true DAG.

Caution: Rule 1 needs an arrow *into* B; $B \to D$ points out. No rule fires, and the CPDAG is $A - B$, $A - C$, $B \to D \leftarrow C$, whose class is $\{A \to B, A \to C\}$, $B \to A \to C$, $C \to A \to B$ (excluding $B \to A \leftarrow C$, a new v-structure). The true DAG is one member.

Exercise. Apply PC to the five-variable DAG above. $B \to D \leftarrow C$ is found, and Rule 1 orients $D \to E$ ($B \to D$, B, E non-adjacent).

10.1.2.4 Extensions

Conservative PC (Ramsey, Spirtes and Zhang 2006). Split faithfulness into *adjacency faithfulness* (adjacent $\Rightarrow$ no separator) and *orientation faithfulness*: for an unshielded collider $X \to Z \leftarrow Y$, X, Y are dependent given every subset of $\mathbf{V} \setminus \{X, Y\}$ containing Z; otherwise every separator contains Z. Check all subsets of X's and Y's potential parents: Z in no separator gives a v-structure, in all a non-collider, otherwise the triple is marked "unfaithful" (underlined). Robust to orientation-faithfulness violations at modest cost.

Order-independent PC (Colombo and Maathuis 2014). An early false-negative CI test changes later adjacency sets, so PC's output depends on variable order (skeletons vary over 25 random orderings). Remedies: average over orderings, or remove edges only after all tests of one conditioning-set size; orientation uses a majority rule over separating sets (conservative is too restrictive).

Fast Causal Inference (FCI) allows latent confounders. True graph $X \to Y$, $W \to Z$, latent

$U \to Y$, $U \to Z$: the skeleton is $X - Y - Z - W$ and FCI outputs $X \circ\!\!\to Y \leftrightarrow Z \leftarrow\!\!\circ W$, flagging a confounder of Y, Z. PAG edge marks:

$X_1 \quad X_2$ (no edge)	X_1 and X_2 are not adjacent
$X_1 \circ\!\!\to X_2$	X_2 is not an ancestor of X_1
$X_1 \circ\!\!-\!\!\circ X_2$	no set d-separates X_1 and X_2
$X_1 \to X_2$	X_1 is a cause of X_2
$X_1 \leftrightarrow X_2$	there is a latent common cause of X_1 and X_2

10.1.2.5 Summary

With a CI oracle PC is complete (recovers the CPDAG). CI tests: partial correlation (linear Gaussian), Fisher's exact or χ^2 (categorical), kernel- or model-based (Appendix 10.1.8). Relaxing sufficiency or faithfulness is active research.

10.1.3 Score-based approaches

Motivation. Search for the best-fitting graph under a complexity penalty.

10.1.3.1 Why scores?

A single wrong CI test can propagate through skeleton and orientation. Score-based methods score fit and simplicity and search over graphs; hybrids (e.g. MMHC) combine both paradigms.

Why it is possible. Relative to the truth (e.g. $E \to A \leftarrow B$, $A \to S$), an extra arc adds parameters and wrong causal claims; a missing arc cannot be compensated by fitting parameters.

Intuition. Truth $A \to B \leftarrow C$ ($A \perp\!\!\!\perp C$, $A \not\perp\!\!\!\perp C \mid B$); proposal $A \to B \to C$ claims the opposite, so it fits the data poorly, as measured by the log-likelihood.

10.1.3.2 Log-likelihood for DAG models

With $P(\mathbf{V}) = \prod_i P(V_i \mid Pa_{\mathcal{G}}(V_i))$ and models P_{θ_i}, for data $D = \{\mathbf{v}^{(j)}\}_{j=1}^n$,

$$\ell(\theta; D) = \sum_{j=1}^{n} \sum_{i=1}^{k} \log P_{\theta_i}(v_i^{(j)} \mid \mathbf{pa}_{\mathcal{G}}^{(j)}(V_i)),$$

a *decomposable* likelihood: each θ_i is fitted separately.

For the true $A \to B \leftarrow C$ and proposed $B \to A$, $C \to B$:

$$\ell_{\text{left}} = \textstyle\sum_j \left[\log P_A(a_j) + \log P_B(b_j \mid a_j, c_j) + \log P_C(c_j)\right],$$
$$\ell_{\text{right}} = \textstyle\sum_j \left[\log P_A(a_j \mid b_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)\right].$$

With saturated local models, $\ell_{\text{left}}/n - \ell_{\text{right}}/n = I(A; C \mid B) > 0$ (Appendix 10.1.7).

The problem. Against the complete $B \to A$, $C \to A$, $C \to B$, $\ell_{\text{right}} = \sum_j [\log P_A(a_j \mid b_j, c_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)]$ fits *better*: $\{P : A \perp\!\!\!\perp C\}$ is a submodel of all $P(A, B, C)$. Likelihood alone always picks a complete graph (overfitting); parameters must be penalized.

10.1.3.3 BIC

$$\text{BIC} = -2\,\ell(\hat{\theta}; D) + m \log n,$$

m the number of parameters; lower is better. For a linear Gaussian network $X_i = \beta_{0i} + \sum_{X_k \in Pa(X_i)} \beta_{ki} X_k + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma_i^2)$, node i has $|Pa(X_i)| + 2$ parameters, so $m = |E| + 2|V|$.

10.1.3.4 Search and GES

DAGs are super-exponentially many, so exhaustive scoring is impossible; local search moves between states by local moves able to reach any DAG. **Greedy Equivalence Search** (GES; Chickering 2002) uses equivalence classes as states:

1. Start with the empty graph.
2. *Addition phase:* move to the best class reachable by adding one edge, until no improvement. (The result can depend on the representative: adding $B - C$ to $\{A \to B \;\; C,\; A \leftarrow B \;\; C\}$ gives $\{A \to B \leftarrow C\}$ or the chain/fork class.)
3. *Removal phase:* likewise, removing one edge.
4. Return the current class.

Theorem 10.2 (Chickering 2002)

Under faithfulness and as $n \to \infty$, GES identifies the true equivalence class.

Proof [Supplement]. With faithfulness and a consistent score such as BIC, as $n \to \infty$:

(a) adding an edge that creates a dependence present in P, or removing one whose dependence is absent, increases the score (local consistency);

(b) the score is equal for Markov-equivalent DAGs. By (a) and Meek's conjecture (proved by Chickering), the forward phase stops at a class whose DAGs are I-maps of the true independence model; by (a) the backward phase stops only at a minimal I-map, which under faithfulness is the true class. Details: Chickering (2002), Theorem 4. ■

Operators (on CPDAGs). Insert$(u, v, \mathbf{T})$: for non-adjacent u, v add $u \to v$, with $\mathbf{T} \subseteq \mathrm{Ne}(v) \setminus \mathrm{Adj}(u)$; orient $t \to v$ for $t \in \mathbf{T}$; convert to a CPDAG. Delete$(u, v, \mathbf{H})$: remove the u–v edge, with $\mathbf{H} \subseteq \mathrm{Ne}(v) \cap \mathrm{Adj}(u)$; orient $h \to v$, $h \to u$ for $h \in \mathbf{H}$; convert to a CPDAG. GES applies the operator with the best score gain.

Example 10.3

Truth $A \to B \leftarrow C$. Forward: Insert$(A, B, \emptyset)$ gives $A - B$; Insert$(C, B, \{A\})$ $(A \in \mathrm{Ne}(B) \setminus \mathrm{Adj}(C))$ gives $A \to B \leftarrow C$; no further gain. Backward: no deletion helps. Output $A \to B \leftarrow C$.

10.1.3.5 Continuous optimization: NOTEARS and DAGMA

NOTEARS (Zheng et al. 2018) fits a linear SEM (noise not necessarily Gaussian) by black-box optimization. For $X = \epsilon_X$, $Z = aX + \epsilon_Z$, $Y = bX + cZ + \epsilon_Y$,

$$\begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{pmatrix} \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} + \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix} \iff \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b + ac & c & 1 \end{pmatrix} \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix}.$$

With data $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $W \in \mathbb{R}^{d \times d}$,

$$\min_{W \in \mathbb{R}^{d \times d}} \frac{1}{2n} \|\mathbf{X} - \mathbf{X}W\|_F^2 + \lambda \|W\|_1 \quad \text{subject to} \quad h(W) = \mathrm{tr}\left(e^{W \circ W}\right) - d = 0.$$

For a binary adjacency matrix B, B^k counts length-k walks, so $\mathrm{tr}(e^B) = \sum_k \mathrm{tr}(B^k)/k!$ counts weighted closed walks including the d trivial ones: $\mathrm{tr}(e^B) = d$ iff B is a DAG. $\mathrm{tr}(e^{W \circ W})$ is a smooth

weighted analogue, solved by augmented Lagrangian (Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning).

NOTEARS-MLP (Zheng et al. 2020) learns general f: $\min_f L(f)$ s.t. $\mathcal{G}(f) \in$ DAGs, $L(f) = \frac{1}{n}\sum_{j=1}^{d} \ell(x_j, f_j(\mathbf{X}))$, $\mathbb{E}[X_j \mid X_{\mathrm{pa}(j)}] = g_j(f_j(X))$, $\mathbb{E}f_j(X) = 0$, with $k \to j$ iff f_j depends on X_k. Since f_j is free of X_k iff $\|\partial_k f_j\|_{L^2} = 0$, set $[W(f)]_{kj} = \|\partial_k f_j\|_{L^2}$ and solve $\min_{f_j \in H^1(\mathbb{R}^d)} L(f)$ s.t. $h(W(f)) = 0$ (H^1: Sobolev space). For $\mathrm{MLP}(u; A^{(1)}, \ldots, A^{(h)}) = \sigma(A^{(h)} \cdots \sigma(A^{(2)}\sigma(A^{(1)}u)))$, the output is free of u_k if column k of $A^{(1)}$ is zero, giving

$$\min_{\theta} \ \frac{1}{n}\sum_{j=1}^{d} \ell(x_j, \mathrm{MLP}(\mathbf{X}; \theta_j)) + \lambda \|A_j^{(1)}\|_{1,1} \quad \text{s.t.} \quad h(W(\theta)) = 0,$$

with

$$[W(\theta)]_{kj} = \|k\text{-th column of } A_j^{(1)}\|_2.$$

DAGMA (Bello, Aragam and Ravikumar 2022). Matrix exponentials are costly and hard to optimize on large graphs; DAGMA uses the log-determinant penalty

$$h_{\mathrm{DAGMA}}(W) = -\log\det(sI - W \circ W) + d\log s, \qquad s > 0 \ (\text{e.g. } s = 1),$$

zero for acyclic graphs and growing fast near highly cyclic regions. It optimizes $\min_\theta Q(f_\theta; \mathbf{X}) + \beta_1\|\theta\|_1 + h_{\mathrm{DAGMA}}(W(\theta))$ for linear or nonlinear SEMs, faster with competitive accuracy; an optimization device, not a new assumption.

10.1.3.6 Summaries

Score-based. BIC/GES guarantees are large-sample only; search is intractable if the score is not polynomial-time; misspecified local models $P(V_i \mid Pa_{\mathcal{G}}(V_i))$ give wrong graphs; no hidden variables.

Constraint- vs. score-based. Constraint-based (PC, FCI): CI tests, sepsets and Meek's rules; output CPDAG (PAG for FCI); sensitive to CI errors. Score-based (GES, NOTEARS, DAGMA): optimize a score over graphs; better in small samples, needs correct specification. Both assume faithfulness and (except FCI) sufficiency; hybrids (MMHC) combine them.

10.1.4 Parametric approaches

Motivation. Functional-form assumptions orient edges that CIs cannot.

With two variables there are no CIs; $X \to Y$ versus $Y \to X$ needs asymmetries in the joint distribution.

Additive noise model (ANM). $y = f_y(x) + u_y$, $x \perp\!\!\!\perp u_y$. Compare

$$\text{H1: } y = f_y(x) + u_y,\ x \perp\!\!\!\perp u_y\ (x \to y) \qquad \text{vs.} \qquad \text{H2: } x = f_x(y) + u_x,\ y \perp\!\!\!\perp u_x\ (y \to x).$$

Two main cases:

(1) f non-linear;

(2) f linear, u_y non-Gaussian.

10.1.4.1 Non-linear functions

Generically an ANM holds in one direction only; in the other the residual's spread depends on the regressor.

1. Fit $\hat{y} = \hat{f}_y(x)$ and $\hat{x} = \hat{f}_x(y)$ by any ML method.

2. Residuals $\hat{u}_x = x - \hat{f}_x(y)$, $\hat{u}_y = y - \hat{f}_y(x)$.
3. Test independence (e.g. HSIC): $\hat{u}_y \perp\!\!\!\perp x$ suggests $X \to Y$; $\hat{u}_x \perp\!\!\!\perp y$ suggests $Y \to X$; neither casts doubt on the ANM.

10.1.4.2 Linear functions with non-Gaussian noise: LiNGAM

Cramér's decomposition theorem: if a sum of two independent real random variables is Gaussian, both are; by induction, for any finite sum.

LiNGAM (linear non-Gaussian acyclic model). For $y = ax + u_y$ with non-Gaussian errors,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix}, \qquad x = u_x, \quad y = au_x + u_y.$$

Regressing x on y leaves a mixture of u_x, u_y uncorrelated with but (Darmois–Skitovich/Cramér) not independent of y; forward, the residual u_y is independent of x.

The trick: ICA. Independent non-Gaussian sources $s_1, \ldots, s_n$ are observed as $\mathbf{x} = A\mathbf{s}$. Mixtures look more Gaussian than sources, so ICA finds directions maximizing non-Gaussianity, measured by:

- *Kurtosis* $\kappa(y) = \frac{\mathbb{E}[(y-\mu)^4]}{\mathrm{Var}(y)^2} - 3$: zero for Gaussians, positive for heavy tails (Laplace), negative for light tails (uniform); outlier-sensitive.
- *Negentropy* $J(y) = H(y_{\text{Gauss}}) - H(y) \geq 0$ (y_{Gauss} Gaussian, same mean and variance), zero iff y is Gaussian.

ICA returns W_{ICA} with $\hat{\mathbf{s}} = W_{\text{ICA}}\mathbf{x}$ up to permutation and scaling (Hyvärinen and Oja 2000). For two variables, run ICA on (x, y) and find the zero in the mixing matrix.

LiNGAM (Shimizu et al. 2006). $x_i = \sum_{j<i} b_{ij} x_j + e_i$, i.e. $\mathbf{x} = B\mathbf{x} + \mathbf{e}$, $\mathbf{x} = (I - B)^{-1}\mathbf{e}$; let $W = I - B$. ICA gives $W_{\text{ICA}} = PDW$ (P permutation, D diagonal), resolved by the structure:

1. $\mathbf{e} = W\mathbf{x}$, W a row-permuted, rescaled $I - B$.
2. $W_{\text{ICA}} = \text{ICA}(\mathbf{X})$.
3. Permute rows so the diagonal has no zeros (unique, as $I - B$ has non-zero diagonal and is permutable to lower-triangular), rescale to unit diagonal: $W = \text{normalize}(\text{rowpermute}(W_{\text{ICA}}))$.
4. $\hat{B} = I - W$; the causal order makes $\hat{B}$ lower-triangular.

Summary. In an ANM $y = f(x) + u_Y$ with f non-linear or U_Y non-Gaussian, and no latent confounding ($X \perp\!\!\!\perp U_Y$), the direction is identifiable.

10.1.4.3 Score matching (NoGAM)

(Montagna et al. 2023.) For $X_i = f_i(\text{PA}_i) + N_i$ with arbitrary noise, the *score* $s(X) = \nabla_X \log p(X)$, $s_i(X) = \partial_{X_i} \log p(X)$ (unrelated to BIC scores), is estimable by score matching without $p(X)$. It identifies *leaves*; removing them recursively gives a topological order, then extra edges are pruned.

Why leaves are special. With $\log p(X) = \sum_k \log p_k(N_k)$, $N_k = X_k - f_k(\text{PA}_k)$,

$$s_l(X) = \partial_{N_l} \log p_l(N_l) \quad \text{for a leaf } X_l,$$

$$s_i(X) = \partial_{N_i} \log p_i(N_i) - \sum_{k \in \text{CH}_i} \partial_{N_k} \log p_k(N_k)\, \partial_{X_i} f_k \quad \text{otherwise.}$$

For a leaf the regression residual is $R_l = X_l - q(X_{-l}) = N_l$, so s_l is a function of R_l; for non-leaves the child terms remain. Under ANM identifiability,

$$X_i \text{ is a leaf} \iff \min_h \mathbb{E}\big[\{h(R_i) - s_i(X)\}^2\big] = 0.$$

Algorithm (NoGAM). (1) Estimate $\hat{s}(X) \approx \nabla_X \log p(X)$.

(2) For each remaining node, regress X_i on X_{-i}; residuals $\hat{R}_i$.

(3) Predict $\hat{s}_i(X)$ from $\hat{R}_i$; record the MSE.

(4) The smallest-MSE node is a leaf; remove it and repeat.

(5) Reverse the removal order and prune extra edges; no Gaussian noise assumed.

10.1.5 Appendix: GES operators in detail

Insert$(u, v, \mathbf{T})$ adds $u \to v$ ($u \notin \text{Adj}(v)$), orients $t \to v$ for $t \in \mathbf{T}$ ($t \notin \text{Adj}(u)$), and completes to a CPDAG. Requires $\mathbf{T} \subseteq \text{Ne}(v) \setminus \text{Adj}(u)$, $(\text{Ne}(v) \cap \text{Adj}(u)) \cup \mathbf{T}$ a clique, and every semi-directed path from v to u through this set. In the v-structure example, after $A - B$ the move is Insert$(C, B, \{A\})$.

Delete$(u, v, \mathbf{H})$ removes $u - v$ or $u \to v$, orients $h \to v$ and $h \to u$ for $h \in \mathbf{H}$, and completes to a CPDAG. Requires $\mathbf{H} \subseteq \text{Ne}(v) \cap \text{Adj}(u)$ and $(\text{Ne}(v) \cap \text{Adj}(u)) \setminus \mathbf{H}$ a clique. These are Chickering's (2002) operators; XGES reparametrises them (Nazaret et al. 2025).

10.1.6 Appendix: Post-nonlinear causal model (optional)

PNL: for $X_1 \to X_2$, $X_2 = g_2\{f_1(X_1) + E_2\}$, $E_2 \perp\!\!\!\perp X_1$, g_2 invertible, so $E_2 = g_2^{-1}(X_2) - f_1(X_1)$. Find ψ_2, ϕ_1 with $\psi_2(X_2) - \phi_1(X_1)$ independent of X_1; if this works in one direction only, orient the edge. Identifiable generically, with special ambiguous cases (Zhang and Hyvärinen 2009).

10.1.7 Appendix: Scores in detail

Why the left likelihood is larger. For left $A \to B \leftarrow C$, right $B \to A$, $C \to B$, saturated maximized population log-likelihoods are

$$\ell_{\text{left}}/n = -H(A) - H(C) - H(B \mid A, C), \qquad \ell_{\text{right}}/n = -H(C) - H(B \mid C) - H(A \mid B).$$

If the collider is true, $A \perp\!\!\!\perp C$ and $H(A) + H(C) + H(B \mid A, C) = H(A, B, C)$, so

$$\ell_{\text{left}}/n - \ell_{\text{right}}/n = H(A \mid B) + H(B \mid C) - H(A) - H(B \mid A, C) = I(A; C \mid B),$$

positive exactly when $A \not\perp\!\!\!\perp C \mid B$.

Bayesian Dirichlet scores (categorical). BD (Cooper and Herskovits) is the posterior probability of a network; the user must set every α_{ijk}, and equivalent structures may score differently. BDe is score-equivalent via an equivalent sample size α; BDeu sets $\alpha_{ijk} = \alpha/(r_i q_i)$ (uniform over CPT cells, not over graphs). With i variables, j parent configurations (q_i), k values (r_i), Dirichlet conjugate to multinomial,

$$\text{BD}(\mathcal{G}, D; \alpha) = \prod_{i=1}^{N} \text{BD}(X_i \mid Pa_{X_i}; \alpha_i) = \prod_{i=1}^{N} \prod_{j=1}^{q_i} \frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + n_{ij})} \prod_{k=1}^{r_i} \frac{\Gamma(\alpha_{ijk} + n_{ijk})}{\Gamma(\alpha_{ijk})},$$

α_i the equivalent sample size. $\alpha_{ijk} = 1$: K2; $\alpha_{ijk} = 1/2$: Jeffreys; $\alpha_{ijk} = \alpha/(r_i q_i)$: BDeu ($\alpha_i = \alpha$).

BIC (maximization form). Dividing by -2,

$$\text{BIC}(\mathcal{G}, D) = \sum_{i=1}^{N} \Big[\ell_i(\hat{\theta}_i) - \frac{\log n}{2} q_i(r_i - 1)\Big], \qquad \ell_i(\hat{\theta}_i) = \sum_{m=1}^{n} \log \hat{P}_i(x_i^{(m)} \mid \pi_i^{(m)}),$$

score-equivalent; it is the Laplace approximation of log BDeu, agreeing up to $O(1)$.

10.1.8 Appendix: CI tests for causal discovery

χ^2 test (categorical). Tests $P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$ for all z. Example (2008 General Social Survey, main news source by age; credit Yibi Huang):

Age	TV	Internet	Newspapers	Other	Total
18–29	109	92	25	36	262
30–49	272	157	88	63	580
50+	345	59	165	63	632
Total	726	308	278	162	1474

Expected counts under CI are $E(x, y \mid z) = N(x, z)N(y, z)/N(z)$, and

$$\chi^2(Z = z) = \sum_{x,y} \frac{(O(x, y \mid z) - E(x, y \mid z))^2}{E(x, y \mid z)}, \qquad \chi^2_{\text{total}} = \sum_z \chi^2(Z = z),$$

compared with χ^2_{df}, $df = (|X| - 1)(|Y| - 1)|Z|$.

Partial correlation (linear).

$$r_{XY \cdot Z} = \frac{r_{XY} - r_{XZ} r_{YZ}}{\sqrt{1 - r^2_{XZ}}\sqrt{1 - r^2_{YZ}}}$$

for scalar Z (multivariate: recursion or residuals). A drop means Z explains the correlation.

HSIC (non-linear marginal independence). Tests $H_0 : X \perp\!\!\!\perp Y$. With Gram matrices K, L and $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, $\widehat{\text{HSIC}}(X, Y) = \frac{1}{n^2}\operatorname{tr}(KHLH)$; with characteristic kernels, HSIC $= 0$ iff $X \perp\!\!\!\perp Y$. Null by permutation, gamma approximation or asymptotics (Gretton et al. 2005); for conditional independence use KCI.

Kernel conditional independence test (KCI). Tests $H_0 : X \perp\!\!\!\perp Y \mid Z$: with kernels K_X, K_Y, K_Z, regress out Z via $R_Z = \epsilon(K_Z + \epsilon I)^{-1}$, $\tilde{K}_X = R_Z K_X R_Z$, $\tilde{K}_Y = R_Z K_Y R_Z$, and test $T_{\text{KCI}} = \frac{1}{n}\operatorname{tr}(\tilde{K}_X \tilde{K}_Y) = 0$. Replaces Fisher-$z$/$\chi^2$ in PC for non-linear CI, at higher cost and with bandwidth sensitivity (Zhang, Peters, Janzing and Schölkopf 2011).

10.2 Local causal discovery

Motivation. What changes when structure learning reasons over local objects (blankets, neighborhoods, order prefixes)?

10.2.1 The map

Motivation. Organize discovery algorithms by signal and reach.

The global workhorses were PC, FCI (CI tests) and GES, NOTEARS, DAGMA (one global score). Here we go local, along two knobs:

- **Signal**: test independences or optimize a score?
- **Reach**: whole graph or a local object?

	global	local
CI tests	PC, FCI	GS, HITON
CI → score		MMHC
score	GES, DAGMA	BOSS

This lecture fills the local column.

10.2.1.1 One primitive: grow–shrink

- **Grow:** add Y while $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$ (CI) or while the score improves.
- **Shrink:** drop Y if $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$ (CI) or if the score improves.

Algorithm	Local object	Signal
GS	Markov blanket	CI
HITON	one target's parents and children	CI
MMHC	skeleton, then DAG	CI + score
BOSS	order prefix	score

10.2.2 Going local on CI

Motivation. Learn Markov blankets and target neighborhoods by CI tests.

10.2.2.1 The Markov blanket

$$\mathrm{MB}(X) = \big(\mathrm{pa}(X) \cup \mathrm{ch}(X) \cup \mathrm{pa}(\mathrm{ch}(X))\big) \setminus \{X\},$$

parents, children and spouses. Given $\mathrm{MB}(X)$, all other variables are irrelevant for X; it is small yet drives skeleton and orientation (Margaritis and Thrun 1999; Margaritis 2003).

10.2.2.2 GS: grow–shrink the blanket

Grow (permissive): add any Y with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$. **Shrink**: drop Y once $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$. In $X \to C \to G$, grow may add G; once $C \in \mathbf{S}$, $X \perp\!\!\!\perp G \mid C$ removes it. Then find true neighbors and orient colliders as in PC.

Algorithm 4 GS-MB(X)

1: $\mathbf{S} \leftarrow \emptyset$
2: **repeat** find $Y \in \mathbf{V} \setminus (\{X\} \cup \mathbf{S})$ with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \cup \{Y\}$
3: **until** no such Y exists
4: **repeat** find $Y \in \mathbf{S}$ with $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \setminus \{Y\}$
5: **until** no such Y exists
6: **return S**

Correctness sketch. Under faithfulness a parent or child of X is dependent on X given any set, so grow adds it; a spouse becomes dependent once the common child is in $\mathbf{S}$. So $\mathrm{MB}(X) \subseteq \mathbf{S}$ after grow. For $W \in \mathbf{S} \setminus \mathrm{MB}(X)$, $\mathrm{MB}(X) \subseteq \mathbf{S} \setminus \{W\}$ separates X from W, so shrink removes W.

Structure induction. (1) Compute $\mathrm{MB}(X)$ for every X.

(2) $Y \in \mathrm{MB}(X)$ is not a neighbor if $X \perp\!\!\!\perp Y \mid \mathbf{S}$ for some $\mathbf{S} \subseteq \mathbf{R}$, $\mathbf{R}$ the smaller of $\mathrm{MB}(X) \setminus \{Y\}$, $\mathrm{MB}(Y) \setminus \{X\}$.

(3) Orient unshielded colliders by local tests.

(4) Repair cycles and propagate forced directions.

Randomized GS. With large blankets, sample conditioning sets from the smaller blanket and pool tests as evidence: anytime-style, controlled runtime, weaker exactness.

10.2.2.3 HITON: target-focused, not whole-graph

HITON wants only one target T's neighborhood (e.g. feature selection for a disease). HITON-PC($\cdot$) returns a node's *parents and children* (not the PC algorithm) by interleaved grow-while-shrink with small, sample-efficient conditioning sets, run on T and a bounded ring around it. Two pieces: HITON-PC and HITON-MB (Aliferis et al. 2003; 2010).

HITON-MB: two rings, then prune. Example: $P \to T \to C \leftarrow Z$, $C \to G$.

1. Ring 1: PC(T) $\leftarrow$ HITON-PC(T), i.e. P, C.
2. Ring 2: HITON-PC(Y) for each $Y \in$ PC(T); the spouse Z, not adjacent to T, surfaces only here.
3. Prune ring-2 candidates separable from T: G (by C) goes; Z (dependent once C is conditioned on) stays.

Algorithm 5 HITON-MB(T)

PC(T) $\leftarrow$ HITON-PC(T); $\mathbf{C} \leftarrow$ PC(T)
for each $Y \in$ PC(T) **do** $\mathbf{C} \leftarrow \mathbf{C} \cup$ HITON-PC(Y)
for each candidate $X \in \mathbf{C} \setminus$ PC(T) **do**
    if $\exists \mathbf{S} \subseteq \mathbf{C} \setminus \{X\}$ with $T \perp\!\!\!\perp X \mid \mathbf{S}$ **then** remove X from $\mathbf{C}$
return C

10.2.3 Bringing in the score

Motivation. Combine local CI skeletons with scores, and search over orders.

10.2.3.1 MMHC: a hybrid bridge

(Tsamardinos, Brown and Aliferis 2006.) MMPC, the max-min sibling of HITON-PC, finds each node's parents and children; their union is a candidate skeleton, within which a score hill-climb picks a DAG:

$$\text{local PC per node (MMPC)} \;\to\; \text{union} = \text{skeleton} \;\to\; \text{score hill-climb} \;\to\; \text{one DAG}.$$

Sound skeleton restriction, heuristic score-based orientation.

10.2.3.2 BOSS: best order score search

(Andrews et al. 2023.) The state is a permutation π; each v's candidate parents are its *prefix*

$$\text{pre}_\pi(v) = \{w : \pi.\text{index}(w) < \pi.\text{index}(v)\};$$

e.g. for $\pi = (A, C, B, D, E)$, $\text{pre}_\pi(B) = \{A, C\}$.

Scoring an order. The projection $\mathcal{G}_\pi$ picks parents from the prefix by greedy grow/shrink, approximating

$$\text{pa}_\pi(v) \approx \underset{\mathbf{W} \subseteq \text{pre}_\pi(v)}{\arg\max}\ \text{score}_v(\mathbf{W}), \qquad \text{score}_v(\mathbf{W}) = \text{BIC}(X_v \mid X_{\mathbf{W}}),$$

(higher is better), and $\text{score}(\pi) = \sum_v \text{score}_v(\text{pa}_\pi(v))$. By $\text{BIC}(\mathcal{G}, D) = \sum_{v \in \mathbf{V}} \text{BIC}(X_v \mid X_{\text{pa}_\mathcal{G}(v)})$, changing v's parents changes one term; grow–shrink trees (GSTs) cache repeated prefix scores, the key to BOSS's scaling.

Algorithm 6 project($\mathbf{X}, \pi$) and best-move($\mathcal{T}, \pi, v$)

$E \leftarrow \emptyset$
for each v in π **do**
  $\mathbf{Z} \leftarrow \text{pre}_\pi(v)$; $\mathbf{W} \leftarrow \text{grow}(\mathbf{X}, v, \mathbf{Z})$; $\mathbf{W} \leftarrow \text{shrink}(\mathbf{X}, v, \mathbf{W})$
  add $w \to v$ to E for each $w \in \mathbf{W}$
return $\mathcal{G}_\pi = (\mathbf{V}, E)$

$best \leftarrow \mathcal{T}.\text{score}(\pi)$
for $i = 1, \ldots, |\pi|$ **do**
  $\pi' \leftarrow \pi.\text{move}(v, i)$
  if $\mathcal{T}.\text{score}(\pi') > best$ **then** $best \leftarrow \mathcal{T}.\text{score}(\pi')$; $\pi \leftarrow \pi'$
return π

Example 10.4 (Best-move)

Truth $A \to C \leftarrow B$. With $\pi = (A, C, B)$, C is a candidate parent of B, and conditioning on the collider couples A, B: the projection has 3 edges. $\pi' = (A, B, C)$ gives the true collider with 2 edges, the same independences with fewer parameters, so $\text{score}(\mathcal{G}_{\pi'}) > \text{score}(\mathcal{G}_\pi)$ and π' is kept.

Benchmarks. For large, dense models BOSS matches GRaSP's accuracy with far better runtime, compares well with fast GES, PC, DAGMA and LiNGAM, and has run on resting-state fMRI with hundreds to 1000 variables.

10.2.4 Zoom out

Motivation. Choose among the methods and remember what none of them solve.

What to use when.

- Sparse graph, reliable CI tests: GS (full graph) or HITON (one target).
- High-dimensional feature selection: HITON's target blanket.
- Large, dense, score-friendly data: BOSS (or MMHC).
- Hidden confounders: FCI-type methods.
- One effect only: effect-specific local discovery (Entner et al.; LDECC).

Limitations.

- Causal sufficiency: GS, HITON, MMHC, BOSS assume no hidden confounders.
- Faithfulness (or score consistency).
- Finite samples: weak effects and large conditioning sets hurt all.
- Specification and tuning: the local likelihood and penalty set the precision–recall balance.

Locality changes the cost of search, not the assumptions.

Comparison.

	GS	HITON	MMHC	GES	BOSS
Main signal	CI tests	CI tests	CI + score	BIC-like score	BIC-like score
Search object	blankets	target blanket	skeleton + DAG	CPDAGs	orders
Locality	Markov blankets	PC/co-parents	local skeleton	edge operators	prefixes
Output target	MEC graph	variables for T	one DAG	CPDAG	CPDAG
Main risk	CI error	target scope	heuristic orient.	many score calls	model spec.

Chapter 11. Mediation, SWIGs and G-Methods

11.1 Direct and indirect effects

Motivation. How to split a total effect into direct and mediated parts in nonlinear models, where "hold the mediator fixed and subtract" no longer defines an indirect effect.

11.1.1 Preliminaries: potential responses

Motivation. Counterfactual notation, including nested counterfactuals.

In an SCM, $Y(u)$ is the value of Y for unit $U = u$; $Y_x(u)$ its value under $\text{do}(X = x)$, so $P(Y_x = y) = P(y \mid \text{do}(x))$; $Y_{xz}(u)$ its value under $\text{do}(X = x, Z = z)$; $Z_{x^*}(u)$ the mediator value under $\text{do}(X = x^*)$. The nested counterfactual

$$Y_{x,Z_{x^*}}(u): \quad \text{set } X = x, \text{ but set } Z \text{ to the value it would have taken under } X = x^*.$$

11.1.2 Introduction

Motivation. Direct and indirect effects, and the prescriptive/descriptive distinction.

For $X \to Z \to Y$ with $X \to Y$:

- **Total effect:** the change in the distribution of Y under $\text{do}(X = x)$, $P(Y_x = y) = P(y \mid \text{do}(x))$.
- **Direct effect:** measured holding all intermediate variables constant.
- **Indirect effect:** in a linear system, total − direct; in a nonlinear system not measurable by holding variables constant.

Holding the other factors fixed cuts every path from X to Y except $X \to Y$, which defines the direct effect. No choice of variables to hold constant isolates the mediated paths, so indirect effects need another definition.

11.1.2.1 Examples

Birth-control pill and thrombosis. The pill may raise thrombosis directly but lower it by preventing pregnancy. The direct effect is biologically stable and more transportable than the total effect.

Race/sex discrimination in hiring. The legal question is the direct effect: same decision had race or gender differed, all else equal?

Smoking, hypertension and heart disease (Hernán and Robins 2020, Ch. 23). Quitting smoking ($A = 0$) reduces myocardial infarction Y; direct: $A \to Y$, indirect: $A \to M \to Y$ via hypertension M. If a drug removed smoking-induced hypertension, how much benefit would remain?

Drug, aspirin and headache. A drug causes headaches, so patients take aspirin, which may affect the disease or modify the drug's effect. A direct effect

$$P(Y_{xz} = y) - P(Y_{x^*z} = y),$$

with z a specified aspirin level, isolates the drug. In linear systems it equals a path coefficient for every z; in nonlinear systems it depends on z, which must represent the target policy. Hence two conceptualisations.

11.1.2.2 Prescriptive versus descriptive

- **Prescriptive (controlled direct effect):** would an untreated patient improve if treated, with aspirin held at a set level $Z = z$? No individual behavior needed: $P(Y_{xz} = y) - P(Y_{x^*z} = y)$.
- **Descriptive (natural direct effect):** would the patient improve if treated, with aspirin at the level the patient takes untreated? Requires individual natural behavior.

If a patient takes aspirin iff treated and treatment works only with aspirin, the NDE is zero but the CDE (at $z =$ aspirin) is not. The prescriptive form is *pragmatic* (a dose prescribed to all), the descriptive *attributional* (drug or aspirin use?).

Why the average natural direct effect? Would the drug stay effective without the headache side effect? The CDE fixes one aspirin level, but aspirin use varies for other reasons that may modulate effectiveness; the average NDE answers this.

Descriptive indirect effects. Only the descriptive definition transfers: untreated, would the patient recover given the aspirin taken under treatment? E.g. if asking about gender is banned, residual sex preference in hiring is the indirect effect via qualifications, predicted from pre-policy data as an average indirect effect.

11.1.3 Mathematisation

Motivation. Controlled and natural direct effects, and identification of the latter.

X is the control variable, Y the response, Z the set of intermediate variables, $Y_x(u)$ as above.

Definition (Controlled direct effect)

$\text{CDE}_z(x, x^*; Y, u) = Y_{xz}(u) - Y_{x^*z}(u)$, and $\text{CDE}_z(x, x^*; Y) = \mathbb{E}[Y_{xz} - Y_{x^*z}]$, the expectation taken over u.

(Other contrasts, e.g. ratios, can replace the difference.)

Definition (Natural direct effect)

$\text{NDE}(x, x^*; Y, u) = Y_{x,Z_{x^*}(u)}(u) - Y_{x^*}(u)$, and $\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$.

11.1.3.1 The cross-world problem

$Y_{x,Z_{x^*}}$ involves two worlds, $X = x$ (for Y) and $X = x^*$ (for Z_{x^*}), never observed together, even with joint randomization of X and Z. NDE and NIE are unverifiable in principle and need the untestable *cross-world independence* $Y_{xz} \perp\!\!\!\perp Z_{x^*}$ (Hernán and Robins 2020, §23.1; Robins and Richardson 2010).

11.1.3.2 Experimental identification

$\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$ is in general not reducible to terms $P(Y_x = y)$ or $P(Y_{xz} = y)$. (General nested-counterfactual identification came later: Shpitser and Pearl 2008; Correa, Lee and Bareinboim 2021.)

Theorem 11.1 (Pearl 2001)

If there is a set W of covariates, non-descendants of X or Z, with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$, the average NDE is experimentally identifiable:

$$\text{NDE}(x, x^*; Y) = \sum_{w,z} \big[\mathbb{E}(Y_{xz} \mid w) - \mathbb{E}(Y_{x^*z} \mid w)\big] P(Z_{x^*} = z \mid w) P(w).$$

Proof [Supplement].

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid Z_{x^*} = z, w) P(Z_{x^*} = z \mid w) P(w) = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid w) P(Z_{x^*} = z \mid w) P(w),$$
$$\mathbb{E}[Y_{x^*}] = \mathbb{E}[Y_{x^*,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{x^*z} \mid w) P(Z_{x^*} = z \mid w) P(w),$$

using $Y_{xz} = Y_{x,Z_{x^*}}$ on $\{Z_{x^*} = z\}$ and the conditional independence (first line), and composition $Y_{x^*} = Y_{x^*,Z_{x^*}}$ (second). Subtract. ■

Graphical criterion. (a): $X \to Z \to Y$, $X \to Y$, W a common cause of Z and Y. (b): in $\mathcal{G}_{\underline{XZ}}$ (arrows out of X and Z deleted), $(Y \perp_d Z \mid W)_{\mathcal{G}_{\underline{XZ}}}$, which ensures the condition.

11.1.3.3 Markovian models

The condition holds with $W = \emptyset$ when Y_{xz} is independent of all model variables. In Markovian models:

- $P(Y_{xz} = y) = P(y \mid x, z)$, where $X \cup Z$ is the parent set of Y;
- $P(Z_{x^*} = z) = \sum_s P(z \mid x^*, s) P(s)$;
- $P(Y_{x,Z_{x^*}} = y) = \sum_s \sum_z P(y \mid x, z) P(z \mid x^*, s) P(s)$,

with S the parents of Z other than X (or any back-door set). Hence, from non-experimental data,

$$\text{NDE}(x, x^*; Y) = \sum_s \sum_z \big[\mathbb{E}(Y \mid x, z) - \mathbb{E}(Y \mid x^*, z)\big] P(z \mid x^*, s) P(s).$$

11.1.3.4 The mediation formula

For $X \to Z \to Y$, $X \to Y$, with no Z–Y confounding, Y under x with the mediator distribution of x^*:

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]\, P(Z = z \mid X = x^*).$$

Smoking example: MI risk if patients kept smoking ($a = 1$) with the blood pressure of quitters ($a = 0$):

$$\mathbb{E}[Y_{1,M_0}] = \sum_m \mathbb{E}[Y \mid A = 1, M = m]\, P(M = m \mid A = 0).$$

11.1.4 Natural indirect effects

Motivation. The NIE, its relation to the total effect, and its identification.

Definition (Natural indirect effect)

$\text{NIE}(x, x^*; Y, u) = Y_{x^*,Z_x(u)}(u) - Y_{x^*}(u)$, and $\text{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*,Z_x}) - \mathbb{E}(Y_{x^*})$: treatment at the reference level, mediator as under treatment.

Decomposition.

$$\begin{aligned}\mathrm{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}),\\ \mathrm{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x, Z_{x^*}}) - \mathbb{E}(Y_{x^*}),\\ -\mathrm{NIE}(x^*, x; Y) &= -\mathbb{E}(Y_{x, Z_{x^*}}) + \mathbb{E}(Y_x).\end{aligned}$$

Adding the last two lines,

$$\mathrm{TE}(x, x^*; Y) = \mathrm{NDE}(x, x^*; Y) - \mathrm{NIE}(x^*, x; Y) :$$

the direct effect of $x^* \to x$ minus the indirect effect of the reverse transition. In the linear case this is $\mathrm{TE} = \mathrm{NDE}(x, x^*; Y) + \mathrm{NIE}(x, x^*; Y)$ (Appendix 11.1.6).

Theorem 11.2 (Identification of the NIE)

If there is a set W of non-descendants of X or Z with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$ for all x and z, then

$$\mathrm{NIE}(x, x^*; Y) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w)\big[P(Z_x = z \mid w) - P(Z_{x^*} = z \mid w)\big]P(w).$$

In non-experimental studies the NIE is identifiable when $\mathbb{E}(Y_{x^* z} \mid w)$, $P(Z_x = z \mid w)$ and $P(Z_{x^*} = z \mid w)$ are. In a simple Markovian model,

$$\mathrm{NIE}(x, x^*; Y) = \sum_{z} \mathbb{E}(Y \mid x^*, z)\big[P(z \mid x) - P(z \mid x^*)\big].$$

Proof [Supplement]. By composition, $\mathrm{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*, Z_{x^*}})$. Conditioning on W and $Z_x = z$ and using $Y_{x^* z} \perp\!\!\!\perp Z_x \mid W$,

$$\mathbb{E}(Y_{x^*, Z_x}) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid Z_x = z, w)P(Z_x = z \mid w)P(w) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w)P(Z_x = z \mid w)P(w).$$

The same with x^* for x gives the second term; subtract. In a Markovian model with $W = \emptyset$, $\mathbb{E}(Y_{x^* z}) = \mathbb{E}(Y \mid x^*, z)$ and $P(Z_x = z) = P(z \mid x)$. ■

11.1.5 Path-specific effects

Motivation. Effects along any chosen set of paths.

With several mediators ($X \to W \to Z \to Y$, $X \to Z$, $X \to Y$, ...), the effect along selected paths is obtained by feeding each mechanism its inputs under x or under x^*, according to whether the edge is selected.

Definition (Path-specific effect; Pearl 2001)

Let $\mathcal{G}$ be the causal graph of $\mathcal{M}$ and g an edge-subgraph of $\mathcal{G}$ containing the paths selected for analysis. The g-specific effect of x on Y (relative to the reference x^*) is the total effect of x on Y in a modified model $\mathcal{M}_g$ built as follows. Partition each parent set as $Pa_i = \{Pa_i(g), Pa_i(\bar{g})\}$, where $Pa_i(g)$ are the parents linked to X_i in g and $Pa_i(\bar{g})$ the rest. Replace each $f_i(\mathbf{pa}_i, u)$ by

$$f_i(\mathbf{pa}_i, u; g) = f_i(\mathbf{pa}_i(g), \mathbf{pa}_i^*(\bar{g}), u),$$

where $\mathbf{pa}_i^*(\bar{g}) = Pa_i(\bar{g})_{x^*}$ are the values those variables would attain (in $\mathcal{M}$ and u) under $X = x^*$. Then $SE_g(x, x^*; Y, u)_{\mathcal{M}} = TE(x, x^*; Y, u)_{\mathcal{M}_g}$.

$g = \{X \to Y\}$ gives the NDE; g the mediated paths, the indirect effect. This leads to nested counterfactuals like $Y_{W_x, Z_x, Z_{x^*}}$, represented in counterfactual graphs.

Conclusions. Path switching, rather than variable fixing, extends direct and indirect effects to nonlinear models and gives indirect effects an operational meaning. They can be identifiable even when Z is affected by X, and always are in Markovian models.

11.1.6 Appendix: The linear additive relation

Let $Z = \alpha X + U_Z$, $Y = \beta X + \gamma Z + U_Y$, so $Z_x = \alpha x + U_Z$, $Z_{x^*} = \alpha x^* + U_Z$. Then

$$\begin{aligned}\mathrm{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}) = (\beta + \alpha\gamma)(x - x^*),\\ \mathrm{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x, Z_{x^*}}) - \mathbb{E}(Y_{x^*}) = \beta(x - x^*),\\ \mathrm{NIE}(x, x^*; Y) &= \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*}) = \alpha\gamma(x - x^*),\end{aligned}$$

so TE = NDE + NIE; also $\mathrm{NIE}(x^*, x; Y) = -\alpha\gamma(x - x^*)$, consistent with $\mathrm{TE} = \mathrm{NDE}(x, x^*) - \mathrm{NIE}(x^*, x)$. Nonlinear models need not preserve this sum.

11.1.7 Appendix: Counterfactual axioms

Composition. For singleton Y, W and any X: $W_x(u) = w \Rightarrow Y_{xw}(u) = Y_x(u)$ (forcing a variable to its natural value changes nothing).
Consistency. $X(u) = x \Rightarrow Y(u) = Y_x(u)$.
Effectiveness. For all X, W: $X_{xw}(u) = x$.
Reversibility. For Y, W and any X: $(Y_{xw}(u) = y) \wedge (W_{xy}(u) = w) \Rightarrow Y_x(u) = y$.

11.2 Single-world intervention graphs

Motivation. How to put potential outcomes on a causal graph and read counterfactual independences by d-separation, without the extra untestable assumptions of a full structural-equation model.

11.2.1 Potential outcomes and graphs

Motivation. The two languages that SWIGs unify.

11.2.1.1 Potential outcomes

For binary X and Y: $Y(x = 0)$ (or $Y(x_0)$) is the value of Y for a unit assigned $X = 0$; $Y(x = 1)$ (or $Y(x_1)$) likewise. This presumes one version of treatment and no interference (SUTVA). "Potential outcome" and "counterfactual" are synonyms.

Response types for binary Y:

$Y(x_0)$	$Y(x_1)$	Name
0	0	never recover
0	1	helped
1	0	hurt
1	1	always recover

Relating counterfactuals and do-notation. $P(Y(x) = y) \equiv P(Y = y \mid \mathrm{do}(X = x))$, but counterfactuals say more, e.g. $P(Y(x = 0) = y \mid X = 1)$, outcomes of the treated had they not been treated. Under randomization $X \perp\!\!\!\perp Y(x = 0)$ and it equals $P(Y(x = 0) = y)$; observationally it may not.

11.2.1.2 Graphs

A DAG carries the factorization $P(\mathbf{V}) = \prod_{X \in \mathbf{V}} P(X \mid \text{pa}(X))$; e.g. for $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$, $P(A,B,C,D,E) = P(A)P(B \mid A)P(C \mid A)P(D \mid B,C)P(E \mid D)$, and d-separation gives $C \perp\!\!\!\perp B \mid A$, $D \perp\!\!\!\perp A \mid B,C$, $E \perp\!\!\!\perp A,B,C \mid D$.

Edges are direct causal relations, and confounders (e.g. unobserved H with $X \leftarrow H \to Y$) are included (Wright's path diagrams, 1920s). For unconfounded $X \to Y$, $P(x,y) = P(x)P(y \mid x)$ and

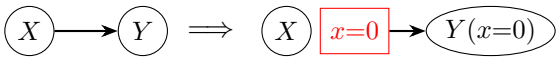
$$P(Y(x) = y) = P(Y = y \mid \text{do}(X = x)) = P(Y = y \mid X = x),$$

$P(y \mid \text{do}(x))$ being the distribution in an experiment fixing $X = x$. No confounding means $X \perp\!\!\!\perp Y(x_0)$ and $X \perp\!\!\!\perp Y(x_1)$; confounding, $X \not\perp\!\!\!\perp Y(x_0)$ or $X \not\perp\!\!\!\perp Y(x_1)$. But $Y(x_0)$, $Y(x_1)$ do not appear on these graphs.

11.2.2 Node splitting

Motivation. Put counterfactual variables on the graph.

Setting X to 0. From $P(X = \tilde{x}, Y = \tilde{y}) = P(X = \tilde{x})P(Y = \tilde{y} \mid X = \tilde{x})$ for $X \to Y$, split X into a random node X and a fixed node $x = 0$; the outcome becomes $Y(x = 0)$:



$X \perp\!\!\!\perp Y(x = 0)$ is now read off, with factorization

$$P(X = \tilde{x}, Y(x = 0) = \tilde{y}) = P(X = \tilde{x})\, P(Y(x = 0) = \tilde{y}), \qquad P(Y(x = 0) = \tilde{y}) = P(Y = \tilde{y} \mid X = 0).$$

The last equation, linking the two factorizations, is the *modularity assumption*.

Marginals represented by SWIGs are identified. $\mathcal{G}(x_0)$ represents $P(X, Y(x_0))$ and $\mathcal{G}(x_1)$ represents $P(X, Y(x_1))$, both identified from $P(X,Y)$ without confounding; $P(X, Y(x_0), Y(x_1))$ is not. $Y(x = 0)$ and $Y(x = 1)$ never share a graph: we assume $X \perp\!\!\!\perp Y(x = 0)$ and $X \perp\!\!\!\perp Y(x = 1)$ but not $X \perp\!\!\!\perp \{Y(x = 0), Y(x = 1)\}$. Hence *single-world* intervention graphs.

Templates. The template $\mathcal{G}(x)$: $X \mid x \to Y(x)$ is a graph-valued function returning a SWIG $\mathcal{G}(x)$ for each x, each a different margin.

Intuition (Robins, VanderWeele and Richardson 2007). Whenever a patient swallows the drug ($X = 1$), instantly give a side-effect-free emetic that expels it before absorption; the recorded outcome is $Y(x = 0)$, observed jointly with the natural choice X.

11.2.3 SWIG construction and properties

Motivation. General SWIG construction and its implications.

Given $\mathcal{G}$ and $\mathbf{A} = \{A_1, \ldots, A_k\}$, form $\mathcal{G}(\mathbf{a})$:

1. **Node splitting:** split each $A \in \mathbf{A}$ into a random node A, inheriting edges *into* A, and a fixed node a, inheriting edges *out of* A.
2. **Relabelling:** each descendant Y of a fixed node becomes $Y(\mathbf{a}_Y)$, indexed by its fixed-node ancestors (e.g. $B(a, \ldots)$, $C(a, \ldots)$, $D(a, \ldots)$); non-descendants keep their names.

Factorization and modularity. $\mathcal{G}$ carries the observed $P(\mathbf{V})$; $\mathcal{G}(\tilde{\mathbf{a}})$ carries $P(\mathbf{V}(\tilde{\mathbf{a}}))$.

- *Factorization:* $P(\mathbf{V}(\tilde{\mathbf{a}}))$ factorizes with respect to $\mathcal{G}(\tilde{\mathbf{a}})$, ignoring fixed nodes.

- *Modularity:* the conditional density of $Y(\tilde{\mathbf{a}}_Y)$ in $\mathcal{G}(\tilde{\mathbf{a}})$ equals that of Y in $\mathcal{G}$ with $\tilde{a}_i$ substituted for each parent $A_i \in \mathbf{A}$ of Y.

Hence $P(\mathbf{V}(\tilde{\mathbf{a}}))$ is identified from $P(\mathbf{V})$.

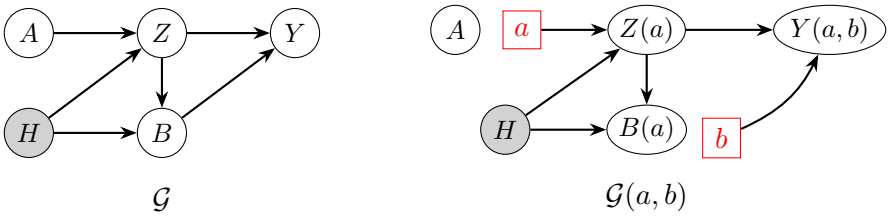
d-separation on SWIGs. If random-node sets $\mathbf{B}(\tilde{\mathbf{a}})$, $\mathbf{C}(\tilde{\mathbf{a}})$ are d-separated by $\mathbf{D}(\tilde{\mathbf{a}})$ and the fixed nodes $\tilde{\mathbf{a}}$ in $\mathcal{G}(\tilde{\mathbf{a}})$, then

$$\mathbf{B}(\tilde{\mathbf{a}}) \perp\!\!\!\perp \mathbf{C}(\tilde{\mathbf{a}}) \mid \mathbf{D}(\tilde{\mathbf{a}}) \quad \text{in } P(\mathbf{V}(\tilde{\mathbf{a}})).$$

11.2.4 A harder inferential problem

Motivation. A SWIG settles a counterfactual independence query.

$\mathcal{G}$: $A \to Z$, $Z \to Y$, $Z \to B$, $B \to Y$, unobserved $H \to Z$, $H \to B$. Does $\mathcal{G}$ imply $Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A$?



Pearl (2009, Ex. 11.3.3) claims the DAG does not imply

$$Y(a,b) \perp\!\!\!\perp B \mid Z, A = a. \tag{3}$$

The SWIG shows (3) holds. In $\mathcal{G}(a,b)$ every path from $B(a)$ to $Y(a,b)$ passes through the non-collider $Z(a)$ ($B(a) \leftarrow Z(a) \to Y(a,b)$, $B(a) \leftarrow H \to Z(a) \to Y(a,b)$), and $B \to Y$ now leaves the fixed node b. Hence

$$Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A \quad \Longrightarrow \quad Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A = a,$$

equivalent to (3) by consistency ($B(a) = B$, $Z(a) = Z$ when $A = a$). (Pearl had used the failure of (3) to reject a claim of Robins.)

11.2.5 Adjustment for confounding

Motivation. The back-door formula in SWIG language.

Graph $L \to X$, $L \to Y$, $X \to Y$; SWIG $\mathcal{G}(x)$: $L \to X$, $L \to Y(x)$, $x \to Y(x)$, so $X \perp\!\!\!\perp Y(x) \mid L$. Then

$$\begin{aligned} P[Y(\tilde{x}) = y] &= \sum_l P[Y(\tilde{x}) = y \mid L = l] P(L = l) \\ &= \sum_l P[Y(\tilde{x}) = y \mid L = l, X = \tilde{x}] P(L = l) && (X \perp\!\!\!\perp Y(\tilde{x}) \mid L) \\ &= \sum_l P[Y = y \mid L = l, X = \tilde{x}] P(L = l) && \text{(consistency)}. \end{aligned}$$

Connection to do-calculus. Factorization and modularity imply all identification results of the do-calculus (Pearl 1995; Spirtes et al. 1993): $P(Y = y \mid \mathrm{do}(A = a))$ is identified iff $P(Y(a) = y)$ is.

11.2.6 Conclusion

Eliminating a false trichotomy. Before SWIGs, the options were graphs without counterfactuals (Dawid), counterfactuals without graphs, or Pearl's NPSEMs with many assumptions untestable even in principle. SWIGs combine both without them.

Summary and extensions. Node splitting links the factorizations of the DAG and the SWIG, whose counterfactual distribution is identified from the observed one. (Not covered: individual- vs. population-level absence of direct effects; partially well-defined interventions.)

References: Pearl (2009); Richardson and Robins (2013), CSSS Technical Report 128; Robins (1986); Robins, VanderWeele and Richardson (2007); Shpitser and Pearl (2008); Spirtes, Glymour and Scheines (1993).

11.3 G-methods for time-varying treatments

Motivation. How to estimate the effect of a treatment sequence when covariates are both affected by earlier treatment and drive later treatment, where regression, stratification and matching fail.

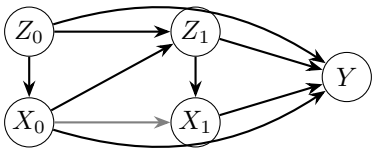
11.3.1 Time-varying treatments

Motivation. Sequential decisions, and why feedback breaks standard adjustment.

Recall the single time point. Back-door adjustment $\mathbb{E}[Y(x)] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]P(Z = z)$; IPW $\mathbb{E}[Y(x)] = \mathbb{E}[\mathbb{1}(X = x)Y/P(X = x \mid Z)]$.

With repeated decisions (medication at each visit, staged policies, adaptive algorithms), the process is history-dependent: action $\to$ state $\to$ action $\to \cdots \to$ outcome.

Causal diagrams. Treatments X_k, history $\bar{X}_k = (X_0, \ldots, X_k)$; covariates Z_k, history $\bar{Z}_k = (Z_0, \ldots, Z_k)$; final outcome Y. Two periods: $Z_0 \to X_0$, $Z_0 \to Z_1$, $X_0 \to Z_1$, $Z_1 \to X_1$, $X_0 \to X_1$, all $\to Y$.



Treatment–confounder feedback: (1) X_{k-1} affects confounder Z_k and (2) Z_k affects X_k. Example (HIV): ART $(X) \to$ CD4 count or viral load $(Z) \to$ next ART decision $\to \cdots$, possibly also $X_{k-1} \to X_k$.

Treatment strategies. *Static*: the same for everyone, e.g. "always treat" $\bar{x} = (1, \ldots, 1)$, "never treat" $\bar{x} = (0, \ldots, 0)$. *Dynamic*: $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$, e.g. "start ART when CD4 drops below 500". Estimands: $\mathbb{E}[Y(\bar{x})] - \mathbb{E}[Y(\bar{x}')]$.

11.3.1.1 Identification assumptions

In a **sequentially randomized experiment** X_k is randomized with probability depending only on $(\bar{X}_{k-1}, \bar{Z}_k)$; observationally, no unmeasured common causes of X_k and Y given $(\bar{X}_{k-1}, \bar{Z}_k)$. An unmeasured U may affect Y and Z_1 if it has no arrow into X_0 or X_1.

- **Sequential exchangeability:** $Y(\bar{x}) \perp\!\!\!\perp X_k \mid \bar{X}_{k-1}, \bar{Z}_k$, $k = 0, \ldots, K$.

- **Consistency:** $Y(\bar{x}) = Y$ when $\bar{X} = \bar{x}$.
- **Positivity:** $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k) > 0$.

11.3.1.2 The fundamental dilemma

Adjusting for Z_1 removes confounding of $X_1 \to Y$ but blocks $X_0 \to Z_1 \to Y$ (and, with $U \to Z_1$, $U \to Y$, opens the collider path $X_0 \to Z_1 \leftarrow U \to Y$). Not adjusting leaves X_1 confounded. Regression, stratification and matching cannot do both.

G-methods (Robins) are consistent under time-varying confounding:

(1) the g-formula (outcome modeling);

(2) IP-weighted marginal structural models (treatment modeling);

(3) g-estimation of structural nested models (effect modeling). All share the same assumptions and handle Z_1 without blocking the path through it.

What you already know. The truncated factorization (Chapter 2) for (Z_0, X_0, Z_1, X_1, Y): from

$$P(z_0) \cdot P(x_0 \mid z_0) \cdot P(z_1 \mid x_0, z_0) \cdot P(x_1 \mid z_1, x_0, z_0) \cdot P(y \mid x_1, z_1, x_0, z_0)$$

drop the treatment factors:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_0, z_1} \mathbb{E}[Y \mid x_0, z_0, x_1, z_1]\, P(z_1 \mid x_0, z_0)\, P(z_0).$$

This is the *g-formula*; sequential exchangeability makes each factor estimable. New here: why feedback breaks standard estimation, and weight products, MSMs, SNMs.

11.3.2 The g-formula

Motivation. Standardize over the time-varying confounder instead of conditioning on it.

Idea (two time points, no baseline confounder). Standardize over Z_1 *given past treatment*, not its marginal:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_1} \mathbb{E}[Y \mid X_0 = x_0, Z_1 = z_1, X_1 = x_1]\, P(Z_1 = z_1 \mid X_0 = x_0).$$

Derivation. Temporally,

$$P(y, x_1, z_1, x_0) = P(y \mid x_1, z_1, x_0) \underbrace{P(x_1 \mid z_1, x_0)}_{\text{remove}} P(z_1 \mid x_0) \underbrace{P(x_0)}_{\text{remove}};$$

under $\mathrm{do}(X_0 = x_0, X_1 = x_1)$ the assignment terms drop, leaving outcome $\times$ confounder model. For $K + 1$ time points,

$$\mathbb{E}[Y(\bar{x})] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{X} = \bar{x}, \bar{Z} = \bar{z}] \prod_{k=0}^{K} P(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1}),$$

with $\bar{x}_{-1}, \bar{z}_{-1}$ empty, $P(z_0 \mid \bar{x}_{-1}, \bar{z}_{-1}) = P(z_0)$.

11.3.2.1 Iterated conditional expectations (ICE)

The sum over $\bar{z}$ needs densities $P(z_k \mid \text{past})$, hard for high-dimensional Z_k. ICE replaces it by backward nested regressions, the data averaging out one confounder per step:

$$Y \xrightarrow{\text{regress}} \hat{Q}_K \to \cdots \to \hat{Q}_0 \xrightarrow{\text{average}} \hat{\mathbb{E}}[Y(\bar{x})].$$

Two time points. (1) $\hat{Q}_1(z_1) = \hat{\mathbb{E}}[Y \mid x_0, z_1, x_1]$;

(2) $\hat{\mathbb{E}}[Y(x_0, x_1)] = \hat{\mathbb{E}}[\hat{Q}_1(Z_1) \mid X_0 = x_0]$, averaging over the empirical Z_1 in the $X_0 = x_0$ group.

Three time points as variable elimination. Eliminate Z_2, then Z_1:

$$Q_2(z_1, z_2) = \mathbb{E}[Y \mid x_0, z_1, x_1, z_2, x_2], \qquad Q_1(z_1) = \mathbb{E}[Q_2(Z_1, Z_2) \mid x_0, z_1, x_1],$$
$$\mathbb{E}[Y(x_0, x_1, x_2)] = \mathbb{E}[Q_1(Z_1) \mid X_0 = x_0].$$

For Q_1, regress the pseudo-outcome $\tilde{Y}_i^{(2)} = \hat{Q}_2(Z_{1i}, Z_{2i})$ (treatments set to (x_0, x_1, x_2)) on Z_1 among rows with $X_0 = x_0$, $X_1 = x_1$.

General recursion. With $\hat{Q}_{K+1} = Y$, for $k = K, \ldots, 0$:

$$\hat{Q}_k = \hat{\mathbb{E}}[\hat{Q}_{k+1} \mid \bar{X}_k = \bar{x}_k, \bar{Z}_k], \qquad \hat{\mathbb{E}}[Y(\bar{x})] = \hat{\mathbb{E}}[\hat{Q}_0(x_0, Z_0)].$$

ICE never models $P(z_k \mid \text{past})$, so it scales to many time points and high-dimensional Z_k.

11.3.3 IP weighting and marginal structural models

Motivation. Re-weight to a pseudo-population with unconfounded treatment.

Idea. Weight each individual by the inverse probability of their observed treatment given confounders, which "breaks" $Z_k \to X_k$ (U never points into treatment).

One time point: why $1/P(X \mid Z)$. Under exchangeability, consistency and positivity,

$$\mathbb{E}\Big[\frac{\mathbb{1}(X = x)Y}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[\mathbb{1}(X = x)Y \mid Z]}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z]P(x \mid Z)}{P(x \mid Z)}\Big]$$
$$= \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = \mathbb{E}_Z\big[\mathbb{E}[Y(x) \mid Z]\big] = \mathbb{E}[Y(x)].$$

Up-weighting rare (X, Z) combinations gives a pseudo-population with $X \perp\!\!\!\perp Z$.

Two time points (no baseline confounder), regime (x_0, x_1):

$$W^X(x_0, x_1) = \frac{\mathbb{1}(X_0 = x_0)}{P(x_0)} \cdot \frac{\mathbb{1}(X_1 = x_1)}{P(x_1 \mid x_0, Z_1)},$$

$$\mathbb{E}[W^X(x_0, x_1)Y] = \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] \frac{P(x_0, z_1, x_1)}{P(x_0)P(x_1 \mid x_0, z_1)}$$
$$= \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] P(z_1 \mid x_0) = \mathbb{E}[Y(x_0, x_1)].$$

The factors undo the assignment of X_0 and of X_1 given (X_0, Z_1). With baseline Z_0, use $P(X_0 = x_0 \mid Z_0)$.

The weight product. Divide by $P(X_k \mid \text{past})$ at every k:

$$W^X = \prod_{k=0}^{K} \frac{1}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)} \qquad \text{(non-stabilized)}.$$

Stabilized weights. For binary treatments

$$\mathbb{E}[W^X] = \sum_{\bar{x}, \bar{z}} \frac{P(\bar{x}, \bar{z})}{\prod_k P(x_k \mid \text{past})} = \sum_{\bar{x}, \bar{z}} \prod_k P(z_k \mid \text{past}) = \sum_{\bar{x}} 1 = 2^{K+1},$$

so the pseudo-population is inflated and rare cells give extreme weights. Use the numerator $P(X_k \mid \bar{X}_{k-1})$:

$$SW^X = \prod_{k=0}^{K} \frac{P(X_k \mid \bar{X}_{k-1})}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

It involves no Z, so validity is kept, and $\mathbb{E}[SW^X] = 1$, with much smaller variance.

Proof (one time point). With $SW = P(X)/P(X \mid Z)$,

$$\mathbb{E}\Big[\frac{P(x)}{P(x \mid Z)}\mathbb{1}(X = x)Y\Big] = P(x)\,\mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z]P(x \mid Z)}{P(x \mid Z)}\Big]$$
$$= P(x)\,\mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = P(x)\,\mathbb{E}[Y(x)].$$

With $Y \equiv 1$, $\mathbb{E}[SW\mathbb{1}(X = x)] = P(x)$, so $\mathbb{E}[SW] = \sum_x P(x) = 1$ and

$$\mathbb{E}[SW\mathbb{1}(X = x)Y]/\mathbb{E}[SW\mathbb{1}(X = x)] = \mathbb{E}[Y(x)].$$

The factor $P(x)$ rescales the $X = x$ arm to its marginal size and cancels in the ratio (and in an MSM).

Proof (general). Let $r_k(x_k \mid \bar{x}_{k-1}) \equiv P(X_k = x_k \mid \bar{X}_{k-1} = \bar{x}_{k-1})$. The observed law is $p(\bar{z}, \bar{x}, y) = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})p(x_k \mid \bar{x}_{k-1}, \bar{z}_k) \cdot p(y \mid \bar{x}, \bar{z})$. For $SW^X = \prod_k r_k(X_k \mid \bar{X}_{k-1})/p(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$,

$$p_{SW}(\bar{z}, \bar{x}, y) = p(\bar{z}, \bar{x}, y)\, SW^X = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r_k(x_k \mid \bar{x}_{k-1}) \cdot p(y \mid \bar{x}, \bar{z}),$$

a proper density (so $\mathbb{E}[SW^X] = 1$) without $Z_k \to X_k$ arrows. For static $\bar{x}$ let $r(\bar{x}) = \prod_k r_k(x_k \mid \bar{x}_{k-1})$. By the g-formula

$$\mathbb{E}\big[SW^X\mathbb{1}(\bar{X} = \bar{x})Y\big] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{x}, \bar{z}] \prod_k p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r(\bar{x}) = r(\bar{x})\,\mathbb{E}[Y(\bar{x})],$$

and $\mathbb{E}[SW^X\mathbb{1}(\bar{X} = \bar{x})] = r(\bar{x})$, so $\mathbb{E}[SW^X\mathbb{1}(\bar{X} = \bar{x})Y]/\mathbb{E}[SW^X\mathbb{1}(\bar{X} = \bar{x})] = \mathbb{E}[Y(\bar{x})]$.

11.3.3.1 Marginal structural models

An MSM models the counterfactual mean as a function of strategy, e.g.

$$\mathbb{E}[Y(x_0, x_1)] = \beta_0 + \beta_1 x_0 + \beta_2 x_1 + \beta_3 x_0 x_1,$$

fitted by weighted regression in the IP-weighted pseudo-population. Variants: saturated (one parameter per strategy); dose–response $\mathbb{E}[Y(\bar{x})] = \beta_0 + \beta_1 \mathrm{cum}(\bar{x})$, $\mathrm{cum}(\bar{x}) = \sum_k x_k$; conditional MSMs with baseline covariates.

11.3.4 G-estimation of structural nested models

Motivation. Model and remove the history-specific effect of each decision.

The blip. An SNMM models the history-specific effect of each decision, the *blip*, with the future at reference (0):

$$\gamma_1(x_0, z_1) = \mathbb{E}[Y(x_0, 1) - Y(x_0, 0) \mid X_0 = x_0, Z_1 = z_1] \text{ (time-1 blip, given history)},$$

$$\gamma_0(x_0) = \mathbb{E}[Y(x_0, 0) - Y(0, 0)] \text{ (time-0 blip)}.$$

From 000 to 111, $\mathbb{E}[Y(\bar{x})]$ climbs $000 \to 100 \to 110 \to 111$ by $+\gamma_0, +\gamma_1, +\gamma_2$; the MSM models the whole climb, the SNMM the steps.

G-estimation: find the blip by removing it. $Y_{\text{removed}} = Y - \psi_1 X_1$ rebuilds the outcome without the late treatment; choose ψ_1 so that Y_{removed} is balanced across X_1 given history (an estimating equation). IPW models treatment; SNMM models the effect; both recover $\mathbb{E}[Y(\bar{x})]$.

In practice: one time point. With blip $\gamma(z;\psi) = \psi x$, $H(\psi) = Y - \psi X$. At the true ψ, $H(\psi) = Y(0) \perp\!\!\!\perp X \mid Z$, so $H(\psi)$ is uncorrelated with X given Z. With $\hat{e}(Z) = \hat{P}(X = 1 \mid Z)$,

$$U(\psi) = \sum_i (X_i - \hat{e}(Z_i)) \underbrace{(Y_i - \psi X_i)}_{H_i(\psi)} = 0 \quad \Longrightarrow \quad \hat{\psi} = \frac{\sum_i (X_i - \hat{e}_i) Y_i}{\sum_i (X_i - \hat{e}_i) X_i},$$

an IV-type ratio with instrument $X - \hat{e}$. Recipe:

(1) fit $\hat{e}(Z)$ (e.g. logistic);

(2) form residuals $X_i - \hat{e}_i$;

(3) plug in. Consistent if $\hat{e}$ is correct; an outcome term $\mathbb{E}[H(\psi) \mid Z]$ makes it doubly robust. With $\gamma(z;\psi) = (\psi_0 + \psi_1 z)x$, the instrument is $(1, z)(X - \hat{e})$.

Two time points. $X_0 \to Z_1 \to X_1 \to Y$, blips $\gamma_1 = \psi_1 x_1$, $\gamma_0 = \psi_0 x_0$; latest blip first.

1. $H_1(\psi_1) = Y - \psi_1 X_1$ is the candidate $Y(x_0, 0)$, with $Y(x_0, 0) \perp\!\!\!\perp X_1 \mid (X_0, Z_1)$. With $\hat{e}_1 = \hat{P}(X_1 = 1 \mid X_0, Z_1)$, $\sum_i (X_{1i} - \hat{e}_{1i}) H_{1i}(\psi_1) = 0$ gives
$$\hat{\psi}_1 = \frac{\sum_i (X_{1i} - \hat{e}_{1i}) Y_i}{\sum_i (X_{1i} - \hat{e}_{1i}) X_{1i}}.$$
2. $H_0 = H_1(\hat{\psi}_1) - \psi_0 X_0 = Y - \hat{\psi}_1 X_1 - \psi_0 X_0$ is the candidate $Y(0, 0)$. With $\hat{e}_0 = \hat{P}(X_0 = 1 \mid Z_0)$,
$$\hat{\psi}_0 = \frac{\sum_i (X_{0i} - \hat{e}_{0i}) H_{1i}(\hat{\psi}_1)}{\sum_i (X_{0i} - \hat{e}_{0i}) X_{0i}}.$$

Telescoping, $\mathbb{E}[Y(1,1)] - \mathbb{E}[Y(0,0)] = \hat{\psi}_0 + \hat{\psi}_1$; outcome terms $\mathbb{E}[H_k \mid \text{history}]$ make each step doubly robust.

11.3.5 Doubly robust estimation

Motivation. Combine outcome and treatment models for two chances at consistency.

The g-formula needs a correct $\mathbb{E}[Y \mid \bar{X}, \bar{Z}]$, IPW correct $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$; AIPW is consistent if either is correct.

Time-fixed review. With $\hat{m}(x, z) = \hat{\mathbb{E}}[Y \mid X = x, Z = z]$ and $\hat{\pi}(z) = \hat{P}(X = 1 \mid Z = z)$,

$$\hat{\tau}_{DR} = \frac{1}{n} \sum_{i=1}^{n} \left[\hat{m}(1, Z_i) - \hat{m}(0, Z_i) + \frac{X_i \{Y_i - \hat{m}(1, Z_i)\}}{\hat{\pi}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{m}(0, Z_i)\}}{1 - \hat{\pi}(Z_i)} \right].$$

If $\hat{m}$ is correct the residual terms have mean zero; if $\hat{\pi}$ is correct they remove the bias of $\hat{m}$.

Time-varying extension. Fit $\hat{\pi}_k$ at each time and outcome models backward (ICE-style); consistent if at each time either model is correct (Bang and Robins 2005). Modern variants: TMLE, sequential regression with cross-fitting.

11.3.6 Practical considerations

Motivation. Dynamic strategies and positivity.

Dynamic strategies. For $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$, $g_k(\cdot) \in \{0, 1\}$. G-formula: simulate forward, drawing Z_k and setting $X_k = g_k(\cdot)$. IPW: weight those who followed g,

$$W^g = \prod_{k=0}^{K} \frac{\mathbb{1}(X_k = g_k(\bar{X}_{k-1}, Z_k))}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

G-estimation: SNMMs handle dynamic regimes via effect modification by Z_k.

Positivity violations. K binary times give 2^K histories, so cells are sparse, weights extreme, estimates unstable. Remedies: stabilized or truncated weights, parametric models.

11.3.7 Comparison of g-methods

Motivation. What each method models, targets and risks.

	G-formula	IPW / MSM	G-estimation / SNM
Key model	outcome + confounder dist.	treatment assignment	treatment + blip function
Target	$\mathbb{E}[Y(\bar{x})]$ (marginal)	$\mathbb{E}[Y(\bar{x})]$ (marginal)	conditional blip effects
Effect modification by Z_k	outcome-model interactions	not directly	directly modeled
Extreme weights?	no	yes	no
G-null paradox?	yes (parametric)	no	no
DR version?	yes	yes (AIPW)	yes

All are consistent under correct specification and the same conditions. (G-null paradox: parametric outcome and confounder models cannot all be correct under the null, so the parametric g-formula may reject a true null.)

Estimand, model, estimator.

Method	Estimand (what)	Model (what must be right)	Estimator (how)
g-formula	$\mathbb{E}[Y(\bar{x})]$	outcome (+ confounders)	standardize / ICE
IPW	$\mathbb{E}[Y(\bar{x})]$	treatment	weighting
MSM	$\mathbb{E}[Y(\bar{x})]$, marginal	*marginal form* + treatment	weighted regression
SNMM / g-est.	blip γ_k (history-specific)	*blip form* + treatment	g-estimation (balance)
AIPW / DR	$\mathbb{E}[Y(\bar{x})]$	outcome or treatment	augmented IPW

Structural models (italic) restrict the estimand; *nuisance* models describe observed data. MSMs are fitted by IPW, SNMMs by g-estimation; only the SNMM targets the history-specific blip, enabling effect modification and optimal dynamic regimes.

The big picture. The g-formula standardizes over Z, IPW/MSM removes $Z \to X$ arrows, g-estimation subtracts blips backward, AIPW combines both models.

Key takeaways.

1. Time-varying treatments need multiple time points and treatment strategies.
2. With confounders affected by prior treatment, regression, stratification and matching fail.
3. G-methods work under sequential exchangeability, consistency and positivity.
4. Doubly robust methods protect against misspecification.
5. Choose by problem structure, available models and positivity.